

Game Design *Therapoetics*:
Autopathographical Game Authorship as Self-Care, Self-
Understanding, and Therapy

by

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Faculty of Information
University of Toronto

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Abstract

An autopathographical game is an autobiographical narrative of illness and disability applied to the creative process of computer game authorship. Autopathographical game design (AGD) during an Autopathographical Game Jam (AGJ) is a healing process for game designers—a *therapeutic* platform for self-care, self-understanding, and therapy. This data-driven theorization examines how thirteen game designers with mental illness, emotional trauma, and disability author autopathographical games during an Autopathographical Game Jam—specifically how they render their autopathographical narratives through the distinct formal properties of computer game design, and how this process feels therapeutic. The first-person experiences informing their design processes include bipolar disorder, anxiety, depression, post-traumatic stress disorder (PTSD), attention/deficit hyperactivity disorder (ADHD), colour-blindness, grief, shyness, childhood memories of being bullied, and insomnia.

This study's findings demonstrate that autopathographical game design engenders four dimensions of poiesis (creative making) that co-constitute AGD-AGJ as a therapeutic process, and when integrated amplify the therapeutic effects on the game designer: *sociopoiesis* (collective making during the AGJ), *autopoiesis* (self-making through a self-reflexive and

introspective praxis), *fabulopoiesis* (re-imagining the self through metaphor and analogy), and *logopoiesis* (reevaluating the self through the articulated ‘calculated making’ of implementation and programming). The integrated therapeutic effects of these four dimensions seem to arise when subjects inhabit three epistemological perspectives during the AGD-AGJ process: the first-person subjective perspective (autopoiesis), the second-person relational perspective (sociopoiesis), and the third-person objective perspective (fabulopoiesis and logopoiesis), which together cater to well-being.

This study’s qualitative methodology is situated in a framework of grounded theory, case studies, discourse analysis, and research-creation. Data collection methods are semi-structured interviews with subjects, video recordings of the two-day Autopathographical Game Jam, and collection of subjects’ processual artifacts, e.g., sketches and doodles. Discourse analysis of interview testimony and processual artifacts is used to theorize on the phenomenological ontology that defines the AGD-AGJ experience for subjects.

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“If you can talk about it, why paint it?”

Francis Bacon (artist)

Chapter 1

Introduction

1.1 Section I: Warrant and analytics

1.1.1 Warrant for this study

Starting from 2012, there began to emerge in the North American¹ independent games scene a body of autobiographical and deeply personal digital games about mental illness, emotional trauma, and disability. This phenomenon electrified contemporary games culture. Some of the most publicized games of this genre include *dys4ia* (2012), Anna Anthropy's autobiographical account of undergoing hormone replacement therapy and living as a transgender woman; *Depression Quest* (2013), Zoe Quinn's text-based interactive narrative about living with depression; *Medication Meditation* (2013), Kara Stone's game examining self-care practices for mental health; and Ryan and Amy Green and Josh Larson's *That Dragon, Cancer* (2016), exploring the Greens' grief over their young child's death from cancer. The significance of this phenomenon in games culture is due largely to the critical discussion it generated among game developers, gamers, and game critics who questioned the legitimacy of these games given that they challenged the fundamental definition of games as playful entertainment. While popular culture commonly designates mental illness, emotional trauma, and disability as serious existential subjects that denote pain and suffering, games and play are framed as fun, escapist, and frivolous activities. The emergent illness game genre disrupts preconceptions about the ontology of games in unexpected and unorthodox ways. For example, *Depression Quest* is described on the author website as follows: "this game is not meant to be a fun or a lighthearted experience" (*Depression Quest*, 2013). Anthropy also claims that some gamers and critics questioned whether her game *dys4ia* was a game at all, despite their admiration for it (Anthropy and Clark, 2014). The highly distinctive character of the illness game phenomenon inspired and informed the initiation of this study.

The illness, trauma, and disability game has been named a *game autopathography*. In doing so this study is inspired by two literary scholars who have developed special terminology

¹ North American in this context refers to United States and Canada.

to refer to the autobiographical illness or disability narrative: medical humanities scholars Anne Hunsaker Hawkins (1999), who originally coined the term “pathography”, and G. Thomas Couser (1997), who first used the term “autopathography”. Against the background of the cultural phenomenon described above, this study examines the creative process of thirteen game designers-developers² as they design autopathographical games during a weekend-long ‘game jam’—The Autopathographical Game Jam— organized by the researcher for the purpose of conducting empirical research for this study. A ‘game jam’ is a social event mobilizing participants to design and/or prototype games over a condensed period of time, anywhere between 24 and 48 hours.

1.1.2 The problematic

Why is this a worthwhile and compelling subject of research? First of all, there are important similarities and differences between the two forms of autopathography, the autopathographical narrative (AN) and the computer game autopathography (CGA). The similarities are that both the autopathographical narrative and the computer game autopathography are rendered into a shareable form, either as a first-person or second-person form, or both. Both also draw on autobiographical dimensions of first-person, lived experiences with forms of mental or physical illness, emotional trauma, and/or disability. Regarding differences, illness and disability are perceived as serious and rather unpleasant topics in the popular imagination, while games are perceived as fun and frivolous. The computer game autopathography in this particular study was meant to be shared primarily with Autopathographical Game Jam (AGJ) participants. This is not typically the case with the

² The terms ‘game designer’ and ‘game developer’ have varied meanings, some of which overlap. One typical distinction differentiates between design and development based on the processes of conceptualization in the former and implementation in the latter. According to this distinction, computer game designers conceptualize the game’s virtual world, the set of game player characters (GPCs) and non-player characters (NPCs), moveable and non-moveable objects, the interaction design, and game architecture—the game rules, player actions, and game behaviors that constitute the computer game as a system of play. Computer game developers engineer and *implement* this architecture through a programming language and may or may not be involved in the design process. This study’s thirteen subjects individually authored their own game prototypes from ‘scratch’ except for two subjects who collaborated on one game. As such, they are involved in both game design and development qua implementation according to the aforementioned distinction. Given this configuration, this study uses the term ‘game designer’ to denote the activities of both computer game design *and* implementation, as well as digital graphic design, animation, modelling, and any other sub-praxes of game design and development.

autopathographical narrative or literary autopathography, which is usually meant to be shared with the general public and is often published for that purpose. Indeed, some of the autopathographical games authored during the Autopathographical Game Jam are in and of themselves rendered in third-person form, i.e., devoid of any first-person references. This is not the case with the autopathographical narrative, which is usually rendered via the “I” pronoun in first-person narration. The linguistic mode of communication deployed in a written narrative using a first-person pronoun is differentiated from the visual and/or performative mode used in a game. For example, during gameplay, the first-person perspective translates into the player’s field of camera vision and movement; hence it substitutes for the “I” pronoun in written narrative. The issue of the use and rendering of the first-person, second-person, and third-person perspectives in computer games as compared with other media and art forms is essentially one of translation. This has implications for this study’s thesis and will be discussed throughout this dissertation.

Secondly, the phenomenon of autopathographical game designers and their authorship base has not been the object of any sustained study, mainly because of its recent appearance on the games scene. Thus, games researchers have privileged either the player in user-experience studies or the game text in humanities-based games criticism at the expense of designer experiences and authorship processes. Media studies have also to some extent over-emphasized reception while de-centering the author and artist, influenced in no small part by Barthes’s *The Death of the Author* (1967) and the espousal of the ‘death of the subject’ by thinkers in poststructuralism and postmodernism³. There are some exceptions in other fields such as philosophy of art and computer science, with theorists and researchers calling for an emphasis on authorship. Art philosopher Susanne K. Langer (1942 & 1953), whose ideas on artistic expression inform and inspire this study, embraces the artist’s standpoint as a foundation for any theorization of the creative arts:

In the first place, philosophy of art should, I believe, begin in the studio, not the gallery, auditorium, or library. Just as the philosophy of science required for its proper

³ Poststructuralist and postmodernist thinkers such as Michel Foucault and Jacques Derrida have problematized subjectivity in their critiques of modernism.

development the standpoint of the scientist...so the philosophy of art requires the standpoint of the artist to test the power of its concepts and prevent empty or naive generalizations (p. ix).

Computer scientists also understand the deep value of studying authorship processes. Scott Turner, for example, argues that in order “to build a computer program to tell stories, we must understand and model the processes an author uses to achieve his goals” (as cited in Wardrip-Fruin, 2009, p. 219).

Given the dearth of humanistic scholarly research on the authorship process, this study aims to unify our understanding of the two forms, the autopathographical narrative and the computer game, by setting out to investigate the creative design process involved in AGD and its function and meaning for game designers. As such, this dissertation charts new territory in media studies and computer game studies that has implications for two main audiences: computer game and media researchers, and health researchers. Finally, the game autopathography and/or autopathographical game design are to be differentiated in this study from the field of *serious games* and gamified health software applications, the last two of which address applications of games in clinical practice, education, and academic research. This is not a clinical, medical, or psychological study; rather, this study mobilizes a humanistic approach to understand the autopathographical game design process as a compelling form of *therapoiesis*⁴ affecting those that engage with it. As gamified applications, serious games can be highly instrumental and didactic in their objectives. But many of this study’s subjects’ conceptualizations of their games demonstrate that autopathographical game design exists outside of these institutional contexts and prescriptive norms. These norms should therefore not be applied to it, given the potential for a heavy-handed, didactic approach that may be employed by *serious games* designers in their quest to gamify for specific instrumental and institutional aims⁵. Therefore, this study’s focus is

⁴ The term *therapoetics* in this dissertation is preceded by Classics scholar Julia Nelson Hawkins (2017) in her recently published book: “Therapoetics After Actium: Narrative, Medicine, and Authority in Augustan Epic”.

⁵ In his seminal work “Serious Games”, Clark C. Abt (1970; 1987) states that “games may be significant without being solemn, interesting without being hilarious, earnest and purposeful without being humorless, and difficult without being frustrating” (p. 10). Many of the autopathographical games authored during this study, as well as autopathographical games on the independent games scene, oppose Abt’s characterization of *serious games*. Some

on the authorship of the game autopathography as a unique poetic form doubling as a therapeutic tool for its designers. This study is not making claims applicable to the *existing* field of serious games per se; perhaps it is making claims that could best serve a re-conceptualization of this field and to the exciting future possibilities of serious games—an area worthy of analysis but beyond the scope of this dissertation.

1.1.3 The analytical objective

The investigative strategy of this study will be to address the following analytical questions:

1. How do study participants integrate two *semiospheres* or realms of meaning-making, namely the ‘autopathographical narrative’ and the ‘computer game’ via autopathographical game design (AGD)?
2. What happens when designers creatively experiment with or transgress the formal properties of these two semiospheres? How does the integration of these two semiospheres affect their properties?
3. One of the findings of the study is that subjects find the integrative process of these two semiospheres therapeutic beyond what may be predicted by the two semiospheres on their own. This finding has thus generated a third analytical question, namely: how is the integration of the two semiospheres therapeutic? I.e., how does this integration impart self-understanding and self-insight; and how does it feel restorative and healing for study subjects?

The central task of this study will be to answer these analytical questions, and at the same time suggest a generalized account based on the specific case of this study’s autopathographical game design (AGD) process conducted during an Autopathographical Game Jam (AGJ) to the general phenomenon of autopathographical game design, autopathographical game designers, and autopathographical games.

autopathographical games are in fact, frustrating, humourless, and hilarious, to be demonstrated in the subsequent chapters.

1.1.4 The Autopathographical Game Jam (AGJ) as a process

During the summer of 2014, study subjects were recruited from the Greater Toronto Area. These were adult-age game designers with mental illness, disability, emotional trauma, or other health-related issues. The subjects were invited to render the autopathographical narrative into a digital game during an ethics-approved, researcher-organized, inclusive game jam subsequently named *The Autopathographical Game Jam*. The two-day, seven-hour per day event that included thirteen game designers took place at University of Toronto's Semaphore Lab (Demo Room) during the weekend of September 20-21, 2014. The AGJ was modelled on the game jams organized by *Dames Making Games* (DMG), a Toronto-based feminist collective whose commitment to "diversity, accessibility, and inclusiveness" (Dames Making Games, n.d.) is its core operative value. The researcher's history of involvement with DMG⁶ was a major inspiration for this study's methodological framework. The AGJ was curated as a welcoming and warm space for game designers to engage with autopathographical game design. All of the tables were stocked with ideating materials such as notebooks, pens, colour markers, chocolate power bars, and whimsical neon-coloured toys. The AGJ space supplied a variety of snacks such as chips, pretzels, pop soda, bottled water, donuts, a catered lunch, i.e., gourmet pizza and burritos inclusive of everyone's dietary needs and preferences, and coffee.

The AGJ comprised a variety of activities including registration of subjects (consent form signing), the researcher's brief presentation of her study and data collection process, subject introductions to the group, 'jamming' (making games), informal interviewing of subjects during the game design process, group discussions, playtesting, socializing, breaks, and lunch. The data collection process consisted of recording and filming the game jam using video cameras and photography and collecting subjects' processual artifacts (i.e., sketches, doodles, some digital assets of the AGD process). After each day of the AGJ, the researcher recorded her personal impressions and thoughts of the AGD-AGJ process in memory notes. Finally, semi-structured, individual interviews were conducted with all thirteen subjects several weeks after the

⁶ The researcher has been a member of the collective *Dames Making Games* (DMG) since 2012, participating in one game jam (*No-Jam*) organized by DMG where she developed a digital game. As a practicing professional filmmaker and artist, she has also participated in DMG's socials where she has presented her own game art to the community.

completion of the AGJ in order to gather more in-depth information about their AGD-AGJ experiences.

1.1.5 The analytical method

This study is data-driven, which means that theoretical analysis is grounded in empirical evidence. Thus, the data drives theorization. The study's theoretical framework draws on diverse disciplines, including game and media studies, literary theory, narratology, semiotics, phenomenology, and relevant scholarship on art and aesthetics. The reader may often ask: why not focus the analysis on the computer games qua media texts? The answer is that in order to understand autopathographical game design, this analysis requires empirical evidence of the game designer's authorship process so as to access the *structure* of the autopathographical rendering in a computer game. Therefore, this study focuses on processes (AGD) as opposed to products or outcomes (the autopathographical games qua media texts). Furthermore, the medical humanities, literary criticism, game studies, and games criticism focus on the *texts* of the literary autopathography in the former and computer game in the latter. These fields have generally not examined the structure of the authorship process behind those texts, which testifies to the existence of a research gap in both fields. The empirical evidence consists of the following: audiovisual (video and photographic) recordings of the weekend-long, 7-hour per day Autopathographical Game Jam (AGJ) event; informal individual and group interviews during the AGJ with thirteen game jam participants—three female and ten male study subjects between the approximate ages of 20-33; semi-structured individual in-depth interviews conducted with the above study subjects a few weeks after the AGJ in order to gauge their deeper impressions of the game jam and AGD process; the collection of their processual artifacts (i.e., informal sketches and doodles, digital assets); and the researcher's memory notes and reflections of the AGJ.

1.2 Section II: Context and findings

To provide context for the subsequent analytical chapters, it will be helpful to present a brief description of the following: The Autopathographical Game Jam (AGJ) as a process; the thirteen study subjects, their 'stories' and their autopathographical games; and finally, a short list of major findings.

1.2.1 Introduction of AGJ study subjects and their autopathographical games

Thirteen recruited subjects participated in the Autopathographical Game Jam, three females and ten males between the ages of 20 and 33 with the following mental health experiences: severe bipolar disorder, anxiety, depression, PTSD (post-traumatic stress disorder and/or emotional trauma), ADHD (attention/deficit hyperactivity disorder), color-blindness, extreme introversion or shyness, social anxiety, grief, childhood memories of being bullied, and insomnia. It is important to mention that four of the thirteen subjects testified that they had had previous experience authoring an autopathographical game about illness or disability in either a solitary or game jam context. Therefore, the AGJ was their second or third experience authoring an autopathographical game.

The names of the following thirteen study subjects have been anonymized for their safety. Their gender identity has nevertheless been retained. Their stories are introduced in alphabetical name order.

1.2.1.1 Subject 1: Alex

Alex is a professional game developer living with bipolar disorder who usually makes arcade-style mobile games. His creative hobbies include writing, poetry, and “stream of consciousness rants” dating back to when he was 14. The AGJ was Alex’s sixth or seventh game jam in the last three years. Alex stated that while in a game jam he is generally surrounded by a community of game developers living with mental illness and mood disturbances, which implies that he is not alone and he can connect with others with similar experiences. During the AGJ, he authored an autopathographical game called *Brain Pains* via the game engine *Unity*, a game of colour-coded nodes and ropes that visually resembles a neural network. He said⁷ that the game is:

⁷ When quoting subject testimony throughout the dissertation, this study alternates between the voice of study subjects in past-tense and the dissertation writer’s implicit voice in present-tense, the latter of which signifies her interpretation of subject testimony. In other words, the following present-tense statements, “[subject name] implies” or “[subject name] suggests” (among other possibilities) signal *the dissertation writer’s interpretation* of what she thinks the subject means. The following past-tense statements, “[subject name] said” or “[subject name] explained”

...an abstract expression of what I'm doing with repressed issues.

The game's concept is to "untangle spider webs in your brain". The game's red nodes symbolize emotional triggers that stand as a metaphor for repressed trauma. The game's premise is based on:

...dealing with therapy stuff without having therapy.

He explained:

My game is about my struggle with not having professional help. I'm on so many wait lists and it's really hard to be trying to diffuse the bombs in your emotional trauma without any tools and without any professional help.

Alex's intention was to create ambience and mood through his game, which does not offer a win-lose outcome:

I kind of like things that are experiences rather than goals.

Alex testified that although *Brain Pains* was his first graphic-based autopathographical game, he has also made some autobiographical text-based games with the *Twine* software that have not been published.

1.2.1.2 Subject 2: Brody

Brody is a software developer who has also, prior to the AGJ, created an autopathographical game about his experiences with mental illness, "a project dealing with my disorder". He shared a candid introduction to the group of AGJ subjects about the topic of his current autopathographical game:

I had a mental breakdown about a year and a half ago and I quit my job and found out I had PTSD and a mix of bipolar disorder and borderline personality disorder.

During the AGJ, he authored a murder mystery game, *You're Invited*:

(among other possibilities), emphasize the subject's distinct voice and the direct claims that they are making at the time of the interview.

(This game is) about scripting, and how intense conversations can be with PTSD and trying to analyze like every damned pattern to everything.

Games have always been a safe refuge from volatility for Brody since his childhood when he was bullied and lived in a dysfunctional home:

I really feel that my experiences as a child, games were just about the only positive experience I had besides my siblings. Because I'd go to school with constant bullying there and then I'd come home, and my Dad was a severe alcoholic...just a nightmare. And so, the only place that I'd felt that I was competent, and I could do things, was in videogames. I was good at them, I could beat games.

Game design is his chosen artistic medium for expression given his fond memories of games.

1.2.1.3 Subject 3: Ellie

Ellie has worked as a web and graphic designer and has previous training in the fine arts. Ellie has always wanted to make a game and the AGJ provided a chance for her to “break the seal”. This was her first game jam, and she designed the interactive narrative about insomnia, *This is Your Awesome Sleep Machine*. She explained the topic of her autopathographical game:

The theme of my game deals with insomnia and the neurological, potentially psychosomatic aspects of insomnia, and how the mind and body are connected.

Ellie's game explored the first-person experience of insomnia as being trapped in an endless cycle of sleeplessness. She harnessed the branching-style format⁸ of the text-based game design software *Twine* to convey the repetitive quality of the insomniac's thought process by looping choices and actions in the game. Ellie said that “*Twine* and insomnia go well together” because this software is great for capturing stream-of-consciousness thought. The gameplay consists of two main choices for the player:

The choice of having an alternative spirit guide to try to help (the player with insomnia to) resolve their insomnia versus just going at it alone for the night. And those two options on the initial screen determine what kind of game they'll be playing, one that's

⁸ Branching storylines present multiple choices for the player, a common format of interactive narratives. That is, the player is repeatedly given choices in the narrative that lead to multiple new choices creating recursive narrative structures.

independent and more traumatizing as the insomnia, versus one that's more therapeutic on the other side.

Ellie's purpose in developing this game is that it will become a "tool to help you achieve some kind of peace."

1.2.1.4 Subject 4: Fred

Fred is an independent game developer, and his contribution to the AGJ is a mobile puzzle game. *Brain Buster* is inspired by a series of traumatic experiences, including looking after a dead man's body on a hiking trip and pulling a drunk driver out of a crashed car. Fred had dealt with his stressful experiences at another general-purpose game jam towards:

...self-therapy/DIY therapy...using game development as a form of coping with emotions.

During the AGJ, Fred wanted to understand how he used computer game design to "cure himself" by making a puzzle game that is also a meta-inquiry into his own healing process. He explained:

The whole idea is that I wanted to mechanically represent how I got over my PTSD, which was through talking to people over the phone.

A few months after the AGJ, Fred produced multiple new iterations of his initial concept, which he tried to pitch to game publishers. He showed the game prototype to a variety of veteran, emerging, and independent game developers, who then gave him feedback on his game. Based on this feedback, Fred said that his emotional trauma was "too abject" to represent explicitly in the mainstream mobile game market that he was aspiring to enter. Nevertheless, his puzzle game was his:

...first potential successful product that's based on a personal experience.

1.2.1.5 Subject 5: John

John is a professional game developer whose "prototypes have been picked up by Sony and Microsoft". He has struggled with severe bipolar disorder since he was eight years old. John's first untitled autopathographical game was a 3D action-based game created using the *Unity* game engine. It explores the highs and lows of mania and depression that constitute his

experiences with severe bipolar disorder. The game also advocates for assisted suicide for individuals with severe and untreatable forms of mental illness. The goal of the player, who adopts a simple 3D character as their avatar, is to “chase the happiness light (orange ball) and just keep moving round” in a space of darkness. John explained:

And the longer you’re going through darkness the more depressed you get, and the slower your character gets, the more depressed he looks and acts. As you’re going, you see these light areas that will have these terrifying-looking light sources on them, and they are pockets of mania that make your character impossible to control. The longer he stays in them, the mood goes up to uncontrollable levels.

According to John, the game player character recognizes that they cannot possibly win this game after they start playing the game, which results in the game character either being killed or committing suicide. John’s game builds a level of discomfort for the game player by taking choices away from them and thus undermining their sense of agency and control.

1.2.1.6 Subject 6: Lukas

Lukas is a physics major who specialized in developing mini-arcade games at the time of the AGJ. This was his first game jam and his first autopathographical game. He appeared to harness AGD as a tool for “self-hypnosis” to help him deal with the social stigma of the ‘loner’ attached to his introverted personality. He explained that he felt pressured by social expectations to be constantly sociable and gregarious, which is difficult for someone like him who is extremely shy:

Because of the social pressure on me—you cannot live alone, you cannot do anything alone. You need to get into a team, you need to get socialized, you need to network, and that’s the social pressure on our life. They (society) seem to reject people who are not socialized well.

The game Lukas created during the AGJ is called *Cooperative Defense*. In this game, he explored how introverts can learn self-acceptance and self-empathy while slowly learning to connect with others on their own terms and at their own pace:

...reducing the stress for working alone at this moment, but not forever.

He elaborated on his game concept, which appears to work with a goal-oriented objective:

I will make an objective for people and try to create the scenario where they can play with friends or they can play alone and by looking at how they get into the objective, I will know if this is OK.

Lukas found the autopathographical game “exciting” in terms of how it departs from the usual pattern of mainstream forms of computer game design.

1.2.1.7 Subject 7: Marco

Marco is a game developer and an exchange student from overseas who authored his first autopathographical game during the AGJ, a 3D experiential game appropriately called

Connections:

My game is about connections because of all my experiences in life. I’m a person that comes from several places and has several connections and I’ve realized that I have always had this problem with crisis and identity and belonging somewhere and always being different in a certain way.

(It’s) an abstract game about creating connections with other objects or people (through) colour.

His game also explores the themes of paradox, surprise, and reversibility in life:

I was bullied by people that I never thought I would be bullied by. There were even times in my life when I was a strong friend with this person and then she suddenly betrayed me and became a bully... Sometimes people that were my bullies became my friends.

Marco’s game, which was authored using the game engine *Unity*, harnesses the themes of movement and fluidity, i.e., collision physics and simulation of gravity, in order to symbolize his experience of being ‘pulled towards’ and ‘pushed away’ by friends—of friends becoming bullies and bullies becoming friends. The game also comments on Marco’s experiences of acculturating into new countries, having an identity-in-flux, and feeling lost at times. The game does not produce a win/lose outcome; rather, it is visually and procedurally abstract deploying “shapes and colours” and “mechanics and physics interactions” to communicate feeling and mood.

1.2.1.8 Subject 8: Martin

Martin is a game developer who at the time of the AGJ was completing a BSc in Computer Science. Martin has loved games since he was a child and wanted to do something

different with his life than “sit in a desk and write reports”. Martin is color-blind and cannot see red:

I am trying to make a game about how it’s hard to be visually impaired and see everybody doing simple stuff so easily and you have to struggle.

Broken Lens was Martin’s first personal puzzle game made using the *Construct 2* game engine and *HTML5* markup language. He explained why he chose the puzzle genre to depict his experiences with colour-blindness as a problem to be solved:

In my case, it was a puzzle because I think it is the easiest one to associate with a problem.

He importantly added that he was interested in more than just framing colour-blindness as a problem; hence the need to create an experience through his game:

I want to incentivize the player to go beyond the problem.

In *Broken Lens*, Martin creatively simulates colour-blindness for the player, mixing colours like a painter, and using hexadecimal notation and colour values to convey how he perceives colour. The game tries to create the first-person experience of what it is like to live in a world where colour-blindness is largely misunderstood. In other words, Martin distinguished between not being able to see red via his perceptual system, and living with the experience of not seeing red while embedded in a world of expectations. Martin explained how the game conveys the feeling of colour-blindness for the player through the mechanics and rules that are used to convey colour:

I’m managing the colours to change the colours of the game and recording, simulating the colour blindness, turning off the red, the green, the blue, so I can confuse the player.

1.2.1.9 Subject 9: Nathan

Nathan is a self-taught freelance game developer with a background in theatre. During the AGJ, he authored the “music and rhythm game” *ADHD Hero*, based on the popular music rhythm game, *Guitar Hero* (2005). Making a personal game was a “big leap” for Nathan. His

game reinvents the ADHD diagnostic tool—the continuous performance test (CPT)⁹—as a musical instrument that is “a commentary on how those (diagnostic) tools can be games”. The inspiration for his autopathographical game, which was developed in the *Unity* game engine was to:

...integrate some of my experiences around internalized self-image, the anxiety, and the bad issues that come with growing up with ADHD and being told that you're lazy.

He elaborated on his experiences growing up as a child with ADHD:

Going through school as a gifted, dual-exceptional, whatever you wanna call it, I did feel like I was gaming the system all the time. I would do everything I could going through school to find a different way to get the answer without doing the homework, without showing my work. If I could find a trick, it wasn't cheating, it was just doing the math in my head instead of on paper. You can train yourself to perform really well on IQ tests. Which means, all the test is measuring is how good you are at taking IQ tests. It's not measuring your IQ and that's...I like doing that.

Nathan stated that autopathographical game design engenders mindfulness for him and enables him to “articulate these experiences”.

1.2.1.10 Subject 10: Sam

Sam is a recent BSc Mathematics graduate and aspiring game developer also living with ADHD at the time of the AGJ. His stealth game *Anxiety Corp.* was developed using the programming language *Javascript* and *Notepad++*, and aimed:

To tell the story of dealing with anxiety in a dead-end office setting.

Sam described how he selected to focus on specific facets of his first-person experiences in his autopathographical game:

Because there's a lot more to ADHD than distraction, I couldn't cover it holistically. So I started thinking about anxiety and one of the main struggles that I have is avoidance cycles. Essentially, when I encounter a stressful situation it will make me less and less

⁹ “Continuous performance tests (CPTs) are widely used laboratory-based measures of neurocognitive processes that are related to attentional control and inhibition”, claim Munkvold, Manger, & Lundervold (2014).

likely to engage in it and then it just causes this downward spiral where I cease functioning.

The goal of the game player character is to avoid non-player game characters who are co-workers in what appears to be a cubicle farm through “one of the game’s core mechanics of stealth”. Sam described the plot of his game:

The main (game) character starts at a new job in an office and they spend their days tiptoeing around everyone around them. Inevitably, they have to interact with their colleagues and their bosses and as they do, people will say insensitive things or be pushy or just do anything, maybe not even anything wrong, but they will trigger the main character’s anxiety. As that anxiety builds, it becomes harder and harder for the main character to do their job and eventually the game will culminate in the main character’s termination or resignation.

The game deploys the metric of “an anxiety meter” to monitor the game player character as they attempt to duck and escape their co-workers who are about to accost them. The more unable the game player is in hiding from his co-workers, the higher the anxiety meter rises.

1.2.1.11 Subject 11: Vicky

Vicky is a game designer living with a highly stigmatized form of mental illness, who at the time of the AGJ was completing her undergraduate humanities degree. The AGJ was Vicky’s third participation in a game jam making an autopathographical game. For Vicky, the autopathography was about “coming out” and “giving a face to the illness”. The last autopathographical games she authored were meant to address her personal struggles with feeling incapable due to her medical diagnosis.

(I wanted to) break the silence for myself. Because of the stigma, people aren’t coming out (with mental illness), and to me it’s a cycle that feeds itself because you don’t see people who are strong who are saying, ‘yeah, I’ve been diagnosed...and I’m not a complete mess, and I’m not a horrible person, I’m a really great student, I have really high marks, I make games, I’m creative and I struggle with this’.

An important goal of Vicky’s previous autopathographical games was challenging the social stigma imposed by medical diagnosis, claiming that “diagnosis is very restraining”. She made the following distinction to explain this claim:

I don’t *have* (a specific mental illness), I’ve been *diagnosed* with it (emphasis added).

During the AGJ however, she wanted to make a game about a new experience: her grief over her father's death from a heart attack. The previous autopathographical games she had authored:

...still don't feel as personal as (losing my father). They feel more clinical.

Ultimately, she discovered that designing digital art graphics about her grieving process—a 'beating heart' animation—was too emotionally difficult because her loss was fresh and raw. She explained:

Staring at the heart for a really long time and listening to the heart beat and then hearing it stop just started triggering me, so I was just like, 'I don't want to do this anymore'.

She abandoned her first plan to delve into her grief during the AGJ, instead choosing to author digital pixel art using the game software *Pix In*:

I just wanted to do something that was fun and made me feel good...

I made a dress up doll of myself. Now I'm just going to make a bajillion outfits and all of these cute characters with funny hair and outfits.

Vicky's decision to explicitly stay away from autopathographical game design can be interpreted as a form of self-care—practicing self-mindfulness and taking the necessary steps to ensure her own safety and well-being during the AGJ despite any pressure to engage in activities as the rest of the group.

1.2.1.12 Subjects 12 & 13: Yasmine and Cole

Yasmine and Cole are the only two subjects who collaborated on an autopathographical game during the AGJ, "a mixture of sandbox and labyrinth" virtual world named *City Explorer*, made with the *Unity* game engine. Cole is a musician, multimedia artist, and animator and Yasmine is a game designer and undergraduate student in art and design. Cole said that their autopathographical game's concept explores the topic of:

...disability as a classification and category that we assign people (and) expanding possible limits.

Yasmine was inspired by her own personal experiences:

When I was younger I had speech delay, which limited me in a social context a lot. So, I wanted to explore that.

The game harnesses music as an invisible ‘character’ leading the player to explore a world of non-player game characters who represent disability and difference. Music becomes an aesthetic form of interaction; Cole said that he and Yasmine explore its possibilities by gauging “how people (could) interact with this song”.

1.2.2 A short list of major findings

Looking at the data, and keeping in mind that our object of analysis is not the resulting games, nor the experience of players playing them, but the design processes and experiences of their authors, it appears that there is a *structural logic* to the AGD-AGJ process. The most important finding of all is that the AGD-AGJ process appeared to be therapeutic, healing, and/or restorative for subjects.

The short list of the major findings is as follows:

1. Autopathographical game design was an overall therapeutic process that appeared to afford self-insight and self-understanding through the act of introspection; the unique formal properties of AGD such as digital design, rule design, abstraction, and implementation; and the social space of the AGJ in which subjects undertook the authorship process.
2. The semantic opposition between the two semiospheres of ‘computer games as fun and frivolous’ and ‘autopathographical narratives as existential and serious’, suggests that their AGD integration was an unorthodox and exciting experience for subjects.
3. Most of the autopathographical games authored during the AGJ took on an *abstract* form of the autopathographical narrative. Subjects visually and procedurally abstracted concrete details of their first-person narratives, diverging from the written autopathographical narrative, which usually took on a much more concrete form in prose.
4. Subjects openly shared and verbally disclosed, in group conversations and interviews during the AGJ, a range of deeply personal (concrete) details relevant to their autopathographical

narratives. However, they did not disclose those same details in the autopathographical games, due to a number of reasons, including self-care and self-preservation from stigma in case the games are ever publicized or disseminated. As a result, there was a significant misalignment between the type and amount of personal information disclosed during the AGJ's face-to-face interaction process and disclosure in the autopathographical game itself. Thus, it is not the game as final output that confirms the autopathography's ontological status as much as the autopathographical process involved in the game's authorship.

5. The autopathographical game was, relatively speaking, a retrospective exploration of first-person experiences with mental illness, disability, and emotional trauma. Most subjects were not exploring a present distressing state (except for Vicky, who briefly tried exploring her grief and decided to stop due to *heavy* emotional discomfort). Through AGD, they were revisiting past emotional or mental states from a rather safe distance. This safe distance did not necessarily preclude the possibility of 'reliving' a past distressing experience during AGD (as Alex testified); rather, subjects were not in the midst of debilitating distress during the AGJ event.

6. Some subjects partly approached their experiences of illness, disability, and trauma as a problem to be solved or understood through the unique expressive medium of autopathographical game design.

7. Finally, the autopathographical games authored during the AGJ will not be published or disseminated publicly, online or otherwise. (Fred's game is the exception. However, his game does not explicitly divulge his autopathographical narrative but is merely inspired by it.) To date, the games have had only one small and private audience: the AGJ audience, which consisted of the thirteen subjects, the principal researcher, and three research assistants.

1.3 Section III. Thesis

Given the data and findings, the following set of claims will be argued:

Claim 1: The aforementioned findings can be understood through the interplay of four identified therapeutic dimensions of poiesis—*sociopoiesis*, *autopoiesis*, *fabulopoiesis*, and *logopoiesis*, which appear to co-constitute AGD's therapeutic effects. The theoretical importance of AGD

praxis ('doing') leads to poiesis ('creative making') in the context of the AGJ. Poiesis as an aesthetic activity appears to engender or contribute to healing; that is, subjects living with mental illness, emotional trauma, or disability are concerned with the poetic (aesthetic) resources of AGD as well as its instrumental resources (i.e. healing), given that the former appears to engender the latter.

- a) *Sociopoiesis*: the social dimensions of AGD that invite and enhance creativity during the Autopathographical Game Jam (AGJ), namely the poetics of the AGJ space as a therapeutic 'community' for collective making, face-to-face interaction, sharing intimate narratives, and achieving communion with others.
- b) *Autopoiesis*: the self-reflective, self-reflexive (self-referential), and introspective dimensions that AGD invites towards self-inquiry and self-care, which enable the *autopathographical* investigation of first-person experiences and 'private inner world' of thoughts and emotions.
- c) *Fabulopoiesis*: the act of the imagination harnessed in the construction of a 'fabulized self' via AGD. The fabulized self is neither self nor Other; rather, it is a figure modelled in the image of a fable or fantasy rendered as a fictionalized account, a virtual world, and/or game character(s) informed by the subject's first-person experiences. The poetics of fabulization in AGD involve the deployment of metaphor and analogy via abstraction to create autopathographical narratives that re-imagine the self.
- d) *Logopoiesis*: the resources of the digital, procedural, logical, and computational dimensions of building a computer game and articulating the autopathographical narrative (AN), which enable the AGD designer to *implement* a game concept, a game narrative, and a rule-based formal system via digital design (i.e. graphic design, 3D modelling, etc.) and software programming praxes. Performing these implementational tasks through computer game design appears to be both therapeutic and promotes self-confidence for individuals struggling with mental illness.

Claim 2: There are some similarities and some key differences between autopathographical writing (i.e. in prose form) and autopathographical game design during an AGJ, the latter of which appears to amplify the therapeutic effects of AGD, of which the following are sub-claims:

2a: The autopathographical narrative requires two dimensions, namely that of autopoiesis and fabulopoiesis, which both stay in the realm of the first-person perspective.

2b: Autopathographical game design crucially requires three dimensions: autopoiesis, fabulopoiesis, and logopoiesis. This requirement enables a shift between the first-person and the third-person perspectives for subjects, who experience fabulopoiesis as an opportunity to re-imagine themselves through substantial use of metaphor and analogy and logopoiesis as a distancing device engendering mindfulness.

2c: Another difference in fabulopoiesis between autopathographical writing and autopathographical game design is that in the latter, the autopathographical narrative is more heavily abstracted at the audiovisual and procedural level than in the former.

2d: AGD during an autopathographical game jam event (AGD-AGJ) crucially involves all four dimensions: sociopoiesis, autopoiesis, fabulopoiesis, and logopoiesis.

Claim 3: Therapoiesis is explained by the interplay of the four dimensions of poiesis and of the first-, second-, and third-person perspectives that they appear to engender.

1.4 Section IV. Analytical chapter description

Prior to the analytical chapters, the chapter “Methodology and Methods” (Chapter 2) provides an overview of the methodology framework and methods used to both collect data with thirteen human subjects during the Autopathographical Game Jam and analyze that data.

The first analytical chapter, “The Foundations of Therapoiesis in Autopathographical Game Design” (Chapter 3) sets the foundation for the subsequent analytical chapters by introducing the two-part foundational structure of the AGD-AGJ process: 1) the interplay between two semiospheres—the computer game and the autopathographical narrative—and how subjects transgress the aesthetic code of each semiosphere via the distinct formal properties of computer game design; 2) the interplay of the four dimensions of therapoiesis (i.e. sociopoiesis, autopoiesis, fabulopoiesis, and logopoiesis)—*the tetrad of therapoiesis*—and how this tetrad maps onto the interplay of the first-, second-, and third-person perspectives accessed through the

AGD-AGJ process. A brief discussion concludes with how the tetrad of therapoiesis connects to Aristotle's framework of knowledge formation: praxis, poiesis, and theoria.

In the second analytical chapter, "Sociopoiesis: The Autopathographical Game Jam as a Therapeutic Community" (Chapter 4), the Autopathographical Game Jam is examined as a phenomenological social space propelling creativity. The AGJ is a small, transient, and intimate 'community' and its resources for collective making, face-to-face interaction, listening, sharing intimate narratives, bonding, and achieving communion comprise the sociopoetics of this importantly therapeutic space.

The third analytical chapter, "Autopoiesis: Self-Making as a Process of Self-Care in Autopathographical Game Design" (Chapter 5) examines the self-reflexive dimensions of self-making, i.e., the processes of self-reflection, introspection, and self-inquiry comprising the authorship of both the autopathographical narrative and game. Introspection through autopathographical game design and narration is the starting point for self-care and self-healing. This chapter also explores autopoiesis as a therapeutic form of author-identity construction via the theoretical concept of the *auctorial posture* to explain the different roles and perspectives embodied by subjects during the AGD-AGJ process (i.e. game designer, narrator, fabulized self, listener, stranger, friend, healer, self-healer, etc.). These auctorial postures connect to differences in the types of narration and forms of explicit and implicit disclosure—subjects *foreground and background* their autopathographical narratives depending on the context (i.e. the AGJ discursive space versus AGD) for purposes of self-care.

The fourth analytical chapter, "Fabulopoiesis: Metaphor and Analogy in Autopathographical Game Design" (Chapter 6) investigates how authors deploy metaphor and analogy to construct the 'fabulized self' in the autopathographical game. The chapter delves into a theoretical analysis of analogy and metaphor in several domains including the fine arts, film, design, and modern science in order to theorize on the ways in which processes of self-fabulization through analogy building, heuristic reasoning, and abstract thinking lead to self-understanding and self-discovery.

The fifth and final analytical chapter, "Logopoiesis: Autopathographical Game Design as Self-Understanding and Self-Therapy" (Chapter 7) examines the articulated digital,

computational, logical, and procedural dimensions of digital design and programming praxis used to implement the autopathographical narrative into a rule-based formal system and software application, which differentiate autopathographical game design from autopathographical writing. As a form of ‘calculated making’—a highly deliberate and self-analytical creative praxis—logopoiesis engenders the third-person perspective, which serves as a distancing and mindfulness device for subjects.

Chapters 5, 6, and 7 address the steady intensification of healing engendered through the autopoetic, fabulopoetic, and logopoetic properties of AGD, the first two of which are undifferentiated from autopathographical narrative and the third of which is differentiated from autopathographical narrative. Chapter 4 on the Autopathographical Game Jam addresses the social and relational sphere in which therapoiesis seems to thrive.

The conclusion (Chapter 8) offers some closing remarks, lists the research contributions of this dissertation, and gives some examples for future research directions based on the insights gained through writing this dissertation.

Chapter 2

Methodology and Methods

2.1 Introduction

This chapter presents the architecture of this study's qualitative (i.e. *grounded theory*, *case studies*, *discourse analysis*) and arts-informed methodology (i.e. *research-creation*) and the deployment of data collection and data analysis methods as tools of research. The use of qualitative methods for data collection such as semi-structured interviews with human subjects and data analysis is justified by drawing on a humanistic theoretical framework rooted in interdisciplinary and multimodal, i.e., audiovisual, textual, discursive, interactive commitments. The chapter also describes the processes of study subject recruitment, pilot study, primary data collection during the principal case study, which is a two-day, seven-hour per day Autopathographical Game Jam, and finally the completion of post-Autopathographical Game Jam interviews with subjects. This study's socially conscious and arts-informed methodology is an effective tool for studying the game authorship process while meaningfully contributing to a community of independent game designers who seek a platform to explore personal forms of storytelling via game authorship. This study is also a humanistic dissertation that draws on some social science methods to collect empirical evidence and to conduct a data-driven theorization of the human-based phenomenon of autopathographical game design. The study harnesses a predominantly structuralist framework for conducting a discourse analysis of the qualitative interviews with subjects and processual artifacts (i.e. sketches, drawings, doodles).

2.2 Recruitment of study subjects

Upon successful ethics approval from University of Toronto's Social Sciences, Humanities, and Education Research Ethics Board (REB) in May 2014, study subjects were recruited from June to September 2014. The recruitment posting adhered to a set of specific criteria: disability, illness, or trauma-identified adult (18-50 years old) game designers, developers, or artists interested in authoring autopathographical digital games in a two-day organized game jam. The recruitment call was distributed through a variety of game and disability oriented communities, listserves, and creative DIY (Do-It-Yourself) computing spaces in Toronto: *Dames Making Games* (DMG), University of Toronto's Department of Computer

Science and Faculty of Information (Semaphore Lab and KMDI), George Brown College Game Development Program, Autism Ontario, Craigslist Toronto, Reddit, IGDA-Toronto (International Game Developers Association), Women in Games, and the Indie Developers Group on LinkedIn. Social media played a significant role in the recruitment process: the feminist gamemaker collective *Dames Making Games* retweeted the recruitment call to the independent game development community in Toronto. Snowballing the recruitment call through the researcher's personal and professional social networks proved fruitful: friends and colleagues involved in Toronto's games research scenes disseminated the call.

The three-month recruitment process initially led to fifteen game designers confirming their participation in the study, but two of them cancelled at the last minute due to personal scheduling conflicts. All game designers who expressed interest in participating and who fit the recruitment criteria were accepted. The final thirteen study subjects were ten male and three female game designers ranging in age from 20 to 33 years old. In their email correspondence, subjects self-identified with the following and sometimes overlapping mental health and disability conditions: bipolar disorder, attention/deficit hyperactivity disorder (ADHD), colour blindness, depression and anxiety, introversion and shyness, insomnia, emotional trauma, post-traumatic stress disorder (PTSD), grief and loss, childhood memories of being bullied, childhood speech delay, and other undisclosed conditions. The subjects have all had active freelance and/or independent careers in software development, game development, and/or art and design practice. The recruitment call was focused on professional or experienced game designers because they have at their disposal a more sophisticated computer game design skill set to draw upon for game creation.

2.2.1 Number of recruited study subjects: justification

The decision to cap the recruitment number at fifteen subjects was made for a number of reasons. The first reason was the need to foster an intimate and comfortable space for social interaction among game designers who are sharing amongst themselves and disclosing highly personal information. The researcher could effectively manage no more than fifteen subjects with a variety of potential accessibility needs in a two-day game jam, given her limited financial and logistical resources as a doctoral student. The commitment to conduct a small-scale, intimate

game jam deviates from game jam norms in the games community. Typical game jams are large-scale events of hundreds or thousands of participants. The second reason to hold a small game jam was to enable qualitative data collection of the authorship processes of subjects, which is not as feasible with larger groups unless the researcher has the financial and logistical resources to hire multiple research assistants and/or hold multiple game jams.

Yet another reason for recruiting thirteen subjects qua interviewees is the argument made by discourse analysis researchers Potter and Wetherell (1987) that ten subjects is a sufficient number for purposes of conducting discourse analysis of interview testimony. They claim that a higher number of subjects does not necessarily ensure a discourse analysis that is of higher quality. They explain more fully:

If one is interested in discursive forms, ten interviews might provide as much valid information as several hundred responses to a structured opinion poll...For discourse analysts the success of a study is not in the least dependent on sample size. It is not the case that a larger sample necessarily indicates a more painstaking or worthwhile piece of research (p. 161).

Potter and Wetherell (1987) add that “the crucial determinant of sample size, however, must be, here as elsewhere, the specific research question” (p. 161). The sufficient number of subjects adopted by this study, also based on the recommendation of Potter and Wetherell, is cited for purposes of justifying in-depth discourse analysis of interview testimony. The theoretical concerns about the number of recruited subjects were satisfied with the data being gathered only from The Autopathographical Game Jam, which means that additional game jams or interviews did not need to be conducted for further data collection.

2.3 Pilot game jam study

Following the successful recruitment of subjects, a four-hour pilot test game jam was conducted with six ‘pilot subjects’—fellow graduate students on September 13, 2014. The pilot game jam study primarily served as a trial and practice run by providing valuable foresight on how to ensure a smooth run for the Autopathographical Game Jam case study. The pilot game jam helped generate ideas on how to create a welcoming, relaxed, fun, and inclusive environment. For example, the researcher/AGJ organizer familiarized herself with the nature of the solicited space, verifying that its configuration and amenities, i.e., location, technology,

tables and chairs, lighting, WIFI access, wheelchair access, etc., met the potential accessibility needs of subjects. The pilot game jam study also allowed her to practice the logistics of data collection—video recording, interviewing, and workflow management such as working with memory cards and file transfer. Finally, the pilot game jam study prepared her for anticipating any unforeseen circumstances and fostered confidence in her game jam leadership and facilitation skills.

The pilot game jam was held in the Demo Room, Semaphore Lab of the Faculty of Information, University of Toronto, the same room that would be used for the Autopathographical Game Jam. There were some key thematic and conceptual differences between the pilot game jam and Autopathographical Game Jam. Pilot subjects were not required to integrate the autopathographical narrative into a computer game, which was the principal activity of the Autopathographical Game Jam. Rather, pilot subjects were invited to make a simple text-based game of any genre. Pilot subjects did not have any game design skills, although a few individuals had some programming and graphic design skills. Finally, the pilot game jam study ran for a total of four hours as opposed to the Autopathographical Game Jam's weekend-long, seven-hour per day duration. A small portion of the pilot game jam study was devoted to teaching pilot subjects the *Twine* open-source software for authoring text-based digital games. Twine was chosen because of its versatility and easy-to-learn game engine based on the HTML markup language, the CSS style sheet language, and the JavaScript programming language. Pilot subjects were encouraged to creatively experiment with the *Twine* software. To practice data collection and documentation, the researcher conducted individual informal interviews with the six pilot subjects while they were developing their games, inviting them to articulate their ongoing design decisions and their experiential process of making a game in a small game jam. These interviews were documented with photographs and video recordings. Finally, pilot subjects were invited to give feedback about the game jam space and critique the researcher's skills, i.e., presentation method, mode of engagement with pilot subjects, facilitation of a relaxed and inclusive environment, and other relevant issues; these group feedback conversations were recorded. The pilot game jam study was a successful effective trial and practice run.

2.4 The ‘Game Jam’ as research platform

This section gives a brief overview of the use of game jams for academic research, followed by a description of the Autopathographical Game Jam (AGJ) as it unfolded during the weekend of September 20-21, 2014. The AGJ’s significance to this study was three-fold: a) the AGJ was a platform for the purpose of data collection; b) it allowed subjects to engage in a meaningful creative process and artistic production of computer game prototypes that also spoke to the researcher’s community outreach objectives c) it was an object of in-depth theoretical analysis. Only a) and b) are addressed here; for c), please read the analytical chapter “Sociopoiesis: The Autopathographical Game Jam as a Therapeutic Community”.

The *game jam* is a recent worldwide cultural phenomenon dating back to 2002, closely tied to ‘maker’ cultures and DIY (Do-It-Yourself) spaces. The emergence of game jams has reinvented game authorship as a *social experience*. Game enthusiasts of all types, including developers, designers, gamers, artists, hobbyists, amateurs, and professionals are invited by an organizer to assemble in a designated space to rapidly prototype games within a set of conditions such as a tight time constraint, working with a specific theme, and possibly competing for an award. Game Jams are not only creative incubators for developing games, but also community-building platforms for expanding social networking and acquiring new skills (Reng, Schoenau-Fog, & Kofoed, 2013; Guevara-Villalobos, 2011).

Game jams generally fall into one or more of the following categories: large-scale, small-scale, inclusive, competitive, social purpose-driven, and research-driven. The large-scale game jam, e.g., *Global Game Jam* and the Toronto-based *TOJam*, is conducted over a period of two to three days, running 48 hours (as with the former, with local sites hosting in over 90 countries around the world¹⁰) or 24 hours ‘round the clock (as with the latter)¹¹. Game jams are typically competitive as testified by several subjects in this study. However, Shin et al. (2012) claim that game jams are often “confused with programming contests”. They suggest that the creative spirit

¹⁰ See more at: <http://globalgamejam.org/faq>

¹¹ See more at: <http://www.tojam.ca/details/overview.asp>

of game jams is more essential than competition: “a game jam is closer to a musical ‘jam session’ and not necessarily a contest.” Shin et al. analyze the untapped potential of game jams to solve pressing social problems in the world. Their case study is the *Fukushima Game Jam* in which the environmental disaster of the Fukushima Power Plant explosions harnesses game development for disaster management and recovery; this is a good example of a social purpose-driven game jam.

Scholarly work on game jams as community-based arenas for conducting design and arts-informed academic research is lacking. Chatham et al. (2013) argue that the recent phenomenon of game jams is opening new avenues for conducting research in the academic community, especially in the Human-Computer Interaction (HCI) field. The authors characterize a rapid-prototyping game jam as a rich collaboration between artists, programmers, designers, researchers, and practitioners. Game jams are great platforms that “inspire novel game ideas and new ways of thinking” (p. 3175). But while Chatham et al. examine game jams as academic platforms for researchers to make games, they do not investigate the possibilities of collaborating with study subjects or non-researchers, as is the case in this study. Chatham et al. also propose data collection methods for documenting design processes, methods such as using photographs and video recording during the game jam, and posting content and updates on social media spaces. Deen et al. (2014) similarly suggest game jams as a viable research method for conducting research through design practice, an untapped realm in the academic community. The authors propose a game jam for designer-researchers addressing specific research questions, providing insight into design processes, and examining the strengths and weaknesses of game jams as research tools via the researcher’s reflections.

2.4.1 The Autopathographical Game Jam: logistics and structure

The Autopathographical Game Jam was a unique research platform for merging the following research activities and commitments: data collection, creative output by subjects, and community building by championing narratives that are either excluded or silenced by mass-scale, commercially-oriented game jams. The choice of a game jam to study the authorship processes of game designers was an intuitive one for the researcher. Her participation in Toronto’s independent game development community is coupled with her extensive and

professional creative practice in the visual arts, filmmaking, web development, and digital design. As a member of *Dames Making Games*, she made her first digital game in 2012 at one of their game jams, “No-Jam”.

The Autopathographical Game Jam was based on two specific game jam models: the inclusive game jam and the research-driven game jam. The existing academic literature on game jams does not address the prevalent cultures of game jams, which may stifle certain kinds of creative exploration and undermine inclusiveness. For example, some of the subjects interviewed during the AGJ testified that typical game jams are highly competitive, aggressive, and hyper-masculinized spaces driven by the pressure to create commercially-viable games. The inclusive game jam model is exemplified by game jams organized by Toronto feminist collective *Dames Making Games*, whose core operative value is “diversity, accessibility and inclusiveness” (DMG website). *Dames Making Games* orients its membership to marginalized members of the game development industry—“our community of women, trans and queer game-makers and their friends” (Dames Making Games, n.d.)—, and cap their game jams at approximately 20-25 participants. DMG is an exemplar of the inclusive and small-scale game jam model that actively addresses representation inequality in the game development industry.

Similar to *Dames Making Games*, The Autopathographical Game Jam represented game designers whose experiences with mental illness, emotional trauma, and disability have been marginalized as legitimate narratives worthy of computer game design. As an enterprise mobilizing members of the Toronto game development scene, the Autopathographical Game Jam was not only a platform for academic research. It was also a social venue for fostering empathetic connections between independent game designers through shared experiences. Subjects emphasized the importance of sharing and connecting with other designers. As the interview testimony demonstrates, the emotional honesty of the subjects in this game jam would not have been possible at a large-scale, impersonal game jam, or at an event that did not address a specific theme such as the autopathographical narrative.

Addressing the inclusive game jam model, a ‘safe’ and ‘inclusive’¹² environment was created for the thirteen subjects to safely negotiate their disclosure of deeply personal and intimate topics in a mutually supportive space. Thus, the principles of inclusiveness set out in provincial policy were used as a guideline. A commitment was made to respect the guidelines of the Accessibility of Ontarians with Disabilities Act (AODA) in order to ensure that the space was accessible to persons with disabilities, and that the subjects were addressed with respect, empathy, and confidentiality. In this regard, three research assistants with basic training in AODA guidelines were hired to aid with the AGJ’s facilitation and data collection. One of the research assistants had additional training in ‘safe space facilitation’ and ‘active listening’, which translated into specialized skills for listening empathetically, if the need arose.

The research-driven game jam is exemplified by the game jam model of Deen et al. (2014), which examines the parameters of doing academic research via the praxis of game jamming. The researcher was a workshop participant in Deen et al.’s (2014) game jam [4 research] workshop as part of the CHI Conference, 2014. She participated in this workshop, and developed her second game (Garner, Wood, Danilovic, Hammer & Mueller, 2014) based on Deen et al.’s model. This model explores the importance of organizers in creating both a functional environment for subjects in a game jam that serves their research purposes, and an aesthetic experience in which the game jam space is infused with mood, atmosphere, excitement, and joy through a careful curation of the game jam space and game jam activities. This model informed the conceptualization of the AGJ in accomplishing the following: informing the curation of an aesthetic experience for subjects, emphasizing the importance of a facilitator that motivates subjects to progress in the game jam, and informing the research process.

¹² An awareness of the problematic theoretical and conceptual terms ‘safe space’ and ‘inclusive’ means having to also accept their inadequacy in embodying the commitments of inclusive design *in praxis*. These terms refer to idealistic yet flawed concepts, belying the sheer impossibility of creating perfectly supportive and fully inclusive environments for a wide range of human needs and variations. The term “safer space” (Kuppers, 2014, p. 6) more accurately represents the concept’s limitations in keeping with the socially aware *intention* behind it.

2.4.2 AGJ schedule and research questions driving data collection

The Autopathographical Game Jam was held on the weekend of September 20-21, 2014 at the Demo Room-Semaphore Lab, University of Toronto, from 10 am to 7 pm each day. Preparation and organization was logistically complex, from devising an expense budget, renting camera equipment, carefully scheduling game jam activities, ordering catered meals, purchasing supplies and snacks, and coordinating data collection among the researcher and three research assistants. Prior to subjects arriving to the designated room on Day 1 of the AGJ, all workstations were supplied with notepads, pens, colour markers, chocolate energy bars, and neon-coloured toys to add a playful touch to the game jam space. Gourmet pizza and burrito lunches were pre-ordered for subjects, inclusive of everyone's dietary needs. A variety of savoury and sweet snacks, beverages, and coffee were supplied and replenished on a regular basis during the game jam.

The AGJ was structured as a set of assorted activities spread out over two days. On Day 1, subjects arrived to the AGJ room between 10am-noon, signing consent forms and getting colour-coded tags to pin on their lapel. The colour-coded tags designated their preference for data collection anonymization based on 'recording type' (e.g., the green sticker was inclusive of audiovisual recording, the yellow sticker was inclusive of audio recording only, the red sticker designated note-taking of interviews only, no video recording allowed, etc.). The second activity of Day 1 was a 10-minute researcher presentation, where the researcher introduced herself and this study's commitments to the AGJ group, followed by an invitation to subjects to introduce themselves to the AGJ group. The introductions served as an *icebreaker* activity designed to create some warmth among subjects who were, for the most part, strangers to one other.

The introductions were followed by the main activities of the AGJ, namely game jamming (game design) and data collection through semi-structured, informal interviews with individual subjects. That is, while subjects were designing their games, the researcher periodically visited each subject at their workstation to interview them about their authorship process, how they were feeling in the moment, and what stage they were at with their game. Catered lunch was delivered to the AGJ room between noon and 1pm, followed by more game jamming in the afternoon. The last activity of the day were the group discussions, which yielded

creative visual research in the form of drawings and doodles on flip charts, but were also meant to bring subjects together to socialize and bond. The group discussions were organized by dividing the thirteen subjects into approximately three groups (3-4 people per group), whereby each group was clustered around one table. The group discussions were based on the researcher's use of five questions, which were projected on the smart screen for groups to discuss among themselves.

The group discussion questions were the following:

1. Any group discoveries or insights in this Autopathographical Game Jam?
2. Can anyone comment on this process of designing a personal game in a public setting such as a game jam?
3. How is a game jam a vehicle for articulating deeply private or personal stories of illness, disability, trauma? Is this method a contradiction, insofar as it involves telling personal stories in a group setting?
4. How is game design an expressive vehicle for challenging social attitudes about health-related topics such as disability, illness, emotional trauma?
5. Is there something transgressive (which violates boundaries) about making a game on a personal health issue?

The subjects were invited to answer each question within the small group, by either writing something down, doodling, or sketching on a flip chart. The three research assistants recorded the group discussions and followed protocol to ensure compliance with the request some subjects made in their consent form to not have their photo taken. Data collection was underway all throughout Day 1. Among the logistical challenges of data collection were the transfer of the audiovisual recordings from memory cards to a laptop, and the deletion of the memory cards once the transfer was finalized. The researcher was responsible for ensuring that all material was transferred correctly. The group discussions closed Day 1 of the AGJ.

Day 2 followed a structure similar to that of Day 1, with some variation. Subjects were invited to continue with their game jamming (game design praxis). Also, group discussions were not held during Day 2. Instead, Day 2 offered subjects the opportunity to visit each other's workstations, socialize more with each other, and playtest and share their games in the game jam space. The aim, with this schedule, was to create variety for subjects, keep boredom at bay, and facilitate good data coverage of the game jamming process, while also counteracting the intellectual, physical, and emotional stress of authoring a personal game in a game jam. The

objective was to create a fun experience and a comfortable space for social interaction while subjects experimented with the autopathography concept.

This research study was working from five specific research questions applied to the data collection process by way of expanding those research questions into two sets of semi-structured interview questions: informal questions during the Autopathographical Game Jam and semi-structured, individual questions conducted post-AGJ¹³. The five research questions propelling data collection were divided into theoretical (RQ 1-2) and empirical (RQ 3-5) research questions:

- RQ1. Couser (1997) defines the autopathography as an “illness and disability narrative”, which is a “distinct subgenre of autobiographical writing” (ix). Given the above definition of the autopathography in literary contexts, what are autopathographical games? What dimensions or characteristics play a role in their being so categorized?
- RQ2. Given that autopathographical digital games have not been explored before in academic research, how can we gain new understanding of autopathographical games through the following analytic perspectives:
- a) Arts-informed disability studies perspectives¹⁴?
 - b) A combined framework of process and procedurality?
- RQ3. (a) How do autopathographical games express and disclose personal stories of illness, disability, pain, and trauma from the point of view of their authors?
- (b) How does the design process for autopathographical games facilitate the author’s creative expression and disclosure?
- RQ4. How can the design of autopathographical games, as a form of authoring the illness and trauma narrative, function as a vehicle for social justice?

¹³ For both sets of interview questions, please see Appendix I.

¹⁴ Given that one of the strongest research findings was the therapeutic dimensions of autopathographical game design, the medical humanities literature, i.e., autopathography studies and illness narrative studies became more fitting and appropriate theoretical frameworks than disability studies for purposes of theorization.

RQ5. What forms of expression—both of first-person autopathographical narrative, and of intimate personal experiences of pain and trauma—are, and can be, provided through the procedural/processual resources of rule-based digital games?

2.5 Methodological framework for *data collection*¹⁵

This dissertation is situated in the following methodological schools of data collection: *case studies*, *grounded theory*, and *research-creation*, and aims to generate data on the autopathographical game design processes and AGJ face-to-face interaction processes of subjects. A case study is an “in-depth examination of a single instance of some social phenomenon, such as a village, a family, or a juvenile gang” (Babbie, 2007, p. 298); it provides descriptive and explanatory insights on specific structures and processes. The Autopathographical Game Jam is a case study describing the authorship processes of a group of game designers authoring autopathographical games.

There are two main elements in the conceptual orientation of the Autopathographical Game Jam as a case study. The first is that research questions and researcher intuitions direct the collection of data; the second is that the emerging data-driven commitments of grounded theory are used to account for the *discovery* of unexpected patterns and themes in observational data. Founded by sociologists Glaser and Strauss (1967), grounded theory is “theory derived from data, systematically gathered and analyzed through the research process” (Strauss and Corbin, 1998, p. 12). Grounded theory is a theory-building qualitative methodology receptive to the complex and irreducible ways in which meaning-making processes unfold in social reality, but it can also describe a “conceptual ordering” of data, not just generate new theory according to Strauss and Corbin. In the context of a theoretically informed project, it is important to note that grounded theory as a methodology is “designed not to be followed dogmatically but rather to be used creatively and flexibly by researchers as they deem appropriate”, they say (p. 13). For example, this study has drawn on principles of grounded theory by modifying the research process according to insights gained *during* the Autopathographical Game Jam data collection

¹⁵ The methodological framework is divided between *data collection* and *data analysis*. *Discourse analysis* as a methodology for data analysis is discussed in the next section.

process. Indeed, certain emergent themes, such as the apparent therapeutic and confidence-boosting effects of programming praxis and implementation (i.e. logopoiesis), and the use of dense symbolism in the authorship of the fabulized self were noted from the data that escaped the researcher's initial intuitions, and did not align with her preconceived expectations prior to starting data collection.

Creative visual research was employed as a two-fold tool towards a) data collection method for gathering *visual* forms of data generated by subjects, i.e., doodles, sketches, and scribbles/writings, and b) as a means of engaging AGJ subjects in group discussions. All but two subjects were authoring games individually. Engaging in interactive activities alleviated the monotony of individual focus and encouraged subjects to leave their computer workstations to interact with each other. The decision to do creative visual research was inspired by the suggestion that a grounded theory researcher draw on creative, divergent, and non-linear thinking to gain new insights through the research process: "making use of multiple avenues of expression such as art, music, and metaphors to stimulate thinking" and "having fun while doing (research)" (Patton, 1990 as cited in Strauss and Corbin, 1998, p. 13). Strauss and Corbin also argue that grounded theory can tap into the processes of phenomena such as AGD that are otherwise inaccessible through other qualitative methods and methodologies: "qualitative methods can be used to obtain the intricate details about phenomena such as feelings, thought processes, and emotions that are difficult to extract or learn about through more conventional research methods" (p. 11).

The third methodological school central to the objectives of this dissertation—to give back to the community it studies through an active and meaningful engagement with creative and artistic practice—is *research-creation*. As an art methodology, research-creation "integrate(s) a creative process, an experimental aesthetic component, or an artistic work as an integral part of a study" (Chapman and Sawchuk, 2012). Chapman and Sawchuk argue that research-creation, also known as "performative research" and "hands-on engagement," is a "critical intervention" into the normative academic research paradigm privileging scientific validity and argumentation as

the only acceptable tools of production of knowledge and truth¹⁶. Moreover, this study's emphasis on the *authorship process* of autopathographical game design connects with research-creation's "strong potential as a form of intervention precisely due to its often experimental, processual nature" (p. 11).

It is worth emphasizing, as stated in the ethics protocol, that the computer games generated during the Autopathographical Game Jam *do not* fall under the auspices of data; otherwise they would be the legal property of University of Toronto, as per their ethics protocol. Rather, they are considered a form of research-creation, which respects the intellectual property (IP) of the author or game designer. In this vein, the Autopathographical Game Jam case study generated twelve expressive games in a meaningful social space.

2.6 Data collection methods

The data collection methods deployed to document the autopathographical game design process during the AGJ consists of three sets of methods: 1) video and audio recordings of qualitative, semi-structured interviews; 2) creative visual research; and 3) collection of authors' processual artifacts.

The Autopathographical Game Jam was conducive to a multilayered approach to data collection due to its small scale, the small number of subjects, and its compressed time span (two days, approximately seven hours each day). The data was voluminous and captured the experiences of study subjects via textual, pictorial, and audiovisual media. The multiple forms of data allowed the study of the structure of the rendering of the autopathographical narrative in a computer game, which was an important commitment of data analysis.

The collected data consists of two sets of qualitative interviews, photographs, video, audio, and study subject drawings, sketches, and doodles, as well the researcher's memory

¹⁶ The Social Sciences and Humanities Research Council (SSHRC) of Canada has legitimized research-creation as a valid form of research practice in support of funding, defined as "an approach to research that combines creative and academic research practices, and supports the development of knowledge and innovation through artistic expression, scholarly investigation, and experimentation" (SSHRC, 2016).

notes/reflections of the AGJ. The interview testimony, originally captured in video and audio recordings and then transcribed to text, yielded approximately 80% of this study's data. The other 20% of data is comprised of processual artifacts such as sketches, doodles, and photographs of the AGJ.

2.6.1 Breakdown of textual data:

- a) Semi-structured, informal interviews with subjects during the AGJ (transcribed);
- b) Recordings of subject conversations during AGJ playtesting (transcribed);
- c) Recordings of group conversations during AGJ creative visual research (transcribed);
- d) Post AGJ semi-structured, individual, in-person interviews with subjects (transcribed). A few weeks after the AGJ, in-depth interviews were conducted with subjects after they had had some time and distance to reflect on their AGJ experiences. In these interviews, subjects were invited to revisit memories of their experiences making their games during the game jam and share any new insights. The time interval between the AGJ event and the post-AGJ interviews also enabled drawing on the AGJ data to inform the semi-structured interviews accordingly. The post-AGJ interviews were all audio-recorded.
- e) Creative visual research artifacts of sketches and doodles on flip charts acquired during AGJ;
- f) Researcher notes, memory logs, and same-day reflections of the AGJ event;
- g) Subject reviewed 1-2 page 'interview summaries'. This interview summary allowed subjects to make any preferred changes to the interview testimony, check for researcher transcription errors, and the like. These interview summaries were conducted via email correspondence.

2.6.2 Breakdown of audiovisual data:

- a) Researcher's photographs documenting the AGJ event;
- b) Audiovisual (video and/or audio only) recordings of AGJ and post-AGJ interviews;
- c) Design sketches and scribbles/doodles in sketch notebooks provided to study subjects;
- d) Creative visual research (Gauntlett, 2005) consisting of doodles, drawings, and writings using flip charts and colour markers. Gauntlett's "creative visual research" methods engage with media audiences in a "process of making a creative visual artifact"; Gauntlett's method was adapted to engage the subject and capture spontaneous insights by subjects as expressed in the form of a doodle, a scribble, a sketch, or a piece of writing;
- e) Some digital game assets: visual files, jpegs, wireframes, etc.

2.7 Data analysis methodology: discourse analysis and data-driven theorization

This section provides a brief overview of the methodology for data analysis used in this study—discourse analysis—which drove theorization. As was stated in the Introduction, the approach aims to conduct a theoretical analysis grounded in data. This study is committed to an understanding of the structural logic of AGD as the primary organizing principle of the design process. Structuralist and formalist inspired analytic frameworks also enabled a perception of how the authorship process breaks down into various structural components, such as: the internal dialogue of autopathographical game designers that accompanies autopathographical authorship (autopoiesis), the use of metaphor, analogy, allegory, fable, etc. in design of game characters and virtual worlds (fabulopoiesis), and the conceptualization of game narrative into rule-based formal system, programming conditional statements, and in general articulating the design process in formal terms (logopoiesis). The chapter on the Autopathographical Game Jam employed a phenomenological analysis of the social experiences of subjects' authorship processes (sociopoiesis).

Prior to conducting data analysis, the data was subjected to a coding process in order to discern emerging patterns in the data. In this study, the coding protocol consisted of a number of 'codes' discerned from the data, including: creative rendering of the autopathographical narrative into a computer game (a.k.a. fabulopoiesis); subjectivity and the game autopathography (a.k.a. autopoiesis); self-expression through game rules and engineering emotions via a game's formal system (a.k.a. logopoiesis); AGD as therapy (applies across most of the dimensions of poiesis); the Autopathographical Game Jam (a.k.a. sociopoiesis); and finally, the transgressive aesthetics of AGD (a.k.a. the foundations of therapoiesis).

Given that this dissertation's principal data consisted of qualitative interviews and recordings of conversations with thirteen study subjects, this study deploys a form of discourse analysis. In discourse analysis, the objective is to explain how things happen (Potter and Weatherell, 1987) through an analysis of the meaning-making processes of subjects found in verbal testimony. In this case, the verbal testimony in interview passages manifested how subjects make sense of their AGD practice. The verbal testimony was distinguished from literary

citations through careful and explicit bracketing of the subject interview passages. In an attempt to understand the AGD-AGJ process, the interview passages were then reframed, questioned, and linked with other relevant data and theory.

Each chapter harnesses an approach to data analysis that pertains to the specific substantive content addressed therein. Chapter 3 on the foundations of therapoiesis in AGD begins with an analysis of the transgressive aesthetics of AGD, followed by a description of the logical relationship of the tetrad of therapoiesis and the first-, second-, and third-person perspectives. Chapter 4 on sociopoiesis primarily harnesses Martin Buber's (1923;1958) theory of dialogue to explore the poetics of collective making and the emergence of a therapeutic community during the Autopathographical Game Jam. Chapter 5 on autopoiesis shifts between a narratological conceptualization of the autopathographical narrative and the game autopathography, drawing on the literature of autopathography studies, autobiography studies, and game studies. Chapter 6 on fabulopoiesis draws on literature on analogy, metaphor, and heuristics to theorize on the construction of the fabulized self—a fantasy figure and imaginary self. Finally, chapter 7 on logopoiesis provides an analysis of AGD as a therapeutic-analytical tool steeped in literature on general computing, logic, and ludology.

2.8 Data generalizability

The deployment of generalizability in this study adheres to a humanities model as opposed to a quantitative social science model typically more concerned with replicability, transferability, and sampling as defining properties of scientific generalizability. The authorship process of AGD, which is applicable to any creative praxis such as painting, film, design, etc., is a meaningful form of structuring human subjectivity, and as such is not immune to generalization. Thus the data can be related to broader experiences of mental illness, emotional trauma, and disability as untapped aesthetic sources of computational creativity in the technological and digital age. In this study, generalizability emphasizes the relationality of research as evidenced in the interconnections, resonances, and dissonances of the interpreted data in relation to broader issues, such as AGD in general and the experiences of autopathographical game designers outside of this study's parameters as well as the Autopathographical Game Jam as a type of game jam. In other words, generalizability bridges the gap between subjects

authoring autopathographical games and AGD in general as a novel meaning-making process and the AGJ in general as a compelling new platform, both engendered by nascent technologies.

Chapter 3

The Foundations of *Therapoiesis* in Autopathographical Game Design

3.1 Introduction

In order to foster *therapoiesis*—therapeutic effects through creative making (poiesis)—the autopathographical game design (AGD) process in the context of an autopathographical game jam (AGJ) seems to possess a two-part foundational structure. The first part of this foundational structure is the dialectical clash of what appears to be two antithetical *semiospheres* in autopathographical game design—the *autopathographical narrative* and the *computer game*—that when integrated produce the transgressive aesthetics of AGD. That is, the integration of these two semiospheres may prompt study subjects to transgress the ‘aesthetic code’—a set of aesthetic conventions’—of each semiosphere through the novel creative grammars offered by the autopathographical game design process. These aesthetic violations appear to galvanize the creative processes of subjects—they initiate AGD action much like a key starts a car engine. In the second part of this foundational structure—the *tetrad of therapoiesis*, this study reveals a structural logic to autopathographical game design whereby subjects render their autopathographical narratives into a computer game, comprised of the therapoetic relationship of four dimensions of poiesis in AGD-AGJ: sociopoiesis (collective making), autopoiesis (self-reflective and self-reflexive making), fabulopoiesis (imaginative self-reinvention), and logopoiesis (self-analysis and self-mindfulness through a calculated making).

This chapter’s data-driven theorization is followed by the visual demonstration of the tetrad of therapoiesis in AGD through two Venn Diagrams, each of which represents a logical relationship in AGD. ‘Venn Diagram I’ demonstrates the relationship of four dimensions of poiesis which cater to healing: sociopoiesis, autopoiesis, fabulopoiesis, and logopoiesis. This first diagram maps onto one other logical relationship: ‘Venn Diagram II’ shows the integration of the first-, second-, and third-person perspectives contributing to healing in specific ways. These two Venn Diagrams diagrammatically depict the emerging theory of game design therapoetics—the AGD process as a form of self-care, self-understanding, and therapy—which importantly connects to Aristotle’s triad of meaningful human activity of *praxis*, *poiesis*, and *theoria*.

This analysis draws on a theoretical framework of Aristotle's (c. 335 B.C.; 1961) *Poetics*, Vygotsky's (1971) *Psychology of Art*, Lotman's (1984; 2005) *On the semiosphere*, as well as theoretical insights from linguistics and semiotics (e.g. Chomsky, 1966 & 1968; Eco, 1989), game studies (i.e. Caillois, 1958/2001; Juul, 2011 & 2013; Sicart, 2015) and autopathography studies (i.e. Hawkins, 1999; Frank, 1995; Couser, 1997; Jurecic, 2012).

3.2 Part I: The 'clash' of the two semiospheres: the autopathographical narrative and the computer game

Autopathographical game design integrates two semiospheres or meaning-making spaces—the autopathographical narrative and the computer game. This integration may also be conceived of as a 'clash'—a dialectical process that energizes the creative design processes of study subjects. Thus, AGD is taken to be semiotic process that involves the game author employing a suite of what are called 'creative grammars'—formal, aesthetic, rhetorical, and/or compositional 'codes' while designing within a frame of interpretive conventions. A frame of interpretive conventions applies to each semiosphere, the autopathographical narrative and the computer game, which the thirteen subjects breach through creative experimentation with form and content. Put differently, they mobilize the creative grammars of the AGD process in order to violate the 'aesthetic code' of both semiospheres. In order to understand the semiotic interplay of form and content in autopathographical game design, this analysis has adapted the concept of the semiosphere, "a semiotic space, outside of which semiosis cannot exist" (p. 205), which was coined by semiotician Juri Lotman (1984; 2005):

We have in mind a specific sphere, possessing signs, which are assigned to the enclosed space. Only within such a space is it possible for communicative processes and the creation of new information to be realized (p. 207).

A semiosphere is essentially a meaning-making space where signs and symbols are interpreted based on a given social, historical, and cultural context. Lotman goes further in his analysis however by assigning a range of meaning-making possibilities to this semiotic space based on dominant meanings, which radiate from a centre to the far edges of that centre, where presumably, they become more diffuse and less concentrated. That is, the semiosphere is bound by a structure comprised of a core and a periphery:

The division between the core and the periphery is a law of the internal organization of the semiosphere. The dominant semiotic systems are located at the core (p. 214).

The semiosphere is a useful theoretical tool for differentiating how AGD designers expand the vocabulary of computer game design through a set of creative grammars that belong to the different semiotic realms of the autopathographical narrative and the computer game. Lotman's concept of the semiosphere echoes cognitive scientists Hofstadter and Sander's (2013) theorization of mental categories based on the human ability for analogical reasoning for "shap(ing) our interpretations of situations..." (p. 257). The analogy that Hofstadter and Sander use to describe an expanding mental category, which maps to a semiosphere in Lotman's terms, is a metropolis with an old town (core) and a suburban sprawl (periphery). The mental category morphs, expanding from its core to its periphery through new analogies, akin to concentric rings in a growing tree. Both metropolises and mental categories evolve with time and use: "they exhibit a structure that is the result of repeated acts of extension, and in the case of categories, each new extension is due to some perceived analogy" (p. 62). As such, categories do not have precise boundaries according to Hofstadter and Sander. Humans are not content with stale concepts; rather, Hofstadter and Sander state that "the human mind is forever seeking novelty" (p. 64). The novelty of the autopathographical game is that it integrates two semiospheres, each of which possesses specific properties shaped by socio-cultural and historical forces.

3.2.1 Semiosphere I: the autopathographical narrative

Before analyzing how autopathographical game design produces transgressive aesthetics via the interplay of form and content, consider the first semiosphere—the autopathographical narrative. The autopathography is the first-person narrative of illness and disability, which is a subtype of autobiography. Hawkins (1999) originally coined the term 'pathography' to refer to the autobiographical illness narrative—an author's narrative describing his/her personal experiences with illness in book-length form. Hawkins describes the pathography as a "form of autobiography or biography that describes personal experiences of illness, treatment, and sometimes death" (p. 1). Yet, pathography as an existential self-narrative is nevertheless one that is ultimately healing for the author: "pathography provides a kind of vicarious support group, offering some guidance in the bewildering and often frightening terrain of serious illness" (p. xiii), states Hawkins. Couser's revision of the term to *autopathography* highlights its important

ontological status as a self-reflective and self-reflexive (self-referential) account of illness or disability. The literary autopathography takes the form of memoir, essay, diary, non-fiction, or fictionalized account (Couser, 1997), and becomes a narrative-telling¹⁷ necessity due to corporeal dysfunction and disease. Indeed “for some, disease annihilates selfhood, or ends life, before they can undertake a narrative...” (p. 5); hence the need for the autopathography as “an agent of recovery as well as self-expression” (p. 4). Medical sociologist Arthur W. Frank (1995) frames the auto-illness narrative as a platform for expressing the silenced voices of the seriously ill: “seriously ill people are wounded not just in body but in voice. They need to become storytellers in order to recover the voices that illness and its treatment often take away” (p. xi). Frank affirms that “stories have to repair the damage that illness has done to the ill person's sense of where she is in life, and where she may be going. Stories are a way of re-drawing maps and finding new destinations” (p. 53). Couser (1997) adds that the autopathography is not the exclusive domain of writers: “...representation of illness and disability may also take the form of self-portraiture, as do the paintings of Frida Kahlo, and audio or video recording” (p. 6). ‘Art autopathography’ (Derby, 2013) is the latest expansion of this first semiosphere to take hold in theoretical circles, where it is framed as a visual interpretation of the autopathographical narrative exploring how the autopathography translates into the pictorial and visual realm.

The autopathography is also analogous to Aristotelian tragedy given an autopathographical author’s objective of sublimating first-person experiences of *pathos*—suffering, emotion, and pain—into an aesthetic form. Russian social psychologist Lev Vygotsky (1971) uses the analogy of ‘grapes to wine’ to illustrate his claim that art is ultimately transformative of its material (p. 243), in the same way that AGD is transformative of the author’s first-person experiences. Vygotsky’s inspiration for this analogy comes from a religious reference: the transformation of water into wine in the Gospel. He suggests that the true purpose of art is to achieve transcendence: “art’s true nature is that of transubstantiation, something that

¹⁷ This dissertation employs the more precise term *narrative-telling* to mean a form of story-telling that subsumes a first-person perspective of a narrator recounting a narrative. This analysis distinguishes between ‘story’ and ‘narrative’ by arguing that ‘story’ qua ‘sequence of events’ is an abstract term, often used to conflate story with narrative by not semiotically accounting for the narrator’s perspective, which ‘colours’ and ‘arranges’ (biases) the story in a particular way. The chapter on *Autopoiesis* provides a deeper narratological discussion of this issue.

transcends ordinary feelings” (p. 243), which also evokes the Kantian sublime. AGD study subjects do not necessarily achieve transcendence through AGD. Rather, the point is to illustrate Vygotsky’s analogy as an effective one for how first-person experiences transmute through the AGD process into something that is different from the first-person experience, yet may also evoke semblance in powerful ways. The human need to alleviate pain and suffering is great, and storytelling is one tool for sublimating pain into aesthetic beauty. Contrary to Susan Sontag’s (1978) critique of the endemic use of the illness metaphor in literary fiction and medical discourse, which she claims fuels the stigmatization of various illnesses, many of this study’s subjects believe in the therapeutic and revitalizing effects of symbolic expression in AGD. They do not shy away from the use of metaphor, analogy, and other forms of symbolism in order to make new meaning and recover lost meaning. Couser (1997) states: “if illness and disability are reminders of our mortality and frailty, narratives of those conditions are testaments to our resilience and vitality” (p. 295). Thus, if serious or debilitating illness evokes the stark meaninglessness of human existence and the nearness of death, then the autopathography and all forms of narrative, art, and poetry are an antidote to nihilistic urges by recovering meaning through symbolic expression.

The autopathography is thus an author-centric medium and a powerful therapoetic tool for the author’s recovery and restoration, in both material and symbolic terms. Thus, Couser claims that “today especially, narratives of illness and disability are helping us recover our bodies and restory our lives” (p. 295). Subject Sam affirmed that “autopathography is way better for the creator (or) audiences that relate to the creator”. However, in their self-reflexive dimensions, the autopathographical games exemplified in this study have not been taken to be a form of self-engulfment. Despite their focus on the author’s therapeutic restoration, autonomy, and agency, the autopathographical games are nevertheless meant to be shared with others in limited ways (addressed in the chapter on *Sociopoiesis*).

3.2.2 Semiosphere II: the computer game

Consider the second semiosphere—the computer game and some of its distinctive formal properties. A brief definition of ‘game’ takes it to be a system of fictionalized conflicts or challenges, inviting voluntary human action in the form of play, and at least ostensibly without

‘real-life’ consequences¹⁸. That is, “an activity is considered ‘play’ when one’s actions do not have significant real-life consequences” (Laurel, 2004, p. 4). The main activity of games is playing them in a “magic circle of play” where special rules apply (Huizinga, 1949; 2002). When players dynamically interact with a game and its aggregate of representational objects or ‘instances’ such as game worlds, characters, narratives, processes, rules, graphics, colors, sounds, etc., games become dynamic systems of “meaningful play” (Salen and Zimmerman, 2004).

We experience games through their aesthetic form in the feelings and pleasures they engender in the game player (Kirkpatrick, 2011). Brody emphasized the primacy of feelings engendered by gameplay and how computer game designers can author for aesthetic effects:

I can tell my story, but I want people to *feel it*¹⁹. And that’s impossible without doing it in an artistic way. Games are my art form.

Brody suggests that he prefers to use the distinct symbolic affordances of game design as opposed to telling his autopathographical narrative, as he believes the latter does not translate his first-person experience as strongly to gameplay²⁰. He is emphasizing one of the distinctive formal properties of computer games as emergent and dynamic systems of play whereby the player embodies a game character and performs a role in a virtual world, hence the player’s ability to “feel the story”. The distinctive formal properties of computer games enable the design of a system of game rules and constraints, the implementation of which in software code via objects such as images, sounds, strings, integers, conditional expressions, variables, etc., structures first-person gameplay experiences. The gameplay experience that the player inhabits

¹⁸ You can get physically hurt by playing a soccer game on a field, presumably by falling down or crashing into another player or being hit by the ball, which are all examples of ‘real-life’ consequences where the laws of physics affect the human body moving in space. You cannot get physically hurt by playing a soccer game on a computer platform, at least not in the same way as on the field (Juul, 2011).

¹⁹ In verbal testimony, Brody lengthened the vowel in “feel” to put emphasis on this word when speaking.

²⁰ Description and narrative-telling do not always translate to action-based gameplay, with some exceptions, such as in the game *Dungeons and Dragons*, which integrates description and narrative-telling into gameplay.

in real-time is unique to the computer game form and there is an artistic way to render this experience that surpasses narrative-telling, as per Brody's suggestion.

Gameplay experience is essential to our understanding of the aesthetic effects of computer games. Computer games are virtual play regimes (VPRs) in that when they are played, the dynamic rules of gameplay become differentiated from the static rules of game design²¹. In other words, gameplay is subject to *emergence*, a complex dynamic process between the player and the game (Salen and Zimmerman, 2004; Juul, 2011). Emergence is responsible for generating different experiences for the player than those perhaps intended by the game designer, which is why playtesting is an important iterative process of game design. Game rules are the conditions, laws, and standards of the game's virtual world that guide or dictate a player's decision-making process. Generally speaking, a rule, whether internal or external to games per se, is a condition which stipulates that if an actor performs an action, then what follows is the consequence of their action, and if they deviate from the designated rule, then what follows is a different consequence. In games, "rules describe what players can and cannot do" (Juul, 2011, p. 55).

Mainstream videogames such as first-person-shooters assign a range of properties and attributes to games, such as agonistic (competitive) play structured by explicit and implicit rules (Salen and Zimmerman, 2004; Juul, 2011). Explicit rules are generally implemented through quantitative methods that measure player progress such as scoreboards (Juul, 2011), health indicators, incentives, rewards, and a win-lose outcome for game players²². Implicit rules come in at least two types: 1) the VPR's formal constraints; and 2) the VPR's 'unwritten rules'. The VPR's formal constraints emerge out of the inherent limits and determinants of a virtual environment; they are the conditions of inhabiting a virtual world—moving an avatar's body and perspective in a given virtual space, picking up virtual objects, clicking on game resources such

²¹ The "mechanics, dynamics and aesthetics" framework of game design proposed by Hunicke, LeBlanc and Zubek (2004) differentiates between the three principal formal properties of games. <https://www.cs.northwestern.edu/~hunicke/MDA.pdf>

²² To make a distinction, the explicit rules are not always *explicitly noted* in the game through a written description or tutorial. Often, players learn or 'figure out' some or all of the explicit game rules through playing the game itself.

as maps, inventories, and so forth. The VPR's unwritten rules are "inferred by the player from the environment", such as the concept that guns kill (Juul, 2011, p. 47). According to Salen and Zimmerman (2004), implicit rules qua unwritten rules are found in every game; these rules enforce etiquette, good sportsmanship, and other implied rules of proper player behaviour (p. 4, Ch. 12). Grimes and Feenberg (2009) argue that in MMOGs such as *World of Warcraft*, community norms come to construct implicit rules qua unwritten rules guiding player behaviour in games: "formal rules are established by the game engine and operators as well as the player community..." (p. 106), and "community norms come to operate as systems of social control that work to discipline, exclude, or otherwise classify players and behaviours" (p. 115).

French sociologist Roger Caillois's (1958; 2001) distinction between *ludus* and *paidia* is fruitful for the purposes of showing the ways in which games visibly or invisibly instantiate the rules and constraints of gameplay. Caillois argues that *ludus* is "calculation and contrivance" (p. 31), imposing a competitive and rule-based structure on play, which is opposed to the less-structured, open-ended, and improvisational *paidia*. In contrast to *ludus*, Caillois believes *paidia* is more predictable, bearing an "impromptu and unruly character" (p. 28). *Paidia* is "tumultuous and exuberant" (p. 31) in a manner similar to young animals playing, or children engaged in make-believe play. Alex invokes this distinction between *paidia* and *ludus* in terms of the type of ambient, experiential gameplay he prefers to design:

I kind of like things that are just sort of experiences rather than goals.

He indicated that the quality of his design experience is diminished when he has to be more rigid and precise:

You have to design a lot more and you have to be more rigid with your structure and the way I work isn't very rigid. I kind of just flow with it, so I end up with these weird ambient things because I don't design rules often.

Alex prefers to be more relaxed when making games, especially within the constraints of a game jam, which may not allow for a more rigidly implemented system of game rules. To "flow with it" in this context means to design a looser structure for the game that is more ambient and atmospheric, hence more influenced by *paidiaic* than *ludic* play. In some sense, Alex's approach to computer game design is itself playful and *paidiaic*; the style of design Alex deploys is to

design for a less defined game structure towards free-form play that is not bound by strict game objectives, rules, metrics, and a quantified win-lose outcome. The outcome of Alex's specific paidiaic game design style, which ensures a moody and atmospheric game, is unorthodox and unconventional, or as he says, "weird". This assumption that unstructured paidiaic play is at odds with the rigid structure of a rule-based game connects to Callois' dichotomy of ludus (games) and paidia (play).

Play and game scholars have criticized Callois's binary classification because games have elements of both: they come in both ludic and paidiaic form. Therefore, as virtual play regimes, computer games, whether ludic or paidiaic, all have game rules (Pearce, 2009). In other words, paidia is still ruled through a variety of implicit and qualitative determinants that function as constraints. The rules that game designers author in paidiaic play are hidden in that they are less visible to the game player, whereas in ludus, these same rules may be perceived as more visible because they are instantiated in an explicit rule that must be met through a scoreboard, quantifiable tracking system, or a win-lose outcome typical of agonistic play. In paidiaic play, there is no scoreboard keeping track of play choices and you may play without 'penalty' for choosing one action over another. Thus, the rule in paidiaic play is less visible because the player progresses by dynamically engaging in a virtual world, moving or creating digital objects, such as in the atmospheric games *Flower* and *Journey*, which offer open-ended gameplay of "exploratory metaphor and symbolism" (Mullaney, 2013).

Games are also ruled (ludic) and unruly (paidiaic) in ways that reveals how some of us perceive rules as absolute, as not being able to be broken or transgressed. Callois's classification of paidia and ludus assumes that play governed by rules in a game is controlled (ruled) whereas play without rules is 'out of control' (unruly, free). This assumption is incorrect; many games with strict and outward-visible, rule-based systems provide opportunities for players to practice rule-breaking in countless ways, including cheating (Consalvo, 2007) and finding loopholes in the game's design to bypass some of the rules (Juul, 2011).

It is useful to briefly compare Lotman's semiosphere to how games scholar Jesper Juul (2011) classically defines a game according to the elements of ludus and paidia. In the following diagram (see Figure 1 below), Juul places a game with "fixed rules" and a "variable outcome" at

the center, which maps to the core of the semiosphere in Lotman's terms. There are two other categories that deviate from the core of the semiosphere: "borderline cases", which occupy the next concentric level of the semiosphere and "not games", which exist at the periphery of the semiosphere. Borderline cases are "open-ended simulations" such as virtual worlds (i.e. *Second Life*) with no goal-oriented objectives, and pen and paper role-playing games with flexible rules; while "not games" represent free-form play and storytelling among other paidiaic activities (p. 43-45).

Figure 1: Jesper Juul's (2011) classification of games: "on the borders of the classic game model" (Figure 2.10, p. 44).



The next distinct formal property of both board games and computer games is that they are at their most abstract level *formal systems* (Salen and Zimmerman, 2004; Haugeland, 1985).

“Many familiar games—among them chess, checkers, Chinese checkers, go, and tic-tac-toe—simply *are* formal systems”, states John Haugeland (p. 48). A game’s formal system consists of authored “expressive processes”, i.e., the process-driven dimensions of games such as player actions, rules, procedures, and gameplay mechanics organized by the game’s “operational logics” (Wardrip-Fruin, 2009). Haugeland suggests that a game’s formal system is “self-contained” because its “legal moves (which are) token manipulations permitted by the rules” are not affected by the world *outside* of the system’s boundaries of play: “it makes no difference to a chess game, as such, if the chess set is stolen property or if the building is on fire or if the fate of nations hangs on the outcome—the same moves are legal in the same positions, period” (p. 50). Game design scholars Salen and Zimmerman (2004) differentiate three ways of understanding a game of chess, in effect accounting for the semiotic dimensions of gameplay: a) as a *formal system*; b) as a *cultural system*; and c) as an *experiential system* (p. 5). For example, chess as a formal system is closed and self-contained in that meaning-making is contained to the game’s “legal moves” as per Haugeland’s point; chess as a cultural system is open if gameplay is influenced by a given social, cultural, and historical context, accounting for the social rules governing game player behaviors as per Grimes and Feenberg’s (2009) point; and chess as an experiential system (closed and/or open system) may account for either the way in which a player experiences their strategic moves in the game (a closed system), or the player’s emotional life and reputation as chess player (open system) as per Salen and Zimmerman’s point (p. 5).

Consider now the aggregate of representational material objects, which are the *multimedia* ‘instances’ in which a game is rendered. Lukas, who made a game about his experiences with extreme shyness at the Autopathographical Game Jam, suggests that the rich possibilities offered by the multimodality of games is the first reason why he was interested in authoring an autopathographical game:

I think the first fact is that games are more interesting than just the picture, the still picture or text alone.

As *multimodal* forms of expression, games have a significant advantage over other media that are not as richly diverse in the modes of communication they offer. The human universal process of meaning-making occurs on a spectrum of modes of communication, from the gestural mode of speech to the more formalized modes of creative writing, music, and visual art. The term

‘multimodality’ signifies the use of multiple modes of communication within one semiotic text. For example, multimedia art, film, and computer games are multimodal in that they involve multiple modes such as text, sound, image, colour, performance, etc. Kress (2010) argues that multimodality is the use of more than one type of signifying mode to communicate meaning most effectively to the receiver. Different modes “(do) different kinds of semiotic work” (p. 1). Each signifying mode has its own distinct affordances: “writing *names* and image *shows*, while colour *frames* and *highlights*; each to maximum effect and benefit” (p. 1) and when integrated, their unified effect can be a powerful form of communication according to Kress.

Multimodality allows AGD designers to express their first-person experiences through multiple modes of communication, e.g., visual, spatial, temporal, aural, procedural, textual, performative, etc. and harness their unified effects to great benefit, both for themselves and for their player-audiences. Ellie, who authored a game about insomnia using the text-based game design software *Twine*, pointed to the benefits of multimodality in games. Ellie argued that *Twine* is:

Like the next generation of journaling...Making journaling relevant in the digital age.

In this quote, Ellie refers to the use of *Twine* to author an interactive diary, which can be played and performed as opposed to simply read. She suggests that the unified effect of *Twine*’s two primary modes, namely the textual mode of writing and the interactive/performative mode of gameplay, are constitutive of a novel diaristic form. According to Ellie, the procedural and performative mode of games, which connects to player choices, vis-à-vis other media such as movies or books is a distinct affordance of games that elevate their self-therapeutic potential:

When you’re playing a game it’s more visceral than a book or a movie because you have choice. You have the power in a game to actually change, to feel that you can change something in your condition in some way.

Both Ellie and Lukas emphasize the general resources of computer game design and how they contribute to autopathographical game design, suggesting that those distinct affordances of interactivity, the ability of the player to choose, the multimodal richness of games that transcend image and text alone, are intrinsically tied to their aesthetic and therapeutic effects.

3.2.3 The two semiospheres: a brief comparison of key similarities and differences

Consider some key similarities and differences between the two semiospheres, the autopathographical narrative and the computer game, in order to understand why their integration leads to aesthetic transgression. Ellie, who designed a *Twine* text-based interactive narrative based on her first-person experience with insomnia, stated the following about games:

Games are traditionally whimsical and an escape, a form of escape from your problems. For Ellie, as for many others, games are perceived as an escape from the uncertainties and troubles of life, whisking players away from the arduousness of living, pain, and suffering. We enter a fictional world where we inhabit a character and overcome the game's fictional conflict. But illness and pain are 'serious' subjects that cause human suffering, hardly purveyors of fun. Thus, there is an assumption that illness and pain deserve to be treated with dignity and sensitivity. It may indeed seem distasteful and perverse to treat illness and suffering frivolously, as 'fun and games'. Here we have a seeming opposition between the two semiospheres of the autopathographical narrative and the computer game. Reflecting on autobiographical art games, Poremba (2007) confirms that "games are not (popularly) expected or encouraged to reflect back on our real lives—as is implicit in the popular phrase 'it's just a game'" (p. 706). Hence, the unorthodox nature of the game autopathography in rendering the gravity of illness, trauma, and disability as a gameplay experience. From an aesthetic perspective, Vygotsky (1971) would call the game autopathography a "juxtaposition of extremes", in this case, the juxtaposition of gravity and frivolity—a dialectical strategy for achieving "affective contradiction" through "contrasting sensations and emotions" (p. 143).

Consider some differences between both semiospheres. The elitist aesthetic distinction between 'high culture' and 'low culture' refers to institutional norms and cultural regimes governing the production of creative works of art and poetry. It is what Rancière (2007) calls "poetic hierarchies"—the subordination of some forms over others through cultural or institutional discourses (p. 76). High culture denotes highbrow and sophisticated forms of art, music, performance, or literature, while low culture denotes popular, lightweight, 'uncultured' forms of entertainment such as 'schlocky' blockbuster Hollywood films and 'mindless' computer

games²³. At present, computer games may not even qualify as low culture in some domains; rather, they are part of “gutter culture” according to games sociologist Bart Simon (2017):

Play, and maybe especially computer gameplay, seems baser than we give it credit for; it is an ugly gutter culture. It is dirty, smelly, crass, stupid, pointless, icky, and banal. It is not noble, it is the culture of last resort. (p. 617)

Simon follows his statement by asserting that “I didn’t say the gutter was without its romance”, referring to the place of tension that videogames occupy in society and culture. Galloway (2006) similarly states that “video games reside today in a distinctly lowbrow corner of contemporary society” (p. 85)²⁴. Additionally, the stigmatization of videogames as violent, addictive, and morally bankrupt has de-legitimized their cultural status, which is hardly a new phenomenon with emerging aesthetic forms. Witness the moral panic that cinema engendered in its first decades branding it a lowbrow cultural form and undermining its seriousness as an art form. American conservative politicians and religious leaders maligned film in the 1930s by calling it a nefarious and corrupting tool of Communist propaganda, which ushered in decades-long film censorship in Hollywood implemented through the Production Code (Cook, 1990, p. 296-300). Violent and addictive videogames still have traction in the eyes of the general public due to their constant media dissemination and the vilification of the first-person shooter (FPS) videogame genre, which accrues rewards for the player based on a number of simulated ‘kills’. The shooter game is thus the subject of moral panic buoyed by long-standing academic research that links violent videogames to human aggression and addiction. Academic research on the negative psychological and behavioural effects of videogames on humans has either been debunked, or at best framed as inconclusive²⁵.

²³ The cultural genealogy of aesthetic categories is an important discussion beyond the scope of this dissertation. See cultural theorist Sianne Ngai’s (2012) book “Our Aesthetic Categories” for more on this dichotomy of high and low culture.

²⁴ The low-brow status of videogames is slowly changing. See Felan Parker’s (2013) work on “indie games”: http://www.digra.org/wp-content/uploads/digital-library/paper_100.pdf

²⁵ Karlsen (2015) claims that “within the research community there is no consensus about the effect of violent digital games on society”, adding that “the overall bulk of researchers currently conducting research on this issue seem to regard the link as negligible” (p. 18).

The FPS game has been unfortunately used to stereotype videogames, which are extremely diverse in their genres, ranging from action-based shooters, iconic platformers such as *Pac-Man* and *Super Mario Bros.*, character-driven RPGs (role-playing games) such as *World of Warcraft*, exploratory art games such as *Flower* and *Journey*, strategy games such as chess and *Solitaire*, to neon-bright puzzle games such as *Candy Crush Saga*. The commercial videogame also stands in stark contrast to the advent of serious games (Abt, 1970) governed by the institutional norms of academia, medicine, and education, the so-called *gamification* of health, education, and research. Videogames in these contexts enjoy higher cultural status due to their instrumentalized nature and the perception that they contribute to the human betterment of society²⁶. Therefore, given the negative social, cultural, and psychological effects associated with violent and addictive videogames, one can also say that videogames are perceived as transgressive in content in their ability to produce fear and loathing²⁷.

The autopathographical narrative is a rather earnest aesthetic form relative to videogames, and straddles both serious literature, i.e., feminist writer Audre Lorde's (1980; 2006) *The Cancer Journals* and popular culture, such as the "therapeutic narratives of self-help and popular psychology" (Illouz, 2008, as cited in Jurecic, 2012, p. 6). Couser (1997) writes that the autopathographical narrative is teeming with potential for being transgressive: "Middlebrook's advanced account of her (cancer) illness illustrates how such narrative conventions may fail to serve the needs of the ailing person —how illness experience may exceed or defy generic conventions and cry out for innovation" (p. 75). Couser adds that, on the contrary, the autopathography fails to deliver on this front; despite the autopathographical narrative's potential for formal innovation, he affirms that in their emphasis on plot and closure, "personal narratives of illness and disability are in general less transgressive and formally innovative than one might expect" (p. 293).

²⁶ For an in-depth and seminal discussion of *serious games*, please see Clark C. Abt's (1970) book "Serious Games".

²⁷ There is a larger discussion of the connection between misogyny and gender representation in popular FPS videogames, which add to the perception of videogames as transgressive, which is beyond the scope of this dissertation. Please see Jennifer Jenson and Suzanne de Castell's (2013) scholarship: "Tipping Points: Marginality, Misogyny and Videogames".

A shared property between the autopathographical narrative and the computer game is that they are cultural phenomena of the post-modern era, which pushed back against institutional biases and elitist distinctions between art, literature, and aesthetic beauty on the one hand and popular entertainment and everyday life on the other, in part influenced by the Dada and Situationist movements (Lievrouw, 2011, p. 35). For example, Hawkins (1999) framed the pathographical narrative as a “modern adventure story”, invoking the book-length personal accounts of illness that emerged after WWII and flourished with the civil rights and liberation movements (Couser, 1997, p. 5). The computer game as a post-modern innovation of computing technology has at times enjoyed a similar cultural status as the autopathographical narrative in its ability to collapse high culture and low culture distinctions. For example, game design scholar Mary Flanagan (2009) compares ‘radical’ forms of contemporary game design to the Dada and Surrealism avant-garde movements of the 20th century, the latter of which are founded on interventionist tactics deployed by artists who “often have social or political goals, and may seek to open up dialogue by transgressing the boundaries between art and everyday life” (p. 11). Flanagan observes that “some artistic intervention (thus) takes the form of performance, parody, simulation, game, activist, and ‘hacktivist’ strategies” (Flanagan, 2009, p. 12). An extension of post-modernism, the twin-duo of the DIY (Do-It-Yourself)²⁸ revolution and online participatory cultures further highlight the crumbling boundaries between high culture and low culture with the advent of user-generated content, blogging, vlogging, hacking, tinkering, maker movements, and game jams, and their connection to confessional modes of self-expression (especially in blogging and vlogging).

Indeed, the autopathographical narrative and the computer game also share the outcome of empowering their authorship base. Anna Anthropy (2012), the game designer of the computer game *dys4ia*, which explores her personal trials with hormone replacement therapy, emphasizes that computer game authorship gives voice to her unique experiences as a transgender woman, such as wrestling with the medical system in *dys4ia*, which are ignored by mass-market

²⁸ For an excellent discussion of DIY game development by female developers, please see game design scholar Emma Westecott’s (2013): Independent Game Development as Craft. <http://journals.sfu.ca/loading/index.php/loading/article/viewArticle/124>

computer games: “I have to strain to find any game that’s about a queer woman, to find any game that resembles my own experience” (p. 2). From the perspective of the alternative and experimental games scene, autobiographical and autopathographical games represent the recent emergence of a game authorship base that reflect narratives sorely missing in commercial videogame production. As a result, the autopathographical writer and autopathographical game designer both harness autopathographical expression towards creative expression, healing, and reclaiming, through poesis, one’s sense of identity. Computer game design has something unique to offer in the realm of aesthetics, engineering, and design that tends towards healing, self-understanding, and self-insight, and which autopathographical writing and other forms of ‘art therapy’ may not be able to offer.

3.3 Transgressive aesthetics: creative experimentation with form and content

Consider the connection between the integration of the two semiospheres and feelings of strangeness, weirdness, and discomfort that this process appears to elicit in subjects. If, as philosopher Galen Strawson (2000) suggests (invoking Damasio, 1999), “emotional phenomena are the fundamental source — the fundamental engines — of self-experience” (p. 43), then it is no surprise that subject Yasmine suggests that the driving force behind a successful game is to draw on the first-person experience as a springboard for artistic creation:

I think the most effective games start with yourself; I find that any kind of medium is just, the more personal you get, (the better).

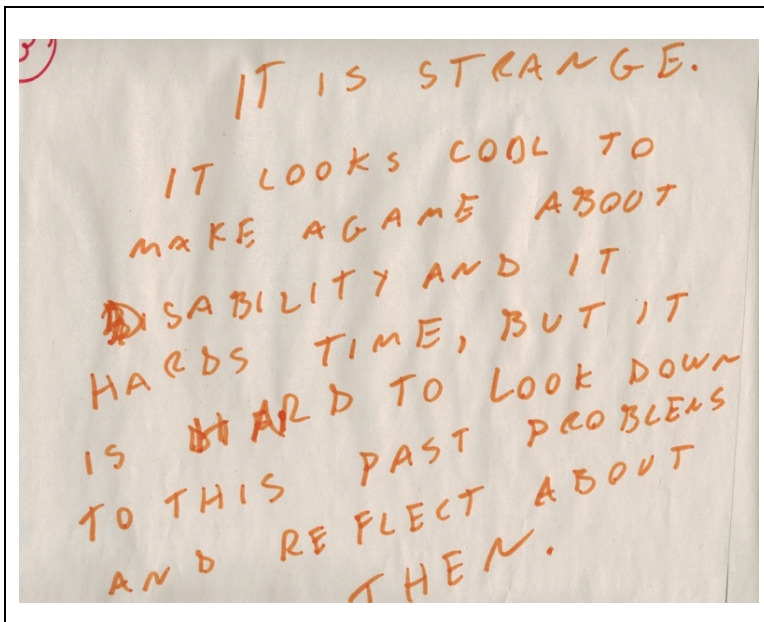
Game designer Anna Anthropy (2012) champions strong subjectivity in videogames as a method for self-actualization. In designing videogames that tell personal narratives about illness, love, and sexuality, Anthropy says that independent game designers have the creative freedom to be “silly and weird and creative and personal”:

What videogames need right now is to grow up. The videogame industry has spent millions upon millions of dollars to develop more visually impressive ways for a space marine to kill a monster. What they’ve invested almost nothing in is finding better ways to tell a story, and in exploring different stories to tell. That’s for us to do: the people who don’t have to sell thousands of copies of a game to break even, who aren’t obliged to fill their games with eighty hours of content, who are beholden to no one, who are free to be silly and weird and creative and personal. (p. 160)

When asked about their experiences designing a computer game autopathography, an anonymous subject connected the AGD process to ‘strangeness’ and discomfort (see Figure 2 below):

It is strange. It looks cool to make a game about disability...but is hard to look down to these past problems and reflect upon them.

Figure 2: Autopathographical Game Jam doodle



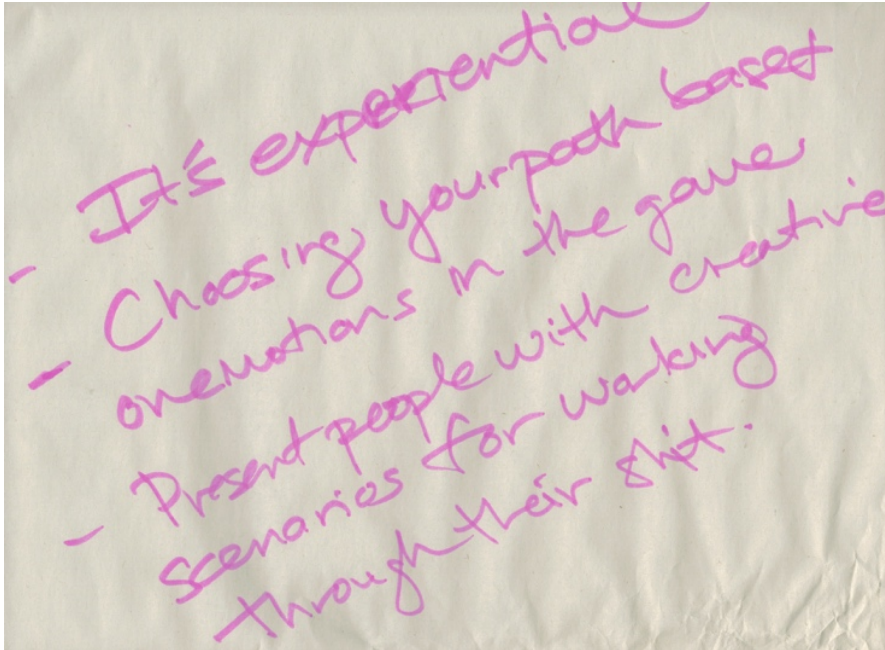
This subject commented on their experience of making a game autopathography, which is unfamiliar terrain to them. It seems that for this subject, authoring a game autopathography is “strange” because AGD praxis involves a process of revisiting past problems that can be distressing and/or uncomfortable. The strangeness of authoring a computer game autopathography juxtaposed with the difficulty of revisiting problematic emotions engenders a tension, which “looks cool”. In other words, AGD seems to hold some kind of mystique due to its potential to produce feelings of alienation and discomfort. Making a game autopathography and having yourself or others ‘play your pain’ may indeed produce unease. Nonetheless, Alex believed discomfort in the context of game design can be good, perhaps even productive, interesting, and richly different. He confirmed:

Discomfort is a good thing.

Couser (1997) echoes Alex’s thoughts on discomfort in terms of the benefits that frustration may add to creativity: “What at first seems to frustrate imagination, however, in the end stimulates it;

herein lies one powerful source of illness narrative” (p. 64). Another anonymous subject wrote that autopathographical game design “present(s) people with creative scenarios for working through their shit” (see Figure 3 below).

Figure 3: Autopathographical Game Jam doodle



The strangeness and weirdness of the AGD process is also echoed by some of the other subjects participating in the Autopathographical Game Jam. Marco used a common metaphor for giving voice to the creative experimentation with form and content that autopathographical game design engenders:

It was out of the box for me.

Fred called the study AGJ space an “odd design space”, which can be interpreted to mean a space that felt odd or strange to him because it mobilizes various design elements pertaining to both the computer game and the autopathographical narrative that are rather unconventional in their integration. Various other subjects confirmed Marco’s and Fred’s thoughts by summoning the unfamiliar terrain (ontology) of autopathographical game design as “exploration”, “doing different things”, and “messing and experimenting”. The creative impulse behind the game autopathography appears to galvanize human creativity, artistic originality, and self-healing.

Alex confirmed that in his computer game design practice, the unusual stands in relation to the usual, the latter of which serves as the default, the ground from which poesis springs forth:

I usually make very simple arcade-y mobile games but I always want to make *weird artistic games* (emphasis added) dealing with mental health and personal issues.

By calling autopathographical games “weird artistic games”, Alex denotes their de-familiarized ontology. In the popular imagination, the dramatic seriousness, abjectness, and discomfort of mental health struggles do not fit with the discourse of games and play as whimsical, escapist, and fun, hence Alex’s suggestion that the game autopathography transgresses boundaries of acceptable game design.

How is the game autopathography an aesthetically transgressive form that estranges and begets feelings of weirdness and discomfort? There are two parts to consider in answering this question: 1) the AGD process is a novel creative language of poesis (creative making); and 2) how the interaction of form and content in AGD structures artistic transgression. Consider first the subject of creative and artistic languages.

3.3.1 Breaking the ‘aesthetic code’: the role of creative languages in creative transgression

As an integration of the two semiospheres—the autopathographical narrative and the computer game—autopathographical game design is a process that employs a *set of creative grammars* for ordering and arranging first-person experiences of the autopathographical narrative into a computer game. It can be construed as a creative language for communicating traumas, feelings, and emotions by rendering the autopathographical narrative into a multimodal form—the computer game. Consider for a moment Cole’s testimony about AGD as a way to invent a new language and communicate what he calls, “this internal, unspeakable thing”:

When you’re creating a game, you’re creating a whole new form of language...For games, the language is action.

Cole suggests that games mediate subjectivity through the non-verbal language of the player’s enactment and performance—inhabiting a game character that has agency to enact change in a virtual environment; thus, in gameplay, the primary mode of communication is action and movement.

It would be helpful to examine some theoretical insights that may shed light on the AGD process in the role of a creative language that can do any number of things, including express unspeakable first-person experiences, generate novel forms of gameplay, and contribute to healing. Linguist Noam Chomsky's (1966 & 1968) theory of generative grammar is premised on "the creative aspect of language use" (CALU)—specifically, René Descartes's original claim that language use is creative-i.e., novel. Chomsky (1968; 2006) claims that language use is 'generative' in a sense of not being "habitual"; and when humans speak, they generate novel combinations of words and sentences based on an innate linguistic competence that is unique to humans. Chomsky explains that novelty as a property of form is not the only condition of the CALU; the other condition is value:

It is the character of a language rather than its form that reflects true "creativity" in a higher sense – in the sense that implies value as well as novelty. (Chomsky, 1966; 2009, p. 75)

In terms of the AGD process, 'novelty' can be conceived of as the unorthodox integration of the two semiospheres—the illness narrative and the computer game—while 'value' points to the deeper meaning or personal significance that this 'transformation', in a Chomskian sense, carries for the autopathographical game designer. The relationship between novelty and value is important; for the AGD designer, the value of authoring a novel game rests in achieving some type of self-actualization through both healing and through the acquisition of new self-knowledge-i.e., self-understanding.

Chomsky also describes generative grammar as language "that makes infinite use of finite means"²⁹ (p. 113), suggesting that creativity (infinite use) is achieved through constraints (finite means). Generative grammar is thus a constrained system of rules generating infinite possibilities, akin to the static rules of a game generating dynamic and novel gameplay. Furthermore, similar to the creative arts, language use is non-instrumental in being "free from stimulus control" and "not serving a merely communicative function" (Chomsky, 1966; 2009, p. 65). The distinction lies in that language is still an instrument, but one used "for the free

²⁹ Chomsky is informed by German linguist Wilhelm von Humboldt in this regard.

expression of thought and for appropriate response to new situations” (p. 65). In fact, Chomsky (1966; 2009), inspired by German Romantic poet and philosopher A.W. Schlegel, draws a strong comparison between language use and the creative and symbolic arts: “art, like language, is unbounded in its expressive potentiality” (p. 68).

Chomsky’s contemporary antecedent is the philosopher Ludwig Wittgenstein (1953; 2009) who considers language use ever-evolving, much like an organism: “to imagine a language means to imagine a form of life” (p. 11e). Wittgenstein similarly suggests that language use is a creative activity generating infinite possibilities: “there are countless kinds (of sentence)” and “new types of language, new language-games, as we may say, come into existence” (p. 14e). In the following quote, Wittgenstein analogizes the creativity of language to town-formation, demonstrating that language-games that serve to merely communicate simple orders, do, in fact, expand and evolve into complex and rich languages, through their similarities— their “family resemblances”— to other language-games³⁰. These newly-formed language-games can be used towards a variety of novel objectives, including doing science and mathematics:

Don’t let it bother you that [some language-games] consist only of orders. If you want to say that they are therefore incomplete, ask yourself whether our own language is complete—whether it was so before the symbolism of chemistry and the notation of the infinitesimal calculus were incorporated in to it; for these are, so to speak, suburbs of our language. (And how many houses or streets does it take before a town begins to be a town?) Our language can be regarded as an ancient city: a maze of little streets and squares, of old and new houses, of houses with extensions from various periods, and all this surrounded by a multitude of new suburbs with straight and regular streets and uniform houses. (11e)

Ironically, the definition of ‘language’ provided by a Google Dictionary search frames it as *conventional* as opposed to creative, perhaps evoking a Wittgensteinian sense of a “primitive language” of giving simple orders (p. 6e): “the method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way”³¹. It may be true

³⁰ Wittgenstein’s analogy also resonates with two previously discussed concepts (on page 61): Lotman’s semiosphere (1984; 2005) and Hoftsatter and Sander’s (2013) analogy for the expanding mental category— a metropolis with an old town (core) and a suburban sprawl (periphery).

³¹ <https://www.google.ca/search?q=Dictionary#dobs=language> Retrieved January 23, 2018.

to some extent that languages for *everyday use* may be more bound to linguistic rules than artistic languages to aesthetic rules because clarity of communication, practical concerns, and function take precedence. Artistic languages such as creative writing and poetry that experiment with rhythm, tempo, meter, and the use of rich and unusual metaphors allow for a greater range of creative experimentation that may not be subservient to functionality. As will be discussed soon, modernist poetry distorts ordinary language, making it ‘dysfunctional’ in order to create an aesthetic effect³².

Consider next what some scholars of aesthetic theory have said about so-called ‘artistic languages’ and other non-linguistic modes of creative expression. Analytic philosopher Nelson Goodman (1968) defines a “language of art” as a “non-verbal symbol system”, distinguishing it from ordinary language. Film semiotician Christian Metz (1974) affirms that film’s communicative power rests in the filmmaker’s “creation”. He makes the distinction that, “to speak a language is to use it, to ‘speak’ a cinematographic language is to invent it” (p. 101). Metz seems to suggest that film language is ‘invented’ and perhaps even mistakenly assumes that natural language is not invented but simply used. As such, the filmmaker invents new vocabularies or what Chomsky (1966; 2009) would likely call “general abstract schemas”—new symbols and new symbolic relationships. Langer (1942) on the other hand argues that linguistic symbols are more fixed and less fluid than non-linguistic symbols. Non-linguistic and non-discursive symbols, such as those belonging to artistic languages, do not have fixed connotations; rather, connotations are more unstable and malleable than those of linguistic symbols³³, she states. Perhaps Langer is gesturing to the fact that artistic languages (given their multimodality) are more variable and allow for a greater range of possible meanings than natural languages.

³² To push this argument further, we can conceive of thinkers such as Jacques Derrida as the ‘James Joyce’ of philosophy in his attempt to radically push the boundaries of philosophical writing and thinking—a worthy analysis beyond the scope of this dissertation. For more on this topic, please see Gary Gutting’s (2011) work, *Thinking the Impossible: French Philosophy Since 1960*.

³³ Langer (1942) refutes the assumption that music is a language, arguing instead that it is a notational system: “music is logically speaking, not a language because it has no vocabulary” (p. 194).

Next, it is important to briefly take up the issue of how aesthetic form is relevant to the concept of a creative or artistic language. In her treatise on symbolic logic, Langer's (1967) general definition of form is that it is a *configuration* of the elements or components of an object: "the way the constituents are put together" (p. 32). Vygotsky (1971) defines aesthetic form as the *outcome* of a process of ordering and arranging: "the result of the constructive subordination of certain factors to others, rather than of their fusion into one" (p. 220). This process of formation has at least two steps in Vygotsky's terms: step one is *selecting* some details while *omitting* others; step two is *emphasizing* some details while *suppressing* others, sometimes through their ordering. Vygotsky adds that an art form's aesthetic effects on the audience depends on the creative way in which its elements are ordered and arranged:

For instance, the sounds a, b, c, or the words a, b, and c, or the events a, b, and c, totally change their significance and emotional meaning if we present them in a different order, say, b, c, and a; or b, a, and c; etc. (p. 149)

The process of arranging and recombining symbols, which connects to Chomsky's (1968; 2006) "creative aspect of language use" (CALU) also relates to Aristotelian poesis, the poietic act of 'making' a song or a tragedy. In the introduction to Aristotle's (c. 335 B.C.; 1961) *Poetics*, Francis Fergusson claims that "the action of perceiving, passing from ignorance to knowledge, is near the heart of tragedy, and the masters of that art all know how to 'arrange the incidents' in such a way as to represent it on the stage" (p. 18). Similarly, Russian formalist Vladimir Propp's (1928; 1968) narrative theory claims that fairy tales are composed of an underlying structure responsible for generating meaning through the order and arrangement of narrative units (i.e. characters). For example, the arrangement of character archetypes propels the narrative forward to its resolution; the function of a character is arranged in relation to other characters they serve. The crucial intervention Propp makes into narratology is to disavow the thematic significance of the story or characters, which is replaceable (paradigmatic), and extol the character's (syntagmatic) function in relation to other characters in logically organizing the narrative. If a character's function is to help the protagonist in surmounting a fictional challenge—a character known as "the helper" in Propp's theory—then this character's function is formative for the protagonist's journey and necessary to the resolution of their predicament.

To breach familiar arrangements—violate form—is to break with established stylistic and formal conventions, or codes, of artistic expression (Eco, 1989). For semiotician Umberto Eco, the code is the structural pattern underlying the rules of language use, “a system of rules that would involve a fixed number of elements and that would exclude some combinations while allowing others” (p. 56)³⁴. To draw on Eco, the following sculpture analogy will illustrate how creative transgression occurs when an existing aesthetic code is modified. The formal properties of an aesthetic form consist of two kinds—material and semiotic: in the first kind, in the materials or elements out of which an artistic work is fashioned, i.e., its physical properties, and in the second kind, at the level of the work’s adherence to an existing aesthetic code, i.e., its relation to the semiotic and symbolic³⁵ form and function of the artistic work, medium, or genre. The distinction between material and semiotic properties can also be tied to linguist Louis Hjelmslev’s (1943; 1963) differentiation between expression—“speech/sound” (akin to material properties) and content—“thought/concept” (akin to semiotic properties), whose connection in the realm of ordinary language generates a linguistic sign (Taverniers, 2008, p. 368)³⁶. To return to Chomsky (1968; 2006), his theory of *deep and surface structures* echoes Hjelmslev’s differentiation³⁷. Chomsky proposes that “each language can be regarded as a particular relation between sound and meaning” (p. 15). Sound maps to Hjelmslev’s “expression” because it is material-i.e., phonetic in the case of speech or textual in the case of writing; meaning maps to Hjelmslev’s “content” and thought/concept. Chomsky’s surface and deep structures inform each other in the creation of form (syntax); they also map to ‘sound/signal’ and ‘mind/semantic interpretation’, respectively:

³⁴ In this particular context of the argument, the use of Eco’s ‘code’ should not be confused with software code or computer languages per se, although the two are of course, related.

³⁵ The term “symbolic” is connected to the concepts of style and rhetoric in argumentation and the creative arts/design including literature, film, graphic design, dance, etc. On a more basic level, symbolization is a higher-order thought process.

³⁶ Taverniers points out that Hjelmslev’s “content-expression” opposition is parallel to structuralist Ferdinand de Saussure’s dyadic model of *signifié* and *signifiant* as comprising the linguistic sign.

³⁷ Chomsky’s theory of deep and surface structures builds on the work of philosophers of language, Antoine Arnauld and Claude Lancelot, who authored “General and Rational Grammar: The Port-Royal Grammar” in 1660.

The grammar of a language is a system of rules that determines a certain pairing of sound and meaning...The deep structure contains all information relevant to semantic interpretation; the surface structure, all information relevant to phonetic interpretation. (p. 111)

To understand the ways in which the material and semiotic properties interact, consider for a moment a clay sculpture versus a marble sculpture. Clay and marble have different *material* properties. Clay is soft and supple, easily manipulated by hand; marble is hard and made of a type of stone. Let us suppose an artist uses clay to sculpt a human head. They may use a combination of their hands with chisels to shape or mold the clay. To sculpt the human head, the process of clay formation will differ from the process of marble formation; in the marble stone, the artist will not be able to use their hands as tools because the marble is too hard to mold using one's hands. Rather, the artist will need stone cutting, grinding, and sanding tools to cut and shape the marble. These material differences are likely to influence the semiotic properties of the artwork, but precisely how they will do so depends on contextual, historical, and cultural factors as well as technological advancements in tools.

For example, consider fingermarks. The material properties of clay facilitate leaving fingermarks in clay, so this is readily possible from a material point of view. But it may be that the reigning culture of clay sculpture at a given point in time, the "aesthetic regime" in Rancière's (2004) sense, does not expect fingermarks and treats them as marks of amateur or incompetent work; this is an example of an existing 'aesthetic code' that artists are expected to follow. In such a context, leaving fingermarks in a finished clay sculpture might be unexpected and transgressive, a violation of reigning aesthetic norms, but perhaps usable by an artist wanting to intervene and make an artistic point. Suppose that over time, this intervention establishes a 'new normal', a new aesthetic code, so that fingermarks become *characteristic* of clay sculpture. But now imagine a second scenario: an artist makes a marble sculpture *with fingermarks*, i.e., they contrive to inscribe fingermarks into the marble, in which fingermarks are pretty infeasible, through some type of novel technological tool. The audience would be baffled knowing full well that marble stone does not facilitate leaving the artist's fingermarks in it. In this case, the bringing of a (now) material property of clay sculpture into marble would be another sort of transgression, disrupting the aesthetic code and deviating from cultural expectations. Therefore, the fingermarks in clay and marble may be both expected and unexpected, depending on the

material and semiotic properties of the form at a given historical, social, and cultural period. This illustrates the possibilities of creative grammars combined with the material resources of technological tools (i.e., clay, stone, hands, chisels, etc.) to violate the existing aesthetic code³⁸ and usher in a new one.

Similarly, Ducrot and Todorov (1982) suggest that the term ‘poetics’ is partially defined as “the choice made by an author among all the literary possibilities (on the order of thematics, composition, style, and so on)” (p. 78). Poetics also “refers to the normative codes constructed by a literary school, a set of practical rules whose use is obligatory” (p. 79). When Ducrot and Todorov speak of the “normative codes constructed by a literary school”, they suggest that literary praxis is governed by its own aesthetic code. This aesthetic code might be conceived of as the core of Lotman’s semiosphere, in a sense. What the above theoretical insights by Chomsky, Wittgenstein, Goodman, Langer, Metz, Aristotle, Vygotsky, and Ducrot and Todorov demonstrate is that poesis qua creative language mobilizes the creative form’s resources (affordances) in a given semiosphere in order to violate or redefine the aesthetic form. As an act of ordering and arranging content, AGD poesis (creative making) can be conceived of as a creative or artistic language that is practiced according to an existing aesthetic code, but also (re)invented according to the game designer’s impulse to imagine a new aesthetic code comprised of new symbols, new symbolic relationships, and new creative vocabularies of computer game design. Recall Wittgenstein’s (1953; 2009) town formation analogy, which speaks to the shifting and expanding creative boundaries of ordinary language; and Chomsky’s (1968; 2006) claim that the interplay between deep (i.e. semantic) and surface (i.e. phonetic) structures in ordinary language use contributes to “grammatical transformations” (p. 93). Thus, these new computer game design vocabularies include ways of choreographing game player actions and performances that previous aesthetic codes have not considered (to be discussed soon).

³⁸ Two sources inspire this discussion of marble stone and clay sculpture: Brian Cantwell Smith’s (1997) “One Hundred Billion Lines of C++” and Michel Serres’s (2008) conceptualization of ‘soft and hard’, ‘software’ and ‘hardware’—the softness of language and meaning and the hardness (materiality) of the natural world (i.e., rocks, pebbles, etc.), as well as their reversibility—software is ‘hard’ too when a program is executed.

Consider next some theoretical reflections on the act of transgression and its deployment in creative forms of expression. Transgression is constitutive of the human creative impulse. Georges Bataille (1962) describes transgressive acts in terms of “wild impulses” that “confer an immediate satisfaction on those who yield to them” (p. 41). Transgression can be enjoyed in games and play but is necessarily prohibited in the realm of “rational behaviour”. According to Bataille, transgression is thus opposed to reason and rationality, such as in the world of work, for a number of practical reasons including the violence and disorder produced by the transgressive act. Transgression exists on a continuum ranging from mild to moderate to extreme acts such as murder and other socially taboo acts. In the case of AGD and other forms of artistic or creative praxis, transgression is achieved at the level of expression or, more precisely, in terms of violating the existing aesthetic code of an art form, such as computer games. Artists both follow and digress from the ‘aesthetic code’ via creative violations existing on a spectrum from radical and modest. Indeed, Eco argues that when violating a code, one hopes to discover a new code that is nevertheless connected to the existing one, and thus that the new code both “upholds and violates” the old code (p. 196). Violating the existing code means de-familiarizing existing creative grammars. In fact, de-familiarization and estrangement are key concerns of the Formalist school of thought. Literary theorist Terry Eagleton (1996) explains that according to The Formalists,

‘Devices’ included sound, imagery, rhythm, syntax, metre, rhyme, narrative techniques...and what all of these elements had in common was their ‘estranging’ or ‘defamiliarizing’ effect. What was specific to literary language, what distinguished it from other forms of discourse, was that it ‘deformed’ ordinary language in various ways (p. 3).

Roman Jakobson (invoking Viktor Shklovsky) urges Russian poets to “make the ordinary strange” and experiment with the form of language in order to transform meaning (Bruner, 2004, p. 696). Eagleton further explains the reasons behind the creative experimentation with language:

Under the pressure of literary devices, ordinary language was intensified, condensed, twisted, telescoped, drawn out, turned on its head. It was language ‘made strange’; and because of this estrangement, the everyday world was also suddenly made unfamiliar. In the routines of everyday speech, our perceptions of and responses to reality become stale, blunted, or, as the Formalists would say, ‘automatized’. Literature, by forcing us into a dramatic awareness of language, refreshes these habitual responses and renders objects more ‘perceptible’ (Eagleton, 1996, p. 3).

For the Formalists, ‘making things strange’ prevents “automatized perception” (Eagleton, 1996) or “automatic readings” (Bruner, 1986, p. 22)³⁹. In a similar vein, Langer (1953) suggests that art jumpstarts the imagination by shifting perception from the ordinary: “the artistic symbol, qua artistic, negotiates insight, not reference” (p. 22). In contrast to Eagleton and Langer who emphasize the role of the strange and nonreferential in literary or artistic praxis, Goodman (1968) accounts for the important role of the referential in artistic experimentation. Goodman thinks that artists invoke the referential (the familiar) in order to transform audience perception; in pursuing the act of “remak(ing) the world”, the artist relies on conventions or “old habits” in order to bring forth novelty and “force unaccustomed associations” (p. 33)⁴⁰.

By encouraging game-makers to “put weird shit into (their) game”, Anthropy (2012) echoes the above claims, including the strangeness of AGD invoked in the aforementioned doodle, Alex’s desire to design “weird, artistic games”, and Jakobson’s, Eagleton’s, and Bruner’s points about ‘making the ordinary strange’. In his discussion of “modding”⁴¹—a violation of game form that artists inflict upon popular videogames—Alexander Galloway (2006) calls it a “countergaming”, a practice mobilizing its own aesthetic code in the “formal grammar of oppositional cultural production” (p. 109). He suggests that modding is another form of artistic transgression and ‘making strange’. Game designers such as Anthropy and this study’s subjects effectively de-familiarize the existing conventional formula of commercial-based videogames.

³⁹ In the same vein, we can conceive of Jacques Derrida making philosophy ‘strange’ through his deconstructive prose in order to jumpstart the imagination.

⁴⁰ Goodman’s (1968) quote: “In representation, the artist must make use of old habits when he wants to elicit novel objects and connections. If his picture is recognized as almost but not quite referring to the commonplace furniture of the everyday world, or if it calls for and yet resists assignment to a usual kind of picture, it may bring out neglected likenesses and differences, force unaccustomed associations and...remake our world” (p. 33).

⁴¹ Galloway (2006) explains: “What is a video game ‘mod’? It is a video game that has been modified or otherwise hacked by a user or group of users. A video game may be modified in three basic ways: (1) at the level of its visual design, substituting new level maps, new artwork, new character models, and so on; (2) at the level of the rules of the game, changing how gameplay unfolds—who wins, who loses, and what the repercussions of various gamic acts are; or (3) at the level of its software technology, changing character behavior, game physics, lighting techniques, and so on.” (p. 107-108).

It is worth considering the contrasting theoretical insights of Vygotsky and Aristotle on the violation of the aesthetic code through estrangement. Vygotsky (1971) argues that artistic experimentation, which amounts to a deviation from cultural expectations, is key to the production of aesthetic effects, engendering a “remolding of man” (p. 259), effectively, a transformation in human perception. In his *Poetics*, Aristotle (c. 335 B.C.; 1961) frames artistic expression as mimetic, whereby the objective is not to imitate by making an exact copy of nature, but to derive a likeness imbued with artistry and poetry informed by nature’s organic properties such as order, balance, harmony, and moderation (p. 108). Clearly influenced by Marxist dialectics, Vygotsky claims that the classical idea of form and content in harmony is wrong; rather, form and content are in a dialectical relationship from the start:

For centuries, scholars of aesthetics have told us of the harmony of form and content...and now we suddenly discover that this was an error, that the form may be in conflict with the content, struggle with it, overcome it (p. 160).

He goes on to claim that emulating nature can be aesthetically limiting, arguing that great artists intuitively grasp that the artistic engineering of artifice facilitates a novel perception of the world. In opposition to Aristotle’s naturalistic bias, Vygotsky argues that art is most effective as artifice, when artists exaggerate or stylize specific features in a creative work in order to reframe objects in a new light. In his analysis of Shakespeare’s *Hamlet*, a play that integrates the template of the Greek tragedy, Vygotsky starts by saying that the play’s success had befuddled earlier drama critics. He argues that the play is successful because in his characterization of Prince Hamlet as a passive anti-hero, Shakespeare both follows and unexpectedly deviates from the tragic hero formula of Greek drama—the aesthetic code of tragedy. He believes that artists strategically and/or intuitively depart from naturalism in order to reimagine an art form and reinvent its creative grammars, which transforms human perception of the world. He affirms: “without new art there can be no new man” (p. 259).

It appears that Vygotsky’s position on ‘art as artifice’ challenges Aristotle’s position on ‘art as nature’. However, this is an incorrect assessment. If we read Aristotle closer, we might discover a different interpretation, one that may account for a dialectical relationship between form and content. Aristotle suggests that poetry is not only mimetic as “a happy gift of nature” (p. 87), but that it *also* de-familiarizes, presumably nature and the natural, in “a strain of

madness” (p. 88). “For sameness of incident soon produces satiety, and makes tragedies fail on the stage” (p. 108). He may be implying that *both* the naturalistic and artificial dimensions of tragedy matter, the latter of which elicits a transformation in perception: “in the one case a man can take the mold of any character; and in the other, he is lifted out of his proper self” (p. 88). According to Aristotle, a lack of variety—“sameness of incident”—in poetic works stifles the tragedy by introducing staleness through the repetition of familiar dramatic elements. As such, the creative devices of alienation, estrangement, and artifice may be part and parcel of Aristotelian mimesis.

If we take all of the above theoretical arguments claiming that artists might combine the familiar (the natural) with the strange (the artificial) towards aesthetic effects—to evoke delight, awe, and other emotional responses in the audience or player, then to rephrase Langer, the artistic symbol negotiates insight into the unknown through both familiarity and strangeness. In the case of AGD as a subjective praxis generating self-insight, the author harnesses their familiarity with computer game design and first-person experiences on the one hand, and on the other, their lack of familiarity with the AGD process as a creative language, one that they have to ‘invent’ as they go along. Therefore, autopathographical game designers, much like artists, poets, and game modders, violate the existing aesthetic code of a creative form such as computer games to ‘make it strange’. Most importantly, the phenomenological effects of the aforementioned Formalist, artistic, and game design devices of ‘estrangement’ and ‘weirdness’ appears to be to prevent stale and habitual forms of perception and produce heightened awareness, wonder, and even discomfort and unease. An important point regarding the last phenomenological effect—the feelings of discomfort that the AGD process sparks in study subjects. The feelings of discomfort appear at 2 levels in the framework of therapoiesis according to subject testimony: 1) at the level of the AGD process produced by the clash of the two semiospheres, as discussed in this chapter; and 2) at the level of the social sphere and the collective making processes of the Autopathographical Game Jam, which appear to produce minor discomfort in some subjects when they verbally disclose their autopathographical narratives to the group. This second level of discomfort will be discussed in the *Sociopoiesis* chapter.

3.3.2 Autopathographical game design as creative transgression

Given the theoretical insights laid out above, consider next the *specifics* of how autopathographical game design (AGD) deviates from the established ontological terrain of autopathographical narratives and computer games. To reiterate, the distinct formal properties of the two semiospheres—the autopathographical narrative and the computer game—are the following. The autopathographical narrative is monomodal in its purely textual form. As per Couser's (1997) claims, the autopathographical narrative is a conventional (i.e. ordinary, typical) form that is much less formally transgressive than one assumes. The formal properties of computer games include both their material and semiotic properties. Their material properties relate to their physical nature as computer software programs running on a physical computer and multimodal virtual play regimes (VPRs) consisting of audiovisual data (i.e. images, sounds) and processes (i.e. algorithms, rules, metrics, actions, and behaviours structuring gameplay). The semiotic properties of computer games relate to their ontological status as fictional worlds of conflict and formal systems of game rules (a closed system) situated in a cultural-historical-social context—a semiosphere (an open system)—structuring their *telos* (purpose), which is to escape from life's difficulties, be entertained, and possibly engage in transgressive, morally dubious behaviour such as simulated violence. Autopathographical game design is transgressive in at least two ways: first, in defying the primary semiotic property of computer games, which is the near-universal assumption that games are fun and escapist, and that the serious topic of illness, trauma, pain, suicide, etc. does not belong in games and play; and second, in invoking the morally-maligned cultural status of videogames as an infantile platform for the expression of a serious and existential topic—illness, pain, trauma, disability. As Ellie said:

In (the instance of AGD), you're completely redefining what the notion of the game is.

John alluded to the presently-evolving ontology of computer games:

I think it's time that games didn't necessarily mean fun because it doesn't have to be anymore.

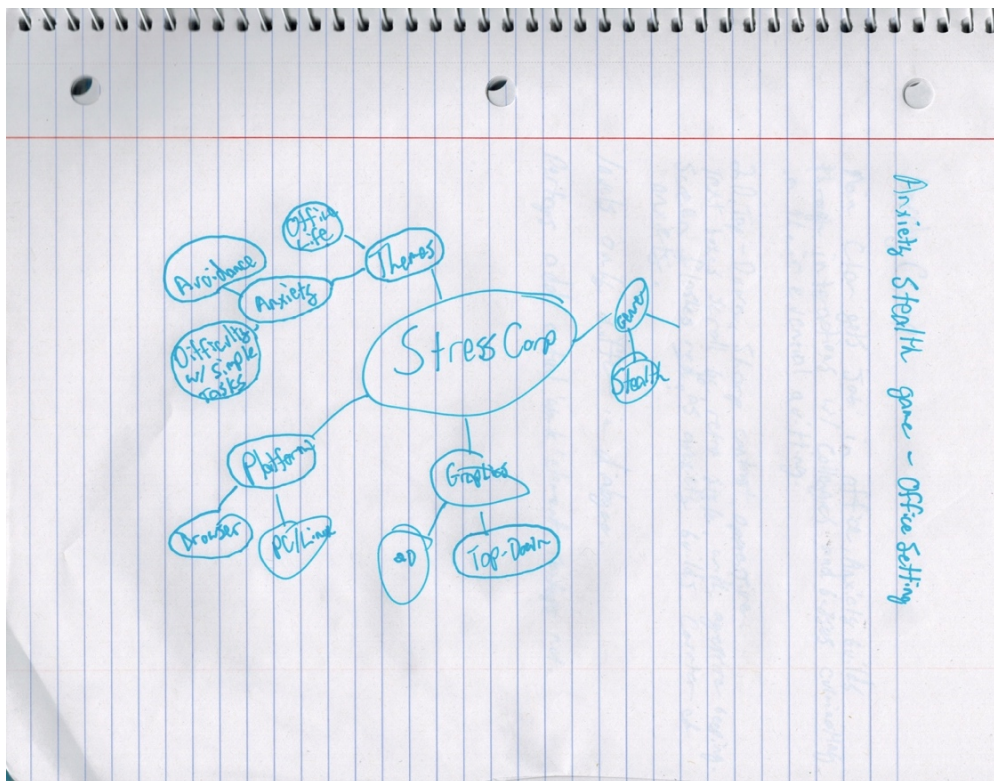
For example, Sam's autopathographical game *Anxiety Corp.*, (see Figure 4 below) is based on his experiences with ADHD and social anxiety. Sam deploys humour transgressively via the implementation of metrics in the form of an “anxiety meter” tracking the player's unwanted

confrontations with his office co-workers “in a dead-end office setting”. Sam’s game is a biting social critique of corporate culture and office work environments that the player embodies through *punishing* gameplay. Sam described his game:

The main character starts at a new job in an office and they spend their days tiptoeing around everyone around them. Inevitably they have to interact with their colleagues and their bosses and as they do, people will say insensitive things or be pushy or just do anything, maybe not even anything wrong, but they will trigger the main character's anxiety. As that anxiety builds it becomes harder and harder for the main character to do their job...

If you’re spotted by a co-worker, they walk up to you, you have a panic attack and game over.

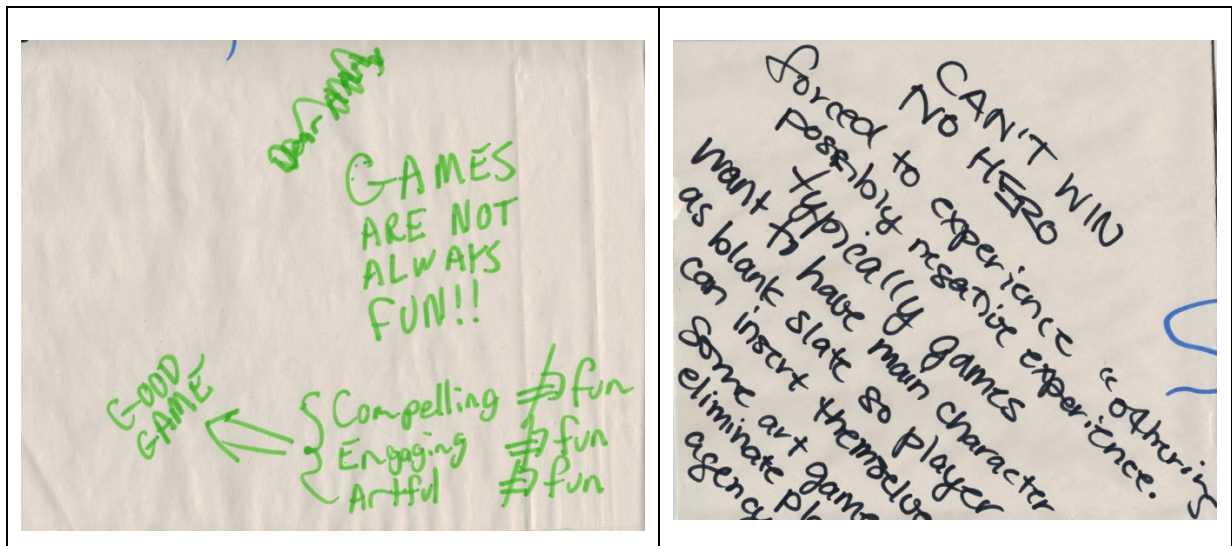
Figure 4: Sam’s design sketch of his game autopathography, *Anxiety Corp.*



“Games are not always fun”, doodled an undisclosed subject during the Autopathographical Game Jam. Furthermore, the subject qualified a good game as “compelling”, “engaging” and “artful”, which does not equal “fun” (see Figure 5, image on left). Another subject listed some attributes of “art games” that transgress the aesthetic code, such as: “can’t win/no hero”, “some art games eliminate player agency” and the player is “forced to experience

‘othering’ and a possibly negative experience” (see Figure 5, image on right). The cultural stereotype that ‘games are fun’ does not account for the plurality of ways in which games are not fun: “the discourse of play as fun succeeds in making invisible or silencing play that is not fun” (Grieshaber and McArdle, 2010, p. 8).

Figure 5: Autopathographical Game Jam doodles



John’s view that games do not need to be fun and these two doodles suggest the following: games that are frustrating, serious, negative, unwinnable; that do not reward the player, that withdraw player agency; that omit a hero/protagonist (recall Vygotsky’s emphasis on Hamlet as anti-hero), that are not player-friendly—these games are exciting to both design and play. These doodles also suggest that there exist multiple ways to design games. John’s, Nathan’s, Sam’s, and Martin’s autopathographical games are not player-friendly, are not enjoyable to play, and do not offer incentive-driven gameplay. Some of these games also offer a lose outcome instead of a win-lose outcome. This is a major difference between the autopathographical game and the autopathographical narrative in regard to their propensity to please the audience.

Autopathographical narrative writers may believe that pleasing the audience with hope and promise is an important objective, which contrasts with the frustration that an autopathographical game designer may seek to produce in the player. Jurecic (2012) gives the example of Reynolds Price’s memoir *A Whole New Life*, which deals with his recovery from spinal cancer: “he presents an *inviting narrative* (emphasis added) that makes it possible for the reader to face his

suffering, not turn away” (p. 57). Some AGD designers author for the opposite sentiment. For example, in playtesting Nathan’s game, Fred told Nathan the following:

Fred: The sign of a good game (is) when it can cause frustration when it’s working well...

Nathan: When you get mad at the game, when you lose and you know it's your fault, that’s a good game. When you lose and you blame the game, not so good.

Fred: Yeah, man (laughing).

Juul (2013) echoes the ‘games are not always fun’ proposition in suggesting that authoring ‘fun’ games is not always the best aesthetic strategy for ensuring pleasurable gameplay. Juul argues that some game players are subject to a phenomenon called the “paradox of failure”. In other words, some games do not produce pleasurable feelings. Rather, they produce pleasure by generating negative affect such as discontent, frustration, and rage. Juul suggests that game players actively court failure, at least some of the time, because the more frustrating the gameplay, the higher the feeling of reward in winning. Vygotsky (1971) similarly claims that the psychological effects of art derive from a combined generation of negative and positive affect: “art (which rather than repelling us, attracts us even as it provokes unpleasant feelings) is bound to be associated with mixed feelings and emotions” (p. 212). In other words, as discussed, Vygotsky believes in the “juxtaposition of extremes” in art to generate aesthetic effects.

Vygotsky also claims that art is an “uneconomical” enterprise, which resonates with the connections drawn above between games and art: “the poet intentionally attenuates the cause of the action, arouses our curiosity, plays with our ingenuity, makes our attention run to and fro; in other words, he wastes our strength and energies to the extent required by his work of art” (p. 204). As an inefficient means of achieving objectives that games scholar Bernard Suits (2005) gives voice to, gameplay engenders struggle and pain to achieve a winning outcome. The inefficient means of goal-oriented gameplay, art, or poetry is strikingly similar to the struggle and pain of everyday life, which is why the poetic, ludic, and paidiaic realms, given their relative safety from physical and existential negative consequences are a powerful human experience.

To extend on Juul’s paradox of failure, consider creative experimentation in AGD that attempts to build empathy in the player by engendering rage, frustration, and negativity. Commenting on the illness narrative, Jurecic (2012) quotes philosopher Martha Nussbaum who

argues that “narrative imagination is an essential preparation for moral interaction” (p. 58), which means that storytelling builds empathy and compassion by crafting an ethical relationship between the story character and reader. That is, autopathographical narratives are assumed to build empathy between character and reader.

Some of this study’s autopathographical games appear to be antithetical to empathy-building given that they are intended to build discomfort and rage in the player. Nathan’s game *ADHD Hero* integrated “de-motivational stuff” in order to produce feelings of aggression and rage that are meant to communicate the experiential dimensions of living with ADHD. He explained:

It’s one of those games that you maybe want to keep playing because failure is immediate. But the next replay happens so fast. I think it’s something that you want to keep doing and improving at and getting better at. But, on the other hand, you could quickly get very frustrated and just smash your laptop or whatever, and that’s kind of the point. I wanted to have these motivational or de-motivational messages or prompts come up in the game, like ‘how did that make you feel, when you got one wrong, did you take your time more’.

Nathan suggests that the game’s de-motivational messages are meant to provoke or infuriate the player to think about their sense of self-worth and competency—how they feel about themselves when they repeatedly fail in playing the guitar qua ADHD continuous performance test in his game. The game’s point is to build awareness by provoking anger, rage, frustration—all negative emotions that may galvanize the player to think more mindfully about a foreign experience.

Similarly, Martin’s platform puzzle game *Broken Lens* attempts to simulate his experience of being unable to see red. Martin described his design decision to intentionally perplex the player:

I’m managing the colours to change the colours of the game and recording, simulating the colour blindness, turning off the red, the green, the blue, so I can confuse the player to don’t know which one is which, like here to turn off the purple or the blue so it gets confusing.

Martin’s strategy for inciting the player’s unease is to use a form of perceptual subterfuge: manipulate the colours in the game, which instructs players to sort objects into colour-coded boxes by hampering the player’s ability to properly view a certain colour. He implemented game

rules based on the hexadecimal (colour) codes in order to deceive the player. This strategy ought to frustrate the player and perhaps induce a lose outcome. Martin's strategy is meant to communicate to the player his subjective experience of colour-blindness, which interferes with his ability to see colour properly and which has caused him distress growing up. He is communicating his experiences by way of controlling the player's emotions and sense of competency.

To better understand the aforementioned examples, it is helpful to turn to some additional theoretical perspectives from game studies. The question of computer games as empathy-building tools is a nascent topic in contemporary game studies. For example, Belman and Flanagan (2010) argue that games are unique platforms for empathy-building because they "allow players to inhabit the roles and perspectives of other people or groups in a uniquely immersive way" (p. 5). In their outlined game design principles for empathy building, they argue that the game ought to explicitly encourage players to take on an empathetic role. They suggest that designers make games that provoke mindfulness in the player in order to "intentionally try to infer the thoughts and feelings of people or groups represented in the game (cognitive empathy), and/or...prepare themselves for an emotional response, for example by looking for similarities between themselves and characters in the game (emotional empathy)" (p. 10). Their prescriptive and didactic approach to empathy-building in games does not apply to these particular AGD examples.

The AGD designers do not encourage the player to take on an empathetic position in their games, as per Belman and Flanagan's principles; AGD designers may even discourage the player to play the game or dampen their desire to keep playing. Games scholars Linderoth and Mortensen (2015) and Miguel Sicart (2015) provide another account of the use of transgressive aesthetics in gameplay. They extend Juul's theory of 'the paradox of failure' by claiming that games can be cruel and manipulative towards the player through forms of "dark play", "cruel play" and "controversial play behaviors", such as deception (p. 4-5). Sicart examines "abusive game design...when the game plays the player" (p. 4). Considering that play is a voluntary activity consigned to a temporary space and time, abusive game design may be too harsh a term, but Sicart's intention is to understand the manipulative tactics game designers use to deceive the player.

For AGD designers, creative tactics of manipulation and disruption of the player's self-assuredness, especially of those experienced players who might feel smug about their dexterity and prowess with videogames go towards imagining rebellious ways of crafting the player's ethical response in relation to the author's first-person experience. Sam explained:

You can force (the player) to have an experience closer to the one that you have while making it feel as if they are choosing their experience themselves (laughs)...I think games can be absolutely terrible and very manipulative.

The objective of this form of 'abusive game design' in AGD is to deceptively awaken in the player an awareness, a consciousness, of an unfamiliar experience and consequently build a relationship with that player. John's game about his experience with bipolar disorder and his thoughts on assisted suicide also forces the player to 'enter his shoes' by recreating John's anger and resentment towards his predicament. The game shores up discomfort through the infinite loop of no-win and no-reward characterizing the disease for John, which involves resignation, frustration, de-motivation, and feelings of inability to conquer the illness, all simulated in the game for others to experience:

I mean, what's the natural reaction to feeling cheated or feeling like you're doing something over and over again? You're going to not want to do that thing anymore, right. The natural reaction is to say, 'Oh, you're forcing me to do this, I don't want to do it anymore.' Well, how do you think I feel when I go through this illness? I don't want to do it anymore.

John implies that he wants to recreate his own feelings of resentment in the player in order to jolt them to think more deeply about living with the ravages of bipolar disorder—what it is like to live with an intractable illness and why assisted suicide might be a humane option for bipolar disorder sufferers. John was intent on communicating this harsh message to others who are not familiar with the devastating fatigue and exhaustion induced by severe mental illness. John added that the fatalistic design of his game autopathography is highly unorthodox even compared to the other games designed during the Autopathographical Game Jam, which he framed as more optimistic:

It was even a little bit uncomfortable when other people were playing (my game) just because (laughing) I have such a harsh view in a lot of ways. I looked at a lot of the other games and people had still tried to make them fun, a lot of them were still kind of cutesy and mine was so dark and it was like wow, I am clearly very cynical.

Deceptive tactics may be used by AGD designers to lure the player into an existentially dark space that they may not want to otherwise inhabit in order to build a relationship between the game player and game author. This relationship may not be necessarily empathetic, but it is one that the game player is forced to occupy for the duration of the game, as claimed by Sam. In this regard, Juul's theory of the 'paradox of failure' and Sicart's theory of abusive game design may be an unusual method for building relationships with players via the transgressive aesthetics deployed in AGD. Vygotsky (1971) also understands the power of transgression of artistic norms by artists and poets. He would respond sympathetically to AGD designers given his view that "the author does not conceal the facts of life but exposes them with the brutality that makes us realize the full impact of the truth upon which his short story is based" (p. 153).

To summarize, when AGD designers transgress the boundaries of the two semiospheres, they may break the aesthetic code by expanding the boundaries of a semiosphere (either one or both). What was once a periphery is absorbed by the dominant core and a new semiosphere arises. Lotman confirms that the borders of the semiosphere are redrawn and a new creative language is formed:

This reconstruction of language already lost to the system, in which system the given text would have brought meaning, practically always brings the creation of a new language...(Lotman, 1984; 2005, p. 217)

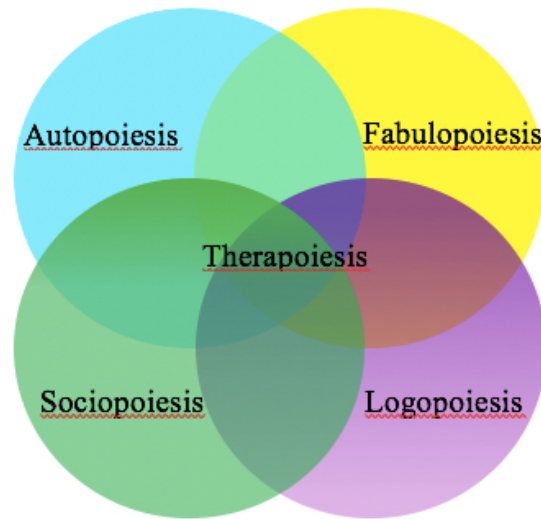
The 'clash' of the two semiospheres—the autopathographical narrative and the computer game—not only expands one or both semiospheres but also paves the way for new ontological possibilities in the relationship between the game designer and player.

3.4 Part II: Towards a theory of computer game design therapoetics

3.4.1 The tetrad of therapoiesis

The AGD process as a form of therapoiesis is defined by a structural logic. That is, autopathographical game design during an autopathographical game jam (AGD-AGJ) can be framed as a tetrad—a logical relationship between four types of poiesis that upon integration contribute to therapoiesis. This relationship can be represented in the following way:

Figure 6: Venn Diagram I—The Tetrad of Therapoiesis



Sociopoiesis builds communion and shared intimacy among a small group of thirteen game designers authoring autopathographical games together during an Autopathographical Game Jam (AGJ). The AGJ invites subjects to make new friendships, safely share their autopathographical narratives, socialize, and inspire each other. The nourishing sociality of the AGJ as a space for ‘collective making’ feels therapeutic for subjects and propels their creativity.

Through autopoiesis, the autopathographical game designer reflects on their first-person experiences as they construct an autopathographical narrative. AGD as a self-reflective (introspective) and self-reflexive (self-referential) ‘making’ process allows subjects to confront their autopathographical narratives with strength, grace, and vulnerability. Autopoiesis also orients subjects towards practices of self-care, such as being cautious with disclosure in the games in order to protect themselves from stigma. The self-reflexive processes of autopoiesis also reveal the heterogeneity of ‘the subject’ whereby subjects inhabit various roles or *auctorial postures* during the AGD-AGJ process such as study subject and interviewee, game designer, game jammer, narrator, listener, stranger, friend, etc.

Fabulopoiesis enables subjects to re-imagine themselves as fantastical and whimsical game characters and fabulize themselves in the game autopathography via densely symbolic devices such as metaphor and analogy, which contributes to self-understanding. Fabulopoiesis is

an analogical mapping process where select elements of first-person experiences with mental illness, emotional trauma, and disability (the source) are analogized to a computer game (the target) through abstraction, the authorship of which is a *structurally-articulated autopathographical narrative* (SAN) through an *intelligibility framework* (IF) that structurally organizes the autopathographical narrative through the distinct formal properties of games (i.e. graphics, audio, rules, player actions and behaviors, processes, etc.). Any game concept and concept art involved in the early stages of fabulopoiesis is first rendered in design sketches and drawings prior to using any software or implementation tools.

Finally, logopoiesis is the implementation of the structurally-articulated autopathographical narrative (SAN) into a playable game and functional software application. Subjects use various software tools such as graphic design, modelling, and music composition software to render elements of the SAN into 2D or 3D graphics and audio before importing these elements into a game engine such as *Unity* where they implement their SAN through programming. This process enables the subjects to formalize their autopathographical narratives as a formal system of game rules and constraints, which translate into player-game actions and behaviours. Engineering the computer game autopathography activates the logico-analytical resources of computer game design towards self-understanding and self-therapy. In other words, logopoiesis is any type of ‘making’ activity that focuses on the digital, computational, logical, mathematical, and procedural properties of computer game design. This includes modelling and designing digital graphics in 2D or 3D, and designing gameplay bound by explicit rules and measurable player progress (metrics), i.e., a quantifiable win/lose outcome, quantified incentives and rewards, player statistics, health bars (i.e. Sam’s anxiety meter), and so forth. Logopoiesis as a digital praxis enables subjects to view their autopathographical narratives analytically—in discrete (clear and distinct) form—and gain a fresh self-perspective. Programming, debugging, and refactoring, which includes the powerful role logical reasoning plays in these praxes is a highly therapeutic and restorative process for several reasons discussed, including boosting the self-confidence and self-assurance of subjects with mental health struggles⁴².

⁴² The order in which various design processes actually happen is not necessarily reflective of the order in which they are discussed. For example, one subject starts composing music before implementing the game’s formal system

3.4.2 The multi-personed perspective in the AGD-AGJ process

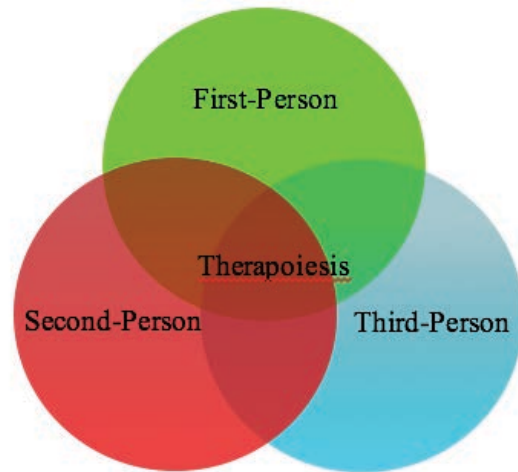
The interplay between the four types of ‘making’ described above, which constitutes the tetrad of therapoiesis, importantly maps onto three epistemological perspectives that subjects appear to adopt towards self-insight and healing. As argued in the *Sociopoiesis* chapter, the first-, second-, and third-person perspectives cater to therapoiesis through their *co-emergence* during the AGD-AGJ process. Simply put, subjects can ‘share themselves’ via the second-person perspective (sociopoiesis), they can ‘be themselves’ in the first-person perspective (autopoiesis), ‘re-imagine themselves’ by shifting between first and third (fabulopoiesis), and ‘understand and distance themselves’ by moving into third (logopoiesis).

For example, sociopoiesis invites subjects to inhabit the second-person perspective through the ‘collective making’ processes of the Autopathographical Game Jam. The second-person perspective enables subjects to enter an intersubjective space of sharing their intimate narratives with each other in order to build communion. In affording self-reflection and self-fabulization respectively, autopoiesis and fabulopoiesis both harness the subjective, first-person perspective to reinvent the self through AGD. In inviting self-analysis, logopoiesis offers the advantage of systems thinking, computational thinking, and problem-solving, all of which play a role in engendering a strong form of the third-person perspective, given its strong reliance on logical reasoning to render a computer game as a functional software program. In this capacity, the third-person perspective serves as a distancing device for subjects to diminish the potentially burdensome effects of the first-person experience mental illness, emotional trauma, or disability. The third-person perspective enabled by logopoiesis also serves as a distancing device for conducting autopoiesis and understanding one’s narrative and experiences more objectively.

The three epistemological perspectives shaping AGD therapoiesis can be represented in the following Venn Diagram:

(discussed in the *Fabulopoiesis* chapter). Other subjects presumably start by implementing (programming) the game’s formal system using placeholders and then import graphics into their game engine to replace the placeholders. The graphics may be designed separately in modelling software or rapidly prototyped as “primitive graphics” (e.g. simple geometric shapes and objects) in the game engine itself. Sometimes designers download pre-made graphics (game assets) from online sources and import them into their game engine.

Figure 7: Venn Diagram II—Epistemological Perspectives of AGD Therapoiesis



Consider some brief definitions of the first-, second-, and third-person perspectives to put the above diagram in context. The first-person perspective is subjective, embodied, and irreducibly phenomenological (Merleau-Ponty, 1945/2002; Varela and Shear, 1999), designating a human subject experiencing the world from their self-reflective, self-aware, and situated perspective. Objectivity is often thought to be an essential feature of the third-person perspective and constitutive of the observer position as it is deployed in language and logic; as such, the third-person perspective is objective, investigative, and analytical (Varela and Shear, 1999). However, in the phenomenological conceptualization of embodied consciousness, the first- and third-person perspectives are not separate: the third-person perspective is constructed via “individual observation and experience” (Varela and Shear, 1999) and as such, “objective understanding is always partial” (Nagel, 1986). The second-person perspective accounts for the interpersonal realm whereby the subject is confronted with the Other in a mutual encounter. According to Varela and Shear (1999), the second-person perspective does not lose its point of “critical distance”. In other words, the second-person perspective is a relationship where two persons interact with each other and extend into one another without fusing into each other. In the second-person perspective, there exists some distance between the subject and the Other that accounts for a difference in identity and that structures personhood.

In this dissertation's analysis, the first-, second-, and third-person perspectives are to some extent, cleaved apart and classified according to the tetrad of therapoiesis in order to tease out the distinct therapeutic effects of each one and of their integration in the AGD-AGJ process.

3.4.3 AGD-AGJ as praxis, poiesis, and theoria

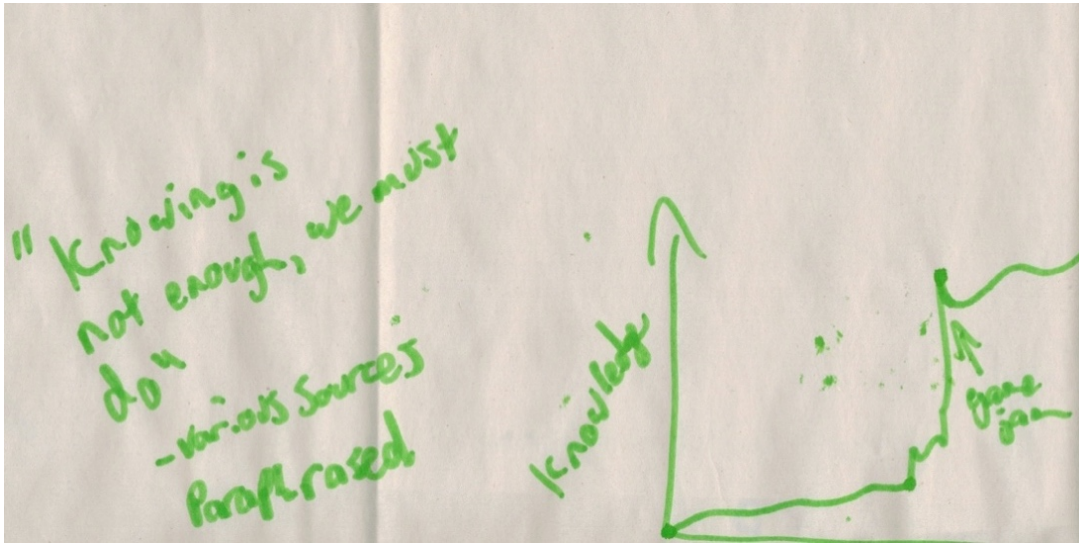
The tetrad of therapoiesis and the multi-personed perspective in AGD-AGJ importantly connect to Aristotle's (c. 335 B.C.; 1961) triadic framework of human knowledge formation—*praxis* (πρᾶξις), *poiesis* (ποίησις), and *theoria* (θεωρία). As defined by Aristotle, praxis begins with human action rooted in motivation and intentionality to act; “in *praxis*, the motive is ‘to do’ something”. Poiesis is “the motive to ‘make’ something”, and theoria is “the motive to ‘grasp and understand’”, a “contemplation of the truth” (Fergusson, p. 10-13). The Greek etymology of the word *poiesis* is *creation* engendering a *transformation* of the world. According to expressive arts therapists Knill, Levine, and Levine (2005), what made poiesis therapeutic for the ancient Greeks is its ability to bring order to disorder: “the formal perfection of their art was an antidote to their knowledge of the chaos of existence” (p. 34).

Consider next some subject testimony that reflects the importance of situating creative expression within a larger framework of action and knowledge building. Sam revealed that one of the primary values of a game jam for designers with mental health struggles is that this sociopoetic platform galvanizes the autopathographical game designer into action (*praxis*):

I can sit and think forever but a game jam really just makes me sit down and do something.

It appears that for Sam, who is making a game about his experiences with ADHD, action or praxis is the lifeblood of creation, of poiesis. Sam suggests that in order to generate self-knowledge, one must engage in an activity and not merely sit and ideate. Another subject affirmed Sam's point about the significance of praxis to knowledge building with a doodle that says: “knowing is not enough, we must do” (see Figure 8 below):

Figure 8: Autopathographical Game Jam subject doodle



To phrase it differently, knowledge on its own is not enough, it must be built and cultivated through skill-building action (praxis). For example, the Socratic method was verbally enacted in performance before the Athenian assembly by Socrates; as such, he was not merely ideating, he was dialoguing with others, communicating his ideas in a dynamic exchange, ideas which evolved, via process, among a live audience *in situ*. In the AGD-AGJ process, designers perform the deed by recording one's ideas down in drawings and sketches, structuring and organizing them through game design, sharing and discussing them with others, implementing and outputting them—all products of a *creative praxis*. Thus, 'knowing' means generating that knowledge through action, and inherent in action is a process of state change, i.e., a morphology, an idea taking shape through a medium of communication. In AGD, this knowing takes the form of self-knowledge and self-insight engendered through the skill-building action of AGD praxis.

Building self-knowledge also connects to what Dreyfus and Kelly (2011) define as the essence of poiesis: a "cultivating", "craftsman-like" and "nurturing practice". Thus, autopathographical game designers learn how to understand—recognize and discern aspects of themselves in the autopoietic process of AGD. Indeed "the task of the craftsman is not to *generate* the meaning, but rather to *cultivate* in himself the skill for *discerning* the meanings that are *already there*" (p. 247). Dreyfus and Kelly's point suggest that praxis and poiesis generate within the creative practitioner the ability to discern and recognize an existing truth in a newly

crafted form. Similarly, Hawkins (1999) explains that the autopathographical narrative engenders insight through the writer's process of organizing the narrative into form: "pathography articulates the hopes, fears, and anxieties so common to sickness, organizing them into a coherent whole and suggesting by example ways of thinking and acting" (p. 11).

Aristotelian praxis is also directly tied to the poet's motivations for writing the play, "the vision, or inspiration, which moves the poet to write or sing" (p. 8), as Fergusson (1961) says. Praxis propels the acquisition of knowledge through a performance of intentionality. AGD is thus the actualization of the intention to create a game. But action is incomplete without its full realization: "when Aristotle says 'action' (*praxis*) in the *Poetics* (c. 335 B.C.; 1961), he usually means the whole working out of a motive to its end in success or failure" (p. 9), claims Fergusson. For example, Jurecic (2012) suggests that pain (in its many forms) becomes a creative impulse that galvanizes sufferers to write the autobiographical illness narrative. In AGD-AGJ, the creative process of AGD leads to a creative output—a game prototype. In other words, Aristotle's concept of poesis is both process-driven through praxis and outcome-driven, generating a creative work. Motive is also related to what Aristotle calls the "rational purpose" that is reflected in the work of art. "The arts reflect not only rational purpose but movements-of-spirit of every kind" (Fergusson, p. 10). That is, the creative arts have their "rational purpose" reflected in the artwork viewed as the artist's undertaking of creative praxis towards a particular *telos* (purpose). Thus AGD-AGJ is a creative process whose *telos* reveals itself to be self-care, self-understanding, and therapy. Thus, Aristotle's triadic approach to knowledge creation dovetails with the motives of autopathographical game design, namely to restore the self via 'doing' (praxis), 'creative making' (poiesis), and 'self-understanding' (theoria).

Autopathographical game design allows subjects to organize their autopathographical narrative (*mythos*) as a formal system (*logos as a principle of order*). In some sense, autopathographical game design recovers "the classical concept of poesis in which rationality dominates" (Knill, Levine, and Levine, 2005, p.11) that postmodern researchers, such as the preceding authors doing expressive arts therapy, have disavowed. In presenting the tetrad of therapoiesis structuring the AGD-AGJ process, this dissertation attempts to update Aristotle's classical theory of poetics for the contemporary digital era. The fine structure of this AGD-AGJ

process is revealed in the next four chapters, each of which is dedicated to each dimension of poiesis.

Chapter 4

Sociopoiesis:

The Autopathographical Game Jam as a Therapeutic Community

4.1 Introduction

The therapeutic nature of the rendering process of autopathographical game design might suggest that it would be undertaken alone or in private, but subjects made their games in the presence of others in the designated space of the study Autopathographical Game Jam (AGJ), with its unique conditions and properties. To reiterate, a game jam is an event where game developers and designers, hobbyists, amateurs, and professionals assemble in a designated space to author digital games under a variety of limitations, such as a tight time constraint and a theme informing game authorship.

It is important to differentiate between autopathographical game jams as a type and this particular study's AGJ as an instance of the AGJ type. Hence, I will refer to this particular study's AGJ as 'the study AGJ' throughout this chapter in order to differentiate it from the autopathographical game jam as a *type* of game jam. The study AGJ brought together game designers who candidly shared their experiences with mental illness, emotional trauma, and disability through social and verbal interaction processes. As a form of sociopoiesis—a *collective making*—autopathographical game design becomes rooted in the ethical acts of sharing, communication, and reception. As such, the study AGJ qua sociopoiesis becomes a part of an overall therapeutic healing process integrating the other poietic forms. This healing process encompasses the first-person experience that is clouded in ambiguity and indeterminacy, the second-person perspective that is relational and dwells in the presence and impact of the Other—another human being—, and the third-person analytical perspective that distills and abstracts first-person experiences in order to impart self-clarity. To loosely invoke Nagel's (1986) point about "straddling the subjective-objective gap", which calls for "something" (p. 40) to structure the optics of the first-person, immersive perspective with the third-person, observational perspective, you need a bridge to extend over both perspectives to structure self-consciousness

spatially⁴³. That bridge is the second-person perspective, which in this study is instantiated in the intersubjective process of sharing spoken and unspoken pain and struggle.

In this regard, Martin Buber's existential philosophy of *I-Thou* relations and his communitarian ethics are useful tools for analyzing the unique properties of the study AGJ as a strong communal space. Using a Buberian matrix to understand the implications of the study AGJ for this type of autopathographical game jam, and in spite of the focus on the second-person perspective, this chapter demonstrates that the first-, second-, and third-person perspectives that engender healing through AGD co-emerge in the communal space of the study AGJ. Buber's framework is also valuable for differentiating between a space where game jam participants are present with others and yet still feel alone and disconnected, and a space where they feel deeply connected to strangers and buoyed and nourished by the quality of their connections. Buber's theoretical enterprise is thus generative as opposed to critical: "Buber's dialogue opts for a constructive hermeneutic focused on future possibilities, not present shortcomings" (Arnett, 1986, p. 52). To further draw on Arnett, Buber's dialogical theory is helpful to distinguish between a loose association of people without a cohesive structure and the meaningful social space of a community where interaction nourishes its members and creates a sense of belonging⁴⁴.

The human process of communication and reception is an important factor in structuring AGD as a therapeutic project. With varying degrees of openness, most of the subjects shared their experiences with mental illness, emotional trauma, and disability, which are received with empathy and non-judgment. Disclosure of intimate details is balanced with lighthearted

⁴³ Nagel's (1986) original quote: "I suggest that the concept of the self is open to objective 'completion' provided something can be found which straddles the subjective-objective gap. That is, the concept contains the possibility that it refers to something with further objective essential features beyond those included in the psychological concept itself—something whose objective persistence is among the necessary conditions of personal identity—but only if this objectively describable referent is in a strong sense the basis for those subjective features that typify the persistent self" (p. 40).

⁴⁴ Interactional linguist Catherine Kerbrat-Orecchioni (1997) clears the confusion that "di" and "dia" present in the word 'dialogue': "in Greek, dia- actually means 'through' (cf. *dialyse*, *diaspora*, *diachrony*), not 'two'" (p. 3). Buber's (re)conception of dialogue as a phenomenon of mutuality and of I-Thou that stays "in-between" persons is fitting because he appears to emphasize "through" over "two", as discussed in this chapter.

conversation, laughter, and food nourishment, which helped subjects feel connected to each other and also fueled the AGD process. Subjects shared their personal experiences with great candour during the study AGJ event, revealing personal narratives which they did not disclose in their autopathographical games. AGD in the context of the game jam platform is unlike isolating and inward-looking authorship forms such as autopathographical writing and its more informal relative, writing in a diary⁴⁵. One of the significant differences between autopathographical game design (AGD) conducted during an AGJ and autopathographical writing is that in the former participants engage in a collective making process.

As nascent worldwide phenomena whose first appearance dates back to 2002, game jams have reinvented game authorship as a social experience. Game jams are part of maker cultures that also include DIY (Do-It-Yourself) hackathons and tinkering workshops geared towards individuals who are passionate about exploring the expressive affordances of technology. As “innovation sandboxes that stimulate explorative experimentation outside of daily communities”, game jams are non-hierarchical, disruptive spaces where professionals and amateurs freely collaborate (Musil et al, 2010, p. 185). While the principal objective of game jams is the development of playable computer games, game jam scholars observe that socializing, bonding, and relationship building emerge as significant activities in this context. Guevara-Villalobos (2011) lists, among other things, “moral support within indie communities” as an important game jam aspiration. Game jamming thus transforms into a “social activity” where social interaction, skill acquisition, and meeting new people who share a common passion sometimes upstage game design praxis (Reng et al., 2013). Furthermore, game jams inspire “new ways of interacting” (Chatham et al. 2013, p. 3176). Therefore, human interaction at a game jam can meaningfully co-determine the game authorship process as in this particular study.

The objective of this chapter is to recognize the emergence within the study Autopathographical Game Jam of a transient therapeutic community propelling creativity, based on the expression of shared experiences. The study AGJ imposed a structure of limits and

⁴⁵ Personal blogging and vlogging are online and digital forms of diary writing, but the process of blogging or vlogging itself is presumably a solitary experience as opposed to a collective experience.

constraints upon subjects to create in this emergent space. The study AGJ's unique social properties and conditions, which construct the AGJ as a therapeutic community and which engender the integrated first-, second-, and third-person perspectives, can be summarized as follows:

- a) A site of enunciation where subjects openly disclose personal experiences; self-introductions are galvanizing for configuring the study AGJ as an empathetic design space;
- b) An intersubjective space for sharing experiences and deeply connecting with strangers, concretized in the small-scale, niche game jam through face-to-face interaction and jammer engagement in collective design processes;
- c) A welcoming space and a safe community that emotionally supports subjects as they engage in creative experimentation;
- d) A sociopoietic space whose parameters or restrictions structure the AGD-AGJ experience for subjects and propel their creativity. These parameters include the call for careful self-presentation, the urgency of the study AGJ's time-limit, and stress and pressure to complete the game prototype, which manage the emotions and design expectations of subjects. Emotional restrictions such as feelings of vulnerability are paradoxically liberating because they mitigate expectations of shame and embarrassment.

4.2 Intersubjectivity and the second-person perspective

It is helpful to engage existentialist thinker Martin Buber's (1923;1958) ideas to analyze this configuration of autopathographical game authorship that occurs in a communal space with distinct conditions and properties. Buber's ideas have influenced the fields of ethics, religious studies, and communication theory and have been applied in therapeutic, medical, and educational contexts to re-humanize asymmetrical relationships based on utilitarian and transactional dynamics, i.e., teacher-student and doctor-patient.

Maurice S. Friedman (2002), Buber's principal biographer and translator, argues that Buber's philosophy has cogent implications for epistemology, i.e., the theory of knowledge (p. 190) that also apply to the study AGJ. He says that Buber's theory of 'I-Thou/I-It' extends the traditional epistemological concern with the subject-object relationship by proposing a new way of knowing, "one from which the I-It, or subject-object, relation is derived" (p. 192). Buber's ideas have fruitful implications for emergent cultures, communities, and practices embedded in digital and networked media in ways that extend beyond the scope of this dissertation.

Both the first-person and third-person perspective adhere to Buber's "I-It" form. However, the second-person perspective requires a mutuality of encounter that is akin to an "I-Thou". Buber affirms the constitutive power of the I-Thou relation: "I become through my relation to the *Thou*; as I become *I*, I say *Thou*" (p. 11). Moreover, "*I-Thou* is characterized by mutuality, directness, presentness, intensity, and ineffability...it is only within this relation that personality and the personal really exist..." (Friedman, 2002). According to Buber, the mutual encounter of I-Thou is not reducible to any form of organized knowledge, such as the first-person perspective, mental representations, concepts, and categories. "True beings are lived in the present, the life of objects is in the past", Buber says (p. 13). Friedman explains: "the knowing of the I-Thou relation takes place neither in the 'subjective' nor the 'objective', the emotional nor the rational, but in the 'between'—the reciprocal relationship of whole and active beings".

Linguist Émile Benveniste (1971) similarly argues that the I-You pronouns, when uttered during the enunciation act by interlocutors, are not reducible to "unique beings". Pronouns are the material traces of the person's essence, their being, which cannot be captured or reduced to a simple signifier. In this regard, it is important to mention that "Thou and You are not the same...Thou immediately brings to mind God" (p. 14), states Walter Kaufmann in his Prologue to Buber's book, *I and Thou*. 'Thou' is infused with a divine presence, an important commitment of Buber. Furthermore, Kaufmann evokes the poetic-spiritual significance of the word 'Thou'. This poetic-spiritual dimension structures the interpersonal realm as sociopoietic, which is relevant to the study AGJ:

Thou is a preachers' word but also dear to anticlerical romantic poets. Thou is found in Shakespeare and at home in the English Bible, although recent versions of the Scriptures have tended to dispense with it. (p. 15)

Friedman (2002) explains that for Buber, *I-Thou* dwells *in between* persons and the world, while *I-It* resides *within* an individual (p. 65). Togetherness and mutuality are *I-Thou* because they stay in-between, and thus within the interpersonal. For Buber, togetherness dwells in the divine moment when persons connect with each other during conversation, listening, and responding, as study AGJ participants do. 'I-Thou' is thus present at the immaterial, *sacred* level yet is nevertheless materially-mediated through the processes of verbal and non-verbal human interaction. By contrast, *I-It* is not a lived relation; rather, it is subject-object knowledge, and as such belongs to the sphere of experiencing and using. *I-It* is subjective, mediated, indirect, and "comprehensible and orderable" (Friedman, 2002, p. 65). Through its subjective dimensions, *I-It* is analogous to a form of the first-person perspective, while in its comprehensible, orderable, and objective dimensions, *I-It* is analogous to the third-person perspective. Despite the Cartesian implications of *I-It* residing within the individual, from an ethical perspective, epistemology ought to account for a degree of second-person or third-person inaccessibility to the private, subjective realm; that is, an individual's intimate thoughts ought not to be subject to surveillance by the Other and forced to be shared, a notion that is implicit in some aspects of modern philosophical thought⁴⁶.

Given the mutual character of *I-Thou*, more nuanced distinctions need to be made between subjectivity and intersubjectivity. Subjectivity is a character or property of thought, experience, and statement in which self-awareness constitutes human identity. Subjectivity is intrinsically relational and situated in the environment. Linguist Catherine Kerbrat-Orecchioni (1980) argues for a distributed understanding of subjectivity, arguing that the subject "is not an homogeneous and monolithic psychological entity, but a complex object, autonomous and determined at the same time, where characteristics combine that are simultaneously social,

⁴⁶ "...in the aftermath of Wittgenstein, it has become a ruling orthodoxy that thought and language are inescapably public, socially mediated phenomena, and hence that there is something deeply wrong with the very idea of 'Cartesian privacy' – of solitary, introspective access to the truth" (Cottingham, 1996, p. xxxvii – p. xxxviii, *Descartes: Meditations on First Philosophy: With Selections from the Objections and Replies*).

individual and universal...” (p. 178). Intersubjectivity is constituted when humans mutually share similar private and subjective experiences (Velmans, 1999, p. 304). According to Velmans, intersubjectivity is a mutually negotiated and agreed-upon description of shared experiences, including “a shared language, shared cognitive structures, a shared world-view or scientific paradigm, shared training and expertise, and so on” (p. 304).

The autopathographical narrative that subjects have in common configures the study AGJ as a space for actively sharing this narrative. When asked what was most meaningful to him about the study AGJ, Martin said:

I think the thing about making a game about personal stuff is sharing.

Martin suggests that the act of sharing intimate and potentially painful experiences was meaningful to him in the context of relating to others during the study AGJ. The second-person perspective is normally understood as directedness towards the Other. The second-person perspective is an “intermediate mediation” between the first-person and the third-person perspective (Varela and Shear, 1999). The second-person as mediating agent is also rooted in emphatically yet critically inferring the intentionality of the Other, according to Varela and Shear. More importantly, the second-person perspective mobilizes features of the Other’s experience that resonate with the first-person experience but does not lose its critical distance. Varela and Shear (1999) explain:

In the second-person position, one gives up explicitly his/her detachments to become identified with the kind of understanding and internal coherence of his source. In fact, that is how he sees his role: as an empathic resonator with experiences that are familiar to him and which find in himself a resonant chord. (p. 10)

As such, the second-person perspective is self-experience extending into the Other based on common and shared features, avoiding a descent into solipsism or self-confinement. Yasmine appeared to invoke the strengths of the study AGJ space for being together with people who share her experiences and perspectives:

I like having a space where I’m with like-minded people.

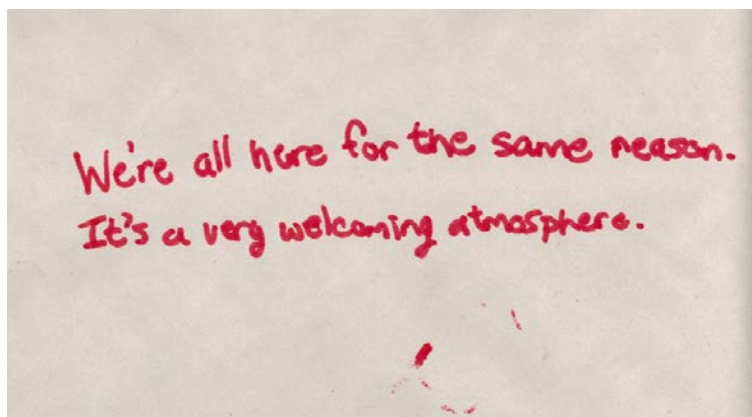
Varela and Shear echo a similar type of meeting, which engenders a level of distance (objectivity) rooted in the intention of finding commonality:

This empathic position is still partly heterophenomenological, since a modicum of critical distance and of critical evaluation is necessary, but the intention is entirely other: to meet on the same ground, as members of the same kind. (p. 10)

Members of the same kind can be persons who may share certain perspectives or experiences. Critical distance in this case emerges when study AGJ subjects take space from private thought by engaging with the Other, being aware of others' presence, and interacting with others. Varela and Shear continue: "such encounters would not be possible without the mediator being steeped in the domain of experiences under examination, as nothing can replace that first-hand knowledge" (p. 10). Study AGJ subjects share this domain of experiences under examination, as they have come together to author game autopathographies based on the shared experiences of mental illness, emotional trauma, and disability. Study AGJ subjects sought "intersubjective validation" (Varela and Shear, 1999) from each other, in this case the validation of a painful experience. Indeed, the study AGJ is structured as a safe and welcoming community in which to share personal experiences. As one jammer wrote on a doodle:

We're all here for the same reason. It's a very welcoming atmosphere.

Figure 9: Autopathographical Game Jam doodle



This dynamic establishes a safe environment where members of a group take comfort in knowing that the shared attributes of mental illness, trauma, and disability, which are negatively judged

according to most social norms, will not be judged negatively by the established norms of the game jam.

To briefly return to Chomsky's (1968; 2006) views on the "creative aspect of language use" (CALU), sharing the autopathographical narrative through narrative-telling (i.e. verbal interaction), through openness to others (i.e. non-verbal interaction), and through the collective making processes of designing autopathographical games together contributes to the emergence of a 'shared creative grammar' of AGD. Thus, the AGD process in the role of a creative/artistic language for ordering and arranging first-person experiences is partially generated through the intersubjective (i.e. social and relational) dimensions of the autopathographical game jam. In its interpersonal and communal dimensions, sociopoiesis qua second-person perspective becomes a feature of the tetrad of therapoiesis in enabling the collective and communal development of AGD grammar.

Intersubjectivity also undergirds a non-binary conception of the first-person perspective as extending into a community of others, which imparts the ability to see beyond and above oneself, and thus encompass a level of objectivity—a detached and analytical perspective. Donna Haraway (1988) invokes the intersubjective position of "situated knowledges" as objective markers of truth, challenging the simplistic assumption that the first-person perspective is an atomized self: "situated knowledges are about communities, not about isolated individuals. The only way to find a larger vision is to be somewhere in particular" (p. 590). Philosopher Donald Davidson (2001) echoes Haraway by stating that communities are effective tools of knowledge transmission: "a community of minds is the basis of knowledge; it provides the measure of all things" (p. 218). Davidson argues for an intersubjective triangulation process of knowledge creation: "triangle composed of a person, his society, and the shared environment" (xvii). The acquisition of third-person knowledge is "not based on a progression from the subjective to the objective; it emerges holistically, and is interpersonal from the start", he states. Third-person knowledge qua 'objective truth', whether in the form of "all propositional thought, whether positive or skeptical, whether of the inner or of the outer", is attained through "communication with others" in a "shared world of objects in a shared time and space" (p. xvii). Intersubjectivity and relationality are thus structuring devices for knowledge creation. In the case of

autopathographical game design in the context of the study AGJ, intersubjectivity structures the act of sharing among subjects, which is experienced as therapeutic.

With regards to game authorship, the game autopathography plays a role as a subjective, objective, and intersubjective entity within the confines of the study AGJ. As aforementioned in the previous chapter, Salen and Zimmerman (2004) state that computer games can be conceived of as both closed and open systems of play: “what makes a system open or closed is the relationship between the system and the context, or environment, that surrounds it” (p. 5). In the study AGJ, the computer game autopathography can be conceived of as an open system extending beyond the boundaries of the subject’s inner world into multiple subjectivities present in the game jam environment. John said:

(The study AGJ) is not really a public setting when we know we’re all here for the same reason.

The open borders stop at the study AGJ; the study AGJ is protective of its shared ‘object’, the autopathographical narrative, given that the majority of subjects have chosen not to disseminate or publish their games online. As such, the study AGJ is a liminal space, which is neither fully public nor fully private.

4.3 The Autopathographical Game Jam as a site of discursive enunciation

One of the key elements that configures the study AGJ as an intersubjective therapeutic space, i.e., as a place where subjects feel supported and safe to express themselves, is the pattern of discourse that it both nourishes and catalyzes. As Benveniste (1971) has argued, language is the key ingredient to building intersubjectivity. We can understand this point by exploring the verbal enunciation processes during the study AGJ. The study AGJ is a site of enunciation where subjects’ candid disclosures in their self-introductions configure the game jam as a meaningful event for subjects, helping build the feeling of kinship. A site of enunciation is a specific context in a given time and space where utterances (*énoncés*) constitute the communication process between subjects. Ducrot and Todorov (1982) call the site of enunciation a “speech situation” whose variance in the use of pronouns (I, You) and deictic words (here, now, etc.) constitute it. The site of enunciation and the *énoncés* (utterances) materially structure and determine one

another, which is why linguistic study focuses on “the imprint of the process of enunciation in the utterance” (p. 324). Discourse (language-in-use) plays a significant role in configuring the study AGJ as a therapeutic space where most subjects feel supported and safe to express themselves. Benveniste has demonstrated that discourse is an “encounter and recognition of subjectivities” (Ducrot, 1989, p. 164)⁴⁷. For Benveniste (1971), the proper subject emerges in the real-time conversation between two speakers addressing each other as “I” and “you”; that is, subjectivity is constituted relationally in the process of speech interaction during the event of live communication with a respondent. Thus, the site of enunciation and the interaction it enables are essential to the formation of subjects.

The surprisingly candid self-introductions on the morning of Day 1 of the study Autopathographical Game Jam were an icebreaker that galvanized emotional bonding among subjects. The thirteen subjects were invited to introduce themselves to the group without pressure. Subjects spoke up about living with complicated mental illnesses, emotional trauma, and other personal adversity:

The introductions were amazing, like that was one of my favourite parts of the game jam, said Ellie.

I really enjoyed when each person stood up and told their problems, stated Martin.

And Yasmine explained:

Just hearing (participant) stories, I just felt so touched at what people were sharing. And they didn’t have to at all but that’s what they did and that’s what they expressed through their games and (there) was just so much meaning to it.

The subjects found the introductions unexpectedly moving due to the fact that subjects did not know each other, as they were then just strangers in a room. Derrida (1985) accentuates the sense of *hearing* in the construction of subjectivity through verbal interaction, to build the

⁴⁷ Original quote in French: “...(Benveniste) est amené à faire apparaître, dans la structure linguistique même, une référence constante au discours, et à la situation fondamentale du discours, *la rencontre et la reconnaissance des subjectivités*” (p. 164).

intersubjective space; he says that we cannot subjectify ourselves without the presence of a respondent to *hear us speak*. Ellie elaborated on this importance of hearing others, which had an impact on her:

I really enjoyed hearing other people share. That for me I felt was very meaningful because these people, I don't know them, and they're sharing extremely personal details of their life and that for me was very, very moving.

Subjects chose words and methods of description to emotionally connect with each other. In order to 'link up' with others, participants used emotion-laden words in their introductions that establish common ground: *struggle, hardship, suffering, trouble, growing up, self-therapy, releasing stress, supporting yourself, and balancing*. This is an example of a form of explicit emotional bridge-building using affective discourse to build a connection. In her revision of Benveniste's (1971) subjectivity theory based on personal pronoun use, Kerbrat-Orecchioni (1980) argues that it is a mistake to think that only personal pronouns structure subjectivity in language. Rather, it is any word—noun, verb, adjective, adverb—, that has an *affective* or *evaluative* dimension, regardless of whether personal pronouns are used or not in an utterance. For example, the first sentence Vicky used to introduce herself to the group is:

Hello everyone, I'm Vicky. I'm a student at U of T, *struggling* through my undergrad.

Martin echoes Vicky in his description of his experience:

I am trying to make a game about how *it's hard* to be visually-impaired and see everybody doing simple stuff so easily and you have to *struggle*.

Several subjects stated that their experiences growing up with various conditions were especially taxing. Martin continued:

I'm colour blind, not a big issue for most people that live with it. It *troubled* me a lot when I was *little*.

Nathan stated that with his game:

I am hoping to integrate some of my experiences around internalized self-image and the anxiety and the bad issues that come with *growing up* with ADHD and being told that you're not motivated or that you're lazy.

Marco also invoked his earlier years "growing up", explaining his experiences of being bullied:

I suffered a lot of bullying, a lot of misunderstanding, which impacted me.

Some common ways that subjects describe their experiences are explicit and candid. Some subjects were more implicit. In the explicit group, there are several subjects who did not mince words about their multiple diagnoses and overlapping experiences:

I'm Brody, I used to be a software developer, I had a mental *breakdown* about a year and a half ago and I quit my job and everything and found out I had PTSD and a mix of bipolar disorder and Borderline Personality Disorder. I went my whole life without knowing this...

John was similarly forthright:

I *suffered* from depression and bipolar disorder starting from age eight. I've been hospitalized seven times before I was age ten and I've suffered from bullying as well as that.

Fred, who directly likened himself to Brody, finding common ground, said:

I'm an independent developer much like Brody, I also *suffered* through PTSD.

Fred was similarly graphic in his retelling of his story, which involved “staying with a dead body for three hours after (a man) died” on a mountain top and subsequent events that compounded his stress. Vicky invoked her struggles with panic attacks and mental illness by disclosing that she has already dealt with these topics through game design by making a “few games sort of in this vein”. Ellie, who was making an interactive narrative, a text-based *Twine* game about insomnia, also stated:

My own personal experience with this has been through anxiety, depression symptoms throughout the course of my life.

In the implicit group, there are several subjects who chose to be indirect without using explicit words. Without disclosing that he has bipolar disorder, Alex added:

I usually make very simple arcade-y mobile games but I always want to make weird artistic games dealing with mental health and personal issues.

Lukas stated that he has joined this game jam for the following reasons:

I came to this game jam trying to...understand what kind, what feeling, what's my purpose on gaming (in order to) *release stress*.

Some subjects emphasized in the group introductions that the objective for their game design is to find ways of coping with, managing, or finding a viable therapeutic alternative for their discomfort and pain. Sam, who has ADHD, stated that his game is:

...going to explore *balancing*; balancing *functioning* at work and *supporting* yourself versus staying mentally healthy and sane really.

Fred emphasized that his game is meant to be a form of “*self-therapy*” and a substitute for professional therapy that he has had trouble obtaining:

I figured that if I didn't do something to deal with the *stress*, I would probably *suffer* really bad mental illnesses. Unfortunately, the wait lines for getting diagnosed and properly treated are unrealistically long. So, I figured that *self-therapy* would be the best way to go about that...

There were a couple of subjects who chose not to say anything about themselves in the group introductions. The point was not to force any one style of introduction, but to invite subjects to choose whether or not to share, depending on their comfort level. Ellie said:

I kind of got emotionally involved in this setting in the couple days that I was there, certainly, so I thought that was really cool.

In conclusion, by sharing intimate autopathographical narratives with their fellow stranger jammers, subjects became emotionally immersed in the game jam space. This proved to be therapeutic in ways to be explored next.

4.4 The Autopathographical Game Jam as an intersubjective space

Interpersonal relationships during game jams are structured mostly through face-to-face interaction.⁴⁸ Consider the following aspects of the data that are in need of an explanation. The study AGJ builds a community of support based on face-to-face interaction and the act of sharing

⁴⁸ There are exceptions—some game jams are mediated by online technology; in other words, they occur in online contexts.

intimate experiences, the quality of which promotes healing. Martin affirmed the necessity of an intense connection in designing a game autopathography in the company of others:

You really need to have deep connections to work on such personal problems.

In her own introduction, Ellie echoed Martin's thoughts, suggesting that deep connections are even more moving when they occur among strangers:

Privileged to be here and it's amazing to me to hear all of your stories. It's just really courageous that people are sharing these deeply personal things with people they don't know.

The above testimonials are all the more remarkable given that Marco experienced the study AGJ environment, a space where strangers come together, as organic, comfortable, and also as being a part of him:

(The study AGJ) felt like an extension of my mind. It felt so natural, it didn't feel like work.

Marco's testimony about the study AGJ challenges the assumption that spaces with strangers are alienating as opposed to inviting. The subject testimonials suggest a generosity of openness on the part of subjects and a willingness to share with others, which builds a sense of communion among participants. Entering into a close mental and emotional bond, study AGJ subjects harness the interpersonal realm as a sociopoietic tool of creation during game jams. Alex, discussing an experience at a previous game jam, suggested that the social and interpersonal properties of game jams are highly motivating and inspiring for finishing a prototype:

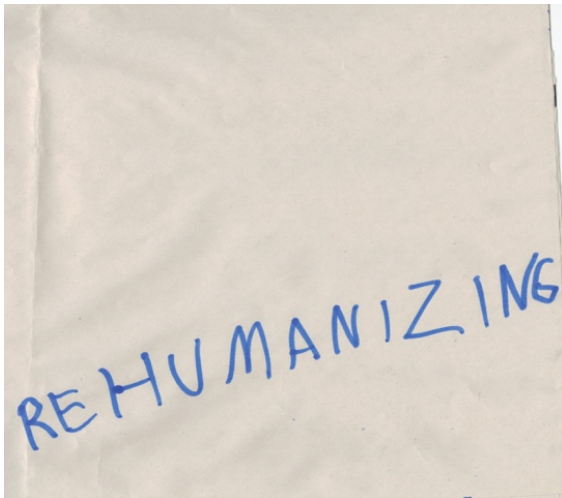
I made this awesome little game and it was super fun and really inspiring. I get super motivation powers during game jams.

Game jam scholars Deen et al. (2014) confirm Alex's sentiment by emphasizing the importance of creating mood and atmosphere at a game jam, or what they call the "Game Jam feeling" to motivate participants in their game design praxis (p. 27).

We can explain the above phenomenon by deploying Buber's I-Thou philosophy, which speaks to the deep connections established between a group of strangers at a game jam who are

sharing meaningful narratives amongst each other. Buber's philosophy of dialogue focuses on the lived and unmediated relationships between humans and the world; I-Thou is not limited to humans and spans animals, trees, natural objects, and God (Friedman, 2002). Buber's I-Thou framework also importantly accounts for a spiritual dimension in the interpersonal realm, a 'divine presence' that fosters communion or kinship among participants based on feelings of mutuality, reciprocity, and empathy. David Bohm (1996) invokes the concept of communion of the early Christians who "had a Greek word, *koinonia*" (κοινωνία), that means a "kind of participation" that not only sustains coherence within the group but could also build wholeness and "make a new change in the individual and a change in the relation to the cosmic" (p. 54). Friedman (2002) explains that "the primary word *I-Thou* can only be spoken with the whole being. The primary word *I-It* can never be spoken with the whole being" (p. 65). It is this fullness of the I-Thou that explains the fact that the AGJ is "rehumanizing", as one subject writes in a doodle (see Figure 10 below), which speaks to this quality of interaction as a lived, concrete relation constituting wholeness.

Figure 10: Autopathographical Game Jam doodle⁴⁹



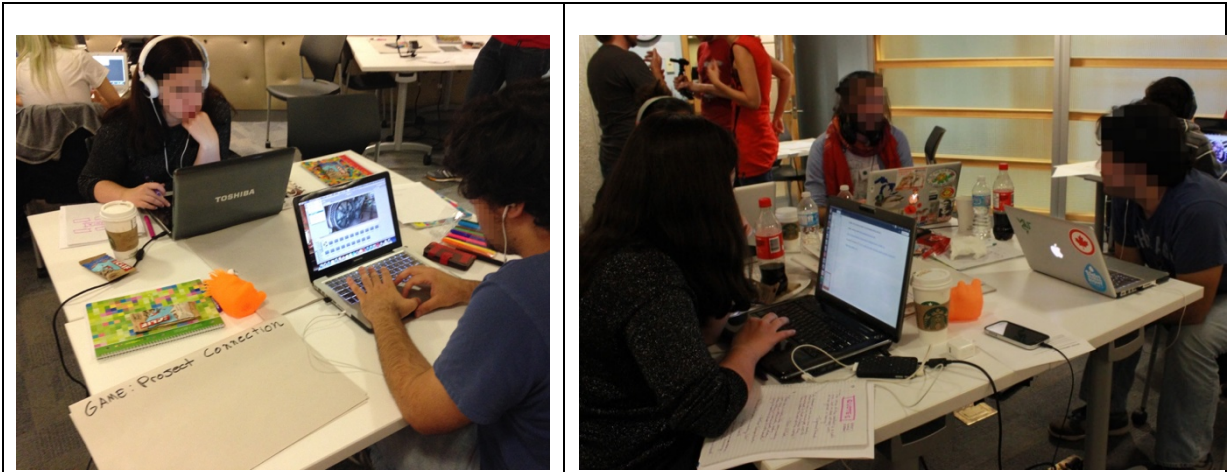
⁴⁹ The doodle is a response to the researcher's question posed to subjects during the study AGJ: "Can anyone comment on this process of designing a personal (autopathographical) game in a public setting such as a game jam?"

4.5 The Autopathographical Game Jam: the shift between first-, second-, and third-person perspective

It is important to elaborate on some distinctions of the study AGJ as a relational space where the shift between the first-, second-, and third-person perspectives is constitutive of mutuality in Buber's sense. Buber does not romanticize the I-Thou relation, which is subject to unpredictability, uncertainty, and even disorder. As Friedman (2002) says, "the I-Thou relation...is not an unqualified good. In its lack of measure, continuity, and order it threatens to be destructive of life" (p. 68). Buber believes that mutual relations involve "distancing and entering into relation" (Berry, 1985, p. 43), which suggests that all three perspectives—first-, second-, and third-person—play a role in mutual relations. Pure mutuality is unsustainable in its intensity, which must be quelled through measured detachment; hence, the *I-Thou* alternates with an *I-It* relationship (Scott et al. 2009). This detachment is both psychological and existential. In its existential dimensions, study AGJ subjects are able to come together and move apart in the game jam space. They have their autopathographical game design workstations and their headphones for some privacy, as well as sitting proximity to each other, if they wish to interact. Interaction can be emotionally and intellectually draining; thus, the ability to alternate between coming together and moving apart in the same space is a form of self-care, providing balance (see Figure 11 below). Ellie explained the flexibility of the study AGJ's spatial configuration:

The environment was very well set up so that you could work alongside people that you had never met before and kind of collaborate but also have your own space as well, which was great.

Figure 11: Autopathographical Game Jam Day 1 photographs. Subjects used headphones for some privacy (photo on left) while also interacting with others (photo on right).



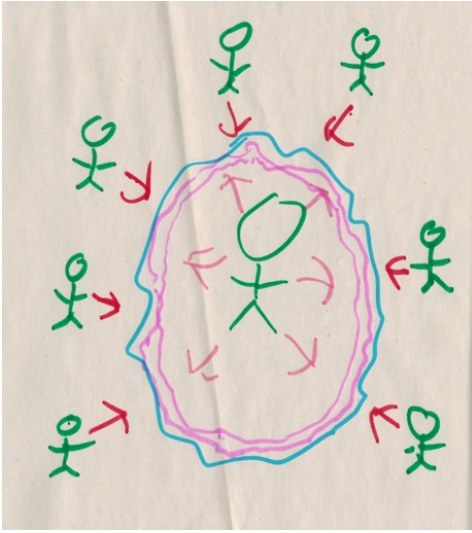
Martin alluded to the strong presence of mutual relations—the I-Thou—addressing his regret and frustration at not having related more to others and shared his story during the study AGJ:

The only thing I regret, not about the game jam but about myself—I didn’t relate it very well, I didn’t talk with the other participants, just like ahhhh. I wish I had talked more and shared a bit more of my experience.

Martin’s testimony suggests that he did not engage as much as he would have preferred in hindsight; he kept to himself, effectively distancing himself from others in the study AGJ space. His AGD praxis was a more solitary event than he would have preferred. It seems that he would have liked to share more, interact more, and relate more. This longing to share more of his autopathographical narrative or share more deeply, expressed as a sigh—“ahhhh”—speaks to the study AGJ as a space where I-Thou and I-It are co-present, sometimes in potential and sometimes actualized. Indeed, it is important that this space allowed for the choice to retreat into oneself, practicing I-It in conjunction with I-Thou, when needed.

The balance between Buber’s I-Thou and I-It can be represented in a configuration of the two modes as co-constitutive of each other, which an anonymous subject invokes in a doodle, drawing stick figures with bi-directional arrows. The doodle shows one large stick figure surrounded by uneven circles. The large stick figure points towards smaller stick figures which point back to the large stick figure (see Figure 12 below).

Figure 12: Balance between I-Thou and I-It—study AGJ doodle



The doodler drew a large stick figure in the middle, which alludes to the integrity of the individual and the private, subjective experience (I-It) in relation to the world. He/she drew several uneven circles around the large stick figure, which may allude to the permeability of subjective boundaries allowing the subject to come into relation with the Other. The bi-directional arrows signify mutual relations and intersubjectivity, which are distinctive properties of the study AGJ space. Buber believes that relational ontology of the human starts from personal relations and extends to the individual, not the other way around. “I-Thou and I-It relations are not secondary to a prior ontological reality composed of separately existing entities who encounter one another...Buber claims that relation precedes and makes possible ontological claims” (Scott et al., 2009, p. 5). Thus, we become an individual through our relationship to the Other. Similarly, according to Benveniste (1971), we become subjectified through the use of I-You pronouns in conversation. Buber’s is an ethical enterprise to show that we balance the second-person perspective, i.e., our relations to Others, with our individual needs that may comprise the first-person and third-person perspective. It bears repeating that all of the ‘personed’ relations arise out of relationality; as human beings, we are not atomized individuals living in a ‘vacuum’: we always exist in relation to Others and the world around us. As such, our relationality is the precondition for the possibility of self and Other. The implication of Buber’s analysis is that the first-, second- and third-person perspectives co-emerge during the study AGJ,

as shown through data collected therein. To put it another way, the interplay between the first-, second- and third-person perspectives become the scaffolding which upholds the therapeutic landscape of autopathographical game design during an autopathographical game jam.

Buber's differentiation between 'individual' and 'person' is most presciently exemplified in his ideas on inclusion and empathy. In order to explain the intersection of inclusion and empathy as pertaining to the study AGJ, we will adopt Friedman's interpretation of Buber's notion of inclusion, which he explains as follows:

...experiencing the other side means to feel an event from the side of the person one meets as well as from one's own side. It is an inclusiveness which realizes the other person in the actuality of his being, but it must not be identified with 'empathy', which means transposing oneself into the dynamic structure of an object, hence, the exclusion of one's own concreteness...(as cited in Berry, 1985, p. 44)

Inclusion appears to be I-Thou; it appears to be multi-perspectival and perhaps straddles one's individual perspective as well as the Other's, while empathy and feeling appear to be I-It because they stay within the individual. Empathy might be an approach of gaining insight into the Other's subjectivity by imagining inhabitation of their subjective state. With empathy, one might seek to temporarily 'enter into someone's shoes', substituting one's own perspective with that of the Other in order to imagine the Other's first-person perspective. Martin described what amounts to empathy practiced during the AGJ:

When I went to see the games, I can see how they transcribe and how they make the games look like. I can see, 'oh I have this problem dealing in this certain way and it really helps to understand better, not your problem, but the problems of others.

He appears to be alluding to the study AGJ, which helps one to understand the problems of others by inhabiting them for a moment while taking the focus away from one's own problems. In its relational aspects of listening to others describe their experiences during the self-introductions and seeing how those persons translate those experiences in the game, the study AGJ thus builds understanding of the personal narratives of others. The function of empathy is to infer the Other's first-person perspective. Because one does not have direct access to it, one must arrive at an understanding of it based on incomplete evidence, as in Martin's case, who suggested that he formed a mental model of how another subject's verbal description of their experiences maps onto a game (he used the word "transcribe"). What is impactful about Martin's

statement is its relevance to Nagel's (1986) point about finding a modest balance between complete self-absorption and destructive detachment: focusing on someone else's "problems", which are not dissimilar to your own, may bring some clarity about your own problems, as Martin demonstrated. "Sometimes, in the philosophy of mind but also elsewhere, the truth is not to be found by travelling as far away from one's personal perspective as possible", confirms Nagel (p. 27).

In consonance with Nagel, Buber's concept of inclusion suggests that human agents in a relationship do not give up their individual perspective. Rather, they integrate it with the Other's. Buber does not think that inclusion is always possible, nor that it is actually productive and beneficial in all contexts, especially those relationships rooted in asymmetrical dynamics, i.e., teacher-student and doctor-patient. Both inclusion and empathy are productive, although they can be used to achieve different objectives: to understand the Other by temporarily 'blocking out' your own first-person experience (empathy), or to include the Other's suffering into your own perspective, while not having to 'give up' your own (inclusion). Therefore, the first-, second-, and third-person perspectives all dwell to varying degrees in the epistemological processes of inclusion and empathy. To include the perspective of the Other while retaining one's first-person perspective is constitutive of the second-person perspective. To understand the Other's perspective by blocking one's first-person perspective through empathy engenders the third-person perspective; that is, the act of empathy may play the role of a distancing device from the first-person perspective. As such, the study AGJ as a sociopoietic space of collective making encourages the co-emergence of the multi-personed perspective through both inclusion and empathy.

4.6 The Autopathographical Game Jam: structure of intersubjectivity

There are two levels of sharing in human relationships that speak to the formal and semantic properties of human interaction during the study AGJ:

- a) an integrity to the *form* of the relationship between participants that is not dependent on the content or words being shared, but rather on the properties of the relation- i.e., open vs. closed and available vs. not available;

b) a level of intimacy achieved whereby participants share experiences as *content* (stories, anecdotes, etc.), either through words, or by sharing their games with each other.

The above dimensions can also be understood using a Buberian matrix⁵⁰. According to Buber (1923;1958), the primary words I-Thou and I-It “do not signify things, but they intimate relations” (p. 3). In the human sphere of interaction, relation in a Buberian sense is form (relation), not content (meaning, signification), although both are required for the communication process to unfold. “What humanity is can be properly grasped only in vital reciprocity”, Buber (1992) says (p. 77). Being open and receptive to others is a property of relation, or form, that may not necessarily pertain to the contents of verbal expression.

In *Paths in Utopia*, Buber (1949;1958) posits the discovery of the divine in the mundane as the criterion for identifying which basic human intersubjective qualities constitute a community. As Fischhoff states in the book’s Introduction, Buber’s community was “not an ideal type”; rather, “it was an empirical type of society with certain marked features, notably a serious and constant concern with the relationship of the divine in the manifestations of routine living”. “Buber’s entire sociology is of a piece with his philosophy and theology in placing central stress on the concept of the community (*Gemeinschaft*)”. In *Between Man and Man*, Buber (1947; 2002) makes the following important distinction about the essence of community: “But community, growing community (which is all we have known so far) is the being no longer side by side but *with* one another of a multitude of persons” (p. 37). This criterion constitutes Buber’s communitarian ethics⁵¹. For Buber, it is “being *with* one another”—mutual availability and

⁵⁰ It must be said that this relationship between the form and content of human interaction was first noted by the Palo Alto communication researchers Watzlawick, Bavelas, and Jackson (1967), who, influenced by Gregory Bateson, proposed the existence of two levels of communication in their study of schizophrenia at the Mental Research Institute in Palo Alto, California: the content (report) and the relationship (command). According to these researchers, the relationship aspect of communication, “being a communication about a communication is, of course, identical with the concept of metacommunication” (p. 53). This chapter is not deploying their framework as a complimentary analytical tool to Martin Buber’s matrix for the reason that, despite their ground-breaking work in this area, the researchers rely on computational terms such as ‘data’, ‘information’, ‘report’, and ‘command’ to analogize human communication to an input-output model, which is inadequate for a phenomenological understanding of human interaction processes.

⁵¹ Buber’s theoretical work on communities is inspired and informed by his observations of Jewish Hasidic communities and his involvement in the Zionist and Hasidic movements (Friedman, 2002).

openness—that constructs a community, rather than intimacy per se. The latter pertains to the *content* that is being shared, such as when someone discloses unspoken painful experiences, while the former implies that the *form* of the relationship structures content. Buber (1949; 1996) states:

It is not a matter of intimacy at all; this appears when it must, and if it is lacking, that's all there is to it. The question is rather one of openness. A real community need not consist of people who are perpetually together; but it must consist of people who, precisely because they are comrades, have mutual access to one another and are ready for one another. (p. 144)

A study AGJ subject echoed Buber's distinction with a doodle stating: "listening to other people's stories helps facilitate more openness" (see Figure 13 below); the accent of this doodle lies on listening to others as opposed to speaking.

Figure 13: Study Autopathographical Game Jam doodle

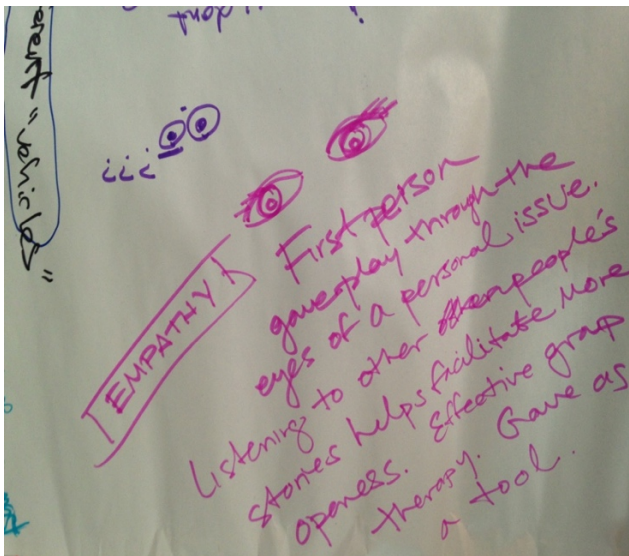


Image: "Empathy (in side-box). (Drawing of eyes). "First person gameplay through the eyes of a personal issue. Listening to other people's stories helps facilitate more openness. Effective group therapy. Game as tool."

Buber says: "basic movement of the life of dialogue is the turning towards the other" (as cited in Anderson and Cissna, 2012) emphasizing the centrality of form to dialogue building. Anderson and Cissna offer an example of what they claim constitutes the beginning of the spatial dialogic act: simply turning your body towards someone to speak or listen to them. There is

more: “when people turn not merely with their bodies but with their essential being, with openness and responsiveness, listening to the other, a dialogic partnership may flourish” (p. 135). In other words, the person can be present for the other, physically and relationally in the space to acknowledge the other person, listen, and respond; this may mean taking the effort to witness someone work on their game, nodding or making eye contact, forming a positive facial gesture, and smiling; all of these gestures constitute openness⁵². Demonstrating openness without the necessity of words, a subject wrote in a doodle: “it’s fun to see how other people work”. Several participants doodled ‘eyes’ as symbols of empathy, acknowledging others’ presence in the game jam space through the act of looking, noticing, and observing with interest and curiosity (see Figure 14 image on right). During the study AGJ, interaction can be found in such simple activities as subjects entering the AGJ room and saying hello, visiting others at their workstations to watch them working on their games (see Figures 14 and 15 below), having a conversation, laughing together, playtesting each other’s games, and so on. John stated what he enjoyed most about the study AGJ:

I really enjoyed going around at various points and seeing what other people were doing. That was the best part for me.

The prior recognition that they have something in common, namely a pain waiting to be shared, predisposes subjects to become open to sharing during the study AGJ. Marco confirmed:

(Study AGJ participants) know what (the game jam) is about and they know that there’s a purpose. So, it’s something meaningful and I think everybody is on a very open, respectful level.

The distinction inherent in this quote is that Marco is talking about being receptive to others’ act of disclosure as opposed to the contents of the disclosure itself, the caveat being that reception may not occur unless there is prior recognition of the nature of the content to be disclosed. Implicit in his statement is that being receptive to disclosure is meaningful because it involves listening to someone, not just pretending to listen, the latter of which is conceivable.

⁵² Watzlawick, Bavelas, and Jackson (1967) also suggest that the relationship (what this analysis calls *form*) of communication can be manifested through verbal and non-verbal ways, including gestures such as smiling: “‘This is an order’ or ‘I am only joking’ are verbal examples of such communications about communication. The relationship can also be expressed nonverbally by shouting or smiling or in a number of other ways” (p. 53).

Figure 14: Study Autopathographical Game Jam doodles⁵³

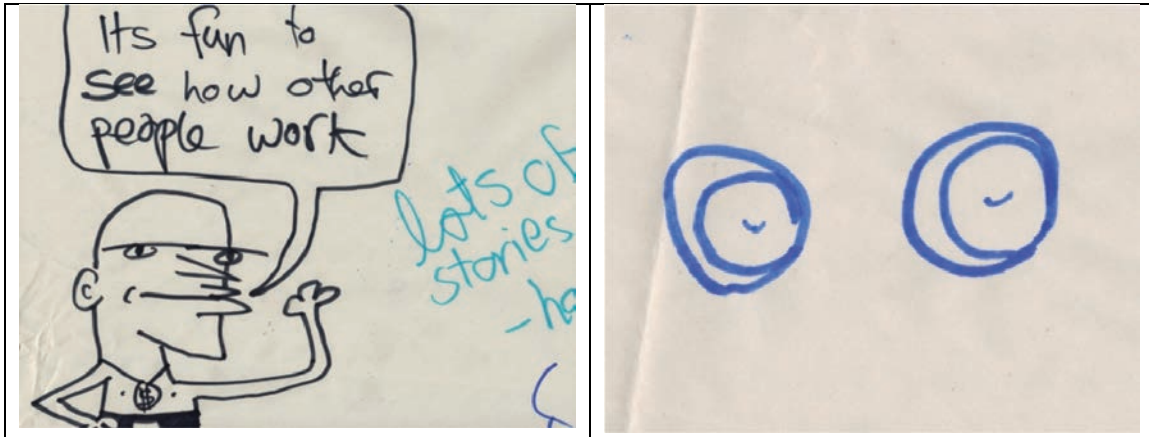


Figure 15: Two study AGJ participants (sitting down and far right) and two research assistants watching a participant explain his game



To invoke Buber's conceptualization of community as a divine presence in the mundane act of living or doing, consider the following examples. Connecting with others at the study AGJ on a mundane level occurred after the Day 1 self-introductions. In a sense, content (meaningful

⁵³ Researcher questions to group: (image-left) Can anyone comment on this process of designing a personal game in a public setting such as a Game Jam? (image-right) How is a Game Jam a vehicle for articulating deeply private or personal stories of illness, disability, trauma? Is this method a contradiction –telling personal stories in a group setting? How so or how not?

information) unleashed form (relation, openness). Intimacy followed by mundane conversations seems to demonstrate acceptance by the community of AGJ subjects. Laughter, levity, and ordinary conversation may demonstrate to subjects who courageously disclosed their experiences that they have been accepted into the group. For instance, a conversation between Fred, John, and a third subject focused on dressing up in a pirate costume for Krispy Kreme Donuts in order to get a free doughnut. Through levity and humour, this simple conversation forges a connection between subjects, which can do any number of things, including relieve tension and whisk away subjects from their momentary silos of autopathographical game design praxis. Thus, mundane conversations become deeply meaningful because they demonstrate acceptance.

The spatial dimensions of the study AGJ space are important in this regard. Yasmine confirmed that the configuration of the study AGJ physical space encouraged being open and receptive with others:

I felt like how the set-up (of the study AGJ) was—I could just talk to various people, I find it easier as a communication tool.

Interactions also build meaningful connections among subjects over other shared mundane aspects: technical troubles and foibles of game development; time pressure to finish the game prototype—“shit takes time”, “need more time!”, “I wish could experiment more”; stress engendered by the creative process of poiesis—“artist frustration”; lack of skills—“I need to learn more”; and the challenges of autopathographical game authorship—“it’s hard to express yourself” (see Figure 16 below, images on left and right).

Figure 16: Study AGJ subject doodles reflecting researcher's question to group: any group discoveries or insights in this game jam?



4.7 The AGJ: the emergence of a welcoming intersubjective 'safe space'

We can extend Buber's concept of inclusion to explore the AGJ as a creative experimentation space where subjects safely share their autopathographical narratives with each other, with minimal emotional risk. It is important to first mention several notable similarities and differences between the study Autopathographical Game Jam and typical game jams, which subjects also talk about in their testimony. First, the study AGJ space appears to take on therapeutic dimensions given its configuration as a supportive community: a subject implies that the study AGJ is analogous to "group therapy" and some subjects either labelled the study AGJ a "safe space" or suggested through their testimony that it was a safe and welcoming space. Therapy support groups in their designated institutional, clinical, or medical contexts are spaces where an "exchange of suffering forms a community of healing" (Levine, 1992, p. 57). But the study AGJ diverges from therapy support groups in certain ways. For example, the study AGJ is a non-clinical space and does not medicalize subjects. Another distinctive feature of the study AGJ is that it is a non-commercial space allowing subjects to escape what Marco called the "commercial bias" of typical game jams. Given that the study AGJ invited subjects to engage in autopathographical game design in a collective space that fostered inclusion, this framing may have implicitly suggested to subjects that autopathographical game design might become a

restorative praxis. The theme informing the study AGJ, the autopathography, may have also signified to subjects that the quest of the game jam was self-therapy, given that the prefix ‘auto’ in autopathography invokes the self and ‘patho’ evokes suffering, disease, and feeling.

What AGJ spatial properties enable sharing among subjects and constitute a therapeutic community? An important part of the answer is that the study AGJ was an emotionally-supportive environment where subjects used game design to explore uncomfortable and stigmatized content, i.e., first-person experiences with mental illness, emotional trauma, disability, and other forms of emotional adversity. Maker cultures have allowed forms of creative exploration that diverge from commercial game design, the latter of which standardizes production according to market-driven competition. Some study AGJ subjects who have previous game jam experiences testified that large-scale, commercially-oriented game jams have some notable strengths, but also significant weaknesses stemming from competitive hierarchies and lack of freedom of expression.

Consider subjects’ repeated references to feeling ‘safe’ during the study AGJ. For example, Nathan and Marco described typical large-scale game jams in Toronto as “exclusive”, “aggressive”, and “macho” environments, “often not welcoming spaces for lots of people”. One study AGJ subject wrote that they consider non-autopathographical game jams to be “intimidating” spaces in which to author a game autopathography (see Figure 17). Nathan stated the following:

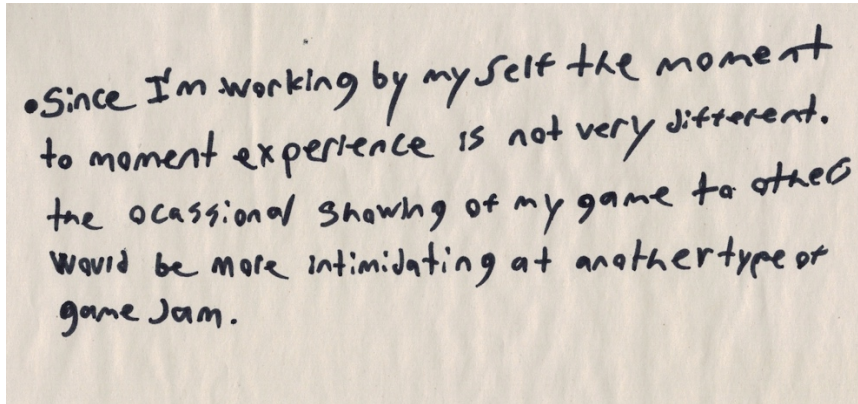
When you’re at a big game jam like TOJam, you don’t feel as comfortable or as safe doing a personal game, and I think there should be more jams like this one.

Nathan testified that the study AGJ helped him to feel “safe” to draw on his childhood experiences with ADHD:

The game jam you created was very welcoming and inclusive...It was a safe space⁵⁴, everyone was making personal games, everyone was talking about their experiences, and I don’t think I would have seen almost any of those games at a normal game jam.

⁵⁴ In study subject recruitment materials, the AGJ event was framed as an inclusive and accessible space given that subjects were asked in email correspondence to notify the researcher of their accessibility needs, dietary restrictions

Figure 17: Study AGJ doodle⁵⁵



Nathan described typical game jams as highly competitive and hyper-masculinized events:

Game jams often have this very macho energy...it just becomes the, 'I don't want to say a dick measuring contest', but do you know what I mean? It's—who can stay up the latest, who can sleep the least, who can eat the least, let's make video games—and so it becomes this bro culture.

The study AGJ allowed for that “creative magic” in Marco’s words, one of the constitutive ingredients of sociopoiesis, that is absent from the typical game jam as a competitive and aggressive environment. Marco echoed the above sentiment by adding that many game jams favour the production of technically superior, commercially-bound games in an “aggressive” (competitive) atmosphere at the cost of creative exploration and ideation:

(Game jams are) more aggressive environments. It's—win, win, win, have a better idea, it's not working, make it work, go fast, you have 48 hours don't stop, go, go, go. There's nothing wrong with them. I think that's fun, you learn how to technically challenge yourself, like that was amazing...But at the same time, you lose a lot of that creative magic, you lose that spark—let's say that they have a bias for being commercial and not just being a game.

and other preferences. All of those requests were duly met (please see the *Methods and Methodology* chapter for more).

⁵⁵ Researcher question addressed in study AGJ doodle: can anyone comment on this process of designing a personal game in a public setting such as a Game Jam?

Sam also claimed that the study AGJ stimulated his creativity, correlating his positive experiences to feeling safe:

I just got this boost of focus and motivation that was unparalleled, which was wonderful and I felt like it was a really safe environment.

Sam referred to the opportunity for authoring a game about his experiences with ADHD during an AGJ; he would not be able to feel safe at another type of game jam directly speaking to those first-person experiences. Cole confirmed Nathan and Marco's sentiments by implying that the study AGJ was successful because it was structured as a non-competitive environment. Competition hierarchizes competitors based on assessing merit according to a set of criteria. The study AGJ did not involve competition and performance evaluation based on any set of criteria, which could have placed unnecessary pressure on subjects:

I think it's nice because...even though we're all in it together, it's not like a competition, because everyone has something different to present or to bring to whatever side of development that they're on.

When asked about her experiences with the study AGJ, Yasmine said that she personally prefers collaborative to competitive environments for both playing and making games. She equated lack of competition with inclusivity:

I personally prefer having it more inclusive. I love the inclusion of games and that even in multiplayer games I prefer co-op than something competitive.

Shin et al. (2012) support the testimony of Marco, Nathan, Cole, Sam, and Yasmine that competition may undermine freedom of expression by claiming that game jams are misunderstood concepts and “confused with a programming contest”. Their case study is the Fukushima Game Jam, which harnesses game development for disaster management and recovery. The authors stress that game jams can be powerful tools for solving pressing problems that have affected a community and suggest that the creative spirit of game jams takes precedence over competition: “a game jam is closer to a musical ‘jam session’ and not necessarily a contest,” —a testament to the presence of strong sociopoietic currents in non-competitive game jam spaces.

It is important to disentangle the political connotations of the ‘safe space’ from the interpersonal meaning of the safe space as a community of support. This will ensure a deeper

understanding of how and why the study AGJ was perceived as both safe and therapeutic by subjects. A safe space is a designated environment, a temporary refuge, where members can express themselves without the risk of emotional or psychological injury in the form of harsh judgment, shame, or abuse. The politically controversial concept of the safe space has recently taken on feverish dimensions in polarizing the political landscape of higher education⁵⁶. So-called ‘safe spaces’ have become caricatures in the media, closely tied to discourse on trigger warnings, the liberal left, and social justice. According to Moira Rachel Kenney (2001), the safe space as an explicit discursive designation for a specific location with unique properties and conditions has feminist and activist roots, emerging in the women’s movement in the late twentieth century. The Roestone Collective describes feminist safe spaces as transient and niche communities geared to women, whereby “spatial separation for varying durations can enable marginalized people to develop an important sense of community and express themselves in ways that might not otherwise happen” (p. 1362).

In their contemporary sociopolitical evolution, common discourse characterizes safe spaces as geared to marginalized groups including women, LGBTQ, and people of colour. The study AGJ falls under this designation given that it was directed to subjects who either identify as disabled and/or experience mental illness and emotional trauma on a spectrum of low to high intensity. However, the term “safe space” is an “overused but undertheorized metaphor” (Barrett, 2010, p. 1). Indeed, feminist and disability studies scholars argue that spaces can be more or less safe, they can be “safer”, but never fully and completely safe (Kuppers, 2014; The Roestone Collective, 2014). In fact, Kenney claims that safe spaces can become places of conflict (p. 24). Thus, it is important not to romanticize safe spaces as utopian paradises where participants escape the vicissitudes of life, an often-invoked caricature. To strip away their politically ideological (e.g. activist, feminist) underpinnings, this analysis uses “safe spaces” as a generic term to denote the phenomenon of communities of belonging and trust that exist in a wide variety of contexts. For example, the phrase “safe space” is sometimes co-opted to designate

⁵⁶University of Chicago has recently banned safe spaces on its campus, arguing that they shut down free speech and prevent “critical thinking”: <http://nymag.com/thecut/2016/08/university-of-chicago-bans-trigger-warnings-safe-spaces.html>

formations that support human behavior unrelated to feminist or liberal roots, i.e., a safe space for white supremacists to express their extremist views⁵⁷.

The politically motivated safe space emerges from, and is a subtype of, a common formation satisfying the basic human need to openly and trustingly share what is disallowed, thwarted, or suppressed, whether by the person who is sharing who may self-censor due to fear of judgment and/or shame, or by others. Therefore, from a phenomenological standpoint, safe spaces qua communities of belonging free from injury exist on a spectrum that invokes their context-dependent meaning. Safe spaces are nevertheless bound by a variety of context-dependent social, material, and discursive restraints. A dance club is architecturally built at the material level as safe to dance without injury. A game is typically a safe space for players to fail and lose because it does not usually come with the real-life consequences of failure and loss. Confiding to someone is a space of safe interaction because we start from the assumption that the confidante will keep your secret ‘safe’. Romantic and family relationships ought to be spaces of safety, but they not always are, because our ability to trust a partner or family member is sometimes compromised. Evoking team-play as a safe space dependent on member loyalty to the team, Erving Goffman (1956) explains the unwritten rules of a team co-operating to avoid betrayal: “(team members) must not voluntarily betray the secrets of the team, whether from self-interest or principle” (p. 135). Safe spaces, irrespective of their context, are both a matter of *universal human necessity*, as in architectural spaces and intimate relationships, and *human niche necessity*, as in the examples of team play, the feminist-based safe space, and the white supremacist safe space. They are ultimately places of inclusion.

Buber’s concepts of empathy and inclusion are a fitting lens for understanding how the study AGJ becomes an intersubjective safe space for subjects. Recall that for Buber, inclusion means integrating the first- and second-person perspectives—accommodating both the subject’s and the Other’s first-person perspectives—in order to build intersubjectivity in the form of

⁵⁷The term safe space is used in this article to refer to spaces safe for the expression of white supremacy. <http://wagingnonviolence.org/feature/alt-right-safe-space-racism-college-campuses/>

shared understanding and mutuality. In the case of an AGJ as an intersubjective space, inclusion means making space in self-awareness for one's own emotions to co-exist with the emotions of Others, which is not an easy task. Inclusion thus entails the prior recognition of the possibility of emotional release, i.e., that affect may become transparent through AGD and in the act of sharing the autopathographical narrative. Another distinction not mentioned before between empathy and inclusion is as follows: inclusion means accepting in the social encounter the potential for the existence of both controlled and disruptive emotions as well as positive and negative emotions. Empathy means 'entering into the shoes' of the Other at the expense of *suppressing* one's own emotions. It is a temporary moment of 'giving up' one's perspective and emotions in order to understand the Other's. Buber's concepts of empathy and inclusion are appropriate to the analysis of a therapeutic AGJ given the deeply intimate experiences informing autopathographical game design. Empathy and inclusion, however, may undermine competitive spaces that are based on a meritocratic ranking of game players. A competitive AGJ is not undoable or impossible, and indeed Buber would argue that you cannot prevent all forms of conflict or tension; but competition would force autopathographical designers to compete with each other, which may add unnecessary and additional tension to the game jam and alienate designers from each other given the highly subjective nature of the autopathographical game. Buber's concepts of empathy and inclusion put pressure on competitive spaces given that the latter seeks to validate the superiority of one perspective over all others; the study AGJ was not assessing subject superiority of skills or artfulness; rather, it was a space that attempted to nourish subjects' skills and artfulness.

It is also possible to theorize on the manifestation of inclusion and empathy in the study AGJ via the theoretical concepts of the face-threatening act (FTA) and the face-enhancing act (FEA) through researchers who have studied human interaction processes, including Goffman (1967), Brown and Levinson (1978), and Kerbrat-Orecchioni (1997). Goffman's (1967) original concept of "face-work" relates to "the actions taken by a person to make whatever he is doing consistent with face" (p. 12). Goffman's explanation of "face" is as follows:

Face is an image of self delineated in terms of approved social attributes—albeit an image that others may share, as when a person makes a good showing for his profession or religion by making a good showing for himself (p. 5).

Saving face is thus one way in which a person might conduct themselves in a social encounter in order to maintain their self-image vis-à-vis others who will perceive them as that person desires to be perceived. As Kerbrat-Orecchioni states, face acts depend on politeness, in that “as soon as one wonders about the degree of politeness or impoliteness of any statement, one comes across the idea of face or something similar” (p. 11). The face-threatening act (FTA) was introduced by sociolinguistic researchers Brown and Levinson (1978) to describe face acts that threaten the social and emotional status, integrity, and autonomy (among other things) of interactors via speech acts or gestures (i.e. verbal and non-verbal forms of communication) such as orders, criticism, insults, evaluation, judgment, disagreements, and so forth. These face threatening acts appear to introduce tension, discomfort, and antagonism in an interaction, however small or slight. One example of an FTA is “emotion leakage, non-control of laughter or tears” (Brown and Levinson, 1978, p. 68). Kerbrat-Orecchioni’s theoretical concept of the “face-enhancing act (FEA)” extends on Brown and Levinson’s FTA, accounting for face acts that “are used to *enhance* these faces, like wishes, thanks or compliments”, that “constitute *first and foremost* flattering behavior toward the other person, that is, an anti-threat” (p. 13). Kerbrat-Orecchioni importantly expands face-work to include a non-paranoid model that actively improves and/or intensifies the self-image of an individual in a social encounter.

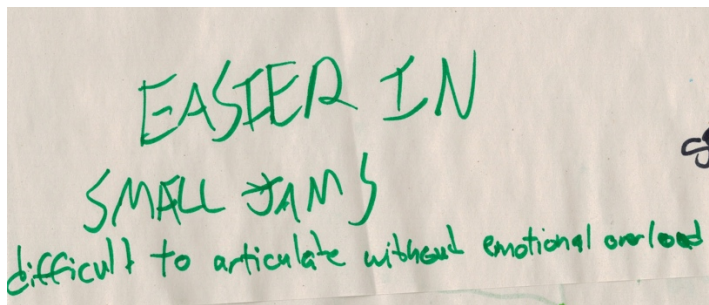
Buber’s concept of inclusion, as a realm of accepting *both* controlled and disruptive emotions can mean integrating FEA and FTA together—whereby they co-exist and are equanimously accepted by subjects of the study AGJ. In some sense, Buber’s notion of inclusion and empathy as imperfect and non-romanticized concepts put pressure on interactional linguistic theory in that they offer a space where face-work can integrate both face-enhancing acts (i.e. the generation of positive emotions) and face-threatening acts (i.e. the generation of disruptive emotions), yet still not undermine the integrity of the relationship between interactors.

Consider for a moment Buber’s communitarian ethics to complete the notion of the safe space and situate it in its historical context. The theoretical concept of the Bund (alliance, league), originating from German sociologist Herman Schmalenbach, is a helpful model in this regard. The Bund, according to Hetherington’s (1998) assessment of its contemporary significance, is rooted in “...affective-emotional solidarity...a wholly conscious phenomenon

derived from mutual sentiment and feeling...” (p. 89). Hetherington marks the other properties of the Bund as elective, unstable, and transient, and sees it as a type of community formation that relies on face-to-face interaction (Wilken, 2011, p. 36). Game jams are transient spaces that last anywhere from one day to a few days; the study AGJ similarly lasted one weekend. Buber’s communitarian ethics are founded on the Bund model. Bunds according to Buber are “small, personal, authentic communities” where relationships “bond people to one another, instead of being merely instrumental or exchange-based” (Etzioni, 1999, p. 154).

In this regard, the study AGJ as a small-scale intimate space with a total of thirteen subjects is important because the number limit of the group decreases the potential for conflict or disruption, according to Goffman (1956). Goffman claims that one way of ensuring that people do not act improperly is to limit the size of the group or team (p. 140). Similarly, a study AGJ subject wrote in a doodle that AGJ is “easier in Small Jams: difficult to articulate with (and) emotional overload” (see Figure 18). Larger groups are more prone to conflict and thus undermine the safety of subjects in articulating the autopathographical narrative.

Figure 18: Study AGJ doodle⁵⁸



In this regard, study subject Martin suggests that small-scale, intimate game jams allow subjects to express themselves in ways that are not possible at bigger jams; he thus ties smaller jams to connecting with others on a deeper, more meaningful level. He said:

⁵⁸ Doodle addresses the following researcher question to group: Can anyone comment on this process of designing a personal game in a public setting such as a Game Jam?

When you are dealing with big numbers it's hard to connect with each one as a person. Usually bigger jams have bigger groups like 6, 7, 8 people working on the same game. It's harder to make a (game about a) personal experience with a lot of different people. You really need to have a deep connection to know what you are doing, for work on such personal problems, it's better to have smaller numbers with small groups.

Alex echoed Martin's thoughts:

It was a nice experience to be in a smaller jam because most of my jam experiences are with TOJam and Global Game Jam that are, you know, 800 people in one giant building. So in a smaller, more intimate jam, it's kind of nice to have that and to be able to be around people who are all designing games about mental illness and illness in general.

Sam also agreed:

I think this specific game jam, this environment, has enabled me to explore this topic effectively. However...in a larger or more impersonal game jam I wouldn't be able to tackle the same subject matter as comfortably or as effectively.

Opposing the assumption that the Bund is an idealistic, utopian enterprise, Buber (1949; 1996) warns that the concept of this community "must be guarded against all contamination by sentimentality or emotionalism" (p. 134). He argues that the community is always subject to forces of hardship, which instill in it the divine spirit: "community is the inner disposition or constitution of a life in common, which knows and embraces in itself hard 'calculation', adverse 'chance', the sudden access of 'anxiety'" (p. 134). This analysis will soon turn to an examination of the dimensions of the study AGJ as a community of support that can nevertheless be anxiety-inducing and 'unsafe', confirming that the safe space is an imperfect term, and moreover that it is imperfect and non-utopian in its praxis of lived and concrete relations.

One of the assumptions about safe spaces is that they are therapeutic spaces, something which feminists have challenged in the context of feminist discourse, given the worrisome implication that the function of therapy is to cure female pathology, i.e., 'hysteria'. Kathy Sarachild, a founder of a 1970's radical feminist organization, argues that feminist safe spaces are consciousness-raising groups, not therapy support groups for women:

The purpose of hearing people's feelings and experience was not therapy...The importance of listening to a woman's feelings was collectively to analyze the situation of women, not to analyze her. The idea was not to change women, not to make "internal" changes except in the sense of knowing more... (as cited in Kenney, 2001, p. 24)

Sarachild fails to recognize that when women gather in a feminist safe space to express first-hand experiences of injustices due to gender, which they could not voice aloud in other environments, they may experience catharsis or a therapeutic effect that has nothing to do with medically-sanctioned therapy or psychoanalysis per se. Couser (1997) for example calls the disability rights movement a “communal consciousness” that is therapeutic (p. 209), because collective expression enables the release of suppressed frustration and anger. The disability rights movement depends on a coalition of individuals who mobilize to collectively voice frustrations with a system that has excluded persons with disabilities from participation in all walks of life. The study AGJ authored games did not make political statements and they were not meant for online dissemination in this regard. Nonetheless, it must be noted that some of the games articulate social and cultural critiques. John’s game autopathography *Untitled* argues for access to assisted suicide for individuals with severe mental illnesses such as untreatable bipolar disorder. Nathan’s game *ADHD Hero* is a critique of the medical system’s pathologization of individuals diagnosed with ADHD. And Sam’s game *Anxiety Corp.* is an implicit critique of corporate cultures and their norms for social etiquette. The above examples demonstrate that critique as a strategy of protest can have therapeutic dimensions in its empowering role of giving voice to legitimate complaints.

The designation of the study AGJ as a thematically oriented game jam based on subject experiences with mental illness, emotional trauma, and disability may have steered it towards a collective consciousness that was experienced as healing because it facilitated the communication and reception of shared values and experiences. Ellie enjoyed the study AGJ because for her:

The environment was conducive to us being creative and feeling supported.

Another subject wrote in a doodle that they perceived the study AGJ space as “effective group therapy” (Figure 19 image on right). Another subject echoed this sentiment by adding that the AGJ as a “medium adds artistry and potential anonymity to the process” (Figure 19 image on left), invoking the therapeutic benefits of the AGJ as an artistic platform for self-expression and the safety of anonymity it affords—not being pressured to share the games with the wider public.

As a matter of fact, Hacking (1999) writes that support groups engender positive moods in participants, which is correlated to healing:

The prognosis for breast cancer patients who participated in support groups and who had a somewhat optimistic attitude to their illness was, in one large sample, dramatically better than the rate for patients who were resigned, depressed, and who did not work at achieving a better mood (p. 110).

Figure 19: Study AGJ doodles⁵⁹

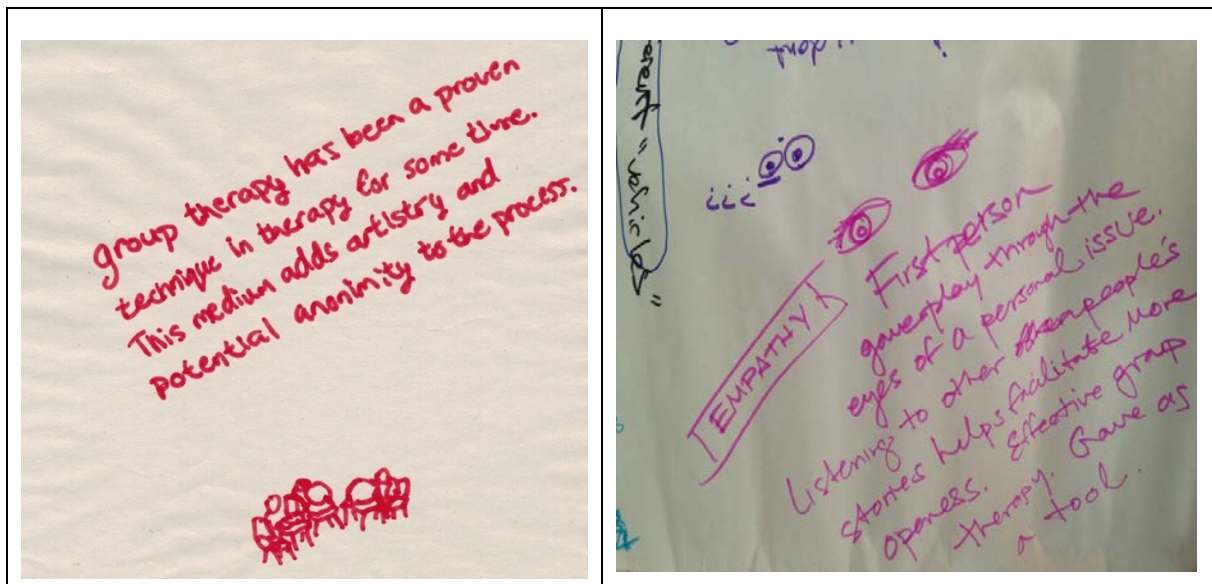


Image-left: “Group therapy has been a proven technique in therapy for some time. This medium adds artistry and potential anonymity to the process” (with a drawing of people huddled working on chairs).

Image-right: “Empathy (in side-box). (Drawing of eyes). “First person gameplay through the eyes of a personal issue. Listening to other people’s stories helps facilitate more openness. Effective group therapy. Game as tool.”

In this vein, curation processes structure and determine safe spaces by defining relationships between inanimate objects, discourse, and human participants, including the researcher’s ethical commitments to the study subjects. Fred invoked the inherent power

⁵⁹ Doodles respond to researcher’s question to group—how is a Game Jam a vehicle for articulating deeply private or personal stories of illness, disability, trauma? Is this method a contradiction since it involves telling personal stories in a group setting?

asymmetry of the researcher-study subject relationship, emphasizing the researcher's responsibility in giving back to the community being studied. He also referenced academic collaboration between researcher and subjects, challenging the one-sided relationship of the I-It in which the study subject's output (such as a game) is simply an object of the researcher's needs:

It's nice to see academics actually reach out to game developers and use them as a way of creator, a way of getting supporting arguments for their theories. Because I know way too many academics who don't do that other than just go off (writing) articles and it makes me really angry.

In other words, the study AGJ is curated with respect not only to the subjects, but also to the organizers and researchers who must ensure that the space adheres to a set of criteria pertaining to the safety of subjects. Indeed, "safe spaces can be created through the introduction, removal or rearrangement of objects" (The Roestone Collective, 2014, p. 1359). For example, the study AGJ space was furnished with a variety of objects: whimsical toys, energy bars, colourful sketchbooks, color markers and pens, gourmet lunch, freshly-brewed coffee, and snacks (see Figure 20).

Figure 20: Photographs of desk workstations before study AGJ subjects arrive (image-left) and gourmet pizza lunch (image-right).



Ellie confirmed that the arrangement of the study AGJ space was positively generative for her own creative process:

Every table was, (laughing) every table was fully stocked with all the little things that you might need for ideating and mind-mapping.

The research study's objective was to curate this physical space in order to create an atmosphere of warmth and comfort, hoping to minimize associations to clinical research spaces as well as nervous tension or anxiety. At the material level, the curation of the study AGJ physical environment helped subjects feel physically good when they were fed good food that is inclusive of everyone's dietary needs and restrictions. At the symbolic level, inclusive food catering became meaningful to them, because it demonstrated recognition of and sensitivity to their material needs. Having said that, the curation of the space was not ideal. When subjects were asked what could be improved about the study AGJ, they offered many suggestions, including: notifying subjects to bring headphones in order to minimize distraction from noise (John), providing sugar and energy drinks to physically fuel the design process (Martin), choosing a game jam space with windows (Alex), extending the game jam running time, possibly to an all-nighter to fuel creativity through some added pressure (Yasmine, Martin, Alex), and scheduling short presentations of games-in-progress (Martin).

Repeating the study AGJ with a different set of subjects, the configuration of the space might pose different challenges for different individuals, and a higher degree of conflict or tension might arise. Such challenges cannot always be foreseen, and moreover they are not inherently bad; they constrain perfect reproducibility, which is where existential phenomenology's theoretical strengths lie for analyzing safe spaces as 'lived spaces'. Therefore, the game jam qua safe space is never fully safe nor is it reproducible. This speaks to its limitations, but also to the relevance of Buber's theory of concrete, lived relations which are not reducible to quantification and decomposition imposing order and control. Indeed, lived spaces are always subject to unpredictable forces that are beyond the control of the facilitator and participants. This analysis delves into some of those unpredictable forces in the next section.

4.8 The AGJ as a sociopoietic platform: parameters propelling the creative process

The study AGJ was defined by various parameters or restrictions, which played an important sociopoietic role of propelling the creative processes of subjects. These parameters included the need for careful self-presentation and impression management (Goffman, 1956), the game jam's time-limit, which pressured subjects to rapidly prototype their autopathographical games, providing a "sense of urgency and accountability" to the creation of a game, and creating feelings of vulnerability. These parameters may have motivated subjects to manage their emotions and steer their design expectations. Subjects emphasized how game jams are galvanizing, especially for those with mental health issues; they motivate participants to take up making and perhaps completing a prototype, thus invoking the advantages of creation in a collective space and the power of sociopoiesis in any game jam experience:

Game jams in general give that sense of an accountability and urgency, which for making a game in general regardless of subject matter is immensely helpful, especially because of my personal weaknesses which are—motivation and getting to the actual making.

Through these parameters, the AGJ platform may have imposed a socially-mediated structure on the autopathographical game design experience that boosted the self-confidence and self-assurance of subjects. Sam explained how these parameters positively shaped his autopathographical game design experience:

I thought it was really wonderful that you had the combination of a game jam, which is a high stress environment but you get a lot of productivity out of people because everyone's in it together and everyone's stressed out, and everyone wants to put together a product that they can show at the end. So, you get this huge amount of productivity, but then you get a whole bunch of people who have suffered and who are dealing with that suffering in the same room and it worked so well together because you have two intersecting qualities.

Marco supported Sam's assertion that the stress of game jams can be productive given that they are ultimately work spaces and should not simulate the comfort of one's own home, because:

Comfort zones are horrible for building. Being in a comfort zone and being comfortable in your work zone—they're completely different. A comfort zone is a place where you shouldn't work. It's your home; you're with people, like you're in a different situation.

And working in an environment is work, like you need that space of work; you need to be in a state of mind that this is work.

Indeed, Sam affirmed:

Having that sense of urgency and accountability to something is just invaluable for getting things (done).

Alex also underscored the stimulating aspect of the study AGJ environment:

Game jams are great because you're *forced* to do and doing is very helpful for mental illness, when you make something in a jam it's really reaffirming to be like, look, 'I made this thing, it's done, people played it'.

The study AGJ time constraint also propelled creative thinking according to one subject, who argued that time limits and deadlines engender lateral or divergent thinking:

It *forces* the creator to create under *pressure* which for me allows for lateral thinking. I feel that this method of lateral thinking is necessary when trying to communicate such an abstract idea (see Figure 21 below).

Alex also explained why game jam deadlines can be very motivating for his design process:

Jams are great because you have a deadline and you *push* yourself to do something in a short amount of time...Whereas normally, a game like this, I work on this a few hours today, then maybe tomorrow, then I'll kind of lose steam...I've made a lot of little prototypes but because I'm not a very stable person, it's kind of hard to finish things...

Figure 21: Autopathographical Game Jam subject doodle⁶⁰

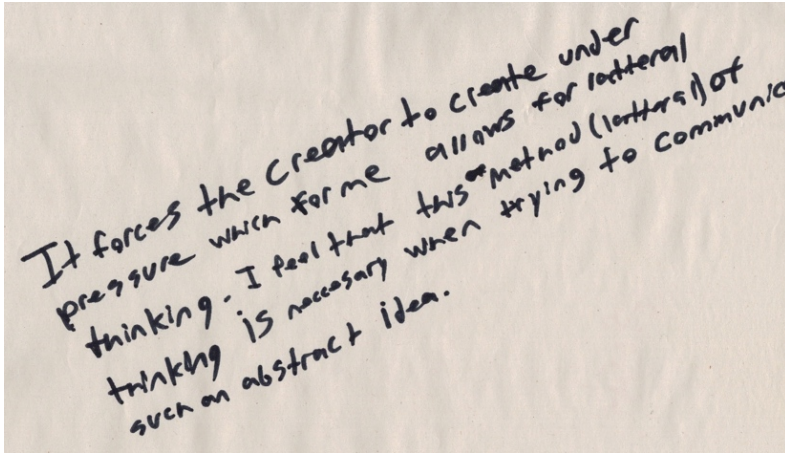


Image: “It forces the creator to create under pressure which for me allows for lateral thinking. I feel that this method of lateral thinking is necessary when trying to communicate such an abstract idea”.

The key words in the above testimonies are the positive forms of ‘force’, ‘pressure’ and ‘push’ invoked by subjects that the game jam deadline enacts in order to motivate designers with mental illness to complete a prototype and stimulate creative thinking, which may also lead to effective communication of their ideas.

The daunting task of exhibiting or demoing⁶¹ a personal game during a game jam can also contribute to feelings of self-pressure. For Fred, “when exhibiting your personal games, expectations are *high as hell*”; he added that his “personal expectations” placed pressure on him to succeed in the task: “you just kind of *beat yourself up*”. Yasmine found the adrenaline rush of the “*crunch time thing*”⁶² of game jams productive for her own creative output, because under

⁶⁰ Doodle responds to the following researcher question to group: How is a Game Jam a vehicle for articulating deeply private or personal stories of illness, disability, trauma? Is this method a contradiction –telling personal stories in a group setting? How so or how not?

⁶¹ To ‘demo’ is to demonstrate a functional software program or a playable game to an audience.

⁶² There is a big difference between the pressure to complete a game over a game jam’s two-day period, which can feel energizing to game designers and continuous pressure over longer periods of time, the latter of which may negatively affect designers’ wellbeing (as per article claims below). A cultural analysis of the game development industry’s so-called “crunch culture” is beyond the scope of this dissertation. Please see Jason Schreier’s (2017)

that stressful constraint “most of my work or my productivity increases”. She said that she prefers the all-nighter game jam because she finds the pressure energizing (she called it “*charging on the game*”) for her creative process:

This game jam’s an exception but having somewhere to like just stay an entire night, for instance, like charging on the game, like I don’t mind.

Martin also expressed desire for a game jam with a dramatically longer time span, perhaps because the game jam platform fosters an environment where an accumulation of a flow state propels creative output (Csikszentmihalyi, 1990).

The stress and pressure of the two-day time constraint of the study AGJ motivated subjects to work fast and complete a playable prototype, which felt good and boosted self-confidence. As the above testimony shows, some subjects appeared to express a desire for *more* stress and pressure, which demonstrates that they were not fearful of this form of discomfort—in fact, they welcomed it. However, the level of discomfort of the game jam’s time-limit is nevertheless regulated; for example, they were not given enough time to add the ‘meat on the bones’ or details on their minimalist game prototypes because this constraint simplifies their design problem space. Giving game jam participants more time during a game jam may prove to be too stressful. Polya (1945; 1957) claims that going into details too soon can be detrimental to solving a problem: “if you go into detail you may lose yourself in details. Too many or too minute particulars are a burden on the mind” (p. 76). As such, the game jam platform enables the designer to churn out a quick prototype, which can be a way to perceive the general form of the problem, akin to a sketch or outline in drawing or writing. Game design scholar Emma Westecott (2013) also argues that games prototyped in Do-It-Yourself and game jam contexts are “sketches” in design form that need further development.

Participating in a game jam may also hold designers accountable to others. Awareness of others and the necessity to perform in front of others imposed a structure on study AGJ subjects that helped them keep up appearances and steer their emotions. As such, the design process was

latest article in The New York Times for more on this phenomenon:
<https://www.nytimes.com/2017/10/25/opinion/work-culture-video-games-crunch.html>

influenced by subjects being observed by others, for example with cameras recording their actions. This influence can be understood in terms of Sartre's suggestion (1943; 1992) that human awareness is a structuring device for intersubjectivity based on a reciprocity that bridges subjectivity between two persons, as the awareness of being observed by the Other as Object allows you to see them as Subject. According to Catalano's (1974) reading of Sartre's analysis of "the look" in *Being and Nothingness*, this revelation rests in the subject's *pre-reflective* recognition that the subject (the "I") exists as a "foreign self seen by the other". (p. 163). Sartre (1943; 1992) says: "it is in and through the revelation of my being-as-object for the Other that I must be able to apprehend the presence of his being-as-subject" (p. 256). It is this apprehension of mutual observation that constitutes the Other as subject for both parties: "I experience the other as an essential modification of my own consciousness", Catalano says.

The way in which the study AGJ group facilitated its own structuration can also be understood in terms of Goffman's (1956) theory of impression management and self-presentation. From Goffman's point of view, when inhabiting a role, a group member becomes a disciplined performer in order to maintain peace and avoid disruption in the group. For example, in the context of a social performance, "expression-control" is one way to enforce the rules of social behavior; expression-control disciplines 'performers' in a space through "the management of one's face and voice" and display of appropriate affective response" (p. 138). Goffman states that "the disciplined performer is also someone with 'self-control.'" (p. 137). It appears that the study AGJ subjects controlled their own social performances in order to finish their computer games without emotionally 'breaking down' in front of others. Vicky explained that this form of self-control allows her to exercise her DBT (Dialectical Behavior Therapy) skills, which presumably builds resilience:

Through the process of revisiting the experience and emotion, things will come up but you don't want to emote that too much, you don't want to burst out into tears in the middle of the room. You know, it can be very draining. And then I tend to get pretty intense at jams and I can get frustrated, especially with Twine because it always breaks and crashes and it's always a mess. So, then you're dealing with that and then at the same time you're also dealing with your emotion that's coming up about the piece, so it's a really good exercise in using DBT skills and doing emotion regulation.

Goffman explains how, through expression-control, team players can build skills to trust themselves and be trusted by others (p. 138), thereby contributing to the cohesion of the group.

Vicky's comment seems a good example of this process of securing group cohesion, which characterized the entire study AGJ. To draw on Buber again, empathy (as opposed to inclusion) towards the Other means suppressing your own perspective and emotions in order to respect the emotional integrity of the Other—which is what Vicky practices at the study AGJ when she manages her own emotions.

4.8.1 Emotional restrictions of the study AGJ: discomfort and vulnerability

This section takes up the topic of discomfort as an emotional restriction engendered through sociopoiesis that is nevertheless fruitful for the individual designer's creative process. Subjects testified to feeling vulnerable in their *anticipation* of embarrassment or shame, which appeared to propel the creative process precisely because this feeling did not become actualized—it stayed in the realm of *potential*. As previously argued, study AGJ subjects are not thrust into an emotionally harmful space rampant with tension in the way that they might be in other game jams. However, the potential to stigmatize or shame the designer authoring a game autopathography still existed in latent form—in this context, Brown and Levinson's (1978) face-threatening act (FTA) becomes a marker that affects how subjects view themselves during the study AGJ. Thus, anticipating judgment by others can feel threatening and add pressure. Alex commented on the potential for receiving contemptuous reactions at other game jams:

(The AGJ) is incredibly empowering and gives you a lot of personal strength to know that there are other people that go through stuff, so I think it's amazing and also just the environment of being there, you're probably a little safer to be creative about these things because you're not going to have some douchey guy come by and say—'what game are you making'?

In line with Buber and Kenney, Depraz (1999) also believes that intersubjectivity is not necessarily free from the potential of confrontation, opposition, and conflict (p. 108). Religious scholar Seymour Cain (1996) discusses how Buber's relational ethics are different from the common perception of conflict:

The dialogical stance does not foreclose the possibility of conflict or division. In fact, it includes it as real possibility, that which we see actualized almost every day. But it steers us away from the demonization of the other person, nation, religious community, or socio-political party, from the dehumanization of our adversaries, of those who differ with us. (p. 136)

If we look at the study AGJ through the above lens, we can see that outright conflict or rampant tension did not arise. This lens helps us understand a critical but subtle dimension of the study AGJ. But as per the Buber/Cain/Depraz/Kenney analysis, this does not mean that there were no tensions at all. Several subjects commented on the presence of feelings of discomfort, though these were not necessarily contrary to creativity. Discomfort is produced through the act of observation and the awareness of being observed by others, to invoke Sartre's analysis of the look again. Observation is not merely tied to sight. According to Catalano (1974), Sartre is not necessarily referring to sight when he discusses observing and being observed. For example, hearing footsteps behind you as you are walking down a street or feeling shame are both self-aware apprehensions and give meaning to 'being seen' by others (p. 160). Although subjects did not give voice to the presence of shame during the study AGJ, they alluded to the potential for shame, hence the usefulness of Sartre's perspectives on this affect. Sartre (1943;1992) speaks of shame as a "mode of consciousness", which "in its primary structure (is) shame *before somebody* (emphasis added)" (p. 221). In other words, shame is invoked through our relationship with others in a social milieu; it is intrinsically a relational phenomenon. Sartre uses the example of someone who makes a vulgar gesture, all alone, and then realizes that they have been seen by someone; this recognition of being seen by another sends an "immediate shudder" in the person making the vulgar gesture. "Nobody can be vulgar all alone!" exclaims Sartre. Goffman (1963) similarly suggests that the awareness of being perceived (observed) by others as different is part and parcel of how stigma works: the "shamed person is ashamed to be ashamed" (p. 155). A person is ashamed of themselves because they are aware of someone else shaming them, or of the potential of society shaming them for what constitutes unacceptable social behavior or social status. "Shame is by nature recognition. I recognize that I am as the Other sees me." (Sartre, 1943; 1992, p. 222)⁶³. To briefly return to the interactional linguistics theories of Brown and Levinson (1978) and Kerbrat-Orecchioni (1997), shaming is also a face-threatening act (FTA), which erodes the self-image and social status of the subject by way of castigating them for inappropriate behaviour.

⁶³ Shame is also structured in relation to the absolute or God, presumably for those individuals who are of religious faith or influenced by religious beliefs to some degree – a discussion beyond the scope of this chapter.

The study AGJ carved out an environment that minimizes the potential for fear and shame, minimizes face-threatening acts, and invites subjects to express experiences that would be stigmatized or rejected at other game jams or collaborative opportunities. Although the study AGJ minimizes the potential for fear and shame, as we have seen, not all tensions were (or could be) eliminated, and you can see these detailed nuances of shame in those tensions that did arise, as for example in John's statement. John confirmed this experience by describing a time when he:

...wanted to do a really emotional, dark game that had a lot of symbolism and sort of interesting psychological elements to it.

He added that the response of his team at the time was not supportive:

And I tried to get a team on board and they were all like, 'no that sounds way too weird, I don't think people would like it.' So (laughing) they weren't into it.

The trepidation of the subjects shows that the potential for tension exists, hence the discomfort and uneasiness, especially on the first day of the game jam, prior to their emotionally moving and candid introductions. Ellie admitted to feeling vulnerable in simply being seen or "exposed" by others at the study AGJ:

Feeling a little bit exposed by the game jam certainly. Again, something you don't really think about before coming in and it's difficult.

She added:

Any time where I have to speak candidly it's always uncomfortable.

Nathan echoed Ellie's feelings of discomfort in the potentiality for shame in the AGJ space by stating that he anticipated to feel a lot more unease, but perhaps was pleasantly surprised that his discomfort was diffused by the group introductions, which made him feel safer:

I thought I would be a lot more uncomfortable than I am. I think the introductions were nice. And I feel safe in this space to talk about that kind of stuff.

The self-awareness of vulnerability invokes the potential for feelings of discomfort and shame emerging in the context of authoring an emotionally-driven game. Sartre's insights about shame, which is an extreme form of discomfort, are helpful for showing that the awareness of being

observed by others instills in subjects the recognition that there is always the potential for discomfort and shame during the study AGJ, which can be restraining. Nonetheless, Martin explained that sharing, accompanied by feelings of trepidation and fear of revealing private experiences, constituted the study AGJ as a therapeutic space. Hence, sharing quells trepidation and fear, which in turn generates *relief* for subjects when they are acknowledged by others, and when shame is thwarted:

It can be very hard at first, you can face a lot of fears. But in the end, you kind of feel relieved of putting these problems outside and sharing. It's a way to relieve the pain — not that it's that bad but you can (pause)... it's like, sharing brings peace.

Alex added that the stakes of authoring an autopathographical game can be high in terms of producing negative emotions, which echoed Martin's reference to fear. Thus, reliving a self-experience can be a risky endeavor in and of itself:

It can be triggering for someone to make something about a personal experience because they have to relive it while they're making it, right, so it's...yeah.

Yet this emotionally risky form of game authorship takes place in a relatively safe environment. There are a number of factors discussed throughout this chapter that mitigate or prevent the potential volatility of autopathographical game design, including feelings of communion and solidarity, acceptance, motivation, and the ability to share openly with empathy. The anticipation of discomfort or shame may restrict the freedom of the designer in some sense, yet this restriction has a silver lining in that it fosters a sense of release when these negative affects do not arise. When shaming or face-threatening acts are undermined, the result can be a form of relief. Thus, these nuances of interaction and anticipation demonstrate how the study AGJ in its emergent qualities diverged from other game jams in meaningful ways.

To summarize, the power of the Buber/Goffman/Sartre/Brown and Levinson/Kerbrat-Orecchioni analysis is demonstrated in the fact that the subtle structures constituting the anticipation of discomfort or shame contributed both to the AGJ's creative power and to its therapeutic power.

4.9 Mutuality is a form of responsibility to the Other that also restores the Self

The Autopathographical Game Jam is a form of sociopoiesis, a collective making which both propels creativity and fosters a therapeutic community for its participants. The act of sharing oneself with another human being has to be accompanied by *self-responsibility* and *responsibility to the Other*, whether in the form of self-control, generosity, openness, empathy, inclusion, or reciprocity. The espousal of these values and responsibilities engenders the co-emergence of the first-, second-, and third-person perspectives, which are the scaffolding of the AGD-AGJ qua healing landscape. These values and responsibilities build critical distance, which is the key factor in the second-person perspective, herein achieved through a variety of structuring devices discussed above: a heightened self-awareness emerging from awareness of the Other; verbal and non-verbal interaction; openness; expression-control; the steering of emotions; and accountability to others. In these activities, the individual ‘remains themselves’, which is a self-responsibility, while simultaneously entering relationships, which is a responsibility to the Other. As Buber explains:

To be fully real the I-Thou relation must be mutual. This mutuality does not mean simple unity or identity, nor is it any form of empathy. Though I-Thou is the word of relation and togetherness, each of the members of the relation really remains himself, and that means really different from the other. (Friedman, 2002, p 70).

These dual responsibilities to self and the Other build a therapeutic community of support for study AGJ subjects who came together as strangers, and who qua strangers bore no responsibilities to one another. Paradoxically, they may find that responsibility to another person, which is sometimes framed as a limitation on the freedom of the self, is also a form of freedom. That is, to influence others in positive ways, such as contribute to their healing, is a freedom from feelings of powerlessness, which may in turn restore the self. The study AGJ can be conceived of as a sociopoietic space that could foster this restorative feeling.

Chapter 5

Autopoiesis: Self-Making as a Process of Self-Care in Autopathographical Game Design

5.1 Introduction

Autobiographical and autopathographical authorship, including autopathographical game design (AGD) are forms of *autopoiesis* or *self-making* that are ontologically defined by ‘auto’—whose origin is from the Greek *autos* (αὐτός) or self⁶⁴. In autopathographical game design, autopoiesis and fabulopoiesis are both narrative-based processes of self-making that take on different dimensions in creative identity construction. AGD autopoiesis is a self-reflexive praxis of looking inward in order to refashion first-person experiences into an autopathographical game. The definition of the word ‘self-reflexive’ is “containing a reflection or image of itself; self-referential” (Oxford Dictionaries, n.d.)⁶⁵. As a process of self-making rooted in self-reflexivity, AGD autopoiesis importantly accesses the self-reflexive praxes of self-reflection, introspection, and self-inquiry. These praxes are intrinsically self-referential in that they involve *the act of self-reference*, which may become a sustained process of referring, indicating, or pointing to oneself in the first-person. Self-reflexivity contributes to a meta-awareness of the process of one’s own making, and in the case of AGD narration during the Autopathographical Game Jam (AGD-AGJ), a “self-conscious narrator” (Oxford Reference, n.d.; Booth, 1952)⁶⁶.

Fabulopoiesis is also a process of self-making, but one that differs from autopoiesis in that it accesses the author’s metaphorical and analogical imagination in order to reinvent the self and ‘break free’ from the act of self-reference and the first-person perspective. This chapter focuses on autopoiesis, i.e., the autopoietic dimensions of autopathographical game design as a

⁶⁴ Source: <https://en.oxforddictionaries.com/definition/us/auto->

⁶⁵ Source: <https://en.oxforddictionaries.com/definition/us/self-reflexive>

⁶⁶ Source: <http://www.oxfordreference.com/view/10.1093/oi/authority.20110803100453429#> The term “self-conscious narrator” is from literary studies and applies to narrators in literary fiction that call attention to the apparatus of storytelling (Booth, 1952). It is a useful term to employ in the analysis of the autopoietic process of authors as narrators.

regenerative process of self-care and self-therapy that contributes to therapoiesis. Autopoiesis is thus an ontological imperative of autopathographical game design without which this creative praxis would lose its pivotal status as a self-reflexive authorship form; without autopoiesis, AGD is nullified and falls apart as a tool, a praxis, and a distinct therapeutic dimension of AGD-AGJ.

The focus on narrative in this chapter is significant because constitutive of autopoiesis as a narrative-based process is *the act of narration*. This act of narration is *autopathographical narrative-telling*, which involves study subjects constructing a *foreground or background autopathographical narrative* through the AGD-AGJ process. Foregrounding and backgrounding the autopathographical narrative comes in three forms:

- 1) Subjects typically foregrounded their autopathographical narratives in discourse—testimonials to the researcher and others during the Autopathographical Game Jam.
- 2) Subjects typically backgrounded their autopathographical narratives in the game itself, omitting identifiable references to themselves—factual and identifiable details such as their legal names, first-person pronouns, and other autobiographical details from their lives (i.e. persons, specific events, etc.). This narrative backgrounding was a result of subjects’ desire for self-care, humility, and self-protection from stigma (‘saving face’). Aided by the formal properties of computer games, those narratives were backgrounded when they became subsumed under game rules and formal constraints; they became further backgrounded through the analogical and metaphorical construction of the fabulized self.
- 3) There was no access to the degree to which subjects foregrounded or backgrounded their autopathographical narratives during the self-reflexive process of autopoiesis, i.e., subjects telling and re-telling their own autopathographical narratives to themselves while making their games. There was only confirmation that this self-reflexive process indeed took place (as will be discussed).

There is one more important aspect to the narrative-based process of AGD autopoiesis in the context of the Autopathographical Game Jam (AGD-AGJ). AGD-AGJ imposes a multi-role and multi-perspectival structure upon the process of autopoiesis in its self-referential dimensions. We may assume that there are two perspectives involved in autopoiesis—the *autopathographical*

author and *autopathographical narrator*—but there are others, if we look more closely. The AGD ‘author’ is thus not a monolithic or unified authorial figure; rather, when we look deeper into the structure of the autopoietic act of identity construction during the AGD-AGJ process, we can see that study subjects mobilize a variety of roles and perspectives: game designer (author), game jammer, autopathographical narrator, speaker, listener, stranger, friend, game player, sufferer, self-healer, healer, and research study subject. Through AGD-AGJ, the autopathographical author activates these roles and perspectives in constructing a heterogeneous author identity. The above roles and perspectives will be read through French theorist Ruth Amossy’s (2009) theoretical concept of *auctorial posture* used to distinguish the multiple hidden positions, perspectives, and roles that shape author identity, such as the author’s public and private personae, author as subject, character, and narrator in their creative work, as well as the author’s readers, audience, and critics—all of whom shape author identity. These auctorial postures also overlap the three epistemological perspectives discussed in the previous chapters: first-person (subjective, affective), second-person (relational, intersubjective) and third-person (objective, detached) perspective. Thus, auctorial posture is also constituted via the multi-personed epistemological perspective of the autopathographical game designer. In AGD-AGJ, these auctorial postures contribute to the autopoietic processes of self-care and self-therapy.

The above analytical issues will be tackled in the following parts:

Part I: How ‘narrative’, which gives structure and meaning to subjectivity, maps to autobiography, autopathography, computer game, and in the case of AGD, autopathographical game.

Part II: Autopoiesis is a self-reflexive praxis of creative identity construction in AGD involving the sustained acts of self-reflection, introspection, and self-inquiry; this self-reflexive process may be construed as narrative-based if we consider subjects as telling their autopathographical narratives to themselves while making their games.

Part III: By adopting specific *auctorial postures* towards self-care and self-therapy, subjects deploy careful disclosure and self-reference in AGD-AGJ in order to foreground or background their autopathographical narratives, depending on the narrative-telling domain, i.e., AGJ verbal testimony vs. AGD proper.

The following theoretical matrix is deployed: Aristotle's *Poetics* (c. 335 B.C.; 1961 & 2006), Goffman's (1956 & 1974; 1986) theories of frame analysis and self-presentation, Foucault's (1988) *Technologies of Self*, narratology and rhetoric (Propp, 1928/1968; Nöth, 2007; Bal, 2009; Amossy, 2009), autobiography studies (Eakin, 1985), autopathography studies (Frank, 1995; Couser, 1997; Jurecic, 2012); and game studies (i.e. Juul, 2011; Bizzocchi, 2007; Anthropy and Clark, 2014).

5.2 Part I: Autopoiesis and narrative

5.2.1 Narrative and subjectivity in autopathographical game design

Narrative and subjectivity are ontologically co-constitutive insofar as narrative is intertwined with subjectivity as a mode of cognition and a mode of perception (Eakin, 1985; Bruner, 1995). Autobiographical scholar Paul John Eakin (1985) affirms that autobiographical expression, whether in the form of an "I" uttered in speech or a full autobiographical novel, is a particular realization of the human will to live: "the power of language to fashion selfhood is not only successful but life-sustaining, necessary to the conduct of human life as we know it" (p. 191). Eakin gestures to the possibility that subjectivity and narrative are co-constitutive in that both structure each other: "the structure of experience itself must be in some sense narrative; it is the narrative form of experience itself that generates the stories that structure it" (p. 132). Eakin is drawing on the work of philosopher Stephen Crites who states that "the form of active consciousness, i.e., the form of its experiencing, is in at least some rudimentary sense narrative" (as cited in Eakin, 1985, p. 131). But also, according to literary scholar Barbara Hardy and Crites, "the structures of narrative reflect, and are derived from, the structures of consciousness" (Eakin, 1985, p. 133). Therefore, narrative gives subjectivity structure and meaning through language and symbolic form, i.e., linguistic and non-linguistic expression (Langer, 1942; Eakin, 1985).

In order to understand the therapeutic effects of AGD autopoiesis, this analysis unpacks how 'narrative', which gives structure and meaning to subjectivity maps to a range of creative modes of expression such as autobiography, autopathography, computer game, and autopathographical game design. Consider first some study subjects who testified that they are

not expressing their personal narratives in their game autopathographies, yet nevertheless gestured to the role of ‘story’ or ‘narrative’ in autopathographical game design.

For example, Marco stated unequivocally that he is not telling a story in his autopathographical game:

Because the game is abstract I’m not telling a story...

Alex implied that integrating a story in the design of his autopathographical game *Brain Pains* was challenging, presumably because he needed specific game design skills and (material or other kinds of) resources that may not have been accessible to him:

I’m making something abstract that doesn’t really have a story. I was going to put a story but I can’t think of how to do it, so it’s taking apart a brain sort of thing.

Alex’s choice to exclude story in his experiential autopathographical game of moveable ‘brain nodes’ appears to be based on practical reasons: it was easier, it took less time, and it was more manageable given the time-limit of the AGJ.

John claimed that games can indeed integrate strong narratives using a “cinematic storytelling approach”, but that under the rigid temporal and logistical constraints of the AGJ environment, he, like Alex, also eschewed a robust narrative in order to rapidly prototype the computer game; hence justifying his choice for an abstract “symbolic” approach focusing on the game rules and formal constraints to drive the gameplay:

It’s hard to do in a game jam type setting because I feel like for a true autopathographical game you’d need much more story, you’d need much more sort of like a cinematic storytelling approach and that’s almost impossible to do in a couple of days. I’m sure it can be done, but in my case I went much more symbolic.

Despite the time limits of the AGJ, which appear to undermine subjects’ ability to integrate their autopathographical narratives into their games, John implicitly suggested that his game might have a story, but that this story is not identifiable as belonging to *him*.

You would be able to tell that it’s about bipolar disorder but you wouldn’t be able to tell my *personal* story (emphasis added).

Alex articulated something similar to John that seems paradoxical on first impression—a desire to make “non-narrative games” that nevertheless “tell a story”:

I’m really interested in trying to make (games) that are completely abstract and non-narrative but will tell a story, will tell a theme.

Alex’s striking point appears to entertain the possibility that you can either foreground a narrative or background a narrative through computer game design. In other words, he likes to design games that *thread* narrative elements and/or render them into abstract form, which suggests that he prefers to find ways of ‘backgrounding’ a narrative in the game. It makes sense if we consider the distinct formal properties of computer games, i.e., game rules and formal constraints through which the story might be abstracted, a story that *will be told*, perhaps in a different form than a narrator narrating a sequence of events that an audience listens to, such as through the game player’s performance. Marco, John, and Alex’s testimony supports one of this dissertation’s claims that AGD subjects appear to foreground their autopathographical narratives by narrating them through verbal testimony (during the AGJ and to the researcher) while backgrounding them, i.e., symbolically rendering them through the game’s rule-based system. As such, the autopathographical narrative morphs into different forms when it is rendered and implemented into the autopathographical game (as will be demonstrated in the *Fabulopoiesis* chapter). This rendering of the autopathographical narrative is determined not only by the formal properties of computer games and by subjects’ creative design decisions, but also, as this chapter will show, by subjects’ objectives towards self-care and therapy.

It is important to theorize on what Marco, John, and Alex might mean by ‘story’ and how narrative and the act of narration (narrative-telling) might manifest in autopoietic forms of authorship such as autobiographical and autopathographical writing, autopathographical game design, and autopathographical narration/performance during the Autopathographical Game Jam.

5.2.2 What is a narrative?

Consider some formal properties of narrative, irrespective of mode and medium, in order to first trace similarities and differences between narrative and game structure and subsequently between autobiographical/autopathographical narrative and autopathographical game design. At an abstract level, a story is a sequence of chronological events organized into a coherent structure

with actors driving the action. At a more concrete and phenomenological level, a story becomes a narrative when told in specific ways by a narrator, i.e., an agent recounting the narrative to an audience. Structuralist film scholars Bordwell and Thompson (2008) define narrative as “a chain of events in cause-effect relationship occurring in time and space” (p. 75). They add that “the set of all the events in a narrative, both the ones explicitly presented and those the viewer infers, constitutes the story” . In effect, “the total world of the story action” constitutes the *diegesis* of the narrative, including objects, events, and characters the audience assumes occur off-screen (p. 76).

Barthes (1977) argues that stories or narratives are everywhere around us irrespective of mode, form, medium, or genre: “narrative is present in myth, legend, fable, tale, novella, epic, history, tragedy, drama, comedy, mime, painting (think of Carpaccio's Saint Ursula), stained glass windows, cinema, comics, news item, conversation” (p. 79). Juul (2011) quotes literary and film theorist Seymour Chatman who argues that, “narratives are indeed structures independent of any medium” (as cited in Juul, p. 48). AI researchers Mateas and Sengers (1998), who study the use of narrative techniques in AI systems, argue that narrative is often used to describe an array of conceptual objects with a “family resemblance” (invoking Wittgenstein) such as a philosophical worldview, a social group’s sense of historical identity, and a storyteller’s account (p. 8). They suggest that act of telling narratives involves organizing human experiences into narrative form and ordering events into familiar narratives. Familiar narratives are genres such as fables, myths, legends, tragedy, comedy, etc. that deploy a set of recognizable narrative conventions and motifs.

Bordwell and Thompson use ‘story’ and ‘narrative’ interchangeably explaining that “a narrative is what we usually mean by the term story. Yet, they also use the term ‘story’ to differentiate between story and plot (discourse): “the term plot is used to describe everything visibly and audibly present in the film before us...(including) “all the story events that are directly depicted” (p. 76). This analysis differentiates between story and narrative based on the presence of an agent telling or recounting a narrative, adopting the terms *narrative* and *narrative-telling* to emphasize the primacy of the autopathographical narrator to this study.

One of the primary ontological dimensions of narrative is its temporal basis and chronological organization (Eakin, 1985). An investigation of the logical structure undergirding narrative shows that an analogy can be drawn between narrative logic and procedurality understood as the system of rules in a game; narrative logic propels the plot in a narrative, and procedurality propels the gameplay when players engage with those game rules. Aristotle's *Poetics* (c. 335 B.C.) similarly frames story in terms of a cohesive structure whose parts are causally and temporally connected: "a story is not a series of events isolated like data points, but an unfolding motion that is continuous and one (1452a 15), no matter how much it may change or even reverse its direction" (Sachs, 2006, p. 5). Bordwell and Thompson confirm that "a random string of events is hard to understand as a story" (p. 27). The plot that characters engage in logically drives the dramatic arc of a narrative, which consists of the set-up, confrontation, and resolution that apply to all narrative forms including literary fiction, theatre, and film. The dramatic arc can thus be mapped to the "skeletal framework of a story" comprising the Aristotelian *logos*, i.e., speech, talk, dialogue (Sachs, 2006, p. 73). Therefore, the plot, sometimes represented through dialogue and other times through, for example, the action of a car chase scene, equates to the logical structure propelling the narrative forward, which is similar to the system of game rules propelling gameplay. Propp (1928;1968) also implies in his seminal formalist work on narrative that the binding agent of a sequence of events in fairytales is a type of *logos*, a logical ordering of narrative events operationalized through the story characters. For Propp, the narrative structure is comprised of a stable pattern, a so-called 'narrative grammar'⁶⁷ which is a set of limited, recurring, and constant character functions that carry the plot action towards its logical resolution, starting from the conflict as the story's premise.

As such, narrative cannot exist without a causal and temporal logic (*logos*) organizing its structure. We may not conceive of narrative as 'logical' per se, because the high emotional appeal (*pathos*) of its dramatic arc camouflages the logical ordering of plot points. From the perspective of the author, this is how we 'engineer' a narrative as a system of logically connected

⁶⁷ Please see Herman (1995) for more on "narrative grammar". Also, Gervás (2013) for more on how Russian formalist frameworks of narrative are used as the basis for creating the automated generation of stories in the computational realm. <http://drops.dagstuhl.de/opus/volltexte/2013/4156/pdf/p106-gervas.pdf>

plot points; for a narrative to engage the emotions or generate meaning for the reader/viewer, it should make logical sense, even if the type of diegetic logic pertains to the unique fantasy world of the story. Hollywood films that employ the common dramatic arc or “three-act structure”—the “paradigm” of setup (Act I), confrontation (Act II), and resolution (Act III) (Field, 1979)—are a good example of ‘engineering’ narratives by way of engaging a standard dramatic template that has proven to be successful. Similarly, an Aristotelian syllogism is the standard template behind a three-part logical argument consisting of two premises and a conclusion.

In contrast to classical drama or narrative, experimental non-narrative films do not adhere to a narrative structure of causal logic. They are conceptual at a high level of abstraction, and logical from an associational or analogical level. Bordwell and Thomson (2008) differentiate between narrative and non-narrative approaches to film form, arguing that aesthetic style can become a formal system (p. 304) in and of itself. In other words, film’s audiovisual and temporal elements such as cinematography, colour palette, editing (i.e., rhythm and tempo of cutting between shots), and sound are deployed as organizing principles in non-narrative film.

5.2.2.1 Narrative and games: formal tensions

Consider next how the form of spectator-based narrative and interactive media converge and diverge in order to understand how narrative might map onto the structure of games. The ontological debate about how story, plot, characters, and narrator map onto games has simmered on and off in game studies⁶⁸. Witness this debate’s re-emergence in games scholar and philosopher Ian Bogost’s 2017 provocative article, “Video Games Are Better Without Stories” in *The Atlantic* claiming that “to use games to tell stories is a fine goal, but it’s also an unambitious one”⁶⁹. A game designer himself, Bogost is merely arguing on behalf of creative experimentation

⁶⁸ This ontological debate has been framed from both ludological and narratological perspectives. Game scholar Gonzalo Frasca (2003) states that the narratology-ludology debate “never took place” (p. 92).

⁶⁹ <https://www.theatlantic.com/technology/archive/2017/04/video-games-stories/524148/>

in game authorship⁷⁰, that game designers should not be pressured to map conventional narrative structures to gameplay presumably because they might limit the creative imagination. Bogost implies that game designers should be encouraged to aesthetically experiment with the distinct formal properties of games and refrain from a rigid devotion to narrative form.

The ways in which narrative maps to games is determined by the distinct formal properties of games, such as game rules. Some game scholars have argued that story and/or narrative are not intrinsic to games (Crawford, 1984; Jenkins, 2004; Bizzocchi, 2007). Jenkins (2004) asserts that “not all games tell stories” and that “games may be an abstract, expressive, and experiential form, closer to music or modern dance than to cinema” (p. 119)⁷¹. One reason “games do not necessarily involve story” (p. 2) according to Bizzocchi relates to the way in which the narrative arc of spectator-based media (i.e., film and literature) maps awkwardly onto a game system. Once you introduce gameplay emergence and start playing a story, you frustrate the trajectory of a fixed narrative arc, states Bizzocchi. For example, playing the game’s “storyworld” (Bizzocchi, 2007) entails physically moving and colliding with game objects and characters, which disrupts the narrative temporal order and troubles stable patterns underlying narrative structures. Nonetheless, Jenkins (2004) confirms that “many games *do* have narrative aspirations” (p. 119). Given that game players manipulate game characters and their perspective of the game world by moving their avatars and directing their gaze at objects in real-time via user inputs, Bizzocchi (2007) suggests that the game character is the meaningful link between gameplay and narrative: “characters (heroes, villains, player-avatars, and non-player characters) live the enactment of the game and the story” (p. 5). If a game does not develop its protagonists (heroes/heroines) and antagonists (villains) to propel the dramatic conflict, such as in the narrative-driven RPG (role-playing game) genre, then *character development*, a meaningful

⁷⁰ The ludology-narratology argument has analogously ‘played out’ in the early 20th century debates about the ontology of cinema in terms of how dramatic structures of novels and theatre translate to the cinematic medium, given that many films have been and still are cinematic adaptations of novels and plays.

⁷¹ The games scholars listed in this paragraph do not seem to differentiate between the concepts of “story” and “narrative”. Please see a discussion of the difference between the two in this chapter’s section on “auctorial posture in autopathographical game design”.

feature of narrative form, is absent and has to be compensated for by the player's imagination, such as in token-manipulation games (i.e. chess).

Consider next how rules organize structure in both narrative and games. Both spectator-based narratives and games employ a logical structure, a system of rules, undergirding the form to propel it to a resolution in a temporal sense. For example, in her book *Computers as Theatre*, Brenda Laurel (1993) connects human-computer interaction and Aristotelian drama based on the structural similarities between interactive media and dramatic theatre accounting for how the computer can harness the causal logic of drama to scaffold interaction (e.g. gameplay). In narrative, various elements such as plot, characters, themes, motifs, and a dramatic arc join forces to propel the narrative forward. In a game, the system of rules propels the gameplay forward, uniting with emergent behaviors at the gameplay level to create meaningful gameplay experiences, or what Salen and Zimmerman (2004) call "meaningful play". Janet Murray (1998) has also maintained that storytelling and interactive media merge most expressively when they try to simulate human behavior through rules: "the computer can be a compelling medium for storytelling if we can write rules for it that are recognizable as an interpretation of the world" (p. 73). Juul (2011) similarly frames games as a combination of narrative fiction and "real" game rules, which encapsulates his ontological argument that games are "half-real" as a blend of reality and fiction:

To play a video game is...to interact with real rules while imagining a fictional world and a video game is a set of rules as well a fictional world.

Juul understates the ontological issue that videogame play is ostensibly free from the vagaries of 'real-life' consequences that "real rules" may engender; player death and injury in a videogame is symbolic and does not endanger the game player's physical body in the same way that a game of soccer or football might endanger the player's physical safety.

Another similarity between narrative forms and games is the use of quantitative devices in both: *metre* in poetry, fiction, and music and *metrics* measuring the player's progress in games, both of which quantitatively organize aesthetic form. Literary fiction and poetry employ metre to organize form and keep the reader's attention by evoking musical tempo and rhythm in the written or spoken word. For example, iambic pentameter in the literary and dramatic

masterpieces of Shakespeare, Milton, and Chaucer is a “metrical art” deployed to aesthetically organize fiction and drama (Wright, 1988). Games on the other hand rely on a variety of metrics externalized as a quantifiable win-lose outcome and quantified incentives and rewards represented in score boards and game character health indicators (health bars), all of which monitor and measure the game character’s progress. And the role-playing table-top game *Dungeons and Dragons* converts the player’s description of characters and actions during the narration process into a quantitative form of narrative generation. Commenting on the death of *Dungeons and Dragons* creator Gary Gygax, The New York Times writer Adam Rogers (2008) suggests a reason why that game captured the imagination:

Once you’ve broken down the elements of an invented personality into numbers generated from dice, paper and pencil, you can do the same for your real self.

Using a ‘character sheet’, players can measure the qualities of a game player character in order to aid gameplay and even use it as a blue-print for themselves.

A difference between the spectator-based narrative and a game is that the system of rules in a game is externalized and fluid at the level of material engagement (interactivity), whereas the system of rules in a spectator-based narrative is internalized and fixed at the level of reception. Thus, material engagement and reciprocal cause-effect are not accessible in the spectator-based narrative; the artifact can have an aesthetic (i.e. physical, emotional, cognitive, intellectual) effect on the reader or viewer, but the artifact itself cannot be affected by the audience reaction. In other words, when audiences watch a film or read a book, they cannot physically affect the book qua material artifact; they cannot physically delete, move, or rewrite a paragraph in a book passage⁷², or re-edit and restructure a movie scene⁷³. But in digital and analog games, they have the ability to interact with material objects, restructure the order of

⁷² The exception here is the practice of annotation –writing on a book’s page or margins, including highlighting, underlining, striking through a sentence, etc.

⁷³ The exception here are Internet and digital media cultures that offer participants the ability to plunder and remix existing media, create their own edited versions or mash-ups and upload them online on such platforms as YouTube. This issue is a matter of context, of course. You cannot refashion a film in a movie theatre, but digital platforms allow you to plunder media and re-arrange the outcome for your own creative satisfaction.

events and affect the ensuing outcome of the game. There is an exception here which relates to human conversations where two or more people are verbally telling stories to each other. It is possible for interlocutors to change or modify their respective stories in the act of conversation, because human beings can affect each other on a material level, i.e., physically, intellectually, and emotionally.

One ontological assumption about games in relation to the *act of narration* is that stories cannot be told in a game; they can only be played (Simons, 2007). Eskelinen (2004) distinguishes between narratives and games based on the assumption that a game is played, and not told: “luckily, outside theory, people are usually excellent at distinguishing between narrative situations and gaming situations: if I throw a ball at you, I don’t expect you to drop it and wait until it starts telling stories” (as cited in Simons). There are exceptions to this assumption that playing a game is diametrically opposed to telling a narrative; the act of telling narratives and constructing stories via a narrator-player position can indeed become an intrinsic part of gameplay, such as in the role-playing game *Dungeons and Dragons*. This role-playing game heavily relies on players ‘telling’ a narrative and ‘describing’ their characters and actions in order to play the game, according to Dormans (2006): “Pen-and-paper role-playing games tend to be more told than enacted. In these games, one plays a character by describing what she does and what she says”. Nevertheless, Simons argues that:

Games studies scholars have already admitted that the boundaries between games and narrative are not very clear-cut, that narrative often plays a significant role in games (as expository introductions, as ‘non-playable’ parts in the form of cut-scenes⁷⁴, as background information, or as report-after-the-fact...)

Finally, alternative game designers Anthropy and Clark (2014) emphasize that narrative is not taken seriously enough in mainstream game design. Anthropy is one among a number of significant game designers who catapulted the digital interactive narrative, such as the choose-your-own-adventure story to public attention. Digital interactive narratives invite players to

⁷⁴ A cut-scene is a short non-interactive sequence in a video game that interrupts the gameplay or serves to set up the game action. Cut-scenes offer narrative exposition and information about the game characters, sometimes serving as a form of character development for building an emotional connection for the game player. Cut-scenes cannot be played, but they can be skipped over if the player wants to return to the gameplay.

engage with multiple story choices and sets of authored possibilities where the outcome is not fixed, but fluid and contingent on player choices. Anthropy and Clark especially underscore the importance of personal narrative in games, putting focus on the author qua narrator: “until we learn how to tell *real stories* in games, ‘story’ is always going to mean ‘cutscene.’” (p. 8). Anthropy and Clark suggest that narrative, as we typically perceive it, can be effectively translated in games using so-called ergodic elements—or what Noah Wardrip-Fruin (2009) calls “expressive processes” (i.e. player-game actions and behaviors)—resulting in novel emergent experiences. Their impassioned plea to re-center the personal narrative in videogames has a purpose: to disrupt videogame tropes and stereotypes commonly found in the first-person shooter (FPS) genre.

Anthropy and Clark’s statement gives voice to a tension that exists in both game studies and in autopathographical game design proper. This is the tension between: i) the semantic pull of traditional forms of narrative authorship as envisioned in narratology, i.e., developing meaningful characters and worlds that resonate with author and audiences; and (ii) discovering novel ways of aesthetically experimenting with select narrative elements through the distinct formal properties of computer games. One way this tension is negotiated in AGD is that subjects omit identifiable and factual self-references-i.e., their actual age or legal name, descriptions of actual actors and events that pertain to their autobiographical and autopathographical narratives (i.e. family, friends) instead opting to abstract their autopathographical narratives into the symbolic form of game rules and fabulize themselves through metaphor and analogy. Subjects share those identifiable self-references to some extent when they verbally narrate their autopathographical narratives during the AGJ and in researcher-directed interviews.

To summarize, this analysis proposes an integration of the Aristotelian and narratological models of narrative structure with game structure. Thus, narrative is taken to be comprised of at least the following key ingredients irrespective of mode or medium:

a) temporal ordering of events experienced by actors or agents (“fabula”⁷⁵) (Bal);

⁷⁵ Based on narratologist Mieke Bal’s (2009) definition of *fabula* as “a series of logically and chronologically related events that are caused or experienced by actors” (p. 5); to be discussed in-depth in the *Fabulopoiesis* chapter.

- b) diegesis—“visible and inferred storyworld” (Bordwell and Thompson, 2008) or the game’s “story world” (Bizzocchi, 2007);
- c) characters and narrators (actors, agents, perspectives, roles, functions) propelling the narrative or gameplay (i.e. Propp; Bizzocchi, Anthropy and Clark);
- d) *logos* scaffolding plot and dramatic arc in spectator-based narrative (i.e. Field’s “paradigm” of set-up, confrontation, and resolution) rendered in game form as explicit and implicit game rules and game behaviours (i.e. Laurel, 1993; Murray, 1998).
- e) *pathos* (emotional appeal).

5.2.3 Two forms of autopoiesis: the authorship of the autobiographical and autopathographical narrative

Consider next the similarities and differences between the autobiographical and autopathographical narrative, of which their authorship constitutes autopoiesis or self-making. The ontological significance of autobiography and autopathography in autopoiesis (i.e. self-making or self-authorship) has implications for subjects in how they negotiate truth and fiction in AGD. Despite the well-established theoretical stance that autobiography is a form of fiction, AGD subjects *perceive* the autobiography, of which autopathography is a subgenre or subtype, as revealing of factual truths through the use of identifiable self-references, which causes them discomfort. As such, they select specific ways to narrate real and/or perceived truths about themselves contingent on the domain of narration (i.e. verbal testimony during the AGJ versus through AGD).

Autobiography and autopathography are forms of narrative that inherit properties from biography and fiction. Autobiography “by definition involves *self*-representation” (Couser, 1997, p. 181). Autobiography is “rebirth in narrative” (Eakin, 1985, p. 227) through autopoietic self-creation, which applies to both autopathographical narrative and autopathographical game design. Referred to as ‘life writing’ in literary circles, autobiography scholars Sidonie Smith and Julia Watson (2001) connect the intrinsic value of the autobiography to narrative, emphasizing the self-reflexive nature of autobiographical authorship: “getting a life means getting a narrative, and vice versa” (p. 80). An autobiography is importantly a *sustained* self-description of the narrator recounting or reflecting on their life story. The autobiography is a subtype of the biography, the latter of which is a sustained written description centered around the life of a

person who has achieved public attention, such as a historical or political figure. “Autobiography is the literary expression of the self-determined life”, a representation of identity, autonomy, and agency (Couser, 1997, p. 182). What distinguishes autobiography from non-autobiography is “a narrative’s claim to be a version of the author’s own life anchored in verifiable biographical fact” (Eakin, 1985, p. 185). The autobiographer’s claims to truth and realism have been problematized by autobiography theorists, akin to how documentary film has been recast as a perspectival construct of reality. In a historical sense, autobiography can also be viewed as a “historicized self” or a temporal narrative of self-development and self-actualization, states historian Peter Burke (2011). In a psychological sense of self-construal and introspective meaning-making, cognitive psychologist Jerome Bruner (1995) states that the autobiographical process is “a way of construing experience – and of reconstruing and reconstruing it until our breath or our pen fails us” (p. 161). As Eakin (1985) suggests, the autobiographical act is determined by the collusion of past memories and present needs towards an immediate objective: “the materials of the past are shaped by memory and imagination to serve the needs of present consciousness” (p. 5).

If, as American novelist Mary McCarthy (as cited in Eakin, 1985, p. 55) suggests, autobiography is the ‘quest of the self’ evolving into ‘*making* the self’, then autobiography and autopathography offer a platform for autopoiesis as self-discovery. Eakin (1985) proposes that “self-discovery is inseparable from the art of self-invention” (p. 55), albeit this study makes the useful distinction between autopoiesis as a self-reflexive form of self-awareness and fabulopoiesis as self-re-invention through fantasy and the imagination—both of which appear to contribute to self-discovery and self-transformation. Autopoiesis is also an articulation of inner states of subjectivity into the material artifact of an autobiographical essay, an oil self-portrait, or a computer game autopathography; these inner states are ambiguous, indeterminate, and vague prior to the autopoietic act yet become clarified and distilled through the distinct formal properties of computer game design⁷⁶. Eakin quotes autobiography scholar Janet Varner Gunn for whom the self is “the displayed self...who speaks, who lives in time” through language (as

⁷⁶ This distillation process will be discussed in the *Logopoiesis* chapter, which focuses on the praxis of implementation (i.e. rule-design, programming) as a material and tangible form of self-making.

cited in Eakin, 1985, p. 184). In other words, the self is a material subject made of flesh and bone who ‘speaks’ and ‘lives’ either through the act of enunciation (speech act), or in the case of this study, the enunciative act during the Autopathographical Game Jam and the player’s performance of the autopathographical game.

The formal distinctions between the autobiography and the autopathography are not as clear-cut, and this is especially so in relation to their potential therapeutic effects on the author. Judging by his many references to the autobiographical act as a form of self-therapy, self-recovery, and self-revitalization, Eakin (1985) suggests that autobiography is inherently autopathographical in nature. He confirms that “a good many biographers have affirmed the therapeutic value of the autobiographical act” (p. 24). For example, autobiographical expression had a revitalizing effect for Henry James, writes Eakin: “the autobiographical project emerges in this context as a sign of returning health, demonstrating ‘my vitality...to say nothing of my will to live and to write’” (p. 82). From the standpoint of a formal similarity between autobiography and autopathography, he gives the example of Jean-Paul Sartre’s novellas, *Nausea* and *The Words*, saying that “the plot structure of each of these first-person narratives displays the aetiology of an illness” (p. 156). Indeed, according to philosopher Jean Baillargeon, “Sartre has said that he realized with *The Words* that he had, perhaps through the writing of the book, overcome a kind of neurosis, a neurotic relationship to literature” (personal communication, Sept 4, 2017).

Autopoiesis, whether in the form of autobiographical narrative or autopathographical game design, is also a compelling form of self-care and self-transformation via (re)ordering and (re)arranging one’s inner world. Eakin explains: “too many conflicting emotional interests are involved for life ever to be wholly acceptable, and possibly it is the work of the storyteller to rearrange things so that they conform to this end” (p. 6). Eakin cites Leon Edel’s thoughts on the act of composition as a therapeutic praxis: “Edel believes that the act of composition ministered not only to (novelist Henry James’s) physical recovery but to his psychological health as well” (p. 58). Citing Stephen Marcus, Eakin also mentions the Freudian connection between narrative and mental health: “illness amounts at least in part to suffering from an incoherent story or an inadequate account of oneself” (p. 170). Given these theorizations, it is apt to conclude that the

ontology of autobiography, based on narrative as a mode of subjectivity, is partially yet intrinsically autopathographical.

Couser (1997) affirms that “writing...bids to be an agent of recovery as well as self-expression” (p. 4). He is interested in the autopathographical narrative form of “body writing” that explores the phenomenological dimensions of illness and pain, “what dysfunction feels like and how it alters self-perception”. He claims that autopathographical narrative does not fully depend on familiar narrative structures such as plot, which presupposes a logical ordering of events: “the essay and the pathographic sketch...do not depend on plot...” (p. 294). He gives the examples of Oliver Sacks’s *The Man Who Mistook His Wife for a Hat* (1985) and John Updike’s *Self-Consciousness* (1989): “such writing relies less heavily on chronology and plot; it combines vividness of description, apt and detailed verbal simulation or representation of somatic experience (personal or witnessed firsthand) with speculation about the cultural, political, and epistemological implications of embodiment” (p. 294). Nevertheless, Couser does not account for other types of organizing principles in the autopathographical narrative used to substitute the organizing function of the plot—i.e., characters/narrators/voices/perspectives, rhetorical devices, descriptive or visual imagery such as metaphor and analogy, mood, atmosphere, etc.

As a result of the above theoretical insights, consider this provisional typology (see table below) showing the layers of how autopathographical narrative is foregrounded and/or backgrounded through AGD-AGJ. The narrative component, whether subsumed in the form of

characters, (self)narrator, game rules, plot, dialogue, aesthetic style, etc. becomes an organizing (ordering) principle in the autopoietic process.

Table 1: Autopathographical narration in AGD-AGJ—provisional typology



Backgrounding and foregrounding the autopathographical narrative in the AGD-AGJ process, which serves as an organizing principle of first-person illness, trauma, and disability experiences, can also be conceived of in relation to sociologist Erving Goffman's (1956 & 1974; 1986) organizational theories of human behaviour and interaction in *The Presentation of Self in Everyday Life* and *Frame Analysis: An Essay on the Organization of Experience*. In the first work, Goffman (1956) addresses self-presentation in the "backstage-frontstage" performances of human agents adopting specific types of conduct in order to manage how they are perceived by others-i.e., "impression management". According to Goffman, "a 'performance' may be defined as all the activity of a given participant on a given occasion which serves to influence in any way any of the other participants" (p. 8). Backstage conduct consists of human behaviour that society considers inappropriate in frontstage contexts; examples include improper posture and sitting, being on a first-name basis with others, profanity, playfulness, aggression, and other types of informal and *intimate* conduct:

Throughout our society there tends to be one informal or backstage language of behaviour, and another language of behaviour for occasions when a performance is being presented...In general, then, back stage conduct is one which allows minor acts which might easily be taken as *symbolic of intimacy* (emphasis added) and disrespect for others present and for the region, while front region conduct is one which disallows such potentially offensive behaviour. (p. 78)

In other words, emotional, verbal, and other forms of intimacy are reserved for the backstage realm. During the AGD-AGJ event, the expectation is to see frontstage performances in front of a group of strangers during the Autopathographical Game Jam while backstage conduct-i.e., being emotionally intimate, is left for the introspective AGD process itself. Contrary to this expectation, foregrounding and backgrounding the autopathographical narrative in AGD-AGJ appears to frustrate Goffman's backstage-frontstage dichotomy. For example, subjects foreground their autopathographical narratives during the Autopathographical Game Jam by enacting one dimension of backstage conduct within frontstage performance—disclosing their *intimate* narratives to a group of strangers. To put it differently, during the AGJ event, subjects manifest backstage behaviour, such as intimate narrative-telling, embedded in a frontstage performance—in front of an audience of mostly strangers, to foreground their autopathographical narratives. Moreover, subjects background the autopathographical narrative in the AGD process itself, hiding intimate details and enacting a frontstage façade in the game.

This frontstage and backstage reversal demonstrates that ‘context’ is key, which brings us to Goffman’s (1974; 1986) ‘frame’ theory—a more apt approach to reading how the autopathographical narrative becomes ‘reframed’ in the dual context of AGD-AGJ.

Goffman uses the frame metaphor to discuss organizing principles of human experience and human interaction processes. Goffman’s frames refer to context, background, limits, distinctions, brackets and borders, arrangements, and so-called “primary frameworks”—frames of reference structuring lived encounters according to implicit underlying principles. His analysis, much like in *Presentation of Self*, draws on theatre, film, radio, comics, games, and play—mimetic forms that expand or contract (limit) human attention and structure human interaction. A variety of creative and technical devices structure experience in these domains, such as the theatrical stage separating audience from performers, stage curtains, camera optics, temporal devices (i.e. fade-outs, ellipsis, etc.) and acting techniques (i.e. voice projection, on-stage choreography of actors, etc.). Goffman (1974; 1986) explains why his methodology of the frame is fruitful for understanding how humans structure experience: “...one can learn how our sense of ordinary reality is produced by examining something that is easier to become conscious of, namely, how reality is mimicked and/or how it is faked.” (p. 251). Goffman’s frame is also a transformative act—to frame and reframe a human experience or encounter—and is not always distinct; sometimes the frame is blurry and ambiguous, such as in the case of human and animal play⁷⁷. Frames-within-frames also show the recursive ways in which humans perceive limits and constraints as embedded within other limits and constraints.

Thus, to background the autopathographical narrative in AGD proper is to “mute” certain intimate or expository details of that narrative, much like Goffman describes the ways in which the theatrical stage is arranged with actors taking turns speaking and muting one another’s performances:

One person at a time tends to be given the focus of the stage, front and center. (He will often rise from a chair to take it.) The others onstage, especially those not engaged in talk with the current central person, tend to be arranged out of focus, their actions *muted*

⁷⁷ Goffman is informed by Gregory Bateson’s (1972) *A Theory of Play and Phantasy*—specifically, his discussions of animals playing where the line between play (unseriousness) and aggression (seriousness) is blurred.

(emphasis added), the result being that the attention of the audience is led to the speaker. (p. 140)

Goffman also uses the metaphor of cinematic optics such as focus to describe how individuals reframe action by focusing and refocusing on its spatial and dimensional elements such as closeness and distance:

Any event can be described in terms of a focus that includes a wide swath or a narrow one and—as a related but not identical matter—in terms of a focus that is close-up or distant. (p. 8)

As such, subjects alternate between foregrounding and backgrounding their autopathographical narratives in order to sharpen (bring into focus) and/or blur (bring out of focus) details of that narrative, depending on the context—the AGJ event and/or the AGD process—, both of which converge and diverge in the act of reframing.

As a narrative-based self-making process, autopoiesis involves telling a self-narrative to oneself and/or to others. The specific issue of self-narrator as agent of self-transformation is tackled in the next few sections, which then covers why and how AGD subjects background their narratives for purposes of self-care. Thus, another discussion of this process of foregrounding and backgrounding the autopathographical narrative in the AGD-AGJ process follows in Sections 5.4.2 and 5.4.3.

5.3 Part II: Autopoiesis and self-reflexivity

5.3.1 Autopoiesis as a self-reflexive praxis of narration and self-care in AGD

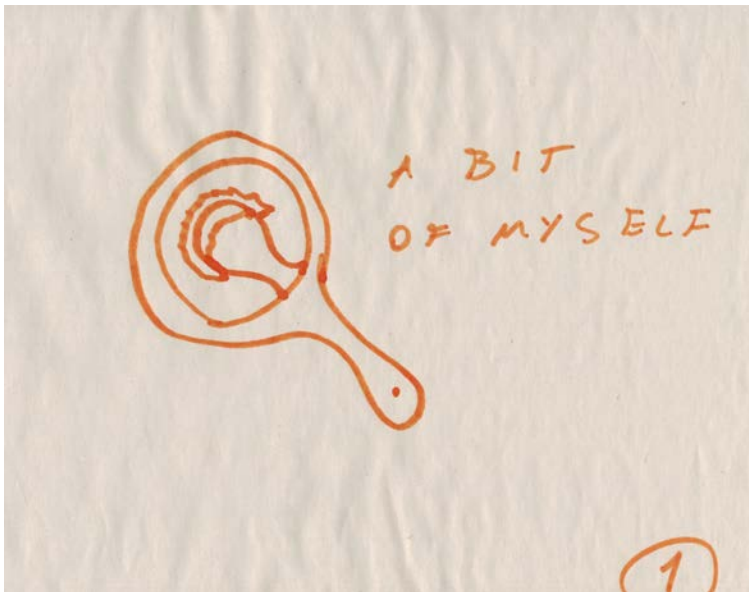
Autopoiesis is a self-reflexive (self-referential) praxis of creative identity construction in AGD, which involves the acts of self-reflection, introspection, and self-inquiry. The self-reflexive dimensions of autopathographical game design are a fundamental element of autopoiesis and its processes of self-creation and regeneration. Consider first the process of self-reflection, which Brody invokes. He states that that autopathographical game design in its autopoietic dimensions:

...involves a level of analysis...where you have to *look at yourself*.

For Brody, ‘looking into himself’ may mean reflecting upon his own anxious thoughts and PTSD triggers, which appear to characterize his first-person experiences with mental illness. Or it may also mean reflecting upon his autopathographical narrative and trying to figure out how to render it in the autopathographical game. In either case or both, autopathographical game design requires the act of looking at yourself and reflecting on your autopathographical life narrative or some portion of it. This requirement is nonetheless welcomed by humans. The act of self-reflection seems to be a fundamental human need for exploring the self, as per Hofstadter’s (1985) point: “part of human nature is to be introspective, to probe” (p. 19).

Consider another example of self-reflexivity in AGD. An unidentified subject’s AGJ doodle shows a drawing of a hand mirror with a reflection of the Self captioned, “A bit of myself”, illustrating the primary ontological status of AGD autopoiesis as a self-reflexive form (see Figure 22 below).

Figure 22: Autopathographical Game Jam doodle—“A BIT OF MYSELF”



The expression of ‘looking into the mirror’ is a common metaphor of self-assessment and identity formation through the act of self-recognition. It is only when we reflect upon ourselves by ‘looking into the mirror’ that we can initiate the complex process of self-transformation that is autopoiesis. To prepare for assessment, which precedes modification and change, we must first recognize what is already there. The mirror is the definitive tool of self-perception and self-

identification, as Lacan's (1949; 2006) psychoanalytic theory of the mirror-stage would attest—the child's realization, through their self-recognition in a mirror, that they have an identity apart from their mother⁷⁸.

For Marco, the mirror as a metaphor for reflecting on the inner self is tied to the emergence of self-dialogue and self-narration inherent in the autopathographical game design process. In its self-dialogical form, the AGD process engenders self-questioning for the AGD designer:

It's like I am interviewing myself inside my mind...it was like getting a mirror and then putting a mirror in front of another mirror so you have infinite reflections. It's like going back and forth non-stop to yourself.

Self-reflection in its self-dialogical form also becomes “infinite” for Marco, but this ontological dimension appears to feel generative for him. Marco's process of “going back and forth non-stop to yourself” echoes, at least poetically, Hofstadter's (1999) “strange loop” which applies to a different context. The Gödelian strange loop “arises in formal systems in mathematics”:

...a loop that allows such a system to ‘perceive itself’, *to talk about itself* (emphasis added), to become ‘self-aware’, and in a sense it would not be going too far to say that by virtue of having such a loop, a formal system *acquires a self* (p. P-3).

Hofstadter adds that this strange loop is nevertheless meaningless in the context of a formal system's limited self-awareness because it consists of “collections of rules for churning out an endless series of mathematical truths solely by mechanical symbol-shunting without any regard to meanings or ideas hidden in the shapes being manipulated” (p. P-3). Indeed, a defining feature of autobiographical and autopathographical authorship is the meaning that the author's self-referential process generates. Hofstadter (1985) confirms that “self-reference—even of the sharp Epimenides type⁷⁹—is connected to profound aspects of life” (p. 19) such as self-generation—

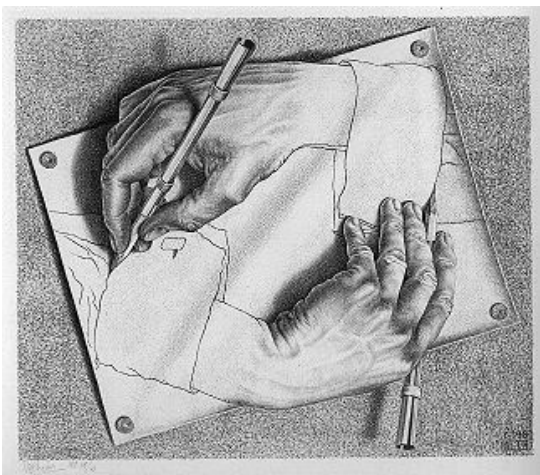
⁷⁸The mirror as a metaphor of identity formation also carries a stigmatized mythological image of narcissism that warns against the potential for excessive self-reflection and self-rumination hidden in the act of ‘looking into the mirror’, which does not apply to this study.

⁷⁹Nöth (2007) explains Epimenides's Paradox which underscores the phenomenon of self-reference as “one of the sources of paradox in language”: Epimenides, a Cretan, utters that ‘all Cretans are liars’, which “creates the famous paradox according to which Epimenides either says the truth and thus falsifies the statement that all Cretans lie,

the act of generating and re-generating the self. In other words, autopoiesis is a meaningful act for the author because it contributes to re-generation and self-transformation.

In its self-referential and self-generative dimensions, AGD autopoiesis is also similar to the hand drawing itself in Dutch artist M. C. Escher's famous drawing, except that Escher's drawing is *representative* of a process of self-making while AGD autopoiesis *is* the process of self-making.

Figure 23: M. C. Escher's drawing



Semiotician Winfried Nöth (2007) states that “the metaleptic element in the picture of the hand that is drawing itself consists in the transformation of the drawn hand into a drawing hand. Escher’s message seems to convey this paradoxical message: ‘I, the draftsman’s hand am drawing a hand that draws itself’” (p. 76). Nöth explains the rhetorical concept of *metalepsis* (μετάληψις) applied to narrative whose fundamental property is self-referentiality:

Metalepsis...is a narrative device by which a narrator begins to participate in the life of the fictitious persons who are nothing but his or her own creation, or vice versa, in which the characters of the author’s own narrative creation begin to enter into a *dialogue* (emphasis added) with the author (as in Pirandello’s *Six Characters in Search of an*

since at least one Cretan is speaking the truth, or he lies, like all other Cretans, but then his proposition would negate itself and mean: “It is a lie that all Cretans are liars”, which by double negation would mean the opposite, namely that no Cretan lies. The source of this paradox is in its self-referential nature: the speaker’s utterance refers to the speaker’s speech act, but the semantic level of the proposition enters into a conflict with the pragmatic level of what the speaker utters and the scope of what he refers to, which is partly what others say, partly what he himself says” (p. 75).

Author). The self-referential nature of narrative metalepsis has its explanation in the enunciative frame: the addresser turns into a self-addressed addressee. (p. 75-76)

Examples of “metaleptic self-reference” (Marie-Laure Ryan as cited in Nöth, 2007) are both Epimenides’s paradox and the picture of Escher’s hand drawing itself (p. 75). Metalepsis was first identified in narratology by Gérard Genette (Pier, 2013) in order to understand how author-of-the-text and author-qua-narrator-in-the-text converge and diverge, especially if there are multiple, hidden, or implicit narrators and/or perspectives embedded in the text⁸⁰ and/or its production: “what is at stake are the forms and degrees of violation of the boundary between the telling and the told” says Pier. In other words, in metaleptic self-reference it may be difficult to draw the boundary between the author that tells the narrative and the narrative that recounts the author’s telling of that narrative. Nöth’s explanation of metalepsis also highlights the author’s *participation* in the life of their fictitious characters, which in AGD could be conceived of as a process of either authoring fictitious game characters and/or inhabiting them through gameplay. Thus the AGD author participates in the life of the game character(s), which are created in the image of the author’s first-person experiences with mental illness, emotional trauma, and disability (a.k.a. the fabulized self)⁸¹.

Self-reflection for Marco may also mean contemplating oneself via a sequence of self-questioning in order to regenerate the self through autopathographical game design. Self-questioning and the self-transformation it engenders entail two things at this point in the autopoietic process: 1) the use of language to communicate with the inner self in order to internally voice, foreground, and/or background the first-person narrative of being bullied in his childhood, in Marco’s case; and 2) the emergence of a ‘mirror-self’ in the act of self-recognition and self-conversation. In other words, to have a mirror-self is to talk to your mirror image as

⁸⁰ Alas, there is no time to cover the use of embedded (nested) voices in narrative such as those in modernist writer Virginia Woolf’s *Mrs. Dalloway*; she nested voices in her novel all the way up to sixth-level intentionality. Please see more on this rhetorical strategy in Lisa Zunshine’s (2006) book: *Why We Read Fiction: Theory of Mind and the Novel*.

⁸¹ The theorization of the connections between metalepsis and participation is a study for future directions that is beyond the scope of this dissertation, which is briefly addressed in the *Conclusion* (please see “inhabiting the fabulized self”).

Marco appears to have been doing in self-dialogue. Alex also invoked the role of an internal narrator in AGD autopoiesis by suggesting that the process of self-reflection in autopathographical game design involves “working with triggers and your mental language” and “working with the way you speak to yourself internally”, which appears to have an objective—to self-heal without any professional help such as psychotherapy:

My game is expressing dealing with trauma and mental issues without any help. So that’s a complex thing because you’re working with triggers and your *mental language*, you’re *working with the way you speak to yourself internally*, what skills you have and don’t have, like if you understand DBT or cognitive behavioural therapy or don’t.

The self-reflexive process for Alex also involved self-assessment and self-reappraisal of his skills, his comfort level, as well as his design decisions pertaining to self-therapy. This process assessed his ability to mobilize previously-acquired skills through professional psychotherapy such as dialectical behavior therapy (DBT) or cognitive behavior therapy (CBT). Furthermore, for Alex, restoring the self through the intense and difficult self-reflexive praxis means directly facing emotional and traumatic fears and triggers, which could induce emotional injury and stop the author’s autopoietic self-generation in their tracks:

It’s not very easy to make stuff about personal experiences that are intimate because if I have to think about how then I have to address the thing. If I have to address the thing then I have to relive the thing (laugh). And that can be really hard, so yeah, like if you can keep it small in scope.

Alex explained that thinking about his autopathographical narrative means “addressing” it, and addressing it leads to “reliving” it, which can prove painful. He was invoking the arduous and draining emotional process involved in self-reflection and introspection, which involves the hard task of ‘riding’ difficult emotions and reliving them through the act of autopoiesis. Yet, Alex seemed to suggest that he nevertheless persevered in this act, despite emotional dangers and potential for failure. The key for him is to keep his AGD process manageable, “keep it small in scope”—a form of self-care.

Briefly turning to Eakin (1985) who emphasizes the invaluable resources of language in the autobiographical construction of identity and self-awareness, we can see that Marco’s and

Alex's self-dialogical processes are implicitly defined by the invisible yet important role that language plays in engendering the act of self-questioning. Language is 'invisible' to others in this case, because Marco and Alex were 'talking to themselves inside their heads', so to speak; nobody else hears them tell their narratives to themselves in that instance of self-dialogue, but only themselves. For example, Eakin cites Hellen Keller's own experiences with the power of internal language in generating self-awareness, and David Bleich, who argues that "without language, it is not possible to distinguish between awareness and self-awareness" (as cited on p. 197). Indeed, Benveniste (1971) confirms that language generates subjectivity: "'it is in and through language that man constitutes himself as a *subject*, because language alone establishes the concept of the 'ego' in reality, in *its* reality which is that of the being" (emphasis in original) (p. 224). Keller, the famous deaf-mute writer, describes the transformational power of language to her sense of identity. For Keller, discourse is neither heard nor spoken using phonetic speech sounds; discourse takes the gestural and tactile form of sign language drawn in the palm of her hand by her teacher Annie Sullivan:

When I learned the meaning of 'I' and 'me' and found that I was something, I began to think. Then consciousness first existed for me...Before that moment my mind was in a state of anarchy in which meaningless sensations rioted, and if thought existed, it was so vague and inconsequent, it cannot be made a part of discourse (Keller as cited in Eakin, 1985, p. 212).

Eakin affirms that in Keller's case, "it is an act of language in which experience is transformed into symbol" (p. 213). But more than that, language, broadly conceived in its many communicative modes including tactile, aural, visual, procedural, performative, and textual, articulates 'identity' as an explicit awareness of the existence of self. Language is an ordering principle of narration, for speaking and for telling, including, in AGD's case, speaking and telling to oneself. In its multi-modal form, language is a structuring device that brings form and meaning to first-person experiences.

Additionally, in the context of autopoietic self-dialogue, Derrida's (1985) claim that the act of subjectification depends on another person *hearing* us speak takes form through two continuous yet alternating conscious states. When one is in dialogue with oneself as Marco and Alex were during the AGD process, the mirror-self emerges because it takes two to interact; there is a mental image construed of another self, a self that is neither self nor Other, that might

aid conversation for Marco and Alex and structure self-awareness. Further down in the process of AGD, this mirror-self becomes formalized as a fictionalized self—a *fabulized self*, which becomes rendered and implemented in the autopathographical computer game.

However, while the autopoietic process of AGD may *feel* “infinite” to Marco in its self-referential and metaleptic dimensions, it may be more recursive than infinite in Hoftsadter’s (1999) sense:

A recursive definition (when properly formulated) never leads to infinite regress or paradox. This is because a recursive definition never defines something in terms of itself, but always in terms of *simpler versions of itself* (emphasis added) (p. 127).

Subjects may define themselves in terms of their AGD creations—game characters, game worlds, and narratives which are de facto “simpler versions” of themselves. Furthermore, autopoiesis may shift into fabulopoiesis at some point during the AGD process. In fact, in fabulopoiesis—the creation of the imaginary figure of the fabulized self—the first-person perspective appears to shift to the third-person perspective; hence the recurrent act of referring to oneself may stop as subjects find ways to ‘break free’ from the self-referential process.

5.3.1.1 *Crafting the self through self-reflexive praxis*

The self-reflexive process in AGD autopoiesis also becomes a recurrent praxis that exercises and hones self-dialogue. Marco explained:

Remember the part that I was saying how I was interviewing myself inside (through AGD)? That’s the part that it looks like I’m interviewing myself all the time. And then the weird thing is that these past couple days I’ve been having flashbacks and it’s constant, and then I’m interviewing myself again like ‘are you sure that was that really meant something? Are you sure?’

Marco’s quote above suggests that the tangible skills of computer game design are not the only type of craftsmanship that subjects exercised and honed during the AGD-AGJ process. They exercised, honed, and/or acquired another set of skills that are much more intangible yet pivotal to building resilience: the willingness, tenacity, and courage to reflect on their weaknesses and flaws, their foibles, discomfort, distress, and pain. To draw on Weizenbaum’s (1976) definitions of hard and soft skills, craft as ‘hard’ and tangible skill in the STEM field (i.e. science, technology, engineering, mathematics, computer science) is accompanied by intangible ‘soft’

skills, which in the autopoietic context are the willingness to reflect on your actions and behaviors, the courage to tell a personal narrative about pain and illness, and the persistence to recurrently question yourself through self-skepticism as Macro questions himself through flashbacks and self-interviewing. And as Alex previously mentioned, honing those soft skills means to bravely face, address, and relive painful experiences despite the presence of fears, but also, and very importantly, to *learn how* to manage and contain those fears in order to prevent them from consuming and destroying you.

This autopoietic process of recurrent self-inquiry that is honed and practiced, like a craft, leads to self-mindfulness. For Nathan, AGD teaches him mindfulness and the skill of articulating his first-person experiences in an autopathographical game. Nathan explained:

For myself, I wouldn't call (AGD) therapeutic but it definitely helped me to be more mindful and be able to articulate these experiences.

Nathan later modified his original statement in the post-AGJ interview to suggest AGD indeed proved to be therapeutic because he was able to learn how to mobilize the acts of (self)-reflection and introspection:

It was therapeutic because I got to be more reflective and introspective and am now more mindful.

For Fred, self-reflection as a praxis and a set of extended activities is mobilized via the iterative design of a puzzle game inspired by another game he authored informed by a series of first-person traumatic experiences. Fred's experiences include looking after a deceased man's body on a hiking trip and pulling a drunk driver out of a crashed car. Fred first tackled his distress at another general game jam "using game development as a form of coping with emotions". During the Autopathographical Game Jam, his objective was to author a game puzzle in order to understand the process that he deployed to "cure himself" in the previous game jam. In other words, Fred designed an autopathographical game at the AGJ informed by his authorship of another game at a previous game jam that dealt directly with his trauma. His iterative design process becomes a self-reflexive praxis based on going back to re-author and revise the same game concept repeatedly.

To ground the self-reflexive acts of self-reflection, introspection, and self-inquiry as acquired skills of self-care and an extended set of self-praxes, consider the framework set out in Michel Foucault's *Technologies of the Self* (1988),⁸² which is informed by Classical Greek hermeneutics of self-care and self-knowledge. A technology is a tool for crafting something concrete and material (*techne*), and the activity and skill required of *techne* defines every use of that technology. A technology of self is a skillful form of identity construction, a work of artistry or craftsmanship of the self (*self-techne*).

According to Foucault, "technologies of the self":

...permit individuals to effect by their own means or with the help of others a certain number of operations on their own bodies and souls, thoughts, conduct, and way of being, so as to transform themselves in order to attain a certain state of happiness, purity, wisdom, perfection, or immortality (p. 18).

Foucault asks, "what is this self of which one has to take care, and of what does that care consist?", to which he answers: "the Self is a reflexive pronoun, and it has two meanings. Auto means 'the same,' but it also conveys the notion of identity". Foucault invokes Alcibiades, the famous Athenian statesman, orator, and general for whom self-care was a principle and an ethos applied to the soul, not simply an activity of taking care of the body as a material substance. Self-care was a perspective on how to live an artful life. Foucault explains that "this theme of taking care of oneself was not abstract advice but a widespread activity, a network of obligations and services to the soul..." (p. 26-27). Self-care also went hand in hand with the Platonic duty of self-knowledge or 'know thyself'. On the benefits of self-reflection as a form of self-mindfulness to prepare for inevitable existential crises, Foucault states:

It was generally acknowledged that it was good to be reflective, at least briefly. Pliny advises a friend to set aside a few moments a day, or several weeks or months, for a retreat into himself. This was an active leisure—to study, to read, to prepare for misfortune or death. It was a meditation and a preparation (p. 27).

⁸² In the context of game design therapoetics, Foucault's concept of 'technologies of self' is a productive theoretical tool for understanding dimensions of self-care, irrespective of Foucault's use of it in a different epistemological context to discuss political and erotic applications of self-care that are of no concern to this dissertation.

In autopathographical game design, self-reflection appeared to be a form of self-care praxis that unfolded during the two-day, weekend-long Autopathographical Game Jam. The seeming paradox of integrating rapid prototyping and socializing with the deliberate and contemplative activity of self-reflection is dissolved if we invoke the ancient Greeks' ideas on self-reflection as an extended set of self-praxes: forms of exercising, skill acquisition, and dialoguing with others. Foucault extends the act of self-reflection to the act of writing:

Writing was also important in the culture of taking care of oneself. One of the main features of taking care involved taking notes on oneself to be reread, writing treatises and letters to friends to help them, and keeping notebooks in order to reactivate for oneself the truths one needed. Socrates' letters are an example of this self-exercise. (p. 27)

During the Autopathographical Game Jam, subjects sketched and ideated, which is a form of self-exercise. Subjects also acquired additional game design skills; as Yasmine confirmed, she can “prototype various skill sets” during a game jam. They also dialogued with their co-jammers and play-tested each other's games. To again invoke the Aristotelian imperative of *praxis* and ‘doing’, there is a distinction between self-reflection as an act of thinking where thoughts are transient and ephemeral, and self-reflection as an act of skill acquisition that consists of writing down and drawing, recording and rereading thoughts, ideating through sketches, dialoguing with others who can offer feedback and critique, and so forth. In the ancient Greek perspective, the praxis of writing is an extension of self-care praxis and exemplifies the ethos of self-care as a form of soul nourishment, because it captures what is fleeting and impermanent, i.e., a thought. The autobiographical or autopathographical act, *autopoiesis*, is thus a form of life praxis. Eakin (1985) states: “the autobiographical act is revealed as a mode of self-invention that is always practiced first in living and only eventually—sometimes—formalized in writing” (p. 9).

As such, self-reflection, introspection, and self-inquiry are also autopoietic processes of self-care in AGD involving a set of extended activities or self-praxes: self-judgment, self-appraisal, self-re-appraisal, skill acquisition, and telling emotionally challenging autopathographical narratives with courage and mindfulness, all towards the goal of self-transformation and re-generation of self. In this courage to explore intense emotions and practice a sustained form of *autopoiesis*, AGD subjects such as Marco, Nathan, Alex, and Fred seem to be rewarded with a heightened sense of mindfulness. But also, and very importantly, their testimony supports the claim that autopathographical game design is not simply an autopoietic act of self-

reflection, introspection, and self-inquiry in thinking about oneself and dialoguing with oneself; AGD autopoiesis is a creative act of self-making that integrates those activities in *self-formation*. Put differently, AGD autopoiesis is an act of generating a narrative of self that teaches about oneself.

5.4 Part III: Auctorial posture and author identity

5.4.1 Auctorial posture in autopathographical game design

In doing self-reflection, introspection, and self-inquiry—autopoietic praxes that can become challenging and overwhelming for the psyche—subjects have at their disposal another resource that might help them practice self-care and therapy. This resource is the possibility for accessing a multi-faceted, heterogeneous author identity through a set of auctorial postures available during the AGD-AGJ event. These auctorial postures aid the autopathographical narrative-telling process for study subjects and enable them to shift between different roles and perspectives during the AGD-AGJ event.

The process of self-making or autopoiesis speaks to the important role of the author's identity in autopathographical forms of expression. However, the concept of author identity or 'author subject' is a fraught one, theoretically speaking, because as has often been claimed, identity and subjectivity are not homogeneous entities. For example, games scholar Cindy Poremba (2007) argues that autobiographical games offer novel possibilities for self-narrative that are also valuable to a postmodernist critique of the unity of the subject (p. 707). Eakin (1985) asserts that "truth and identity in autobiography are plural, and, therefore, so is the autobiographical text..." (p. 36). Barthes (1977) summarizes the difficulty of envisioning a simple, unified author identity: "*who speaks* (in the narrative) is not *who writes* (in real life) and who writes is not *who is*" (p. 112). In this statement, Barthes acknowledges Lacan's similar line of questioning: "Is the subject I speak of when I speak the same as the subject who speaks?" (as cited in Barthes). Similarly, in his examination of the literary autopathography, Couser (1997) differentiates between the narrator as a diegetic agent propelling the autopathographical narrative, and the author who is located in time, real life, and off-narrative (non-diegetic) space.

Consider briefly some tensions of claiming the existence of a plural identity in autopathographical authorship, as the type of identity that Lacan, Barthes, Eakin, and Poremba

would attest to. On first impression, the autopathographical narrative theoretically contradicts the postmodernist ethos of fragmentation and disunity of the subject. In other words, it seems that the motivation behind autopathographical authorship is to recover a unified subject and identity. For example, illness narrative scholar Mike Bury (2001) asserts that “under conditions of adversity, individuals often feel a pressing need to re-examine and re-fashion their personal narratives in an attempt to maintain a sense of identity” (p. 264). The role of voice in recovering a unified identity in autopathographical authorship is also important: medical discourse and medical praxes are often perceived as a silencing apparatus of first-person patient experiences; hence the patient’s need to reclaim their voice through the poetics of telling illness narratives (Couser, 1997, p. 21). Authorship of the autopathographical narrative, whether in the form of written prose or a computer game, recovers the autopathographer’s voice in this regard. Thus, the autopathography may disrupt the theoretical perspective of subject-instability by operationalizing a *recovery* of a robust subject. ‘Feeling whole’ becomes a concern of individuals who are in discomfort and pain to rid themselves of the discomfiting feeling engendered by illness and pain.

Consider yet another contradiction that appears to be present in autopathographical authorship. In order to feel ‘whole’, autopathographers may also need to inhabit *several roles at once* complimenting each other, such as ‘sufferer’ and ‘healer’. As Frank (1995) seems to suggest, the roles of ‘sufferer’ and ‘healer’ may both play a part in constructing the illness narrator’s identity: “the ill, and all those who suffer, can also be healers...through their stories, the ill create empathic bonds between themselves and their listeners” (p. xii). Frank’s point could also extend to sufferers in the role of healing others in addition to healing themselves. Therefore, one way in which AGD subjects invite recovery and construct a robust subject is through the inhabitation of a plural authorial identity that AGD-AGJ appears to invite. When an autopathographical author can play the role of healer to others in the game jam space as well as healer to their suffering self, accessing these multiple roles confer them with agency and autonomy. This multi-role dimension in autopathographical authorship is integrative as opposed to splintering in a schizophrenic sense.

A way to understand the multiple roles and perspectives that constitute the author’s identity, which contribute to self-care and healing in AGD-AGJ is through the theoretical concept of the auctorial posture. In the *Double Nature of the Author’s Image*, Ruth Amossy

(2009) explores contemporary meanings of the authorial persona. She explains that to analyze auctoriality:

...it is necessary to take into account not the real person of the one who signs a literary work, but rather his imaginary figure. It is a discursive image that is elaborated both in the so-called literary text and in its surroundings, in this case in accompanying speeches such as editorial advertising or criticism...

The self-images projected by the writer are not of the same order as the representations made by third parties. It will therefore be necessary to describe and serialize these different types of discursive images, while at the same time positing their close interdependence. It will be assumed that the way in which they intersect and combine influences the reader's interaction with the text, on the one hand, and functions in the literary field, on the other hand. (p. 1)

The author constructs an auctorial persona, or “an imaginary figure” (Amossy, 2009) that is a mosaic of discursive roles and perspectives enacted through a variety of modes, forms, and platforms including speech, the author's writing, narration, author interviews, and through others building the author's image. As Amossy states, “the image of the author is always built by others, professionals, critics, biographers, etc., which put in place a representation of the writer on which the latter has no direct grasp” (p. 3).

To draw on Amossy's auctorial persona or “imaginary figure”, subjects foreground and background their autopathographical narratives by mobilizing a variety of *auctorial postures* in the AGD-AGJ process, through which they perform the narrative-telling: role, e.g., game designer, the fabulized self, game player, jammer, stranger, friend, sufferer, self-healer, healer (of others), and research subject; voice, e.g., (discursive and non-discursive) narrator, speaker, listener; and an epistemological point-of-view, e.g., first-person, second-person, and third-person perspective. These auctorial postures are discursively enacted during the Autopathographical Game Jam, during the post-jam interviews with the researcher, and through the distinct formal properties of AGD. These auctorial postures stimulate the subject's AGD process and become therapeutically expressed through AGD as three epistemological perspectives that co-emerge: the first-person (subjective, affective) perspective, the second-person (relational, intersubjective) perspective and the third-person (objective, detached) perspective.

Auctorial posture is thus operationalized through AGD-AGJ in the following *roles* (a-h) and *perspectives* (i-k):

- a) Subject qua *discursive narrator/speaker*: subjects foreground (tell) their autopathographical narratives to researcher and game jammers during the Autopathographical Game Jam (and post-jam to the researcher).
- b) Subject qua *non-discursive narrator*: subjects background (tell) their autopathographical narratives through the formal properties of autopathographical game design.
- c) Subject qua *game designer*: embodying the full process of autopathographical game design, including *public persona* as game designer.
- d) Subject qua *fabulized self*: subjects deploy metaphor and analogy to background their autopathographical narratives qua fantasy worlds.
- e) Subject qua *'sufferer', 'healer' and 'self-healer'*: subjects discover methods of self-care, self-understanding, and self-therapy through the AGD-AGJ process. Healing others is not something they do consciously—rather, it emerges through sociopoiesis—the Autopathographical Game Jam as a therapeutic community (as discussed in the *Sociopoiesis* chapter).
- f) Subject qua *game player*: subjects play their own games and the games of their fellow AGJ jammers. Game players inhabit the author's *fabulized self* by playing the author's game autopathography.
- g) Subject qua *game jammer* (sub-roles include stranger, friend, and listener): subjects build relationships with other game jammers during the Autopathographical Game Jam, which feels therapeutic (as discussed in the *Sociopoiesis* chapter).
- h) Subject qua *research subject*: subjects are after all participants in an academic research study and this is yet another posture that modifies their performance and self-presentation to the group.

The following three epistemological *perspectives* overlap the above *roles*:

- i) Subject qua *first-person (subjective) perspective*: subjects inhabit their first-person experiences with mental illness, emotional trauma, and disability while designing an autopathographical game.
- j) Subject qua *second-person (intersubjective) perspective*: subjects share their autopathographical narratives with other jammers at the AGJ via the intersubjective activities of socializing and bonding.
- k) Subject qua *third-person (objective) perspective*: subjects inhabit an analytical, objective, detached position afforded by the digital, logical, and analytical dimensions of computer game design, i.e., programming, refactoring the code, debugging (discussed in the *Logopoiesis* chapter).

The concept of *auctorial posture* manifests the different ways in which subjects narrate their autopathographical narratives within the multilayered therapoetic landscape of AGD-AGJ. As previously mentioned, a story becomes a narrative when told in very specific ways by a *narrator*. We assume that a narrative refers to a speaker of some kind—either the author, narrator, character or a configuration of some or all of these roles. A narrative is *spatially structured* via a *situated perspective*, a point of view or “focalization” in Bal’s (2009) sense, embedded in a narrator. For example, first-person narration gives the illusion that it is subjective and self-referential, hence the experience is one of immersion in what the narrator sees. Third-person narration gives the illusion of a detached and analytical position observing the narrative events or characters from a distance. Multiple narrators telling the same story ‘coloured’ in different ways and from different standpoints can give the illusion of pluralism. For example, in the film *Rashomon* (1950), the story of rape and murder is narrated from the distinct perspectives of four different individuals. The point of view of the narrator can be represented in a different mode or medium, such as linguistic pronouns (‘I’, ‘he/she’), cinematic subjective and point-of-view shots or moving camera positions in a game.

Indeed, Bal claims that in a narrative text of any kind, including speech, an agent tells a respondent “a story in a particular medium, such as language, imagery, sound, buildings, or a combination thereof” (p. 5). She explains:

This agent cannot be identified with the writer, painter, composer, or filmmaker. Rather, the writer withdraws and calls upon a fictitious spokesman, an agent technically known as the narrator (p. 9).

Bal is invoking the attitude, stance, and position of the auctorial posture, which can be framed as fictitious because the posture is a construct of the narrator's performance. In AGD-AGJ, subjects occupy the narrator role in this postural sense, playing the roles of not only the game designer and author, but all the other roles listed above—jammer, friend, listener, sufferer, healer, etc. Similarly, Gérard Genette's (1980) most significant revisions of narrative theory is the distinction he makes between mood (perspective) and voice (narrator, speaker) according to Jonathan Culler. In the foreword to Genette's *Narrative Discourse*, Culler explains:

(Narrative theorists) have failed to distinguish properly between mood and voice...between the question who is the character whose point of view orients the narrative perspective? and the very different question who is the narrator?' Thus, if a story is told from the point of view of a particular character...the question whether this character is also the narrator, speaking in the first person, or whether the narrator is someone else who speaks of him in the third person, is not a question of the point of view, which is the same in both cases, but a question of voice...Insistence on the difference between narration and focalization is a major revision of the theory of point of view (p. 10).

Genette is in some sense also accounting for auctorial posture as the difference of voice and perspective that could apply to the same narrator, regardless of whether that narrator is speaking in the first-person or in third-person. Linguist Oswald Ducrot (1980) suggests that utterances show an interplay between the first-person and third-person perspective, which can shift the designation of subject, depending on who is speaking for whom or who is speaking on behalf of whom:

If X speaks on behalf of Y, I call Y Speaker, even if he is not speaking subject. If, on the contrary, X speaks in his own name, he is both speaking subject and speaker⁸³ (p. 30).

Ducrot's theory of discursive polyphony (Ducrot, 1996), which is inspired by Mikhail Bakhtin's literary theory of polyphony gives voice to the dialogic structure of subjectivity, echoing

⁸³ In original French: "Ainsi, si X se donne comme porte-parole de Y, j'appelle Y le locuteur, bien qu'il ne soit pas sujet parlant. Si au contraire X parle en son nom propre, il est à la fois sujet parlant et locuteur".

Genette's distinctions. He accounts for the possibility of multiple viewpoints present within one single speaker's utterance⁸⁴. "Every utterance is a sort of small play, a sort of mini-dialogue" (p. 24).

The assumption is that the language of subjectivity presents a single, unified first-person perspective, and a speaker whose verbal utterances mirror their internal reality. For example, we may conceive of material evidence of the first-person perspective, such as the use of first-person pronouns, as linguistic evidence of a single, unified first-person perspective reflecting the inner emotional state of the speaker. In the same vein as auctorial posture, Ducrot challenges this assumption of first-person unity. Polyphony presumes a plurality of voices within the limits of a single work (p. 34). In his update of the theory of enunciation in discourse, Ducrot designates the realm of the *locutor* (speaker) as semantically responsible for the utterance; this is similar to a narrator telling a narrative to a listener. The locutor is the person designated as the speaker through the first-person pronoun "I". Ducrot (1996) explains that in discourse, one cannot trace the intellectual origin of the author/producer; one cannot trace who first conceptually formulated the utterance, given that people often reproduce existing ideas and judgments. The producer is equivalent to the author, whose origins in the context of verbal discourse are unknown. Ducrot designates another layer to the act of enunciation, the enunciator, which is the point of view of which there may be multiple. The enunciator is the argumentative orientation adopted by the locutor, and it may not necessarily align with the locutor.

Given the above theoretical insights by Amossy, Bal, Ducrot, and others, subjects inhabited at least two narrator roles during the AGD-AGJ: 1) they foregrounded their autopathographical narratives in discourse during the Autopathographical Game Jam and post-AGJ interviews; 2) they backgrounded their autopathographical narratives in their games due to a number of formal, aesthetic, and self-care reasons. The misalignment between these two types

⁸⁴ Ducrot (1996) states: "(The theory of polyphony) leads one to think that each one of our utterances represents a multiplicity of points of view, some of which can differ from the locutor's. For example, in a negative utterance such as 'Peter will not come', we have one point of view according to which Peter will come and another point of view, which disagrees with the first. You remember that I called the origin of those points of view enunciators." (p. 26).

of narration are discussed next—foregrounding and backgrounding the autopathographical narrative—by focusing on study subject objectives of self-care, self-protection, and humility.

5.4.2 Backgrounding the autopathographical narrative in AGD: a form of self-care, self-protection, and humility

The deployment of auctorial posture in the AGD-AGJ narration process creates a secure space where subjects can be in control of their autopathographical narratives. Subjects implement various perspectives and roles to determine how disclosure is configured through autopathographical game design narration and during the Autopathographical Game Jam event. The chapter on *Sociopoiesis* has already discussed how subjects foreground their autopathographical narratives through discourse—via introductions and conversations with the researcher and others during the Autopathographical Game Jam. For example, when subjects embody the auctorial roles of speaker, listener, stranger, friend, and healer, they build meaningful connections with each other and the researcher team. Most subjects felt comfortable foregrounding their narratives during the AGJ by telling their autopathographical narratives in self-referential mode, using first-person pronouns and deploying an emotionally candid degree of disclosure.

Presently, this section addresses how they background their autopathographical narratives in AGD through a disavowal of identifiable self-references, i.e., first-person deixis referring to the speaker (e.g. me, myself, I, etc.) and deictic words that reveal the speaker’s context such as ‘this’, ‘here’, ‘now’, and ‘yesterday’. Eakin claims that “the referential basis of autobiography is...inherently unstable, an illusion produced by the rhetorical structure of language” (p. 186). The self-referential dimensions of autopathographical game design are equally tenuous, although the reasons why AGD authors practice non-disclosure and disavow identifiable self-references are related to a number of aesthetic and therapeutic concerns. Disclosure of autobiographical self-truths and identifiable details of the author’s life is generally a matter of free will and choice. Eakin (1985) cites the “confessional model of autobiography” as “the courage to utter it” (p. 5). In the realm of contemporary digital media cultures, the confessional model translates to ‘over-sharing’ personal narratives on social media and via personal blogging, vlogging, sexting, etc. Online practices and technologies have reconfigured the digital terrain as a confessional space, blurring the boundaries between the private and public realms. In opposition to these practices of

disclosure and over-sharing, this study's AGD designers under-share and camouflage details of their autopathographical narrative—in other words, they background their autopathographical narratives in the game itself.

Backgrounding their autopathographical narratives, just in case subjects decide to disseminate their games publicly in the future, alleviates the fear of stigma or as Ellie states, “saves face” in the critical public eye—a form of self-care for subjects. Backgrounding their narratives is also a way to inhabit an auctorial posture, a persona that hides various autobiographical details about themselves. Autobiographical and autopathographical narrative is often materially marked by the author's identifiable self-references in the text—material traces of naming and self-naming that discloses the author's legally-recognized identity. But also, the type of confessional and revealing narration may prove too unsafe for AGD designers. For example, John inserted only one reference to bipolar disorder in his game using a text title: “Chase happiness. Pockets of mania are a mixed blessing. No escape from depression”. The game was not supplying concrete references to John as the game author or narrator, who, as previously discussed in this chapter claimed that the game is not linking the theme of bipolar disorder to his own *personal* story.

Marco's testimony seems to suggest that he regards narrative form as too self-expository in the moment that a narrator starts to “give a narrative” in the autopathographical text, that is, identify and name persons through *discourse*:

I really enjoy narratives, don't get me wrong, but because this game is a portrait, I feel that giving a narrative — what's the word — didn't feel comfortable, you know? It just didn't feel right.

It is important to note that Marco's statement of “giving a narrative” implicitly suggests an audience that receives that narrative; hence his decision to control what is given and shared in his autopathographical game.

Nathan also stated that words are too vulnerable for him:

It's like I could write a song about my experiences but I wouldn't put lyrics in it because, again, that's too vulnerable for myself.

Indeed, Nathan backgrounded his autopathographical narrative through the AGD process:

(The autopathographical game is) about my experiences but it's not explicitly. Certainly, there's no exposition, it's not telling my story per se...

Nathan echoes the distinctions made by theorist Jay Ruby (1978), who in his discussion of film and video autobiography argues that a distinction is warranted between autobiography and self-reference, without which “all symbolic articulations would have to be called autobiographical” (p. 7). Ruby claims that self-reference is sometimes constituted through the author's use of allegory and metaphor, a “common occurrence in novels and popular song lyrics” (p. 7). In fact, in AGD proper (AGD sans AGJ), the author's identifiable self-references transmute into the fabulized self through the rhetorical devices of analogy and metaphor and through the formal properties of computer games; hence a self-reference, such as the “I” first-person pronoun is converted into the visual and performative analogical equivalent—the subjective point-of-view (POV) shot in film and the first-person camera perspective in games embodied by the player's *moving* position (Galloway, 2006). As an interesting side note, in his *Poetics*, Aristotle (c. 335 B.C.; 1961) critiques the overuse of the poet's first-person voice in poesis, perhaps because he knows that the poetics of fabulization offer a much better substitute for mimesis: “The poet should speak as little as possible in his own person, for it is not this that makes him an imitator” (p. 109).

It must also be noted that at the level of programming and implementation of game rules and formal constraints, the autopathographical narrative becomes further backgrounded via programming language syntax, writing algorithms, and working with inputs/outputs, integers, strings, variables, conditional expressions, arrays, loops, etc. Therefore, the abstraction processes involved in both fabulopoiesis and logopoiesis invite a further distillation of the autopathographical narrative. In a way, the autopathographical narrative fades from view the more it is processed through autopathographical game design.

John, Nathan, and other subjects also background their autopathographical narratives by authoring moody and atmospheric gameplay experiences. Marco stated that the category of

‘poetry’ resonates with his creative vision for his autopathographical game far more than the category of ‘narrative’. He explained how he differentiates poetry and narrative:

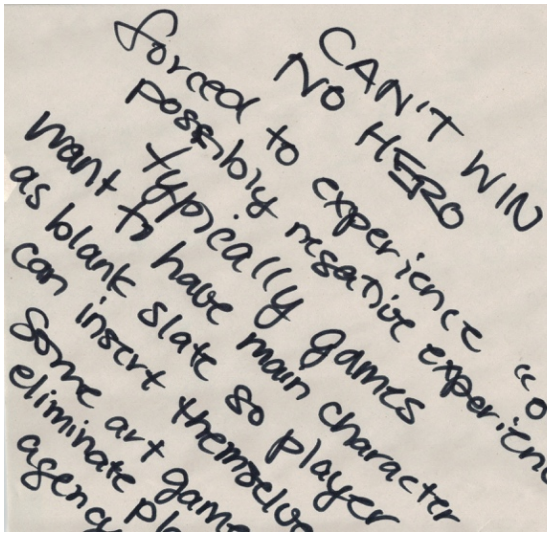
Poetry is all about the feeling, how it sounds, how it impacts you inside even if you don’t know what it really means, but it still impacts you in a certain way.

In other words, Marco invoked his vision of the autopathographical game as a lyrical form closer to poetic verse than to poetic prose based on the phenomenological attribute of “feeling” as opposed to foregrounding a narrative through identifiable and developed game characters, a sequence of events, and so forth. De Man (1979) similarly underscores the potential strengths of autobiographical poetry by asking: “can autobiography be written in verse?” (p. 920). Marco rendered the game autopathography in more abstract and densely symbolic terms in order to alleviate the discomfort of having to foreground his narrative of being bullied in his childhood. In this vein, the multimodality of computer game design was a significant advantage for subjects, allowing them to find ways of backgrounding their autopathographical narratives by harnessing non-discursive modes of expression through images, sounds, and computational processes. When Lukas explained that the multimodality of games is “a rich format” for the designer, he was suggesting that this multimodal richness of games can be used to an advantage in telling narratives in ways that are both interesting and safe for the author. He added that he is a private person who is not comfortable disclosing details about himself publicly:

Because personally, I’m the type of person that don’t want to talk about myself to public.

By omitting narrative elements such as identifiable and developed characters and events, the AGD designer can also create a realm where the game player can “insert themselves” into a “blank slate”. An anonymous subject confirmed in a doodle: “typically games want to have main character as blank slate so player can insert themselves” (see Figure 24 below).

Figure 24: Autopathographical Game Jam subject doodle



This statement suggests that creating an abstract character without character delineation and specificity can also be an advantage in games because the player can inhabit an experience in the game without the additional baggage of role-play that a developed game character with a history and detailed identity engenders. The extraneous material imposed on players in role-playing games (RPGs) can be exhilarating to inhabit, but more abstract and symbolic games might allow for a wider range of meaning-making on the player's part as well as safely camouflage the AGD author's personal narrative. Alex explained why backgrounding his own autopathographical narrative is liberating for him, supporting the benefits of designing characters as "blank slates":

I like things that are implicit because then someone else might get a different experience out of it. If I don't do a great job of making it but it's abstract enough that people don't know they'll just be like, 'oh this is cool'. Does it mean this? And I'll be like, yes, it means that (laughs). It's kind of cheating.

Alex suggests that harnessing ambiguity and backgrounding the autopathographical narrative through non-disclosure may diversify the range of meanings that the gameplay experience engenders for the player yet simultaneously create a universal landscape that anyone can easily slip into and perform.

5.4.3 Backgrounding the autopathographical narrative through autopoiesis: seeking humility and fearing the ‘mark’ of stigma

Some subjects’ decision to background their autopathographical narratives can also be connected to: i) their desire to demonstrate humility, and ii) their fear of how foregrounding might stigmatize their *public persona*, which again invokes how auctorial posture is negotiated in AGD autopoiesis. They suggest that they harbor a sense of humility insofar as they worry that foregrounding their autopathographical narratives in their games gives the impression of narcissism and self-absorption. For example, Brody and Marco converted their autopathographical narratives into game allegories. Brody’s murder-mystery game *You’re Invited* invites the player to inhabit a hyper-attentive sleuth whose task is to predict a killer-to-be at a party of eccentric dinner guests. Marco’s experiential game *Connections* simulates a universe of colliding ‘planets and stars’. Brody and Marco both backgrounded their autopathographical narratives of, respectively, experiences with PTSD and being bullied in childhood, which they readily foregrounded during the AGJ introductions and in interviews with the researcher. Brody explained his reasons to withhold personal information about himself in his game:

I immediately ruled out some of the darker stuff because being dark can be limiting. If I wanted to make a game about my suicidal tendencies, it would almost seem showy, like ‘oh look how much I hurt’ or I’m afraid that it would give that off. So, I cut a lot of that stuff out...

I wanted to avoid *leaking in* my own personal sorts of things...

Brody backgrounded his autopathographical narrative because his story is too bleak and pessimistic to tell in a game. Brody believes that “darker stuff” is limiting because it discloses to audiences the author’s pain and suffering, which is “showy” or exhibitionistic. In Brody’s case, it may be that he fears of being perceived as both self-engulfed and/or nihilistic. Backgrounding the autopathographical narrative in Brody’s case means telling it in some measured and modest way in order to control how he is perceived by the public should the game be released or disseminated.

Further to that, Brody also backgrounded his autopathographical narrative by stripping away the narrator’s explicit voice from his game. Brody implied that he was adopting the

universal third-person perspective for his autopathographical game in order to control how his autopathographical narrative is conveyed in the game:

To understand my personal back-story, it wouldn't be put through with the game. The hope is that (players) would just simply experience what I experience, which is not experiencing what I experienced from the outside perspective, so without the context of 'this is somebody's experience'. I don't have that. I'm experiencing it because it's my experience. I think I subconsciously wanted to avoid leaking in my own personal details. For example, stuff that I'm trying to express about (my ex-wife), I don't want it to be disclosed. I think (my own personal experience) detracts from the game too much if trying to deliver the experience.

Now, if I was trying to tell people about Brody, then yeah, that would be what I'd do, but I'm not trying to tell people about Brody, I'm trying to tell people about my experience.

Brody's testimony above reveals how auctorial posture is negotiated in the dual domains of the Autopathographical Game Jam as a site of enunciation and in AGD proper. In the above testimony, Brody referred to himself as "Brody" in the third-person in order to articulate his desire to background his autopathographical narrative, which consists of the specificity of his identity as Brody. Instead he wanted to tell us about his "experience" without naming and identifying himself—'marking' himself in the game. He uses the term "my experience" to denote a background autopathographical narrative that leaves out identifiable self-references. When he states, "I'm not trying to tell people about Brody", he really means, 'I'm not trying to foreground my autopathographical narrative in the game'. To put it another way, when Brody says, "I'm trying to tell people about my experience", he is invoking the background narrative, hidden under an experiential veil. He is really saying, 'I'm trying to tell people about *what it feels like* to be Brody; and to convey that feeling, he deploys fabulopoiesis— metaphor and analogy to re-imagine Brody as a hyper-analytical sleuth.

Brody also importantly implies that referring to other meaningful persons in his life, such as his ex-wife, could negatively impact their personae. Protecting others from the potential damaging effects of disclosure in autopoietic forms of authorship defers to the needs of others, a modest and humble act that is an alternative to using one's own confessional impulses towards purgation and catharsis.

It is helpful to briefly turn to Thomas Nagel (1986) to appreciate the benefits of humility and modesty, which he frames as an “attitude” in bridging the gap between the subjective first-person perspective and the objective third-person perspective:

The most general effect of the objective stance ought to be a form of humility: the recognition that you are no more important than you are, and that the fact that something is of importance to you, or that it would be good or bad if you did or suffered something, is a fact of purely local significance. Such humility may seem incompatible with full immersion in one’s life and in the pursuit of those enjoyments and goods that it makes possible. It may sound like a form of deadening self-consciousness, or self-denigration, or asceticism; but I don’t think it has to be. It does not create self-consciousness but simply gives it content.

Our capacity for taking an external view of ourselves poses the problem; we cannot get rid of it, and we must find some attitude or other that reckons with it. Humility falls between nihilistic detachment and blind self-importance.

In his book *The View from Nowhere*, Nagel reconceives of humility as a tool for achieving objectivity—a moral imperative in science and philosophy. Humility is the sweet spot, a middle ground between the extremes of the nihilistically detached and the self-engulfed. In the context of Brody’s desire for humility and modesty, this would mean managing obsessive self-rumination and self-absorption on one end and nihilistic detachment on the other, an example of building the second- and third-person perspectives via an orientation to others and the world. According to Nagel, the ability to see objectively, to soberly view oneself as always existing in relation to others, displaces the self as the center of the universe—this centering of self is also a primal and unavoidable impulse of autopoiesis.

Excessive self-centering in autopoiesis may also be detrimental to both autopoiesis as a creative praxis of self-making and to the well-being of the autopathographical narrator. Nagel’s point strikes an important balance for purposes of AGD autopoiesis; humility might encourage subjects to demote the overpowering stature of their subjectivity and ‘turn down the volume’ of self-reflexive rumination that is constitutive of autopoietic praxis. When we are able to view ourselves as but a small inconsequential ‘dot’ in a world of other ‘dots’, this act of humility and modesty subdues our concept of self-importance. This is also fundamental to keep in mind when theorizing on the powers of autopoiesis, which, in its endless self-referential and metaleptic

dimensions is always in danger of slipping into nihilism.

Let us turn to Marco next, who backgrounded his autopathographical narrative in order to avoid identifying himself and his particular world through ‘facts’:

I don’t want to give facts about myself because, I’ll be very honest, I hate games or any content that start giving facts about the creator, like in a very personal level...that annoys me. It just doesn’t feel well. It’s just too personal, don’t mix with that.

When Marco emphatically states that he does not want to give *facts* about himself, he may mean *visible marks*, which link his identity to mental health struggles in the text (game). Stigma sociologist Erving Goffman (1963) would argue, concealment of mental disorder is a form of “stigma management”; “stigma is...an attribute that is deeply discrediting” (p. 3). Social attitudes and norms convey the message to the stigmatized that they must conceal their ‘marks’, which Goffman defines as “signs carrying social information” (p. 45). Marco’s statement demonstrates that he does not want to foreground his autopathographical narrative for a number of reasons. One of these reasons might be severe discomfort, which relates to how observable evidence of his narrative has the potential to become a mark of stigma in the public eye.

Ellie chose to background her autopathographical narrative in her text-based *Twine* interactive narrative on insomnia in order to “save face a little”, meaning that she wanted to retain the dignity of her public persona and also shield herself from the negative effects of social stigma. Ellie’s rational intention can be explained through Goffman’s (1956) theory of impression management about how humans as social actors perform and construct their identities depending on the ‘audience’ watching, which also resonates with Amossy’s concept of auctorial posture. Humans select specific information to reveal depending on the type of social context of reception by enacting front-stage (public) personae and back-stage (private) personae, which in the context of authorship and author identity are types of auctorial posture. “Saving face” in the context of AGD means under-sharing in the critical public eye; non-disclosure is a way to protect oneself from public judgment, especially if AGD designers decide to disseminate their games online in the harsh climate of online games culture. Alex explained why autopathographical game design is challenging in this regard:

It can blow up in your face if people, if the Internet gets a hold of it in the wrong way...

Not only is disclosure of mental illness, emotional trauma, and disability unsafe for public exhibition should subjects decide to disseminate their autopathographical games, but audience reactions, which can be extreme and volatile in online spaces, can induce emotional injury and discredit facets of author identity, such as their public persona.

Therefore, subjects practice autopoiesis by foregrounding and backgrounding their autopathographical narratives in Goffman's sense of impression management and framing experience, an act which is contingent on the context or stage of self-presentation. By regulating and adjusting the degree to which they *compose* their autopathographical narratives in different contexts, domains, and realms of human experience, they 'rack focus'⁸⁵ on their narratives, from sharpening focus during the Autopathographical Game Jam to drastically softening focus in AGD. Foregrounding and backgrounding the autopathographical narrative is a type of social compositional technique allowing subjects to inhabit roles and perspectives—auctorial postures—in order to take care and heal themselves, assert their authorial agency and autonomy, and safely and rationally achieve self-regeneration.

⁸⁵ To riff on Goffman's frame analysis theory, this term is borrowed from compositional praxis in filmmaking. A cinematographer will rack focus when they shift focus from one object in the cinematic shot to another—this shift or rack focus is recorded by the camera.

Chapter 6

Fabulopoiesis: Metaphor and Analogy in Autopathographical Game Design

6.1 Introduction

Fabulopoiesis is an analogical and metaphorical *making* of gameplay, game narratives, game worlds, and game characters based on the author's first-person experiences with mental illness, emotional trauma, and disability. This chapter examines the analogical process of how study subjects, in their role as game designers during the Autopathographical Game Jam (AGJ), render their first-person experiences into a computer game; in effect, how they 'model' a *fabulized self*, a storied reconstruction of themselves through autopathographical game design (AGD). For the purposes of this analysis, it will be useful to understand the development of this model as being an analogy or metaphor of the self. That is, subjects model the fabulized self as an auto-analogical pathographical computer game (AAG) implemented as an abstraction into the game—a third-person account of first-person experiences. In contrast to autopoiesis, which is a self-reflexive form of self-making as discussed in the last chapter, fabulopoiesis is a self-re-invention through the act of the imagination.

The analytical objective is to recognize how the authorship of the auto-analogical pathographical computer game (AAG) resembles and differs importantly from its source—the subjective experience. This analogical process maps properties of resemblance between two objects, the first-person experience (source) and computer game (target). It prioritizes and suppresses information based on what is meaningful to subjects, what they seek to reveal, and the time-limits imposed by the rapid prototyping demands of the two-day Autopathographical Game Jam (AGJ). Analogies are powerful simplification tools and authoring the auto-analogical computer game in the context of the AGJ appears to lead to self-understanding and healing via the processes of self-fabulization and abstraction. The analysis also investigates how subjects organized their decision-making and problem-solving processes in order to author the AAG, and how they employed a design plan for resolving the problem of translating emotional and experiential complexity into a simplified set of graphics and rules in a computer game.

This analysis starts with a general description of the fabulized self, the imaginary ‘self-model’ that the study subject constructs through AGD. With that apparatus in place, some illustrative samples of analogical structures used in AGD are identified. What follows is an examination of analogy theory via examples of analogy and metaphor in different domains: the literary narrative, art and design, film, and modern science. Understanding the relations between how analogy and metaphor are used in the poetic arts and how they are used in modern science is crucial to understanding their use in authoring autopathographical computer games, as AGD involves forms of praxis that blend some of the distinct formal properties of both art and science. Finally, analogy theory is applied to AGD examples in order to describe how subjects convert their first-person experiences into a computer game through analogy and metaphor. The theoretical framework draws on multidisciplinary sources, most notably narratology (Shklovsky, 1925; Bal, 2009), analogy theory from cognitive science (Gentner, 1989; Gentner and Jeziorski, 1993; Hofstadter and Sander, 2013) heuristics (Polya, 1945;1957), autobiography studies (Eakin, 1985) and the medical humanities (Couser, 1997; Frank, 1995; Sontag, 1978; Jurecic, 2012).

6.2 Authoring the fabulized self

In autopathographical game design, subjects recast their autopathographical narratives as fabulized selves through the use of metaphor and analogy. As Eakin (1985) suggests of autobiographical narrative, the outer world is a metaphor for the inner world (p. 84). A fabulized self is a narrativized, imaginary reconstruction that is neither self nor Other, but rather a fantastical figure that integrates select properties of both the self, i.e., a set of personal memories and first-person experiences and an imaginary self. As an autobiographical extension of subjectivity, the act of self-fabulization can be likened to “the art of creating a self which does not exist” (Goronwy Rees, as cited in Eakin, 1985, p. 26). Eakin affirms the performative aspects of autobiography when he invokes self-fabulization: “the autobiographer’s art necessarily involves a talented impersonation” (Eakin, p. 64). The fabulized self is a thus product of the imagination which draws on the subject’s first-person experiences, the stylized impression of which is rooted in metaphor and analogy.

Subjects model the fabulized self as an auto-analogical pathographical computer game (AAG). Put differently, they render the fabulized self through the distinct formal properties of

computer game design as a set of instances—a virtual world, game character(s), avatar, narrative, colours, and sounds—structured through an array of “expressive processes” (Wardrip-Fruin, 2009) such as game rules, scoreboards, metrics, actions, and behaviors that build the gameplay experience.

Given that the construction of the fabulized self takes place in the context of the AGD-AGJ (autopathographical game design during an Autopathographical Game Jam), it is worth mentioning the game’s ontological status as an autopathographical form. Autopathography scholar G. Thomas Couser (1997) and medical sociologist Arthur W. Frank (1995) both invoke the self-fabulization of the author in the illness narrative, alluding to its ontological status as a form of self-truth. Couser first claims that “a narrative in which an individual successfully hides an impairment is neither a disability autobiography nor a narrative of passing” (p. 214). Yet in the same book he also asserts that: “narratives of illness and disability rarely focus exclusively on bodily ailment or dysfunction; rather, they are usually also about other things in addition to, or in place of, those ailments” (p. 102). Frank claims that personal illness narrative authors propose a re-imagined version of lived events, which are just as valid as those authors that stick to more realist personal accounts: “I have read personal accounts I considered evasive, but that evasion was their truth. The more reconstructed the story, the more powerful the truth of the desire for what is being told, as the corrected version of what was lived” (p. 22). In its commitment to the agency, self-determination, and healing of the author, this analysis views autopathography as an imaginary space which ought not to be restricted based on calls for autobiographical realism or disclosure of mental illness, emotional trauma, or disability. On the contrary, the autopathography is well suited to the healing and self-transformative potential of stylization and self-fabulization.

The fabulized self is taken to be a derivative of a set of narratological concepts—*fabula* and *sujet* (or *syuzhet*)—originally coined by Russian formalist Viktor Shklovsky. These two terms, which posit a binary to explain the structure of narrative, are controversial with narratologists, who often define them differently. As Michael Scheffel (2010) claims: “where details are concerned, though, Èjxenbaum, Šklovskij, Tomaševskij, Tynjanov, Vygotskij, and other theorists proceed from markedly different starting points, using the corresponding terms (*fabula* and *syuzhet*) with different, sometimes even opposing meanings in each case”.

For example, Shklovsky claims that “the concept of *sujet* is too often confused with a description of the events in the novel, with what I’d tentatively call *fabula*” (as cited in Schmid, 2010, p. 176). Narratologist Wolf Schmid (2010) claims that for Shklovsky, *sujet* is the “process of artistic construction”—an artistic rendering of events and characters in a narrative that corresponds most closely to ‘form’, while “*fabula* is the material in *sujet* formation” (Shklovsky, 1925, as cited in Schmid, p. 176/177). In a similar vein, cognitive psychologist Jerome Bruner (1986) invokes the universal and archetypal dimensions of the *fabula* by claiming that “the *fabula* and the *sujet* correspond to the timeless and the sequenced” (p. 17). Narratologist Mieke Bal (2009) offers a “three-layer distinction—text, story, *fabula*”, which constitutes the totality of narrative. According to Bal, “events, actors, time, and location together constitute the material of a *fabula*” (p. 8). Bal makes a distinction between ‘story’ and ‘*fabula*’ “based on the difference between the way in which the events are presented and the sequence of events as they ‘occur’ in the imaginative world of the *fabula*. This phenomenon demonstrates that something happens with the *fabula* which is not exclusively language-based”. Bal also claims that “narrative texts differ from one another even if the related story is the same” (p. 6).

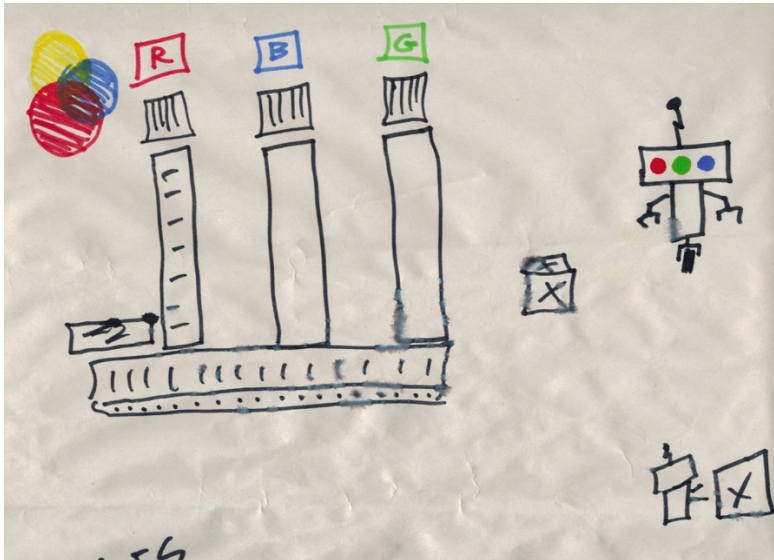
In the context of these definitions of *fabula*, the *fabulized* self is taken to be the imaginary narrative of a game autopathography that is modelled in the image of the author’s first-person experiences with mental illness, emotional trauma, and disability. Bal’s distinction between *fabula*, story, and text is adopted: “a *fabula* that has been ordered into a story is still not a text” (p. 9), which invokes the specific ordering of a *fabula* into a narrative, as opposed to the way in which that narrative ordering is implemented through the formal properties of a text, such as literary prose, a film, or a game. Consider a hypothetical example of a *fabula*: an event from our childhood such as being lost in the supermarket and frantically looking for our mother until we find her. This *fabula* can be ordered into a narrative using linear (sequential) order or non-linear (non-sequential) order, such as when using flash-back and flash-forward techniques to tell the story. This *specific narrative ordering and arrangement* of the *fabula* (being lost in a supermarket) is then rendered in a specific mode or medium. That is, the narrative becomes mediated by the ‘text’ when it is told through a perspective (a narrator) and a mode and/or medium such as oral storytelling, a written account, a comic book, a film, or a game. Thus, the structure of narrative in a given text subsumes a specific perspective (narration), a specific

narrative order and arrangement of events, actions, and actors (the fabula), and a specific mode and medium (the text, i.e., prose/verse, visual art, moving image, game, etc.).

6.2.1 The *Fabulized Self* in autopathographical game design

What is the relationship of the fabulized self to a fable in AGD? The fabulized self is sometimes authored in the likeness of fabled characters. Fables are symbolic representations and exaggerations of human adversity that posit a moral. The fable form finds novel ways to express human struggle, whether in the use of animals (anthropomorphism), or abstract objects (personification) to symbolize human behaviour and intentionality, or in the use of fabled characters to carry symbolic functions in the plot (Propp, 1928;1968). “The fable enriches us with a sense for the various ways that people manage to find comfort in situations that in fact bother them” (Hofstadter and Sander, 2013, p. 118). The use of fabled characters in the game autopathography also invites the designer to alleviate the uncomfortable or abject feelings that first-person experiences with mental illness, emotional trauma, and disability may generate. For example, Ellie’s *Twine* game about insomnia, *This is Your Awesome Sleep Machine*, which employs a “spirit guide” to direct the player to cure their insomnia, might be characterized as a fable albeit without an explicit moral. Consider some archetypal methods of self-fabulization in autopathographical game design through the use of *personification*, a type of metaphor that assigns human characteristics to non-human objects. The fabulized self is instantiated in many forms in AGD: a color-blind robot (Martin’s game); a spirit that helps the player find peaceful sleep (Ellie’s game); an invisible guide in the form of music (Cole and Yasmine’s game); a detective in a murder mystery (Brody’s game); a puzzle representing the author’s thought processes (Fred’s game); and a diagnostic tool qua musical instrument (Nathan’s game). Martin’s *Broken Lens*’ robot is a color-blind game character, who has to sift through color-coded boxes on a factory assembly line (see Figure 25 below).

Figure 25: Martin's preparatory sketch for his game, *Broken Lens*



Marco's simulation of a gravitational force between spheres as planets in a universe is symbolic of friends who become bullies and bullies who become friends. He personified the planets as humans and universes as human social networks to evoke the push-pull dynamics of friendships and their ability to reverse destiny. In the game *Brain Pains*, study subject Alex designed a neural network of movable colored nodes—an analogy of his bipolar brain. Some of these nodes are emotionally “explosive”; Alex personified them as human memories that can be triggered when the player clicks on them. Cole and Yasmine's game used music to direct and guide the player to move towards seeking colorful non-player characters (NPCs) who symbolically stand for “disabled people”. The NPCs are symbolic of Yasmine's speech delay when she was a child, information which is not shared in the game. The music is personified as a character guiding the player in the virtual world. The player hears music, which is akin to a human character in the game because it directs the player where to move inside this virtual world. These brief examples of fabulized selves are personifications of the authors' first-person experiences with illness, trauma, and disability.

Consider the following example as a more detailed assessment of the author's self-fabulization. Software developer and game designer Brody has experienced PTSD as an anxiety-inducing event when interacting with strangers on a daily basis. For Brody, daily conversations,

...always feel like (they are) life or death, there's way too much going on, I need to know way too much for the supposed atmosphere.

In other words, daily conversations for Brody take on a hyper-intense and hyper-analytical dimension, which is a particular aspect of his first-person experiences with PTSD. Brody modelled this dimension of his PTSD experiences—his thought processes that cause him great discomfort in daily conversation—as a murder-mystery game, *You Are Invited*.

I am doing a murder mystery game about how intense conversations can be with PTSD and trying to analyze like every damn aspect of everything.

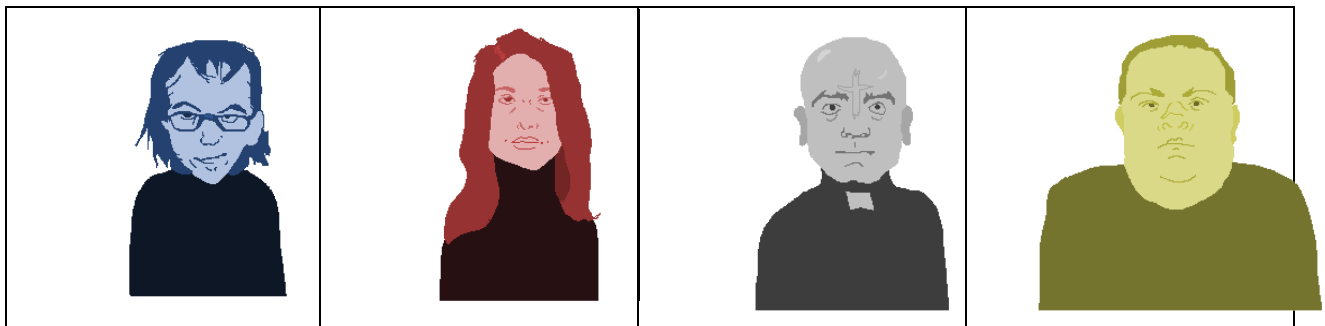
I'm trying to model—the entire game is supposed to represent life.

He personified his inner psychic script of anxious and uncomfortable thought processes as an interactive conversation among the game player character (GPC) and the non-player characters (NPCs):

I wanted to make a game about conversations that are overly intense. You pay way too much attention, you pay more attention than you need to. You have to follow every detail. You need to know before you end the conversation everything that you want to say, etc.

The main game player character plays a sleuth-detective who must find a potential murderer among a set of dinner party guests (NPCs) (see Figure 26 below).

Figure 26: Brody's game *You are Invited*. NPCs (from left to right): Prof, Wife, Priest, Mayor



The sleuth must analyze the dinner conversation among the non-player characters—the professor, the wife, the priest, the mayor, etc.—in order to predict their actions and prevent the murder-crime from occurring:

The (game) premise is that you are invited to this murder mystery dinner. When you get

there, it's revealed that not only is this a murder mystery dinner, but there is somebody there intending to murder everyone at the dinner by the end of the evening, unless you can solve everything and listen to the conversations. So, you talk to people, you have to get the information from them, pay *very* close attention to what they say, capture what they are saying, and then you get it all... You have to do the analysis to get at the information to do the things that I do.

Brody appeared to endow the sleuth-detective character that the player inhabits with various talents and capabilities. The function of a detective is to investigate crimes and discover the truth, which is associated with objectivity. The detective character has an advantage in Brody's game by bearing the hyper-attentive characteristics of Brody's daily experience with PTSD, which in Brody's life takes on negative dimensions. Brody's obsessive thinking, which causes him emotional discomfort, is translated into the game character's hyper-analytical, detail-driven eye for solving crimes. The detective character symbolizes how Brody's obsessive thinking, anxiety and compulsion "to analyze every damn aspect of everything" can benefit society by preventing murder and catching a potential killer before he commits the crime. Brody re-wrote and re-imagined his PTSD from a cognitive and emotional disadvantage in his life to a positive advantage in the game—a great example of poetic justice.

Another of the detective character's capabilities is the ability to view the action from multiple perspectives. Brody made the design decision to author two perspectives that the player embodies via the detective game character: an overhead first-person view of the dinner table and an eye-level first-person view of the dinner table guests. As he says,

The design was going to be split between two first-person views. One that was the table top and one that was the people.

The eye-level perspective symbolizes the human perspective, while the overhead-table perspective symbolizes a more objective and detached position encompassing the whole scene of the dinner party. Through the overhead table player view, Brody is nesting a third-person perspective inside a first-person perspective in order to impart a more comprehensive view of the dinner table action.

There is another aspect to the game—its temporal dimensions and lack of win outcome. Brody explained that in the game,

There is no win or lose. I want (the game) to actually take place over the course of a dinner—so it's only 45 minutes to 1 hour and a half long. There is no (save mode). You are just sitting there in one go.

Brody's objective is to construct a gameplay experience that does not allow the player to take rest from the interactive experience of the 'dinner party', out of a desire to communicate to the player his own difficulty from disengaging from his PTSD-induced anxiety and repetitive thoughts.

Given the design decisions above, Brody undertakes two important tasks through autopathographical game design. First, he turns to the poetics of self-fabulization in order to view his PTSD experience from a positive perspective instead of the negative dimensions that cause him emotional distress. The sleuthing detective character's hyper-analytical eye helps the player predict and prevent a murder from occurring. Finally, the objective for self-fabulizing his personal thoughts as an interactive conversation among a set of game characters is to metaphorically communicate his first-person experiences to medical professionals:

The game works to decouple my experience from me and build something that someone else can walk across and have access to—its purpose. I'm trying to get across a subjective experience. It's not surprising that (psychologists and psychiatrists) are not getting it from my words.

There is a desire on Brody's part to communicate the complexity of his subjective experience with PTSD through gameplay in order for medical professionals to understand him better and perhaps, offer better therapy or treatment. Brody suggests that inhabiting the fabulized self through gameplay may be a more effective method than telling a narrative through words for giving medical professionals access to a subset of his subjective experiences.

6.3 Understanding metaphor and analogy

It is useful to harness analogy theory in order to understand the above examples of the fabulized self that subjects deploy in autopathographical game design. General understandings of analogies can also be deployed to understand one of the most important aspects of AGD: solving the problem of how to render a first-person experience of mental illness, emotional trauma, and disability into a computer game. Thus, in this study's context, understanding study subjects' self-fabulization depends on their use of analogy and metaphor; both are important creative tools for

finding solutions for how to render a first-person experience into a computer game. Subjects are compelled to search for similarities between a facet— or, as Nathan states, a “subset” of their first-person experiences— and the formal properties of computer game design, including the properties of a specific design tool such as text-based game design versus graphics-based game design.

Hofstadter and Sander (2013) suggest that thinking in images consists of employing analogy, metaphor, allegory, fable and model, among other types of analogical reasoning. We can understand analogy broadly as highlighting similarities in the form and structure of different objects and conceptual categories. In other words, “an analogy is a ‘match-up’ produced through the act of mapping correspondences of components from one representation to another” (Chalmers, French & Hofstadter, 1992). Gentner (1989) defines an analogy as a knowledge representation based on systemic similarities between two different objects (using ‘base’ for what is called the ‘source’ in this analysis):

...an analogy is a mapping of knowledge from one domain (the base) into another (the target), which conveys that a system of relations that holds among the base objects also holds among the target objects. (p. 201)

Aristotle declared that “to make good metaphors implies an eye for resemblances” (335 BC; trans. Butcher, 1961). Recognition of similarities in an analogy typically does not involve objects in the same conceptual category, i.e., human and human, cat and cat, chair and chair, because analogies are not based on “shared surface qualities” (Gentner and Jeziorski, 1993). Gentner and Jeziorski (1993) claim that “thematic connections - do not contribute to the analogy”. Rather, modern analogies are based on a “preference for structural consistency and systematicity”, a deeper structural resemblance between two dissimilar objects or conceptual categories. Analogies can be: proportional as the Ancient Greeks asserted, i.e., $8 \div 4 = 2$ is analogous to $6 \div 3 = 2$ —both equations are divided by the number 2, which connects them; structural, i.e., Rutherford’s model of the hydrogen atom is analogized to the solar system on the premise that both revolve around objects and are causally connected—through electromagnetic force in the atom’s case, and through gravity in the case of the solar system (Gentner, 1983); or functional, as when analogies map correspondences between the similar functions of objects such as wings,

lungs, and feathers in birds and fins, gills, and scales in fish, despite material or surface dissimilarities (Itkonen, 2005, p. 9).

Analogies and metaphors deriving from surface dissimilarity are particularly striking to the human imagination. When two analogous objects are different on the surface level, such as the human heart organ and the hydraulic pump yet are perceived as similar in their shared functional essence, this recognition elicits wonder and surprise (Hofstadter and Sander, 2013, p. 135). For example, the provocative philosophical and scientific analogy of the ‘mind as computer’ is based on the alleged structural and functional resemblance between minds (and by extension, brains) and computers as information processing systems. Another instance of analogy-making can be found in the proto-museum and Baroque *Kunstkammer*, or cabinet of curiosities, which offers patrons the opportunity to behold such disparate objects that analogizing becomes a wondrous activity (Bredekamp, 1995). In ‘The Allegory of the Cave’ in Plato’s Republic (380 BCE), Plato authors an elaborate analogy for his epistemology, comparing prisoners in a cave seeing shadows on the wall to human beings mistaking mere appearance for truth and reality. Gentner and Jeziorski (1993) assert that “only commonalities strengthen an analogy”.

While Hofstadter and Sander appear to not differentiate between analogy and metaphor, Gentner and Jeziorski (1993) make a distinction between the two:

We view metaphor as a rather broad category, encompassing analogy and mere-appearance, as well as a variety of other kinds of matches. On this view, analogy is a special case of metaphor, one based on a purely relational match (p. 452).

On Gentner, Bowdle, Wolff, & Boronat’s view (2001), while analogies are based on alignment and one-to-one mapping, some metaphors require processes such as metonymy and phonological mapping (p. 201). For example, consider the pun—“what did the boy cat say to the girl cat on Valentine’s Day? You’re purrr-fect for me!”⁸⁶. The word “purrr-fect” demonstrates a phonological mapping between a cat’s purring (source) and the phonetically similar word

⁸⁶Source of pun: <http://onlinefun.com/puns/11/>

‘perfect’ (target). According to Gentner and Jeziorski (1993), some poetic metaphors are “not bound by the one-to-one mapping constraint and can include mixtures of several bases, as well as thematic and metonymic relations...” (p. 452), such as in the following verse lines (as cited in Gentner and Jeziorski):

“the voice of your eyes is deeper than all roses” (e.e. cummings); or “On a star of faith pure as the drifting bread /As the food and flames of the snow” (Dylan Thomas). (p. 452)

A metonym is similar to a metaphor, but according to Burke (1941) a metonym is a substitute for something that reduces complexity. Burke explains that “the basic ‘strategy’ in metonymy is this: to convey some incorporeal or intangible state in terms of the corporeal or tangible, e.g., to speak of “the heart” rather than “the emotions” (p. 424). Inconsistencies present in the layering of metaphor and metonymy and deviations in an analogy may offer more novelty than the logical, predictive value of commonalities in an analogy. This phenomenon reflects a difference between science, which searches for a common pattern between different objects, and the creative arts, some of which aesthetically privilege deviation from a pattern in order to creatively experiment with form and content. Science is a system of harmony and unity based on our human nature’s ability to “appreciate Order” (Boole, 1854; 2005, p. 311/p. 314). The emotional effects of a creative or poetic work are amplified when viewer expectations are subverted, that is, when the pattern is broken. As per Vygotsky’s (1971) claims, artists and poets lean towards artifice when creating poetic arrangements that emphasize deviations from nature. In other words, the creative arts accentuate the disanalogous properties of objects to elicit wonder and surprise in the audience.

Analogy is pervasive in all spheres of life (Hofstadter and Sander, 2013), and metaphor has been claimed to be “the omnipresent principle of language” (Richards, 1965, p. 92). “Analogy-making itself is a perceptual process” (Chalmers, French & Hofstadter, 1992), while analogical reasoning is a tool for pattern-recognition. Analogies have been claimed to be implicated in insight, revelation, and scientific discovery, and drawing analogies from past experience combines imagination and memory (Danesi, 2002; Hofstadter and Sander, 2013). Humans perceive analogous relations between familiar and unfamiliar objects or situations in order to understand the ‘unknown’, mapping knowledge from a familiar domain based on past experience onto an unfamiliar one (Gick and Holyoak, 1983; Gentner and Jeziorski, 1993). “The

very essence of an analogy is that it maps some mental structure onto another mental structure” (Hofstadter and Sander, 2013, p. 50). The mapping process produces an “interdomain analogy” defined as “the selective exploitation of past experiences to shed light on new and unfamiliar things belonging to another domain” (Hofstadter and Sander, 2013, p. 17).

Consider that the computer game autopathography as genre was an unfamiliar domain to some subjects. Lukas, Nathan, Ellie, Sam, Yasmine, Cole, Marco, and Martin had never made games about depression, anxiety, disability, loneliness, insomnia, being bullied, etc. Lukas confessed that,

...the topic is not something that I had ever thought about before.

The autopathographical game as ‘unknown’ territory prompted him to think differently about computer game design as a frontier of different possibilities:

Making the game autopathography, now I realize it. That’s one exciting thing. That’s something that constrained my idea; I focused only on mini-arcade games. I got constrained by my own career path and my experience and I didn’t know that until now, today. That’s one very interesting thing I discovered when I came here.

Lukas’s quote suggests that AGD is an exciting discovery due to his unfamiliarity with this genre. He perceived this unfamiliarity as positive in the sense of inspiring him to take up something novel in computer game design that would galvanize his creativity.

As such, analogies have at least two epistemological roles. They are logical inferences, contributing to argument and reason, but they are also phenomenologically experienced in a way that contributes to or intensifies understanding and creative thinking. The implications for this dissertation’s subjects are that analogies serve as both intermediate problem-solving tools during the autopathographical game design (AGD) process, and as ‘models’ of self for subjects, not only for them to use in the design, but also—and crucially, for the therapeutic function of AGD—as new phenomenological experiences of self, that adjust, contribute to, or intensify their self-understanding in imaginative ways.

In other words, subjects emotionally experience analogies as flights of the imagination and intellectually deploy them to solve problems and understand the unknown. Analogical

reasoning is a perceptual process in that we imagine or infer hidden structural similarities between two surface-dissimilar objects, yet it is also a cognitive process because we use analogies as reasoning tools to arrive at conclusions about the world and the self via abstraction. Both roles can be seen not only as manifest in, but as fundamentally constitutive of, the AGD process and experience.

6.3.1 The illness metaphor and analogy in the poetic arts

The objective of the following examples of metaphor and analogy is to show the underlying structural similarities and dissimilarities in analogy and metaphor in order to harness them in the analysis of analogical structures deployed in AGD. Consider first a brief example of the illness metaphor and medical analogy as deployed in the literary narrative. Literary autopathographical narratives are drenched in analogical devices and metaphors, which sometimes obscure content through symbolism: “what counts as an illness story is by no means clear”, says Frank (1995, p. 21). Metaphor and analogy infuse the illness narrative with a variety of symbolic connotations such as difference, deviance, misery, challenge, journey, chaos, moral redemption, moral flaw, etc. Susan Sontag (1978) explains how the metaphors of disease are harnessed in both medicine and fiction: in medicine, in order to soften the brutality of severe illness for the benefit of the patient; in fiction, in order to aestheticize illness for the reader. The medical establishment uses metaphor to frame intractable illnesses such as cancer through the “language of warfare”: words such as “invasion”, cancer cells’ “attack” of the body’s “defenses”, and “killing” cancer cells construct a medical ‘war’ on cancer, notes Sontag (p. 64). Sontag argues that the “trappings of metaphor” worsen social stigma by implication that cancer is a horrible disease akin to the vicious and barbarous nature of war. She also compares how cancer and TB are differently portrayed in literature via specific metaphors:

Nobody conceives of cancer the way TB was thought of—as a decorative, often lyrical death. Cancer is a rare and still scandalous subject for poetry; and it seems unimaginable to aestheticize the disease (p. 20).

Illness narrative scholar Ann Jurecic (2012) claims that Sontag’s *Illness as Metaphor* posits an argument that “metaphors arouse our imaginations in ways that overwhelm reason” (p. 68). Couser offers a divergent interpretation of the illness metaphor as a tool for the author’s empowerment while discussing Audre Lorde’s (1980; 2006) writings in *The Cancer Journals*:

“unlike Sontag, who saw cancer metaphors, and the war metaphor in particular, as the patient’s enemies, Lorde embraces the trope of war; she takes pride in identifying her one-breasted self as an Amazon warrior, and she chooses to define her scar as a badge of honor” (p. 52). To draw on Couser, Lorde’s use of the illness metaphor as a tool to positively re-imagine her illness experience is similar to the deployment of metaphor and analogy in autopathographical game design in re-imagining the self through the act of fabulopoiesis.

Hawkins (1999) calls the pathography a “modern adventure story” that re-conceives the illness experience as a thrilling dramatic arc. In regard to re-imagining illness as narrative, consider Frank’s (1997) description of the use of metaphor in structuring three underlying sub-narratives of the illness narrative that appear to function as tools of self-understanding and self-healing for the author: the quest narrative, the restitution narrative, and the chaos narrative. These sub-narratives are analogies and metaphors serving as tools for authors to recover meaning for themselves that is often lost in the process of being sick. In the quest narrative metaphor, illness is a journey of self-discovery for the author. The restitution narrative deploys a mechanistic view of medicine as progress (the “triumph of medicine”) and as having the role of ‘fixing’ the dysfunctional body (Frank, 1997, p. 88). And the chaos narrative analogizes illness to a disorderly mess to be tamed via the organizational properties of storytelling.

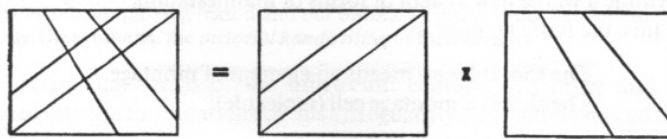
6.3.2 Common analogies in the creative, poetic, and media arts

Sontag’s example of cancer as body invader is an example of a conventional analogy. Hofstadter and Sander (2013) call such analogies “banalogies”, or banal analogies, because they are so overused in discourse that we take them for granted (p. 143). Humans are not content with stale and hackneyed tropes in their search for novelty (Hofstadter and Sander, 2013, p. 64). Novel analogies help the perceiver find connections between disparate concepts, they say, which could serve to provide a deeper understanding of unfamiliar objects. Without novel analogy and metaphor, there is no possibility for artists and poets to imagine different worlds and different ways of being. Consider the following analogies in film, painting, photography, and typography, which are rooted in rhythmic, narrative, and audiovisual properties as well as context: fast-paced editing in a film can be analogized to a fast heartbeat if accompanying a chase scene; if the narrative is in correspondence, a film shot of rain on a window sill can be analogized to human

tears and sadness of a character; in painting, messy brushstrokes, such as in Abstract Expressionist paintings, can be analogized to emotional turmoil; in photography, a steep diagonal composition can be analogized to tension or dizziness; in typography, such as in the context of text written in social media, all capitalized text used for emphasis– ‘ALL CAPS’—can be analogized to a person shouting or showing anger, as the amplification of speech in shouting corresponds to the amplification of text from lower-case (not shouting, regular tone of speech) to upper-case (shouting, irregular tone of speech).

An example of how analogies harness both perceptual and conceptual levels of human experience is provided by the Russian formalist filmmaker Sergei Eisenstein. Eisenstein (1929) analogized Soviet Marxist dialectics to a particular style of film montage that he pioneered in order to affect the viewer through a powerful ideological message. This example is used to show the way in which Eisenstein implemented abstract ideas through both material practice and analogical reasoning, in this case through film editing and cinematography⁸⁷, presaging how AGD designers would convey abstract ideas through their own material practices. The compositional properties of filmic shots are edited so that a left-diagonal shot composition is intercut with a right-diagonal shot composition, producing a ‘clash’ between the two opposing diagonals in the mind of the viewer (see Figure 27 below).

Figure 27: Image taken from Eisenstein’s (1929) essay: “A Dialectic Approach to Film Form”



⁸⁷ Music and audio (i.e., dialogue, sound effects) have a very prominent formalist role in Eisenstein’s films, which are integrated with image in very specific ways to emphasize the rhythm and tempo of the image editing (montage). The discussion of music and audio are omitted for purposes of brevity. Please see Robert Robertson’s (2009) “Eisenstein on the Audiovisual: The Montage of Music, Image and Sound in Cinema”.

This montage technique evokes a conflict that resolves itself, a metaphor for the upheaval of the Bolshevik Revolution and, in turn, class struggle—a fundamental concept of Marxist philosophy. In this example, the construction of a multiple analogy is based on searching for similar structural relations between the following objects: cinematography, editing, conflict, upheaval, struggle, and Soviet Marxism. A multiple analogy is “a comparison in which several sources are likened to a target” (Shelley, 2003, p. xiii). The material processes of editing and cinematography are similar to each other in that editing is constituted as an inter-frame process through the combination and arrangement of two or more shots, while cinematography is constituted through the intra-frame combination and arrangement of visual composition and movement. In the context of Eisenstein’s style, both are harnessed towards the representation of abstract ideas of conflict and struggle. The abstract ideas of conflict, upheaval, struggle, and Soviet Marxism all funnel into the broad ideological objective of Eisenstein’s work. The common “relational predicate” (Gentner, 1982) unifying all of the above material processes and abstract ideas in the context of Eisenstein’s filmmaking style is *opposition*. Given that all of the above objects are analogized in Eisenstein’s work through moving image narrative (film), they are an example of a multiple analogy, which layers one analogy on top of another through higher-order reasoning. In other words, to understand the full breadth and depth of Eisenstein’s ideological message, you need to have some familiarity with the context which his films reference, namely the Russian Revolution of 1917 and Marxist philosophy.

6.3.3 Analogy in modern science

Given that computer game design and development is a blend of art and engineering, authoring autopathographical games involves forms of praxis that invoke some of the distinct formal properties of both art and science. Understanding how analogy and metaphor are used in these domains is therefore important to understanding their deployment in AGD. According to Gentner and Jaziorski (1993), unruly metaphors demonstrate more variation than consistency in form and structure than analogy, especially in regard to the evolution of analogical reasoning in modern science. Metaphors can highlight deviation and difference, as opposed to the “structural consistency and systematicity” prized in modern science (Gentner and Jaziorski, 1993, p. 470). At the same time, rhetorician I.A. Richards (1965) argues that “as (philosophy) grows more abstract (and severe), we think increasingly by means of metaphors that we profess not to be

relying on” (p. 92). Richards is alluding to the deep connections between metaphorical thinking and abstraction, the latter of which is an objective of scientific and logical reasoning; he suggests that they are not mutually exclusive and indeed cannot exist without each other. Bruner (1986) makes a distinction which Richards does not: he states that for the domain of science, analogies in the form of “wild metaphors” and “metaphoric crutches” “help (scientists) get up the abstract mountain” (p. 48). Boyd (1993) confirms that:

...one should expect that when metaphorical language is employed in a scientific context, its function should either lie in the pretheoretical (prescientific?) stages of the development of a discipline, or in the case of more mature sciences, it should lie in the realm of heuristics, pedagogy, or informal exegesis, rather than in the realm of the actual articulation or development of theories. (p. 482)

In other words, scientists depart from metaphors through their formalization of categories, working out a precise and rigorous categorization of objects—a classification, to create systematized, disambiguated theories, hypotheses, and models that can be easily understood and shared within the academic community.

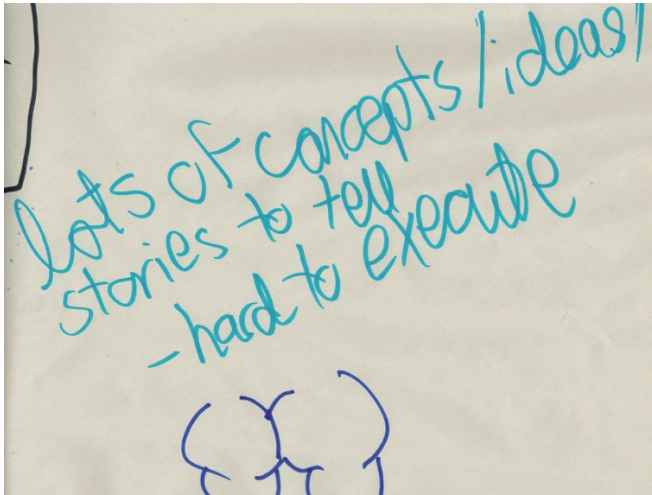
Gentner and Jaziorski (1993) echo Bruner and Boyd, claiming that in science “unruly metaphors play a role in the discovery process, and that some degree of tolerance for loose analogy is important for creativity, as in Polya’s (1945;1957) advice to keep vague analogies but try to clarify them” (p. 476). But Bruner stresses that the scientific objective is to ultimately “convert those dense metaphors into the transparent, frangible hypotheses of science—or into untestable axioms that will generate hypotheses that, with luck, may be tested” (p. 52). In other words, scientists formalize categories, i.e., they work out a precise and rigorous categorization of objects—a classification— to create systematized, disambiguated theories, hypotheses, and models. Analogies in the creative arts and poetry may be retained at the completed stage of a creative work, as opposed to the intermediate stage of a scientific work, differentiating the creative arts from science. The ‘artistic license’ grants artists the right to lock down the creative work at the level of analogy, metaphor, and allegory—a blurrier, greyer, more fluid level of categorization—in order to invoke ambiguity, mystery, and play in audience reception of the text or artifact.

The standardization, systematicity, and methodical rigour of scientific practice may apply to computer game design in its engineering dimensions. That is, the practice of formalization involved in science bears some similarity to the formalization processes involved in engineering the computer game qua formal system of rules and constraints. As such, AGD designers also have to work a precise and rigorous categorization of objects by way of authoring their game as a formal system. As Sam explained, computer game design involves formalization processes that distill a game narrative or concept into rigid numbers and values. This will be discussed in greater depth in the chapter on *Logopoiesis*.

6.4 The role of heuristics in prototyping the autopathographical game: the intelligibility framework (IF)

Through self-fabulization, subjects organize their first-person experiences into an analogous autopathographical narrative (AAN) prior to the structural implementation of this analogous narrative into the computer game autopathography. There are discernible steps in subjects' processes of "analogical transfer" (Gick & Holyoak, 1983), that is, of mapping select information from source (first-person experiences) to target (computer game). In developing the auto-analogical pathographical game (AAG), subjects (i) appear to construct an *intelligibility framework* (IF) for their first-person experiences. The IF is akin to a mental design plan that designers deploy in their planning and preparatory stage in order to solve the problem of how to translate from the source to the target. The IF allows designers to build parameters around the fabulized self in order to more easily implement their autopathographical narratives in the computer game. The IF is the achievement of the resolution between the richness (complexity, ambiguity, indeterminacy) of the first-person experience and the formality (abstraction, systematicity, clarity) of the computer game as a medium of expression. Subjects invoke the difficulty of the analogical transfer process of AGD, and the need for a mental road map, of which the IF is an example, to make their games. As one subject wrote in a study AGJ doodle: "lots of concepts/ideas/stories to tell—hard to execute" (see Figure 28 below). In other words, the subject suggests that it is easy to imagine a scenario for a game autopathography compared to executing this scenario—implementing and formalizing it through the formal and structural properties of computer game design.

Figure 28: Autopathographical Game Jam doodle



In the next steps of the analogical transfer, study subjects (ii) abstract their first-person experiences into the IF by (iii) developing an analogous autopathographical narrative (AAN) that does the following: selects, exaggerates, or simplifies some of the properties of their first-person experiences in the framework's terms, and ignores or suppresses others. This process results in (iv) a structurally articulated analogous narrative (SAN) consisting of a number of game elements that comprise the gameplay system including dramatic arc and game outcome, gameplay characters, virtual world, game rules, and game metrics to measure player progress. Then, study subjects implement the SAN via the processes of formalization involved in digital design (e.g. audiovisual design, 3D or 2D modelling, computer-generated graphics, etc.) and programming.

Autopathographical game design in the context of a two-day game jam presents two general challenges to study subjects in terms of the above logistics of the authorship process. These are the need to adhere to a strict time limit in order to generate a prototype, and the requirement for a mental road map—the intelligibility framework (IF)—for solving the problem of how to implement the autopathographical narrative into a formal system of rules and constraints. Consider first the challenge of prototyping based on the two-day, seven-hour per day time limit of the study AGJ. Subjects did not construe the time limit as negative. On the contrary, some of them considered it a positive attribute for AGD, as per Sam's statement:

I like the fact that (the AGJ has) a tight time constraint, it forces you to really forget about the details for now and really just hammer out a prototype.

The game jam time limit forced subjects to prototype their games via heuristic reasoning. Heuristic reasoning is a problem-solving method that mathematician George Polya (1945;1957) calls the art of discovery, which can also be understood as a type of intelligibility framework for the domain of mathematics. Heuristic reasoning relies on trial and error and analogical reasoning to make decisions, without the guarantee for a successful outcome. The significant role of analogy in heuristics is as an intermediate yet crucial step in any problem-solving endeavor whose final objective is formalization. Polya's heuristic reasoning is provisional, never final, certain, or strict (Dasgupta, 2016). "Heuristic reasoning is good in itself. What is bad is to mix up heuristic reasoning with rigorous proof," states Polya (p. 113). Gentner and Jeziorski (1993) give the example of Kepler, for whom "analogy is a source of hypotheses rather than a guarantor of truth" (p. 474). Polya's distinction between heuristics and rigorous proofs may be conceptualized as the difference between finding a provisional solution to a problem and proving a solution.

Prototyping is a form of heuristic because it serves as an intermediate step in solving design problems. As a heuristic, prototyping is a form of inductive reasoning that involves searching for analogies to solve problems. Prototyping is the essence of design work: "much good design evolves: the design is tested, problem areas are discovered and modified, and then it is continually retested and remodified" (Norman, 1988, p. 142). Prototypes are thus works-in-progress that need further iteration and playtesting. Salen and Zimmerman (2006) suggest that prototyping, iteration and testing is nevertheless a rigorous process: "iterative design and testing is more than just informal play—it is a rigorous design process" (p. 21).

6.4.1 The process of analogical mapping in AGD through the IF

The process of modelling the fabulized self as an auto-analogical pathographical computer game (AAG) is dependent on a structural articulation which involves engineering the autopathographical game as a formal system of game rules and constraints. Autopathographical game design as such requires an intelligibility framework (IF) for thinking about how to implement an autopathographical narrative based on a first-person experience into a formal system qua computer game. As mentioned above, the requirement of an IF is the second challenge subjects face in AGD for solving the problem of how to structurally articulate first-

person experiences with mental illness, emotional trauma, and disability into a computer game. The process of analogical mapping in AGD can be represented in the following way:

1. Source: First-person experiences with mental illness (bipolar disorder, depression, anxiety), psychological and emotional trauma, post-traumatic stress disorder (PTSD), disability (ADHD), vision-impairment (color-blindness), emotional or psychic distress (grief, loneliness, bullying, insomnia);
2. Step A: Intelligibility Framework (IF):
 - a) Analogous autopathographical narrative (AAN): abstraction processes: selection, exaggeration, simplification, and formalization resulting in:
3. Mid-Target: structurally articulated analogous narrative (SAN) (*fabulopoiesis*);
4. Step B Implementation: configure structurally articulated analogous narrative (SAN) into structural properties of computer game design (*logopoiesis*);
5. Final Target: first prototype of computer game autopathography.

Consider the role of abstraction in the analogical mapping process involved in AGD. In describing prototyping as “forgetting about the details”, Sam reflects George Polya’s view that heuristic reasoning involves “ignoring the details”. Both Sam and Polya are actually referring to the specific design task of abstraction. The act of abstraction is a primary conceptual tool in heuristic reasoning. Alex emphasized the key role of abstraction in autopathographical game design:

It’s just about thinking outside the box and abstracting things and seeing ‘why am I doing this, you know’?

Alex’s statement invoked the primary dual roles of abstraction and creative thinking in the AGD framework. According to Alex, the combination of abstract and creative thinking in AGD encourages the designer to question the intentions and motivations behind their design decisions— “why am I doing this”. John also invokes abstraction by stating that he started conceptualizing his autopathographical game on bipolar disorder prior to coming to the study AGJ:

I had a bad depressive phase a couple of weeks ago and I knew I’d be doing this game jam. I was like, I should write down how I am feeling right now. And I wrote down just a couple of points.

He abstracted based on a selection of “three notes of what I was feeling”. In other words, John selected approximately three themes or details to emphasize in his autopathographical game design. Alex also invokes abstraction as a primary tool for prototyping computer games. In a game jam with strict time constraints, Alex designs minimalist game prototypes relying on:

Clean lines, less colours, big heads, googly eyes, cartoony, exaggerated.

Using clean lines hides appropriate line details and using less colours hides colour variation such as shade and nuance. As such, the act of abstraction entails amplifying select information—exaggerating shapes into “big heads” and “googly eyes”—that is, making shapes “cartoony”, while hiding other information by using “clean lines” and “less colours” (including less shades and colour gradation). Thus, the tight time limit of the study’s Autopathographical Game Jam forces subjects to abstract and distil information to essences while overlooking concrete details.

In order to better understand abstraction processes in autopathographical game design, it would be helpful to turn to some theoretical perspectives on this topic. Denning and Martell (2015) define the relationship between abstraction and thought: “an abstraction is a mental model that captures the essential features of a thing and suppresses all other features” (p. 59). Hofstadter and Sander (2013) tie the human ability to think abstractly to intelligence, which they characterize as “sorting the wheat from the chaff” (p. 125). They define abstraction as the “result of recognizing and isolating what different concrete concepts have in common” (p. 288), such as searching for common patterns between first-person illness experiences and the structure of computer games. After abstraction, all that may be left is an object with general properties that share a “common systematic relational structure” (Gentner and Jeziorski, 1993) with other objects. According to Hofstadter and Sander, a key aspect of the process of abstraction is that “each person ‘decides’ (that is, encodes) what a situation includes and doesn’t include, and what its key ingredients are” (p. 161). John Stuart Mill also invokes the embodied cognition model by suggesting that the ability to analogize from first-person experience bridges percepts, which are rooted in observation, with concepts as ideas that are inferred from observation:

...of nature, or anything whatever external to ourselves, we know...nothing, except the facts which present themselves to our senses, and such other facts as may, by analogy, be

inferred from these. There is no knowledge a priori; no truths cognizable by the mind's inward light, and grounded on intuitive evidence" (as cited in Olney, 1972, p. 239).

Mill argues for the notion of *a posteriori* knowledge—observations which are recognized as facts via sense perception afforded by the first-person perspective. Hofstadter and Sander reflect Mill's perspective by asserting that the concrete and actually occurring world is what the human senses perceive in the form of percepts while the abstract world is the world of ideas or concepts. Hofstadter and Sander say, however, that "we all blithely *blur* the concrete and the abstract" (p. 287), presumably referring to the embodied (non-Cartesian) ways in which the concrete and the abstract might inform one another in the first-person perspective; or how humans might analogize by imposing a previous understanding (a concept) on new percepts in the embodied act of perceiving.

Merleau-Ponty (1962; 2002) precedes Hofstadter and Sander by claiming that the concrete and the abstract become indistinct in the first-person perspective, but that this lack of separation happens because human conceptualization, which appears to be achieved through the selective process of abstraction, is always and inevitably embodied and thus non-Cartesian. He suggests that we can *imagine* a Cartesian distance between the concrete and abstract, but that this distance does not *exist* in the embodied act of perceiving:

...between the self which analyses perception and the self which perceives, there is always a distance. But in the concrete act of reflection, I abolish this distance, I prove by that very token that I am capable of *knowing* what I was perceiving, I control in practice the discontinuity of the two selves... (p. 49-50)

Merleau-Ponty's insight can be used as a way to understand the abstraction process involved in conceptualizing the fabulized self whereby subjects cull elements from and de-contextualize their first-person experiences into a third-person account—the auto-analogical pathographical computer game (AAG). Thus, the fabulized self, which is achieved through the embodied act of abstraction, enables subjects to *imagine* a distance between the proper self, i.e. the first-person perspective and the fabulized self, i.e., the third-person abstracted account in the AAG, which can, as Merleau-Ponty says, "control in practice the discontinuity of the two selves".

Therefore, to 'create abstractions' is to critically remove object-specific properties so that the percept (object of perception, observation, or fact) is de-contextualized from the embodied

position and left to its essences or generalized properties. As generalizable, abstractions traverse multiple domains. In the context of computer game design, abstractions applied to the computational domain, as opposed to logical or mathematical abstractions, “perform actions” (Denning and Martell, 2015, p. 59). Yasmine affirmed that if you abstract the first-person experience by de-contextualizing it from the concrete realm of the first-person standpoint, it becomes inclusive of a diverse set of human experiences:

I’d say it felt like our game was a little more abstract...It was a bit of ourselves, but it was mostly a bit of everyone’s experiences. We wanted to try and be as inclusive as possible.

Yasmine and Cole, who collaborated together on their game, preserved some meaningful information about themselves—“a bit of ourselves”—in their joint conceptualization of the fabulized self. Thus their fabulized self culls their first-person experiences in order to generalize from them and universalize them for the player and create an abstraction—an “inclusive” gameplay experience.

Fred confirmed the disadvantages of abstracting too much in the design of his game *Brain Buster*, which can be detrimental to the overall “message” the game is trying to communicate to players:

(When) ideas are far too abstract, the point of the message is lost. I try to find some sort of middle ground, which allows me to communicate that abstract idea in a way that is understandable. Unfortunately, I feel that there are certain aspects of subjects that are so abstract that I don’t think they can be communicated to an interactive experience that’s understandable. But then again, that could just be my shortcomings as a designer.

Fred was selective about which specific properties of his traumatic experience to retain in the game, which he communicates through the “rules of the game” as a set of simulated phone calls a player makes in order to solve a puzzle:

I try to find a way of articulating something that happened to me and then trying to create a flow chart or some sort of algorithm from it. I don’t try to abstract the entire idea into a flow chart, instead what I do is I take an individual component of what happened, such as the stress that happened from PTSD. For example, with this game, how I went about solving (the stress from PTSD) and then creating an algorithm through that and then seeing if there’s a potential for a game move, and then developing it from there.

The concept of “stress” is both an abstract and concrete category. It is abstract and conceptual if we decouple it from stress as a first-person experience—an act of the imagination—and it

becomes concrete if we invoke our embodied experiences of stress through gameplay, such as in Fred's articulated desire to render his concrete experiences of 'calling people for help' in his game. To abstract the experience of stress in order to convey it via gameplay, a designer must find a way to express the embodied experience of stress into an algorithm or a set of rules to be experienced through gameplay. With the above statement, Fred gives voice to the complex process of abstracting his first-person experiences into a formal system of rules and mechanics, which must be then played. As such, AGD's biggest challenge is solving the problem of how to communicate some of the richness of first-person experiences with mental illness, emotional trauma, and disability through gameplay. The use of analogy and metaphor significantly aids this communication process, as will be demonstrated shortly through subject examples.

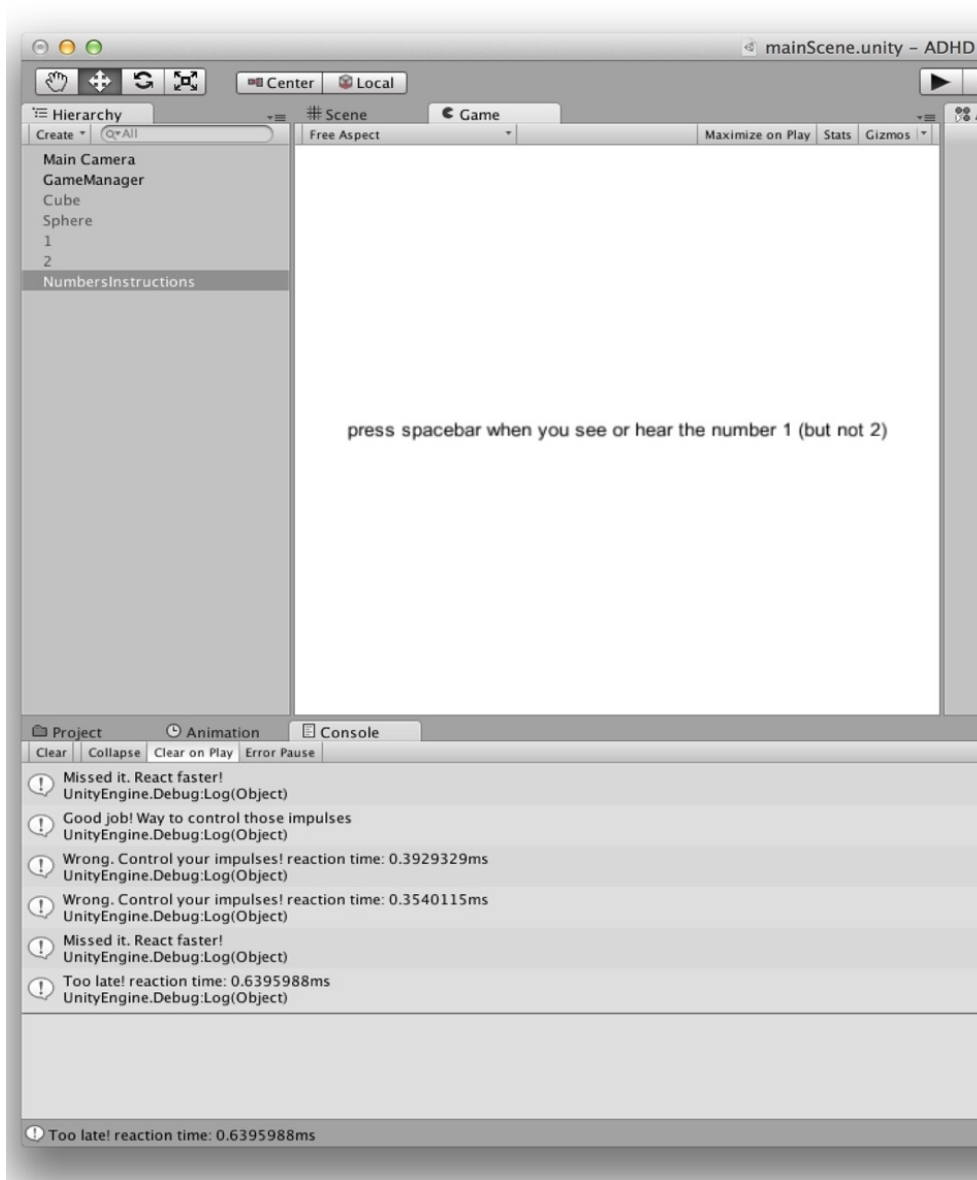
George Polya's (1945;1957) theorization of problem-solving speaks to Fred's account of his intelligibility framework (IF), which gives voice to the slippage between design intentions and the affordances and limits of computer game design. Polya characterizes problem-solving as a set of heuristic practices: abstraction, concretization, analogy, and decomposing and recombining. The design plan can be connected to this set of heuristic practices for re-imagining a design problem in order to solve it, given the material one is working with (in this case, a software program). According to Polya: "we can easily imagine new problems if we are somewhat familiar with the principal means of varying a problem, as GENERALIZATION (abstraction), SPECIALIZATION (concretization), ANALOGY, DECOMPOSING, AND RECOMBINING" (capitals in original) (p. 65). One way is to find an auxiliary route to getting the job done according to Polya. An auxiliary problem is a similar problem, hence an analogical problem used to solve the original problem: "if you cannot solve the proposed problem try to solve first some related problem" (p. 10). Polya also calls this technique "varying the problem". Newell (1981) claims that "generating an auxiliary problem is not a constructive task, but a task of retrieval from long-term memory" (p. 33). Auxiliary problems are thus analogies that draw on past experiences. To decompose means to break down a problem into smaller parts, while to recombine means to take those parts and reconfigure the problem into a new form. Polya describes the decomposition process of breaking down a problem as focusing on a few details that resonate, which is exemplified by John's statement that he focused on "three notes of what I was feeling" for his game autopathography. Polya states that "you decompose the whole into its

parts, and you recombine the parts into a more or less different whole”. (p. 76) Decomposing a problem and then recombining the details into a problem from a newly perceived perspective are “important operations of the mind”, says Polya. In the context of designing the AAG, they play a significant role in subjects’ creative experimentation processes.

Consider some examples of how subjects deploy their intelligibility framework (IF) in discerning intelligible procedures—a design plan—for mapping familiar concepts from their past onto the AGD process. For example, during Day 1 of the study AGJ, Nathan’s initial iteration of his game *ADHD Hero* included a “clinical” visual and procedural aesthetic. Nathan appeared to focus on implementing the game mechanics and rules first in order to scaffold the gameplay, a common design trajectory of most of the subjects authoring game autopathographies. During Day 2, Nathan added the music theme of the popular game *Guitar Hero*, diminishing the game’s clinical aesthetic.

(text continued on the next page...)

Figure 29: Nathan's game *ADHD Hero* as work-in-progress prior to addition of the music theme (capture-screen from *Unity* engine).



The common assumption is that music comes last in multimedia authorship processes such as those of filmmaking and computer game design, and this is for the most part true. Cole's design plan for co-designing his game *City Explorer* was to base his design decisions on his past experiences as an animator and musician. Cole, who collaborated with Yasmine on a joint game, is a musician, multimedia artist and animator with no prior game jam experience. He decided to compose some music first in order to inform his autopathographical game design process (see

Figure 30 below). He explained that music is a meaningful springboard for his ideas because of his comfort level and past experience with music:

I like to start with what I'm comfortable and then start exploring new areas. Part of my process personally, is making music. I like to make a song and then I just go off of that in a lot of ideas and thoughts that can be inside that song that maybe I'm not conscious of...

Figure 30: Cole composing music to inspire his autopathographical game design process



There are a few issues of importance in Cole's AGD process. First, in Cole and Yasmine's game, music becomes an essential cue for the player to discern difference in the game's virtual world. Thematically, their game explores difference in a fictional world of uniformity. Second, working with music composition prior to animating is common to animation praxis, hence a somewhat logical decision on Cole's part. On Culhane's view (1988), some animated films require recording the music first in order to inform the animation process: "in addition to recording the dialogue, in certain types of pictures it is necessary to record sections of the music track before the annotation of the bar sheets can begin" (Culhane, 1988, p. 92). Cole was simply adapting his experiences in animation and music to the AGD process. He was searching for familiarity from past experience, such as the order of composing music first in the animation process, and applying this pattern to the domain of computer game design. The decision to author music as a springboard for the game becomes a decisive factor in influencing the game's thematic concerns, where music becomes an essential character driving the gameplay.

In order to better understand AGD processes such as Cole's through heuristic theory, it is helpful to turn to Polya's problem-solving framework (I will call it 'Polya's Plan'), which can be mapped onto the intelligibility framework (IF) of AGD subjects. Polya (1945;1957) states that "the main achievement in the solution of a problem is to conceive the idea of a plan." (p. 8). He breaks up the problem-solving process into four steps as general rules for solving a problem: 1. Understanding the problem. 2. Devising a plan. 3. Carrying out the plan. 4. Looking back (checking and re-checking) (see Table 1 below). These four steps can be applied to problem-solving in any domain, including "mathematical and non-mathematical, theoretical or practical, a serious problem or a mere puzzle" (p. 2).

Subjects deployed a different problem-solving heuristic in AGD. Subjects harnessed heuristics as a form of improvisation in order to render first-person experiences into a computer game. Alex stated that planning can be restrictive for his creativity:

I find I'm stronger when I don't have a plan. Because I come up with new ideas on the fly.

It is important to emphasize that Alex was referring to less precise and more informal decision-making processes, namely heuristic processes, such as on-the-fly, trial and error, and analogical reasoning that work well with rapid prototyping at a game jam. Autopathographical game design within the context of the Autopathographical Game Jam is thus an improvisation environment conducive to "experimentation and thinking differently" guided by that "spark of magic" (Marco). This does not mean that AGD designers had no plan at all. Rather, it means that they employed heuristics as a form of planned improvisation, a form of creative experimentation that may vary depending on the domain in which it is practiced. In other words, you have to make some kind of plan, even if the plan is minimalist. Even in improvisation techniques such as acting, dance, or exploratory play, you start somewhere— usually from a theme, or a sparse set of movements, or you mobilize a set of objects (during play for example), or you devise a springboard to improvise from, and explore a particular direction.

Mapping Polya's guidelines (left) to autopathographical game design (AGD) (right), we can compare them in the following way:

Table 2: Comparison between Polya's Plan and Autopathographical Game Design's IF

Mathematics (Polya's Plan)	⇒	Autopathographical Game Design's Intelligibility Framework (IF)
1. Understanding the problem: What is the <i>unknown</i>? What is the <i>data</i>? What is the <i>condition</i>?	⇒	1. What is the autopathography (<i>the unknown</i>)? How to map the first-person experience as AAG? (<i>the unknown</i>) How do we convert select information of the first-person experience (<i>data</i>) into a) SAN and b) AAG? What are the <i>conditions</i> linking the first-person experience to the SAN and the AAG: what are the unique properties and limitations of game design during the AGJ (e.g., time-limit); human limitations (e.g., refusal to disclose information, etc.)?
2. Devising a plan: Have you seen the problem in a different form? Do you know an analogous or auxiliary problem (a related problem)?	⇒	2. How to 'model' select information of first-person experience via the IF as SAN and then AAG? Find analogies and disanalogies: e.g., games and illness both offer overcoming challenges (analogy) in different domains, e.g., we can fail in games without consequences; in illness, 'failure' can cost us our health or life (disanalogy). How to model past emotional experiences in a computer game? Ex: Abstract an emotion such as the experience of being bullied into a simulation of a 'push-pull' movement/cycle (being pulled and then pushed away, which creates a feeling of powerlessness and frustration in the player); abstract bipolar disorder to up-down (rollercoaster) movement in a game, also creating feelings of powerlessness. Iteration at the conceptual level (brainstorming ideas) and material level (recording ideas as sketches on paper) is possible.

3. Carrying out the plan: Carrying out your plan of the solution, check each step. Can you see clearly that the step is correct? Can you prove that it is correct?	⇒	3. Implementation of plan as the [system of game rules, narrative, characters, worlds, etc.] via digital design and programming. Software programming is based on deductive logic, i.e., Boolean true/false, forcing designer to ensure program functionality. Coding errors warn the programmer of illogical expressions, improper use of syntax and commands, undefined objects, etc.
4. Looking back (checking and re-checking) (p. xvi).	⇒	4. Playtesting the game prototype, debugging, refactoring the code to make sure it is modular (usable, shareable) and well organized. Iteration at the design implementation level: contingent on difficulty of idea implementation due to tight time-constraint of the study AGJ.

6.4.2 Sketching and drawing as a heuristic in AGD

Abstracting their first-person experiences into a computer game, subjects employ analogies that can be traced to their processual artifacts, i.e., drawings, sketches, flow charts, tree structures, and doodles. These preparatory documents aid their construction of the auto-analogical pathographical computer game. Heuristic processes therefore aid subjects in the successful implementation of their autopathographical narratives through autopathographical game design. They provide subjects with ways to re-conceive their ideas, that is, vary their ideas through the diverse processes of computer game design such as sketching their ideas, audiovisual design, modelling, programming, and so forth. Their objective is to solve the design problem of how to render their first-person experiences with mental illness, emotional trauma, and disability, and how to implement their autopathographical narratives into a computer game—a formal system of rules and constraints. Consider sketching and drawing as an intermediate step in which subjects practice heuristic reasoning. The objective is to quickly generate a variation of ideas through rapid sketches and solve the design problem of rendering their autopathographical narratives in a computer game.

Fred devised a plan (step 2) attempting to see the problem in a different form, analogously:

The only way I'm going to do it is through iterations of making a bunch of ideas really

quickly and figuring out (if) this could translate well to a game.

Fred invoked two different types of iteration, one of which is more immediate than the other; both are ways of varying the problem using a different medium or rendering process. The first type of iteration that is fairly immediate is brainstorming ideas at the conceptual level, which may mean recording ideas on paper as rough sketches and diagrams, which most subjects undertook during the study AGJ. The second type of iteration that is less immediate for subjects due to the game jam's tight time constraint is the implementation of the idea, which requires the use of graphic design software, 3D modelling software, and a programming language. Sketching takes less time than implementing those sketches through the structural properties of computer game design—generating digital art assets through digital design and authoring rules and formal constraints through programming—which are much more rigid formal processes and not as immediate as sketching.

Ellie saw *Twine*, a text-based game design software that is analogous to a personal diary, as being more innovative than digital forms of journaling such as blogging and vlogging:

It is like the next generation of journaling...in the digital age.

She considered *Twine* an excellent format for representing the flow of “stream-of-consciousness” thoughts that plague her during her insomnia. She explained:

Twine and insomnia go well together because insomnia's very much a thought-based experience...it's a really good way of representing (recurrent, obsessive, and intrusive thoughts), like what it feels to be trapped in a cycle.

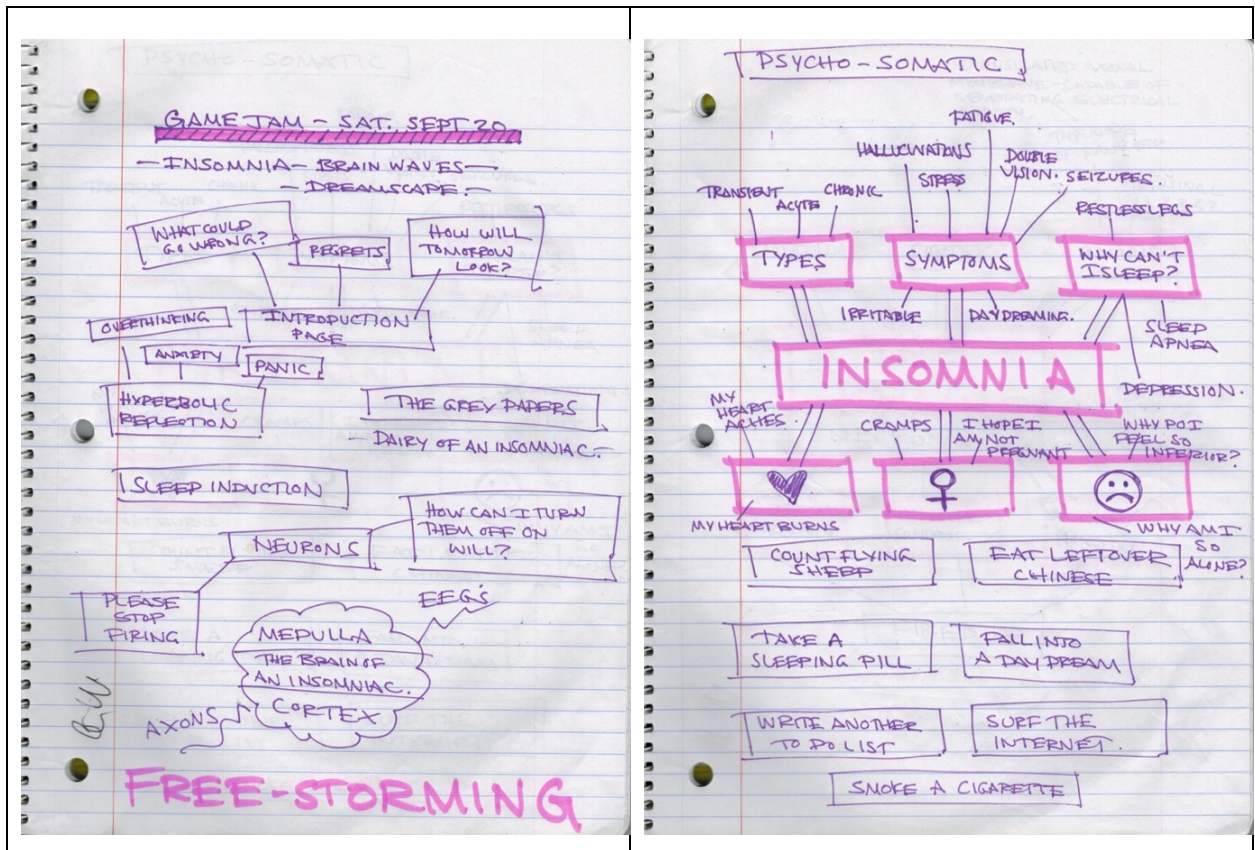
Twine has a branching-narrative interface (see Figure 32) which can easily represent recurrent thoughts by looping back to ‘passages’—a web page in *Twine*'s terms. Ellie's preparatory sketches, which exemplified her provisional narrative concept, illustrate her intermediate conceptualizations of her obsessive and repetitive thought processes accompanying insomnia late at night:

Part of the insomnia experience is a feeling of endlessness. *Twine* is a good way of expressing that because you can loop things really well. Repetitive thoughts are the other thing; when you're suffering from insomnia, it's almost like a form of OCD—the more

tired you are, the more you have these repetitive, negative thoughts. *Twine* is an excellent tool for (representing) this (process).

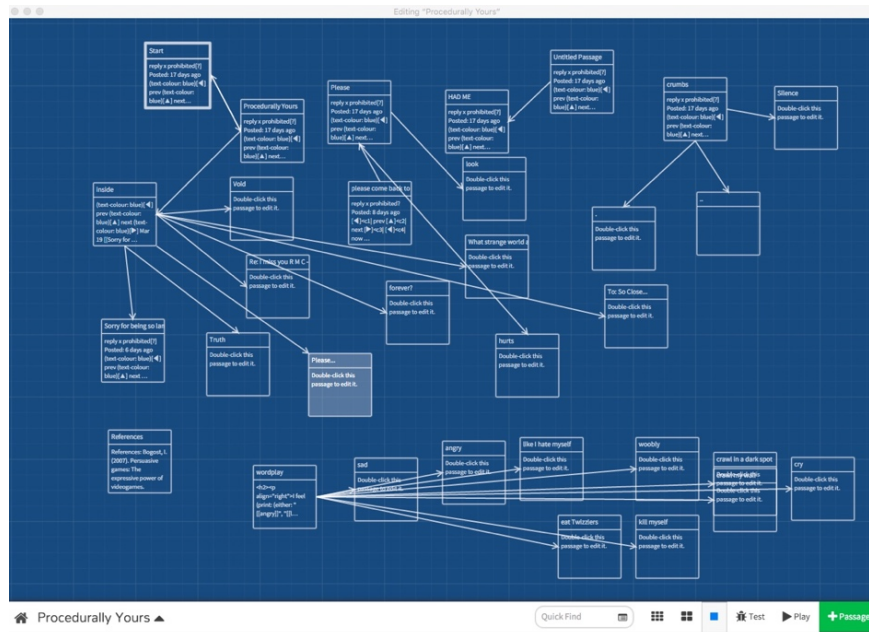
She illustrates her experiences of insomnia as a repetitive thought process in the form of a pseudo flow chart (see Figure 31, image on left) and a tree-like structure (see Figure 31, image on right).

Figure 31: Ellie's preparatory notebook sketches for her game autopathography: *This is Your Awesome Sleep Machine*



For Ellie, seeing an analogy between the *Twine* software system used for AGD and mind processes is a self-discovery, generating wonder and surprise, hence the positive emotions that were generated through autopathographical game design. Ellie later implemented her sketches using the *Twine* software.

Figure 32: Twine software interface



6.5 The *Fabulized Self*: analogical structures in AGD

Consider the fabulized self as exemplified in the analogical structures of the autopathographical games designed during the study AGJ. It is also important to reflect on the therapeutic function of the fabulized self as a new phenomenological experience of self that contributes to the author's self-reinvention, culminating in the implementation of the fabulized self through AGD's logopoetic properties⁸⁸. As a process of abstraction, AGD fabulopoiesis enables subjects to shift between the first-person perspective of 'being themselves' in autopoiesis to 're-imagining themselves' in fabulopoiesis, which helps them understand their lived experiences in unaccustomed ways. Sam affirmed this phenomenon by describing the beneficial effects of abstraction on his self-understanding, which invokes a stronger third-person viewpoint than other creative forms:

(Autopathographical game design) makes me view (my personal experience with anxiety and ADHD) through a simple lens...but if the experience is successfully conveyed it is a

⁸⁸ The implementation through programming praxis is discussed in the *Logopoiesis* chapter.

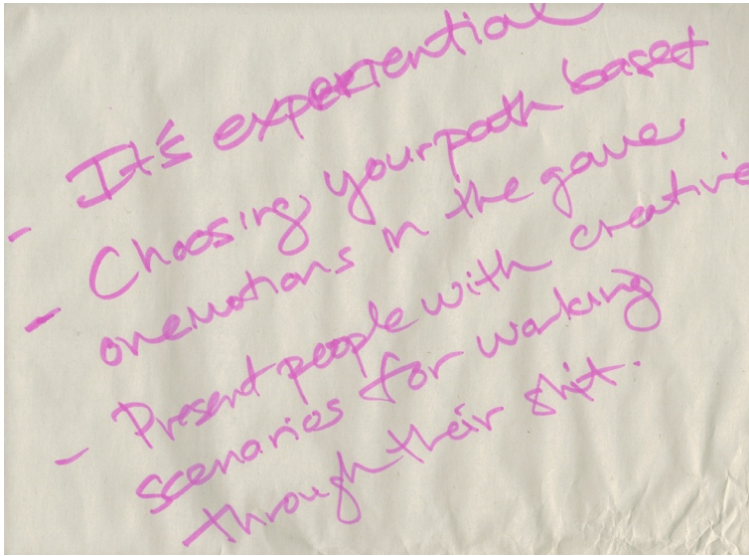
much more objective truth about that experience than say, if I were to write in a journal.

The fabulized self as an analogical model of the author's subjectivity becomes a *filter* in this regard, much like "analogies constitute filters through which the world is seen" (Hofstadter and Sander, 2013, p. 298). For example, authoring different perspectives for the player to embody via the design of the game player character (GPC) and non-player characters (NPCs) allows subjects to impart a new perspective on their first-person experiences.

As previously discussed, the abstraction process in AGD requires distilling select information into essences, i.e., information that is meaningful to subjects. In AGD, emotional memories that have accompanied a first-person experience with mental illness, emotional trauma, or disability are the meaningful information retained in the construction of a fabulized self. For example, Marco pointed out that his AGD objective is to "transmit feelings" to the game player. Hofstadter and Sander (2013) claim that strong emotional memories are tied to the "most central aspects of an event" that a person experiences (p. 171). "Firstly, thought is anchored in the past through analogies, and secondly it is anchored in the concrete world through the body, which has participated in so many experiences" (Hofstadter and Sander, 2013, p. 288). Hofstadter and Sander's statement implies that analogies are rooted in the feeling, sensing human subject who dwells in a material world. Emotional experiences thus play a key role in decision-making processes⁸⁹ and without a doubt are the primary basis of the game autopathography, as exemplified in the doodle statements: "it's experiential" and "choosing your path based on emotions in the game" (see Figure 33 below).

⁸⁹ According to some researchers, the *lack of* emotions hinders decision-making ability: "rational decisions may be impeded not by the presence but, rather, by the absence of emotions" (Pauen, 2006, p. 174).

Figure 33: Autopathographical Game Jam doodle



According to Thagard and Shelley (2001) emotional analogies entail the “transfer of emotions from a source to a target” (p. 344). Emotional analogies harness empathy by enabling people to “understand the emotions of others by imagining their own emotional reactions in similar situations” (p. 335). Consider John, who framed autopathographical game design as a way to coax the player to inhabit facets of his bipolar mind:

(To convert my experience with bipolar disorder via computer game design) is brainstorming and thinking of ways to get the (player) to think the way you’re thinking through game rules and game mechanics... You can’t put them in your shoes but you want to try and lure them into a *similar* situation psychologically.

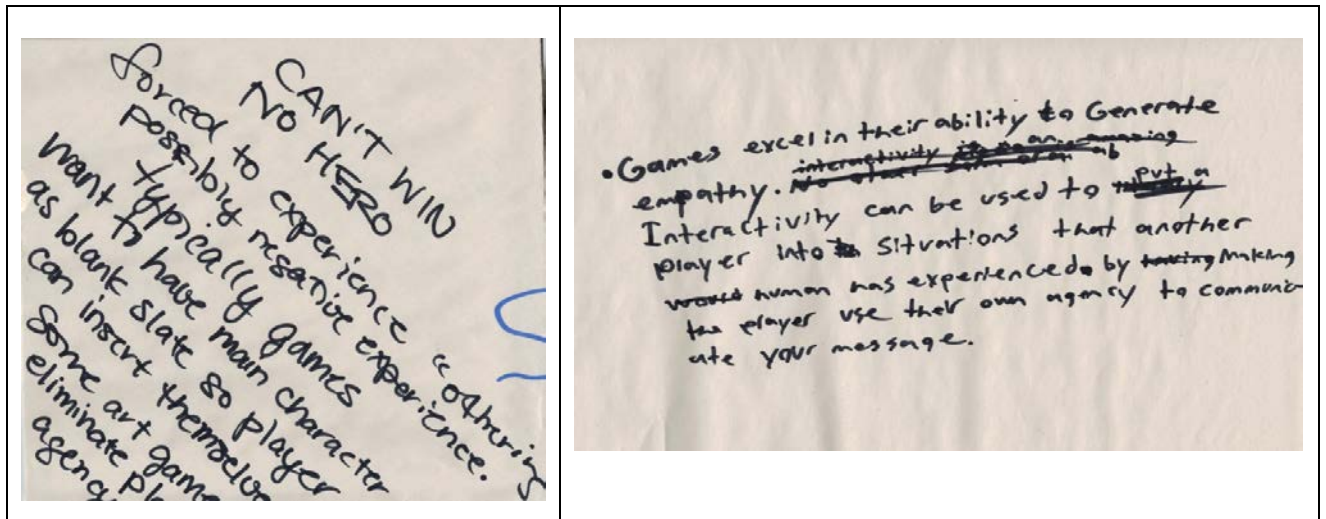
John was hoping to establish an empathetic connection with the player by:

Convey(ing) some of the emotions and trying to make a point about how (bipolar disorder and debilitating mental illness) *feels* for (sufferers).

That is, when John states that his intention is to convey how bipolar disorder feels for him, he is referring to ‘what it is like’ to embody a distinct facet of bipolar disorder—the fluctuation of mania and depression. Harnessing the game rules to that end means engaging with the unique formal properties of games to translate his illness experience, aesthetically, to be ‘felt’ by others through gameplay. This experience may not necessarily be a positive one: as one subject writes in a doodle, a player will be “forced to experience ‘othering’ (through a) possibly negative

experience” (see Figure 34 below, image on left). Analogies are thus mechanisms for building compassion and empathy (Hofstadter and Sander, 2013) that are instantiated as fabulized selves in autopathographical games; as one doodler states, “games excel in their ability to generate empathy” (see Figure 34 below, image on right). Emotional analogies also give rise to subjects’ self-empathy in that they gain understanding of their own emotions by imagining their own emotional reactions through the game’s events or actions.

Figure 34: Autopathographical Game Jam doodles



Consider one type of similarity between games and illness that subjects harness in their authorship of the auto-analogical pathographical computer game (AAG). The illness experience moves sufferers to search for solutions to their illness, in order to cease the pain and discomfort, an intrinsic need of all human beings⁹⁰. Hence, a game’s classic definition of overcoming a challenge such as war enemies in a first-person shooter game invokes a resemblance to overcoming the pain, suffering, and discomfort of illness in the autopathographical narrative, given that, as previously discussed, illness is commonly analogized to a conflict or struggle to be overcome or resolved (Frank, 1995), an adventure story (Hawkins, 1999), and battle or war (Sontag, 1978). The premise—the relational predicate in Gentner’s terms that unifies the

⁹⁰ An interesting side note to this assertion: Bob Flanagan was an American performance artist who used his cystic fibrosis as an integral part of his art performances as a way to conceptualize the analogical similarities between suffering and sexual masochism, illness and sexual practices such as BDSM.

autopathographical narrative and computer game—is the *dialectic of overcoming*. In the design of the AAG, the mapping between game-as-challenge and illness-as-challenge is not just perceived at the conceptual level; it is also implemented at the structural-systemic level of a computer game for players to dynamically engage in a causal structure—a system. Thus, when implemented in computer games, metaphors and analogies are externalized as material systems with which players interact on a material level.

6.5.1 Nathan's game *ADHD Hero*

In his game *ADHD Hero*, Nathan's fabulized self is a medical diagnostic test that doubles, or better yet, masquerades as a musical instrument; now consider the inspiration behind it. Nathan frames his experiences growing up as a child with ADHD as a computer game to be 'won' or 'lost', analogous to a system to be 'gamed', played or overcome, such as the medical system, which diagnosed him. His game is a good example of a poetic act of self-determination, a form of poetic justice. Nathan explained that his computer game trains the player to improve their skills at beating a medical performance test that diagnoses ADHD:

Going through school as a gifted (kid), I did feel like I was gaming the system all the time...If I could find a trick, it wasn't cheating, it was just doing the math in my head instead of on paper... You can train yourself to perform really well on IQ tests. Which means, all the test is measuring is how good you are at taking IQ tests. It's not measuring your IQ and I like doing that.

Nathan's analogy involved a structural mapping between multiple sources and targets that also harnessed metonymy. The first mapping was between his first-person experiences (source 1) and the ADHD medical diagnostic test (target 1), a mapping that is effectively a metonym.

According to Burke (1941), a metonym is a substitute for something that reduces complexity. In Nathan's case, the diagnostic test was a reduction and a material substitute for the complexity and intangibility of his childhood experiences. Nathan stated that his game reflects a "subset of his experience", alluding to an unpleasant part of his childhood when he was forced by parents and doctors to take the ADHD performance test and other similar medical tests. These tests defined his childhood experiences with ADHD:

I did a test like this once, it was the most aggravating thing. I feel like my childhood was represented by being subjected to tests like this.

The second mapping was between the ADHD medical diagnostic test (source 2) and musical instrument (target 2) to be played in order to explore “quote, unquote, failure”. Nathan’s objective was to demonstrate that individuals with ADHD often feel that they cannot adhere to social expectations of normal human behavior. The multiple analogy was set up to ‘resonate’ in the interval between the ability to ‘play’ well both a diagnostic test and a musical instrument; that is, doing well or failing the diagnostic test and getting a musical instrument’s keys ‘right’ (i.e., tonal, consonant) or ‘wrong’ (i.e., atonal, dissonant).

Nathan explained the consequences of failing a question that tests impulsivity and attention span in his game, which corresponds to hitting the wrong keys on the musical instrument: “you get the piano jarring sound if you get it wrong”. Reducing a subset of his first-person experiences—his feelings of “aggravation” from being subjected to testing as a child—to a diagnostic test, which is then analogized to a musical instrument, dilutes the connotation of pathologization inherent in medical diagnosis. Nathan confirmed the playful intervention of his game autopathography:

I really wanted to examine those diagnostic tools and kind of (ask), are they games? And are they gameable? Which they are. Which, I think devalues them a lot as a diagnostic tool.

Nathan reduced the diagnostic test to play and fun. He demoted the seriousness of medical diagnosis to learning how to play a musical instrument, something that can be overcome with training and creative inspiration.

6.5.2 John’s game *Untitled*

In John’s dark and realist game on his experiences with bipolar disorder *Untitled*, the *overcoming* resemblance between the conceptual categories of computer games and severe mental illness was turned on its head. John’s game is unwinnable and there is nothing in the game to be overcome. John’s game autopathography, similar to some autopathographical narratives, is an example that defies formulaic genre conventions and challenges what Couser (1997) calls “the tyranny of the comic plot, the convention of an affirmative ending” (p. 75). John hoped to show the frustration of bipolar disorder which cannot be won like a game or overcome, an analogy that is far too simplistic in his mind:

There is no pause button, there is no way to exit the game. The only way to exit is to kill yourself and die.

Couser (1997) claims that “a comic plot may seem entirely inconsistent with the lives of those whose master status is permanent disability” (p. 183), which applies to John’s severe form of bipolar disorder. Furthermore, Couser suggests that a narrative plot, which by definition is marked by evolution and growth, may also seem incompatible with permanent disability’s unchangeable ontological status and hence seem “unnarratable” (p. 183). This explains why John may have wanted to intuitively subvert the comic plot that defines much of the autopathographical narrative.

Bipolar disorder has no silver lining in lived experience, despite its romantic metaphoric connotations in both literature and first-person memoir, as per Sontag’s (1978) claims that mental illness has, at times, been a commonly romanticized metaphor in literature. Psychiatrist and bipolar disorder sufferer Kay Redfield Jamison’s (1996) book *Touched with Fire* draws a productive link between bipolar disorder and creative genius, thus romanticizing the illness. Jamison’s argumentative move analogizes severe mental illness to a form of creative transcendence—a shared property that is culturally assigned to experiences of both mental illness and spiritual awakening. Sontag (1978) writes: “now it is insanity that is thought to bring consciousness to a state of paroxysmic enlightenment” (p. 36). John defied these common romantic and transcendental tropes assigned to severe mental illness by grounding bipolar disorder in the materialist/physicalist realm of physical disease:

It’s suffering as much as somebody with cancer is suffering.

He was not only refuting the overcoming analogy of games and illness but, as Sontag prefers it, he was also decoupling illness from metaphor. He explained:

In the end you’re always going through the same loop (up and down of mania and depression) and in the game you can turn it off, in life you can’t. The only way to turn it off is, in life, to kill yourself, and for a lot of people that isn’t an option. It’s a wake up call to people.

For John, his powerful message to players was to ground bipolar disorder in physical suffering:

(The game is) more about educating people who don’t have (bipolar disorder).

John is referring to educating people who are under the false impression that bipolar disorder can be overcome either through medication, positive thinking, spiritual experience, or even poiesis itself⁹¹. Jurecic (2012) quotes historian Anne Harrington and medical anthropologist Arthur Kleinman, writer of *The Illness Narratives: Suffering, Healing and the Human Condition*, who both argue that “the physicalist approach to illness falls short” for patients who yearn to entertain more existential questions and who are told that their illness has no meaning (as cited on p. 9). In the case of John’s game, which communicated a hopeless vision of bipolar disorder, John subverted the comic plot common to some autopathographical narratives—an act of subversion that was meaningful to him because it communicated a social message. He confirmed that he “wanted to send a message” (to the study AGJ audience, presumably).

Subjects sometimes threaded the most recognizable or iconic elements of their fabulized selves, elements which are complex when experienced subjectively, yet drastically abstracted through the rules of the game. John based his AAG on his emotional experience in order to communicate the most recognized and prototypical elements of bipolar disorder, i.e., its most iconic and abstract elements, rendering the bipolarity of the illness as an analogy based on an up-down fluctuation. An icon is primarily constructed through analogy or resemblance (a likeness), devoid of causal link to an object: “I call a sign which stands for something merely because it resembles it, an icon”, states Charles S. Peirce (as cited in Pharies, 1985, p. 34). An icon⁹² can be a figurative drawing or painting such as a portrait of a person (real or fictive), or an extreme pictorial abstraction and simplification of a complex object. In an icon, information is reduced to the shared formal and structural properties between the icon and the object it represents. For example, a ‘stick figure’ is an icon of a complex human being—a system of meat, bones, physiognomy, anatomy, self-awareness, and messy emotions, stripped away.

⁹¹ Despite his dark and realist vision, the autopathographical game design process does provide some relief for John, to be addressed in the chapter on *Logopoiesis*.

⁹² From a cognitive standpoint, icons are generally more accessible due to their simplicity of design. That is, they can be recognized faster and more easily than non-iconic objects (Pullin, 2011).

The game's intertitle "Chase happiness. Pockets of mania are a mixed blessing. No escape from depression" laid out the exposition of John's game (see Figure 35 below).

Figure 35: Capture-screen grab of John's computer game, *Untitled*



The game's challenge consists in effusively trying to catch the glowing ball of happiness:

It's just supposed to be chasing that happiness and never quite catching it...balancing the mania and the depression.

Consider John's intelligibility framework (IF)—some potential reasons that may have influenced his analogical mapping process in rendering his fabulized self as an unwinnable game. John's first analogy was a glowing ball, which in and of itself does not correspond to happiness, as it is merely a visual abstraction. But, *chasing* the glowing ball in the context of the game title above, does correspond to happiness—hence the common analogy of 'chasing' happiness in the form of a bright, glowing object. Happiness is often times analogized to a bright and warm colour palette, while melancholia and depression are analogized to a darker palette of blue and black (especially in Western culture). Given that in the popular imagination, when people are happy, they go out in the sun, and when they are sad, they burrow at home and isolate themselves, so goes the conventional bipolar caricature of happiness and sadness. The game's outcome is the second analogy based on a conditional expression that renders bipolar disorder as a fluctuation, symbolic

of the illness' iconic attributes —the mania ('up') and depression ('down'). John explained the rules of the game:

If you go too far down, your character kills himself, if you too far up, you can't control yourself and you die anyways.

The up-down movement in the game was looped to repeat. The up-down movement as an analogy of bipolar disorder mentally shores up another analogy, that of a rollercoaster, commonly attributed to mental disorders. The up-down movement exaggerates the qualities of the illness and abstracts it to its prototypical elemental properties. Although reductionist of the multifaceted qualitative experience of bipolar disorder, the up-down movement implemented in the game rules is a simplified structure that abstracts complexity into an elegant form, much like abstract art distills detailed information to shapes, form, block colours, and composition. The intention of John's use of up-down movement, threaded through the rules of the game, is to negatively affect the player and, like Nathan's game on the ADHD performance test, offer a demotivational message. That is, the player will feel the consequences of not being able to stabilize the fluctuation of going up and going down, no matter how hard they try. The game was thus programmed for a lose-lose outcome and thus no possibility of winning or overcoming, even though the player is not privy to this information. This form of player manipulation, as discussed in the *Foundations* chapter, constitutes transgressive forms of game design—"abusive game design" (Sicart, 2015) and designing games that are not 'fun'. These transgressive game design conventions, so called by contemporary game studies scholars Linderoth and Mortensen (2015), rely on tactics such as deception and manipulation to frustrate players by courting losing (failure) more than winning (success); the intention is to engender a higher feeling of reward in playing the game (Juul, 2013).

6.5.3 Alex's game *Brain Pains*

Consider next an examination of Alex's analogical process for his computer game autopathography *Brain Pains*—that is, his intelligibility framework (IF), which allowed him to construct his fabulized self into an auto-analogical pathographical computer game (AAG). Alex, like John, also suffers from bipolar disorder, depression, and anxiety. Alex's original idea for a fabulized self in his game were the analogies of a "spider web" and a "puzzle", symbolizing the hidden mechanism of his bipolar brain and how he deals with "repressed issues". His first idea

was to design an algorithm for procedurally-generated content (PCG)⁹³ that would integrate ‘triggering’ emotionally positive and negative memories that are ‘randomly generated’ by the game. Authoring and implementing the algorithm in the software code and game engine would require more time and deliberation, therefore he abandoned this idea due to the time limit of the Autopathographical Game Jam:

The original idea was some sort of spider web, but I wanted it to be with narratives in there. It was going to be more like a puzzle... You had to click on them and it’ll pop up a memory and it might be good or bad and just kind of randomly generate that kind of stuff, and you had to choose whether to get rid of it or keep it. I completely scrapped that because I don’t think there’s enough time for me to do it properly and I couldn’t think about a good way to write⁹⁴ it.

Alex eschewed his initial decision to represent his repressed memories in textual narrative form, such as titles or textual descriptions popping up in the game. Alex described his game—which instantiates his fabulized self as personified in interactive mode:

The gameplay is just moving spheres around inside of a thing with some ropes attached to them. The objective is to attempt to untangle as many (nodes) as you can without breaking too many. It’s an abstract representation of what I’m doing with repressed issues.

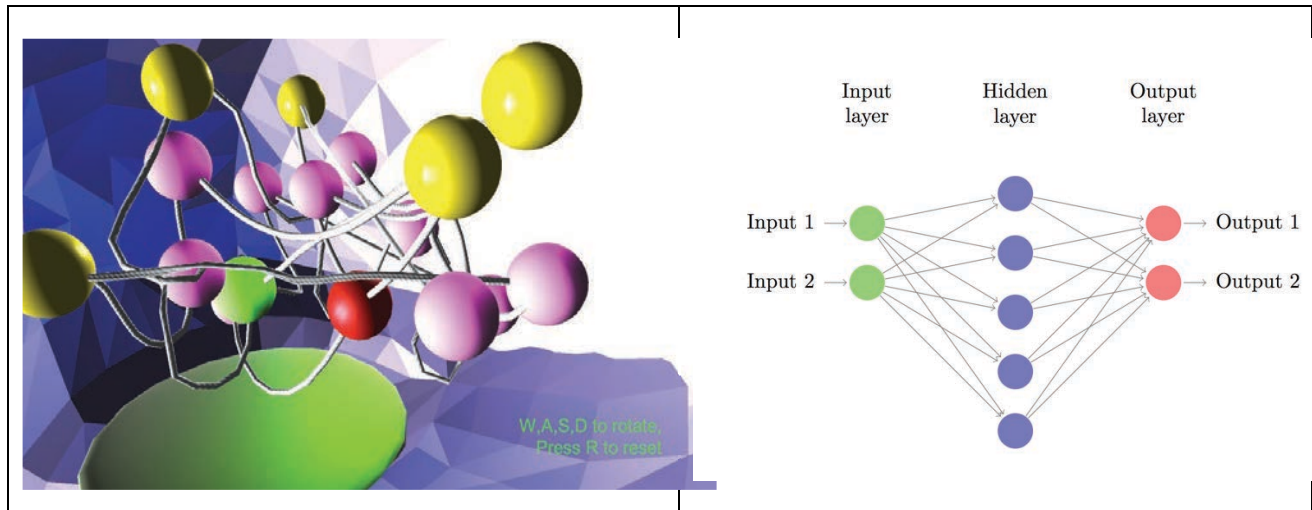
Instead, Alex’s description above implies that he chose the metaphor of a neural network⁹⁵ (see Figure 36 below) which is based on an analogy—a computational model analogous to the human nervous system.

⁹³ “Procedural content generation (PCG) refers to the creation of game content automatically, through algorithmic means. In this paper, game content means all aspects of a game that affect gameplay but are not non-player character (NPC) behaviour or the game engine itself. This definition includes such aspects as terrain, maps, levels, stories, dialogue, quests, characters, rulesets, camera viewpoint, dynamics and weapons” (Togelius et al., 2011).

⁹⁴ When Alex states he couldn’t think of a good way to “write” the game, he is referring to “writing” the game qua software program—programming or scripting a program as opposed to writing text.

⁹⁵ There is no actual mention of a neural network per se in Alex’s interview testimony, however, his description implies something like it.

Figure 36: Still from Alex's game *Brain Pains* (left) and a simple visualization of a neural network (right)



Artificial neural networks or ‘connectionist models’ are defined as “information processing systems, whose structure and operation principles are inspired by the nervous system and the brain of animals and humans” (Kruse, Borgelt, Braune, Mostaghim, & Steinbrecher, 2016, p. 9). If we start from Alex’s original two analogies, a spider web and a puzzle of his brain, and include his decision to modify the analogy to what looks like a neural network, then we have several sources (i.e., Alex’s brain, spider web, puzzle) informing the target (neural network). Invoking again the definition of a multiple analogy, which is “a comparison in which several sources are likened to a target” (Shelley, 2003), we have the following: first analogy: [Alex’s brain $\Rightarrow$ spider web], second analogy: [Alex’s brain $\Rightarrow$ puzzle], third analogy: [Alex’s brain $\Rightarrow$ neural network]. The problem in this case is that there was no structural mapping deployed in Alex’s actual game that would ensure a strong analogy between the different conceptual categories—Alex’s brain, the spider web, the puzzle, and the neural network. There was only an *inspired* surface mapping between the conceptual category of ‘Alex’s brain’ expanding to the metaphors of repressed minds-as-spider webs, confused-minds-as-puzzles, and brains-as-neural networks. As a surface mapping, effectively a metaphor, only select visual properties that accompany this expanded conceptual category are retained.

In other words, despite being a system that can be played, Alex’s game did not function as a *de facto* artificial neural network, even though it visually resembled one. Neural networks are a weak form of artificial intelligence that have a degree of agency in and of themselves to

make a range of autonomous decisions (e.g., solve problems, recognize patterns, learning tasks, etc.). Kruse et al. (2016) describe the function of neural networks and how they work:

They consist of a large number of fairly simple units, the so-called neurons, which are working in parallel. These neurons communicate by sending information in the form of activation signals, along directed connections, to each other (p. 9).

The functions of neural networks qua autonomous agents invoked in the quote above were not incorporated into Alex's rendering; Alex's nodes did not communicate with each other autonomously and did not make autonomous decisions as nodes do in neural networks. It is also important to add that Alex's game was neither a computational model nor a computer game with autonomous agents integrated into the game system, such as NPCs—non-player characters—who are controlled by the software program via artificial intelligence. Alex's game was a 3D visual model, a metaphor rendered at the surface visual level as opposed to the underlying structural-systemic level of analogical mapping. Thus, the mapping between source (Alex's brain) and target (the computational model of a neural network) at the structural-systemic level did not apply to Alex's game.

The primary information that Alex retained is at the visual level of an image of a neural network—a surface mapping in which he retains some of the properties of a representation of a neural network—its nodes, “ropes” (connections) and colours (see Figure 36, image on left). Alex's game used the colours yellow, hot pink, green and red for the nodes; he used one more colour than a typical 2D visualization of neural networks (see figure above image on right)⁹⁶. The red nodes represented Alex's repressed memories that were dangerous to click on, causing the network of nodes to collapse, which signifies an emotional trigger. He explained the symbolism of the colour codes in his game:

When you click on the pink nodes, they drop, the yellow ones are static because they're like cemented memories, it's fragile. So, you can move these around and kind of shift the whole weight of the thing and that red one is like a traumatic memory. So, if you try to affect that one it blows the whole thing up and you lose all of them.

⁹⁶ Alex did not consult this image of a neural network; the researcher is using the image to give an example of a neural network.

Alex analogized the colour red to an emotional trigger, tacitly drawing on the semantic properties of the conceptual category of ‘danger’. He said:

I want some of (the nodes) to be dangerous to untangle because you’ve uncovered some sort of trauma and it’ll explode and knock out some things beside it.

The conceptual categories of a red light, red alert, a stop sign, etc., are semantically defined by the colour red, which universally or near-universally signifies heat, fire, warning, and danger. However, Alex did not disclose the information about what the nodes reference in first-person, subjective terms in the game. Therefore, Alex’s game mapped a *visualization*—an image of a neural network—onto a 3D model embedded in a game engine.

Alex’s game nonetheless still had some structural similarities to neural networks qua systems. Analogically, at the structural-systemic level, Alex retained some similarities whereby the player (human agent) manipulates one node, which systemically affects another node in the network. If neural networks work autonomously, then the manipulation of nodes (taking meaningful actions) is at the level of the neural network as an autonomous agent, not necessarily a human agent/user. He also retained similarities between his game and a neural network at the emergent level of dynamic interaction, *sans* the added probable combinations that may emerge out of neurons communicating with each other, a layer of rich complexity in neural networks that Alex did not include. *Weak emergence* is a property that applies to all analog and computer games, software programs and AI qua systems, in which a multitude of probable combinations can emerge out of a simple set of authored rules in the system: “sometimes complexity can arise quite suddenly and unexpectedly from a handful of simple elements”, state Salen and Zimmerman (2004). Given that Alex’s game appeared to have a few simple rules, they may give rise to rich emergent behaviors that are meaningful for the player, behaviors which are not as complex as a neural network. In Alex’s game, the player can uphold differently combined colored nodes, without allowing them to collapse (the success outcome of the game); the player should avoid entangling the nodes whereby they collapse (the fail outcome of the game). Perhaps Alex’s model can serve as the first iteration of a future computational model that can include more narrative components and more autonomous properties derived from his first-person experiences with bipolar disorder.

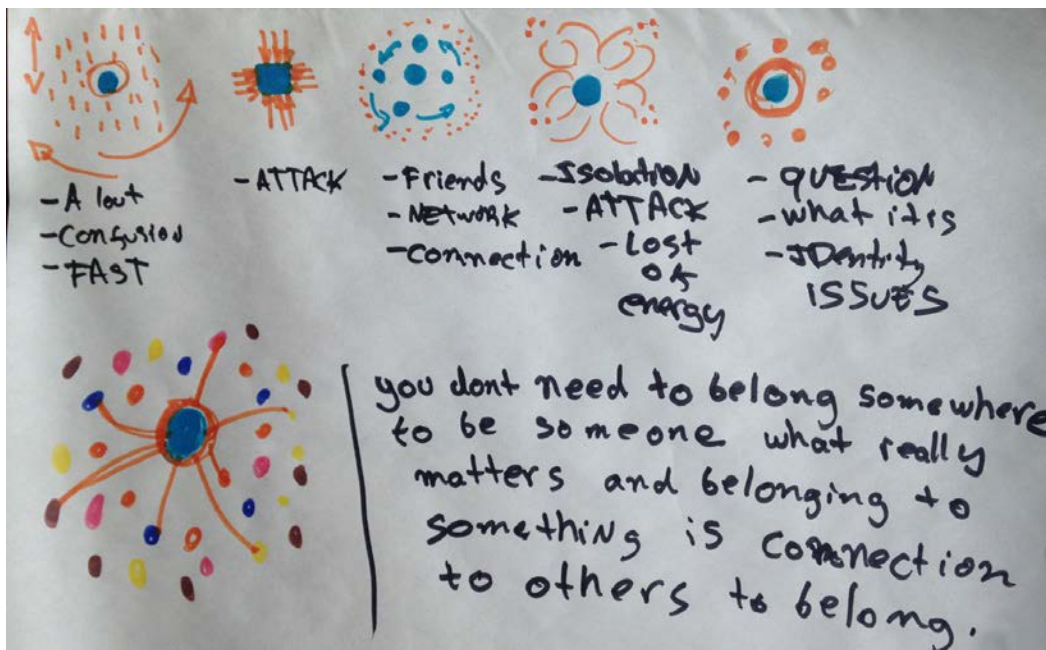
6.5.4 Marco's game *Connections*

Marco's game *Connections* was comprised of a universe of orbiting planets that the player embodies in gameplay—his fabulized self. Marco elaborated on the events that inspired its construction:

I was bullied by people that I never thought I would be bullied by. There (were) even times in my life (when) I was a strong friend with this person and then she suddenly betrayed me and became a bully...sometimes people that were my bullies became my friends.

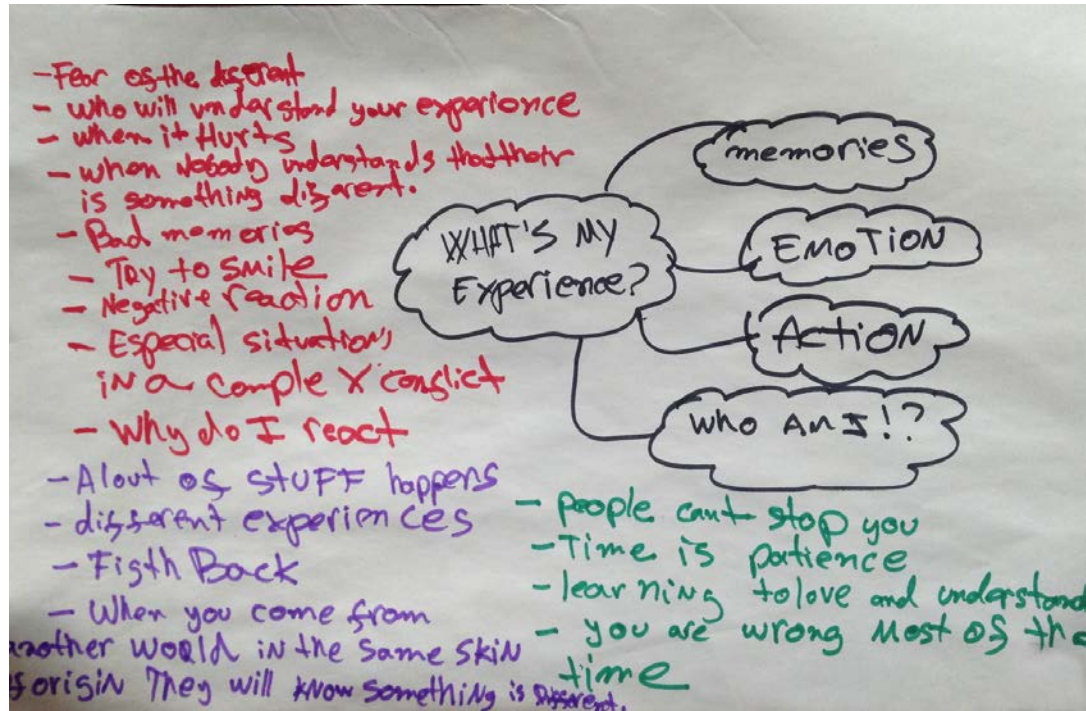
Consider Marco's intelligibility framework (IF), which allowed him to construct his fabulized self—the aforementioned universe of orbiting planets—into an auto-analogical pathographical computer game (AAG). Marco abstracted his autopathographical narrative of being bullied into one simplified conceptual representation—bullies who become friends and friends who become bullies—as the foundation of his exploratory game. Marco then converted this initial abstraction into an experiential, physics-based game, one which uses collision physics to poetically render or personify emotions as gravitational forces (see Figure 37 below).

Figure 37: Marco's preparatory drawings for his game *Connections*



Through these processes of abstraction, Marco omitted the descriptive details of his childhood experience with bullying. Marco noted some of those details in stream-of-consciousness brainstorming style point form notes in his preparatory sketches (see Figure 38 below).

Figure 38: Marco's preparatory drawings for his game *Connections*



From the perspective of visual graphics, Marco's game offered simple lines, shapes, and colours in the form of "universes", "orbits" and planets":

I'm going to make orbits like planets in universes but with colour and with different physical reaction with contact.

Marco stated his attraction to shape and colour:

I love shapes and colours. Since I was a kid, I've had this amazing attraction for shapes and colours.

Marco analogized his experiences of friends becoming bullies and bullies becoming friends into movement and gravity, which he represented through the collision of moving planets in the 3D game. For Marco, the lived, concrete dimensions of the first-person perspective are better analogized to fluid, continuous movement, which is why he:

Wanted the game to feel lived, like you're living in it.

He qualified the emotional dimensions of his first-person experiences into movement:

I'm trying to expose my feelings and ideas through movement.

Marco elaborated on why movement is meaningful to him:

When I'm playing (a game) I can feel how much the programmer, the artist wanted me to move, to feel how I'm moving in a game. Some movements feel like they have more friction, some feel less, some feel a bit faster. When I play a game for me that's very important.

His intention was to design a gameplay experience that does not have a goal-oriented, reward-centric outcome:

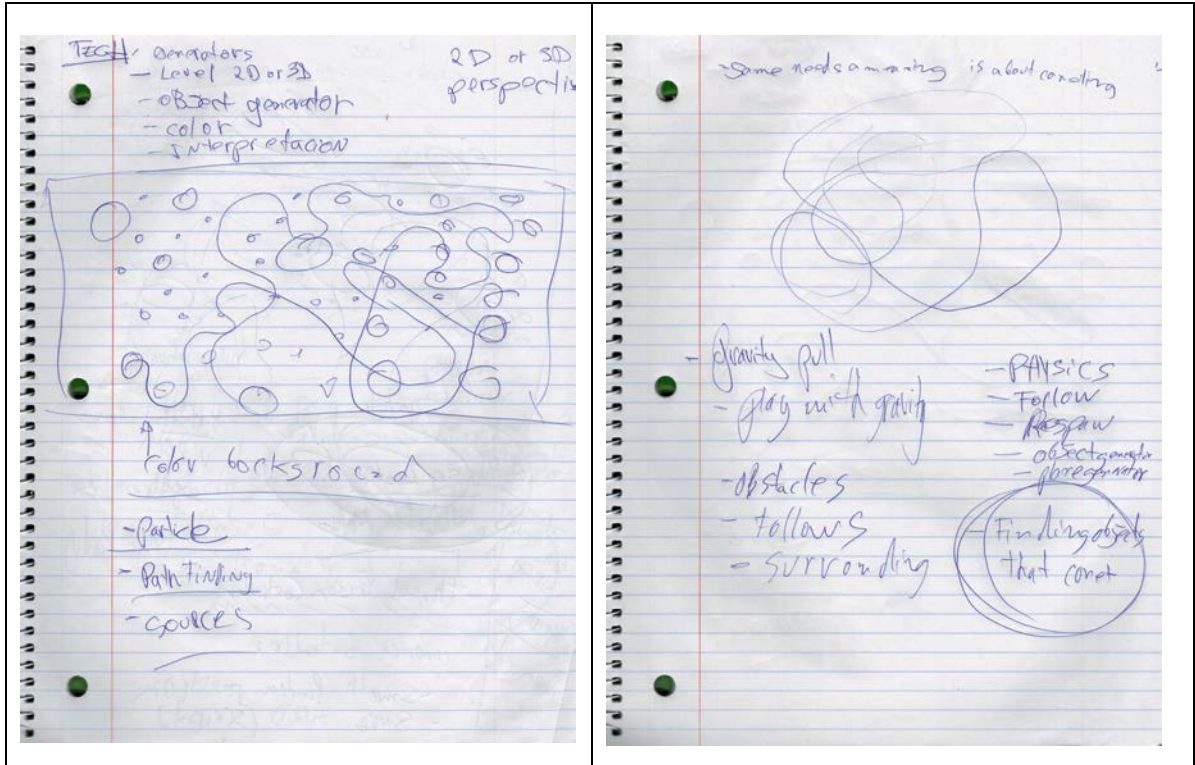
The objective is not to get from point A to point B and you win something or you lose something.

Instead he was more concerned with authoring a gameplay experience that conveys mood and atmosphere:

It is all about the experience that you had inside the game. It's more what the person felt inside of it than (winning points).

Modelling, abstraction, and minimalism in autopathographical game design allowed Marco to abstract his first-person experiences into a simulation of a physics model. Thus, movement in Marco's game abstracts his first-person experiences into a collision detection system through the coordinates of geometrical objects.

Figure 39: Marco's sketch of his game *Connections* illustrating the use of movement in the game



Marco also implied that he views an “emotional experience”—a subjective experience—as a better source for abstractions than figurative (metaphorical, abstract) sources. This brings to mind the common clichéd advice that a creative writing teacher gives to novices: “write what you know”⁹⁷, perhaps because writing from first-person experience carries emotional resonance for the author—it is emotionally meaningful. To say it differently, abstractions as an end product, in the context of Marco’s work, are better informed by first-person experiences than by other abstractions. The range of selection in first-person experiences is higher than in an abstraction:

What I’m trying to express of myself and portray in this game is going to be very abstract because it involves more of an emotional experience than a figurative experience.

⁹⁷ Heller, Z. & Hamid, M. (2014, March 25). ‘Write what you know’ — Helpful advice or idle cliché? *The New York Times*. Retrieved from <https://www.nytimes.com/2014/03/30/books/review/write-what-you-know-helpful-advice-or-idle-cliche.html>

Thus, first-person experience, due to its direct emotional effects on the author, is a driving mechanism behind Marco's abstraction process. For example, Marco said that human connections are most important to him. He saw them best represented in abstract form, due to their essential structural properties:

What really matters to me is making meaningful connections wherever I go. Because I never had this perfect linear (experience) with other people or with groups, so that's how I feel. That's why I wanted the game to be abstract. Because you can't be concrete. Even if you get water that is the most concrete thing in the universe, or light, it's still abstract because it has to react with so many things and that's how I feel. Like I have to react to so many things in so many different ways that everything becomes abstract, everything is a method of concept.

Marco suggested that he cannot render the concreteness—the indeterminacy, complexity, and specificity— of his first-person experience into a computer game because of the multiplicity of ways in which concrete details instantiate themselves in lived, concrete relations. For Marco, the essence of first-person experience dwells in causal relations, which can be better generalized to “wherever I go”. For Marco, movement is mobility, and mobility is freedom. In his testimony, he used water and light to exemplify objects that are both concrete in their materiality, but also abstract in how they react with other objects, invoking their relational properties. The most economical way to get to the form and structure of relationships—the causal relations between objects—is to make an abstraction by hiding information or stripping away the concrete details and ‘facts’ of Marco's autopathographical narrative. Therefore, for Marco, the ‘concrete’ facts of his first-person experiences are not only too revealing and uncomfortable, but they also obfuscate the essence of his fabulized self, which is better rendered as an abstraction—the universal or third-person perspective. In his testimony of the above process, Marco effectively points to his own specific intelligibility framework (IF), which allowed him to make very specific design decisions in order to effectively render his fabulized self into a game.

For Marco, to poeticize is to abstract or strip away descriptive details of his first-person experiences and render his fabulized self into a physics-based game—a third-person account. Marco's design decision allows for the possibility of authoring for players who can insert themselves and inhabit the game world as themselves as opposed to strictly delineated and detailed characters; and in turn, impose their own story upon the game. This minimalist design

strategy, which universalizes human experience, is also easier to pursue given the study AGJ's tight time constraint.

As can be seen from these examples of analogical structures in AGD, subjects render the fabulized self through levels of abstraction and the use of analogy and metaphor as imaginary filters, which allows them to view themselves in a new light and generate a self-reinvention. In Brody's game, the fabulized self is a keenly analytical detective; in Nathan's game, the fabulized self is personified as a method of 'gaming' and beating an ADHD medical test, which has stigmatized him in his childhood; in John's game, the fabulized self is a dark and existential messenger enlightening the unaware of the complexities of a severe disease; in Alex's game, the fabulized self is a brain network whose objective is to 'disentangle' itself; in Marco's game, the collision physics in the game personify the unthinkable developments in his life—bullies becoming friends and friends becoming bullies. Self-fabulization is the act of imagining a fictional self that despite its firm grounding in fantasy influences the self-reality of the AGD designer.

Chapter 7

Logopoiesis: Autopathographical Game Design as Self-Understanding and Self-Therapy

7.1 Introduction

The Greek word *logos* (λόγος) translates to “word”, “reason”, “plan” (Encyclopedia Britannica)⁹⁸, “computation”, and “reckoning” (Perseus Digital Library)⁹⁹. In autopathographical game design (AGD), study subjects rendered their fabulized autopathographical narratives using computer processes. When *logos* (e.g. reason, computation, reckoning, plan) integrates with *poiesis* (i.e. creative making), the result is *logopoiesis* as subjects harness computer game design’s logico-analytical properties to engineer a computer game autopathography, and articulate, in exact terms, the autopathographical narrative. As a digital, computational, and procedural form of poiesis, logopoiesis is a logical, analytical, and calculated *making*, an act requiring mental deliberation and careful assessment creating order. Subjects strongly expressed the perception that computer game design’s logical, analytical, and deliberate nature feels therapeutic and fosters an objective form of self-understanding in the context of AGD. This chapter theorizes on the digital, computational, and procedural resources of AGD as a form of self-understanding and self-therapy based on their ability to engender the third-person perspective in the autopathographical game designer. The third-person perspective yields the standpoint of the objective and mindful observer from which the game designer deploys logopoiesis.

It is true that autopathographical writing and autopathographical game design both involve externalizing the author’s ideas in some structured and ordered form via the logos of narrative-telling. But there is a crucial difference between AGD and autopathographical writing. The difference lies in the use of computer processes in autopathographical game authorship towards engineering a gameplay experience and mathematizing the autopathographical narrative

⁹⁸ Encyclopedia Britannica: retrieved August 2, 2017 from <https://www.britannica.com/topic/logos>

⁹⁹ Perseus Digital Library, Tufts University: retrieved July 8, 2017 from <http://www.perseus.tufts.edu/hopper/morph?l=logos&la=greek#lexicon>

as a formal system of rules and constraints, which translate into player choices, actions, and possibilities. As a form of engineering, AGD logopoiesis requires building a functional software application qua playable computer game. More specifically, the distinct formal properties of computer game design, digitality and procedurality, require the AGD designer to discretely formalize the autopathographical narrative as a digital game and system of formally-expressed rules. This formalization process is implemented through the praxes of digital design, programming, refactoring, and debugging the software code. Formalizing the computer game autopathography entails distilling the autopathographical narrative into a precise and ordered formulation of algorithms, game rules, and formal constraints implemented as various instances of software code, e.g., integers, strings and variables, conditional expressions, loops, and hexadecimal code among other computational expressions executable by a computer. The logical and methodical mindset involved in authoring these computational processes soothe emotional unease, boost self-confidence, and impart self-command. Thus, the shift to a third-person perspective that these types of computational processes facilitate feels therapeutic. As such, autopathographical game design as a craft of engineering the autopathographical narrative cultivates self-insight and healing in ways that differ from autopathographical writing.

The purpose of this chapter is to recognize how the logopoetic and ludic¹⁰⁰ properties of AGD are a viable form of the third-person perspective enabling self-understanding and self-therapy for the author. The following analyses are provided:

- a) ‘Distancing devices’ in the autopathographical narrative, film, theatre, the realist novel, and videogames in order to understand how poetic media engender the third-person perspective in the viewer/reader/player;
- b) How the autopathographical narrative is formalized through digital, procedural, and computational processes and how these processes engender the third-person perspective;
- c) How subjects experience the digital dimensions of computer game design, programming praxis, and the role of logic in programming as therapeutic in AGD.

¹⁰⁰ Ludus is a form of structured gameplay with explicit rules as discussed in the *Foundations* chapter.

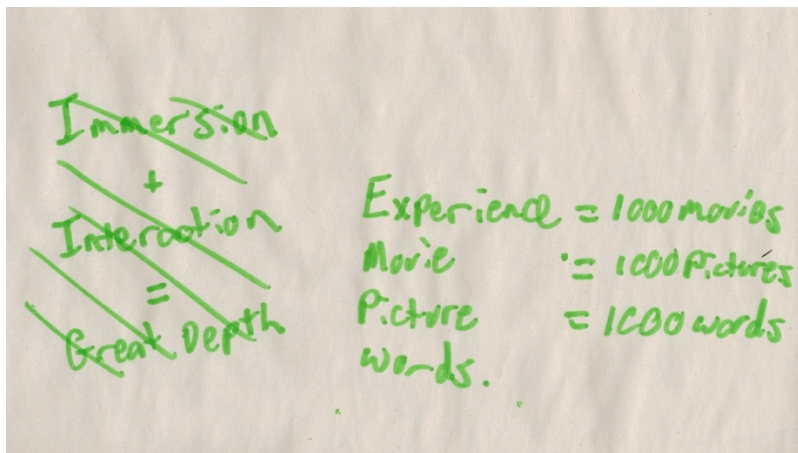
- d) The authorship of formal constraints and game rules in AGD towards symbolic overcoming and self-understanding.

The frameworks informing this chapter are the study of logic, the aesthetics of mathematical notation, and digitality (Boole, 1854/2005; Russell, 1907/1985; Whitehead, 1911/2017; Goodman, 1968; Rota, 1997); computer science perspectives on programming praxis and computational thinking (Iverson, 1979; Abelson, Sussman and Sussman, 1996; Sussman, 2004; Mateas, 2005; Wing, 2006; Denning and Martell, 2015; Dasgupta, 2016; Denning, 2017); digital media theory (Manovich, 2001; Galloway, 2006); Oulipian poetics (Motte Jr., 1986; Berge, 1986); game studies (Caillois, 1958/2001; Juul, 2011; Wardrip-Fruin, 2009); and autopathographical and illness narrative studies (i.e., Hawkins, 1999; Frank, 1995; Couser, 1997; Jurecic, 2012).

7.2 ‘Distancing devices’ in autopathographical narratives and poetic media

The study AGJ doodle below is an ideation of the immersive and interactive properties of gameplay and the rapidly increasing level of immersion that words, pictures, movies, and gameplay engender. The doodler claimed that gameplay experiences are deeply immersive by using the analogy: “experience equals 1000 movies”. In other words, playing videogames enable a significantly higher degree of immersion than watching film, viewing pictures, or reading.

Figure 40: Autopathographical Game Jam doodle comparing immersion between words, picture, movie, and gameplay



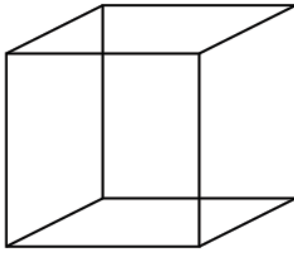
Yasmine also invoked the topic of immersion in her testimony, but she applied it to the realm of making a game as opposed to playing one. She suggests in this quote that the AGD designer ought to find a balance between their role as participant (the first-person perspective) and observer (the third-person perspective) in the making of their autopathographical game:

Don't lose yourself in the game, recognize the game as its own kind of entity but also recognize where you lie within the game as you're making it.

Yasmine evokes the risks of immersion in AGD, prescribing instead for AGD designers to adopt a level-headedness and self-awareness that is mindful of the potentially immersive processes of AGD, which could provoke discomfort and pain.

This section harnesses the above doodle and Yasmine's observation as a springboard for discussing how distancing devices in the autopathographical narrative, film, theatre, realist novels, and videogames engender the third-person perspective in the viewer, reader, and player. 'Distancing devices' are narrative and stylistic techniques whose poetic function is to diminish or disrupt viewer, reader, or player immersion, reactivate disbelief and break the illusion upheld by the media narrative. In doing so, distancing devices engender the objective observer and analytical third-person perspective in the audience or player. It is productive to review the audience-player position in order to illustrate how the logico-analytical properties of technological apparatuses invoke the third-person perspective, which links with the form of objectivity that AGD logopoiesis invites. In the realm of authorship, distancing devices may offer the autopathographical narrator a strategy for generating self-command over intense, painful, or uncomfortable emotions. When autopathographical authors engage with a distancing device in their authorship processes, they engender a perceptual shift between the first-person perspective and the third-person perspective; this shift is analogous to perceiving a Necker cube in two distinct ways. The Necker cube is an optical illusion allowing the viewer to perceptually shift between two different perspectives of the cube, one front-end and one back-end perspective.

Figure 41: The Necker cube¹⁰¹



Indeed, medical anthropologist Marja-Liisa Honkasalo (2001) claims that chronic pain sufferers similarly shift on a continuum between the first- and third-person perspective when depicting their pain: “when representing chronic pain, people move between poles of objectification and subjectification, occupying different positions on the continuum” (p. 10).

In discussing the properties of the objective, third-person perspective, it is important to emphasize its beneficial and constructive properties in the context of autopathographical game design. Detachment and distance are some words used to describe the objective and analytical third-person perspective. In this vein, the third-person perspective may evoke negative connotations to suggest humans who are dissociated, aloof, uncaring, and disengaged from others. However, within the totality of AGD therapoetics, which also integrates the dimensions of autopoiesis, fabulopoiesis, and sociopoiesis, logopoiesis ought not to be dismissed in this way. Logopoiesis invites subjects to perceive their autopathographical narratives in measured and deliberate ways in order to perceive themselves more clearly and understand themselves better. Thus, logopoiesis ought to be considered a positively transformative poetic method of self-knowledge, one that fosters a self-assured and self-possessed self-perspective. Logopoiesis in the context of AGD does not necessarily avoid painful and uncomfortable thoughts and emotions. Rather, it a) can quell negative emotions and thoughts, allowing one to regain self-focus and/or b) can differentiate ambiguous and indeterminate emotions and thoughts via the logopoetic

¹⁰¹ This file is licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license.

properties of AGD in order to refine them, make them clearer, plainer, and more intelligible.

Distancing devices in the autopathographical narrative appear to be an undertheorized area. Autopathography scholars have generally assumed that autopathographical narratives allow the author-patient to re-embody their subjective position after they have been objectified and pathologized through medical praxis, which is a reassertion of the patient's control of the medical narrative and an act of autonomy and agency (Hawkins, 1999; Frank, 1995; Couser, 1997). Frank describes the personal illness narrative as one of "embodiment of these stories: how they are told not just about the body but through it" (p. 3). Frank voices a corporeal poetics narrated through the writer's ailing body given that illness narrators have a strong warrant to symbolically repossess their own bodies from medicine through the act of writing: "modernist medicine claimed the body of its patient as its territory, at least for the duration of the treatment" (p. 10). If "the (medical) chart becomes the official story of the illness" (p. 5), then autopathographical writers offer a compelling counter-narrative to defend their subjective perspective of the official medical story. Couser (1997) argues that medical practices and discourses, such as when doctors treat patients in a dispassionate manner, "reinforce the split between mind and body" (p. 23). Hawkins affirms that patients "often express anger at depersonalizing treatment and sometimes advocate alternative modalities" (cited in Couser, 1997, p. 10). "Personal narratives, especially autobiographical ones, have been one way in which laypeople have responded to the dehumanizing tendencies of medical discourse" (Couser, 1997, p. 34). Consequently, patients produce autopathographical narratives in order to "reestablish their subjectivity in the face of objectifying treatment" (Couser, 1997, p. 11).

Illness narrative scholar Ann Jurecic (2012) suggests that embodiment is a strong concern for many key scholars in the medical humanities: "the theories of narrative set forth in the work of Charon, Frank, and Hawkins respect the irreducibility of the writer's body" (p. 15). Likewise, contemporary phenomenologist Vivian Sobchack (2004), who has written about her own experiences with disability and a phantom limb in her scholarly work *Carnal Thoughts*, encapsulates the turn to embodiment in theoretical accounts of autobiography as filling a meaningful epistemic vacuum: "the contemporary turn to autobiography...can serve as...dare I say, an antidote to objective accounts of the body that don't tell us what we really want to know about our living of it." (p. 20).

But autopathography scholars also underscore resonances between the autopathographical narrative and the objective frameworks imposed on the human body by science, technology, and medicine. For example, Hawkins (1999) alludes to the connection between the pathographical narrative and technological progress in the medical sciences: “it is surely no accident that the appearance of pathography coincides with the triumph of scientific technological medicine” (p. xii). They also mention the autopathographical author’s use of distanciation as a way to cleave distance between the ailing body and the self, a strategy inspired by medical practices: “pure illness narrative tends to disengage the body from the self in the way that medical discourse often tends to do” (Couser, 1997, p. 14). Couser gives an instance of an emotionally-healing “distancing device”, the confessional mode of narration, referring to Gayle Feldman’s autopathography, *You Don’t Have to Be Your Mother*, for whom public disclosure “provides a measure of emotional closure” (p. 69). Jurecic (2012) comments on Stephen Jay Gould, a paleontologist and science writer who turns to statistics as a distancing device to tell his “personal story of statistics” in his book *The Median Isn’t the Message*, where he explores his incurable cancer. Gould represents “how one man’s narrative makes personal meaning of impersonal statistics”, and thus seeks to “humanize statistics” (p. 23). Jurecic (2012) also refers to poet Sarah Manguso’s memoir *Two Kinds of Decay*, which uses “clear, unornamented prose—prose that is clinical in both senses of the word: she observes herself as a patient, and often expresses herself with near scientific detachment... she integrates the precise, functional vocabulary of medicine into her sentences and avoids metaphor” (p. 59).

Consequently, the aforementioned autopathography scholars, given their propensity to privilege the poetics of embodiment, insinuate that distanciation techniques in autopathographical narratives may not always advance the author’s re-subjectification, but instead foster dissociation. Autopathography scholars often frame distancing devices in relation to medical practice and the scientific method, both of which are held responsible for alienating the patient from their body and their sense of identity. Distancing devices as positively generative narrative strategies for the autopathographical writer, which this analysis appraises in the context of computer game design, warrant a thorough investigation in autopathography studies.

Examples of common distancing devices in film and videogames are (ironically) subjective and point-of-view (POV) shots in film, breaking the ‘fourth wall’—an invisible interface upholding audience immersion in theatre and film—, and score cards, maps, and cut scenes, which are non-playable narrative segments in videogames. Galloway (2006) provides a comparative analysis of the cinematic subjective shot and point-of-view (POV) shot, in which the former emulates the phenomenological properties of the first-person gaze and the latter abstracts those same properties. These shots emulate for the viewer the character’s first-person point-of-view, qualified differently in each type of shot. The POV shot is “an abstract shot, an iconographic substitute for the character’s vision” (p. 41) removing the experiential, sensorial realism of the first-person embodied perspective. The subjective shot attempts via a cinematic analogy to convey the phenomenological properties of the first-person perspective in some of its complexity, i.e., the stimuli affecting the sensory system, including the visual, haptic, auditory, kinesthetic, proprioceptive, and interoceptive senses. Consider a subjective shot showing an injured character wobbling while frantically running away from a predator. The camera movement is jerky and the camera focus blurs the image to convey the character’s motion, blurred vision, and dizziness. The subjective shot may be accompanied by an audio track that reinforces the subjectivity the image is intending to convey, e.g., rapid breathing and heartbeat; or a more ‘poetic’ option, such as techno-beat music and industrial sound effects whereby grinding and mechanical noise becomes a metaphor of faltering bodily functions. The analogy is drawn between the jerky, blurry camera image and industrial sound effects and the physical stress and pain that an injured human body experiences.

Galloway (2006) argues that the subjective shot is intended to disrupt the “stabilizing elements in the film apparatus” that engender the contemplative, detached position in the viewer. But while intending to disrupt contemplation and engender immersion, the subjective shot achieves the opposite effect, warns Galloway; as “largely alienating”, the subjective shot reinforces contemplation and diminishes immersion. The more the motion picture camera tries to accurately reproduce the experiential dimension of the first-person perspective, the more it fails. This is demonstrated in the earnest attempt and failed experimental film noir, *Lady in the Lake*, which is entirely filmed using the subjective and POV shots: “*Lady in the Lake* fails not because it doesn’t get it right but because it doesn’t get it wrong enough” (Galloway, 2006, p. 56).

Galloway explains that the film's attempt to accurately and realistically portray the first-person perspective in film by overusing it produces something resembling the 'uncanny valley' in first-person perspective form, which unsettles the viewer.

According to Galloway, the subjective shot does not succeed in effectively conveying the first-person perspective and thwarts viewer immersion in film, except when the subjective shot is combined with a narrative character who is non-human or a machine. Indeed, "the largest number of subjective shots represent the vision of aliens, criminals, monsters, or characters deemed otherwise inhuman by the film's narrative" (Galloway, 2006, p. 50). Moreover, "the merging of camera and character in the subjective shot is more successful if the character in question is marked as computerized in some way" (p. 56). Examples of a subjective shot qua 'computer vision' are film depictions of human-machine cyborgs, e.g., *Terminator: The Sarah Connor Chronicles*, and *Robocop*, whose computerized gaze is overlaid with information processing underway, as when the cyborg-machine scans patterns in the environment, measuring and analyzing objects in their frame of vision, and processing incoming data in order to find and kill their prey.

Figure 42: Examples of the subjective shot qua 'computer vision' in film. Left-image is of *Robocop*, (1987), right-image is of *Terminator: The Sarah Connor Chronicles* (TV Series, 2008-09)¹⁰²



¹⁰² Image sources: <http://www.popmatters.com/feature/hardcore-pov-hardcore-henry-the-pov-shot-and-lets-watch-cinema/P1/>

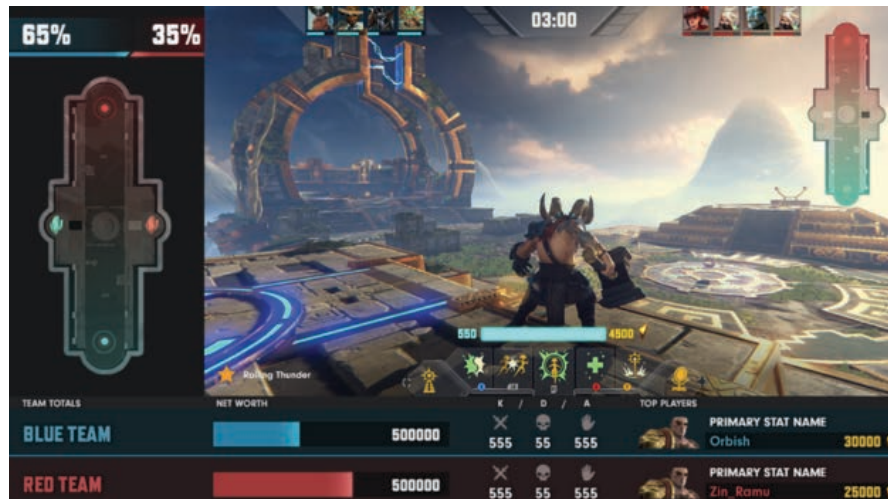
Therefore, the attempt to faithfully render human subjectivity through the camera gaze fails. Galloway suggests that when filmmakers analogize the mechanics of the camera apparatus, which invokes the third-person perspective, to a character such as a machine or cyborg that ostensibly views the world from an objective standpoint, the viewer perceives this analogy as structurally more similar. In this instance, the formal properties of cinematic technology, e.g., cinematography, editing, visual effects, and the semiotic properties of narrative, in this case the cultural expectations of cyborgs and machines as objective entities, conspire to produce a more effective outcome. The integration of the camera apparatus with the cyborg-machine character engenders a more successful subjective shot, given that the technological processes of both become ‘naturalized’ as objective in their synthesis. In other words, the audience intuitively perceives these narratives as if technology is an entity capable of consciousness. And given its capability to ‘reason’ and ‘interpret’ exclusively according to formal rules and procedures, technological consciousness, whether in the form of a cyborg, a machine or camera vision, is by default considered to be objective.¹⁰³

In comparison to film, videogames rely on the active player to structure the first-person perspective through an unrestricted camera position, says Galloway. “The player is the one who controls the camera position, by looking, by moving, by scrolling, and so on” (p. 64) in a virtual environment that can be “repenetrated, zoomed, moved, or reinitialized as is doable in a three-dimensional model”. Thus, videogames are highly immersive for the player. Nonetheless, videogames also come with their own set of distancing devices that thwart immersion for the player, engendering a shift to the third-person perspective and the cool observer, analyzer, or investigator position. Examples of distancing devices include quantified data representations such as score cards, metrics, menus, maps, and visual overlays which require the player to assess and analyze their status in the game. Visual overlays and score cards may show the ‘stats’

¹⁰³ This analysis is not suggesting that technological apparatuses as objective entities are without their biases. The objective standpoint as implemented in many contexts cannot exist without its hidden biases; rather, technological apparatuses referenced in the above context are theoretically objective because they do not *emotionally* engage in the acts of perception and recognition. As such, emotions do not inform their decision-making processes.

(statistics) of the player's achievements or points accrued in the game broken down in numbers and/or percentages.

Figure 43: Example of visual overlays, a score card, and map in the videogame *Breakaway*¹⁰⁴



In fact, the assumed defining features of games, interactivity, and immersion, are contested in game studies. Some game scholars place tensions on the concept of interactivity, claiming that it is a “myth” that games are interactive or ergodic (Newman, 2002), or that games are uniquely interactive in relation to other media forms: “maybe all media forms are immersive or interactive in one form or another, and videogames are just immersive and interactive in particular ways”, states Keogh (2015). Newman (2002) argues that a number of common features in games, such as cut scenes and visual overlays that require the player to analyze or observe the action, frustrate interactivity and immersion¹⁰⁵:

Quite simply, videogames are not interactive, or even ergodic. While they may contain interactive or ergodic elements, it is a mistake to consider that they present only one type

¹⁰⁴ Image from: <http://www.gamefromscratch.com/post/2016/11/04/Amazon-Lumberyard-16-Beta-Released.aspx>

¹⁰⁵ There are many types of interruptions that hamper immersion in gameplay, some of which apply to novice gamers or first-time players. For example, when a player is learning how to play a game, their state of flow becomes interrupted by mistakes and errors due to their lack of skills. They stop and start, trying to figure out where and how to manoeuvre their game character, how to pick up the gun or grab ammunition, how to read a map, how to open up a box of game assets, how to hold their controller or which keys to press. They may stop the game to look for instructions online on how to play the game in Let's Play (LP) videos or instructional videos.

of experience and foster only one type of engagement. Videogames present highly structured and, importantly, highly segmented experiences. Play sequences, from where the idea of the interactivity or ergodicity of videogames derives, are framed and punctuated by movie sequences, map screens, score or lap-time feedback screens and so on.

Newman's statement that "videogames are not interactive, or even ergodic" is unnecessarily radical. But this analysis takes him to be correct in the assertion that there are elements in videogames that may diminish, for a given time, immersion and interactivity for the player. Nonetheless, players who enter 'flow' states¹⁰⁶ while playing games may selectively ignore the descriptive overlays and screens in order to uphold immersion.

Consider the connection between three types of distancing devices: breaking the 'fourth wall' in film and theatre, the cut scene in videogames, and the descriptive passage in realist novels. Audiences generally experience these distancing devices as jarring in 'pulling them out' of the immersive diegetic bubble of the play, film, novel, or videogame, even for a brief time. One type of disruption of gameplay immersion is produced by the cut scene. A cut scene is a short, animated, non-interactive sequence in a video game that introduces and/or propels the game narrative. Cut scenes are story-driven and offer narrative exposition and information about the story and game characters; they also serve character development and help the game player emotionally connect to the game characters. Players may or may not have the option to skip over cut scenes. Some players find cut scenes intrusive because they disturb the immersion of gameplay, although this experience largely depends on context, the length of the cut scene, how well the cut scene develops the game narrative and other factors. For example, an anonymous online discussion forum poster writes: "I hate cut scenes in video games. SO. MUCH."¹⁰⁷. Klevjer (2002) counter-argues that cut scenes do not disharmonize the gameplay experience:

¹⁰⁶ By 'flow', the psychologist Mihaly Csikszentmihalyi's (1990) theory of flow is invoked—a state of optimal mental focus that produces creative satisfaction for people engaging in various activities, including play.

¹⁰⁷ From: <https://www.gamespot.com/forums/games-discussion-1000000/i-hate-cut-scenes-in-video-games-so-much-33278480/>

A cut-scene does not cut off gameplay. It is an integral part of the configurative experience. Even if the player is denied any active input, this does not mean that ergodic experience is paused. (p. 195)

If game players experience cut scenes as disruptive to gameplay immersion, then there is a strong resonance between the videogame cut scene and the descriptive passage in the realist novel, the latter of which Jacques Rancière (2007) discusses. Similar to the cut scene in the videogame, the descriptive passage in the realist novel ‘suspends’ the temporal dynamics of plot and action in the narrative, hence it serves as a distancing device to pull the reader out from the temporal action of the novel. Rancière claims that novelistic description “deprives action of its powers of intelligibility—that is, of its powers of ordered distribution of knowledge-effects and pathos-effects” (p. 121). Rancière suggests that in realist novels, descriptive passages are a method for suspending the cause-effect of the plot action and the story unfolding in time and dampening their emotional effects on the reader.

The ‘fourth wall’ is the invisible interface that separates the performers and actors from the audience in theatre and fiction film, upholding some degree of immersion and suspension of disbelief for the audience. Like videogame cut scenes and descriptive passages in realist novels, authors and performers who break the fourth wall disrupt the diegetic temporal action of the narrative, thus interrupting immersion and engendering distancing. Actors directly face and speak to the audience or introduce action that calls attention to the theatrical or filmmaking process behind the production. This strategy was used to great effect by Bertolt Brecht, claims film and literary theorist Robert Stam (1992):

In terms of acting, Brecht argued for a double distancing — between the actor and the part, and between the character and the spectator — and suggested specific exercises, such as acting in the past tense, and in the third-person, to achieve this distancing. (p. 213)

An example of breaking the fourth wall in Brechtian theatre is when the backstage is intentionally exposed while the theatrical performance is underway, showing the audience stage hands pulling ropes. As Stam observes, “Brecht believed in reflexivity, that art should reveal the principles of its own construction, to avoid the ‘swindle’ of giving the impression that fictive events were not ‘worked at’ but simply ‘happened’” (Stam, 1992, p. 213).

In film, examples of breaking the fourth wall include the Marshall McLuhan sequence in Woody Allen's (1977) *Annie Hall*, when Allen's character Alvy Singer directly addresses the audience and pulls the real McLuhan into the frame to challenge McLuhan's fictitious detractor, and Tom DiCillo's (1995) film *Living in Oblivion*, a meta-inquiry into the filmmaking process via the filmmaking process itself. Thus, breaking the fourth wall in theatre and film is a reflexive device calling attention to the 'meta-data' of the play/film narrative and engaging with it diegetically in order to impart a deeper understanding of the mechanics of the theatrical or cinematic apparatus.

These distancing devices in fiction film, theatre, videogames, and realist novels are primarily structured through description and explanation, and come in two forms: a) qualitative, e.g., novelistic description, game cut-scenes, theatre and film 'fourth wall' disruptions; and b) quantitative, e.g., videogame score cards, calculations, and numerical representations (statistics). Both techniques are instances of *logos* engendering a shift to the third-person perspective in the audience; they require the reader/viewer/player to stop, assess, and analyze both the narrative or play arc presented to them and the technological apparatuses that 'operationalize' the narrative or play arc. Similarly, AGD engenders a distanciation for the author, inviting them to stop, assess, and analyze their own autopathographical narratives and the hidden mechanisms that make them a personal reality.

7.3 AGD logopoiesis: a form of computational therapoetics

The logopoetic affordances of AGD, i.e., their computational, digital, logical, and procedural qualities, can be conceived of as a form of *computational therapoetics* drawing on the concept of "computational thinking" originally coined by Seymour Papert (1980; 2005). Jeannette Wing (2006) emphasizes that computational thinking "is not trying to get humans to think like computers"; rather, it "involves solving problems, designing systems, and understanding human behavior...(including) a range of mental tools..." such as abstraction and logical reasoning. Computational thinking narrowly applies to designers and engineers of computing systems, not their users, claims Peter Denning (2017). Denning differentiates between "users of computational tools" and "computational designers", maintaining that "computational thinking primarily benefits people who design computations and that the claims of benefit to

non-designers are not substantiated”.

This conception of computational thinking is similar to what Michael Mateas (2005) refers to as “procedural literacy”. Even though Mateas’s original intent is to frame procedural literacy as a useful skill for digital media scholars, many of the key skills he singles out below, such as programming and the authorship of computational and procedural processes, relate to the AGD game designers exemplified in this study:

By procedural literacy I mean the ability to read and write processes, to engage procedural representation and aesthetics, to understand the interplay between the culturally-embedded practices of human meaning-making and technically-mediated processes. With appropriate programming, a computer can embody any conceivable process; code is the most versatile, general process language ever created. Hence, the craft skill of programming is a fundamental component of procedural literacy, though it is not the details of any particular programming language that matters, but rather the more general tropes and structures that cut across all languages.

As game developers, programmers, and algorithm designers, AGD subjects fall well within the range of computational thinking and procedural literacy. They are professional or semi-professional game designers and developers who are comfortable with the technical and formal demands of computer game design. They know how to use a variety of sophisticated game engines and game authoring software tools such as the *Unity* 3D game engine to individually author a game. They are skilled and avid programmers; without the software development and programming skills, they cannot reap the benefits of logopoiesis qua third-person perspective. But also, procedural literacy as the ability to program and harness logical reasoning is a positive tool for game designers struggling with mental health issues. The knowledge that one is competent in this form of literacy is evidence to some study subjects that they can be in command of their subjective states and level of creative output, which in turn augments their sense of self-accomplishment. Philip E. Agre confirms the epistemic power of building and engineering inherent in computational forms of authorship: “building things, like fieldwork and meditation and design, is a way of knowing that cannot be reduced to the reading and writing of books” (as cited in Wardrip-Fruin, 2009, p. 101).

Computational thinking and procedural literacy also evoke the *Sapir-Whorf hypothesis*, which proposes that the structure of language influences thought. “Language...affords a symbol

system for *thinking* about something. Language is embroiled with thought itself” (Dasgupta, 2016, p. 67)¹⁰⁸. Subrata Dasgupta also claims that programming languages have not been considered from the position of shaping the mindset of their practitioners:

No one to my knowledge has framed an analogue to the Sapir-Whorf hypothesis for the computational realm but the obsession computer scientists have shown...with programming and other computational languages strongly suggest that some form of the hypothesis is tacitly accepted by the computer science community...the languages of computing (in particular, programming) are intimately entwined with the nature of the computing environment; and that programming languages influence programmers’ mentality (Dasgupta, 2016, p. 67-68).

Compare Dasgupta’s statement with the view of Gerald Jay Sussman (2004), who in the book *Computer Science: Reflections on the Field, Reflections from the Field* echoes Dasgupta’s claim above by drawing on his personal experience as a programmer and practitioner. He also claims that computational thinking has influenced the evolution of human self-perception through, for example, the use of computational metaphors in language. And perhaps, as Sussman suggests in this quote, computational thinking can also be used as a tool for solving personal problems affecting human beings:

I see one of the deepest consequences of computational thinking entering our society in the transformation of our view of ourselves...More telling is the fact that we can describe people and organizations as having “bugs” and that these can be “patched”. Perhaps the most important consequence of computational thinking will be in the development of an understanding of ourselves as computational beings. Indeed, my personal experience as a computer programmer has made me aware that many of my own problems are bugs that can be analyzed and debugged, often with great effort, and sometimes patched. (p. 183)

Sussman’s statement may appear to be unrealistic or far-fetched to some critics, but it is not an unimportant or frivolous reflection about the phenomenological and psychological effects of authoring computational processes through programming praxis. Several of this study’s subjects confirm the above statements to varying degrees by stating that they also find that the praxis of programming, which includes the sub-praxis of debugging software code, is self-therapeutic in

¹⁰⁸ This notion has philosophical antecedents in Herder and Heidegger, an analysis beyond the scope of this dissertation.

that it enables one to feel as if one is “patching” or solving one’s own problems.

Furthermore, Abelson, Sussman and Sussman (1996) emphasize the poetic value of programming languages in extending our perception:

The source of the exhilaration associated with computer programming is the continual unfolding within the mind and on the computer of mechanisms expressed as programs and the explosion of perception they generate. If art interprets our dreams, the computer executes them in the guise of programs! (p. 10)

Abelson, Sussman and Sussman suggest that programming is a poetic form of expression that extends human perception in developing our dreams into fully inhabitable environments. Programming praxis may play the role of a similar prosthesis for AGD study subjects. Consider Goodman’s (1968) point about art forgeries, stating that they depend on our inability to recognize the difference between the art masterpiece and the masterful forgery given that appearances may deceive. Goodman suggests that we need more practice to identify deception by honing our perception to look more attentively: “what one can distinguish at any given moment by merely looking depends not only upon native visual acuity but upon practice and training” (p. 103). Goodman is saying that the acquisition of deeper awareness through praxis extends the limits of our perception. Similarly, the self-awareness that mental illness does not equal one’s identity¹⁰⁹ extends the limits of our self-perception. The ability to recognize difference and become more aware extends the limits of our perception in a Merleau-Pontian (1945; 2002) sense of the blind man’s stick, akin to a technological prosthesis, e.g., a pair of glasses, a magnifying glass, a microscope or a telescope, as Goodman suggests in his analysis.

A Canadian Broadcasting Corporation article by journalist Ieva Lucs (2017) also gives voice to this idea of building recognition and awareness in order to extend self-perception, featuring the game designer Alana Zablocki who has made a game about suicide and her experiences as a psychiatric inpatient. The article suggests that the game design process has allowed the author to differentiate between her social experience of her mental illness from a more objective sense of reality—“this is your illness ... it’s not reality”—, in order to shift her

¹⁰⁹ As the common expression goes: “you are not your mental illness”; that is, mental illness does not define who you are.

perception and alleviate pain. Indeed, “writing the game changed something in Zablocki, she said. She...says she has had relief from her mental health issues” (Lucs, 2017). Programming can thus be conceived of as a prosthetic tool in Goodman’s and Merleau-Ponty’s sense for extending self-perception through self-awareness acquired through praxis. The AGD study subject also seeks to understand their own feelings, emotions, thoughts, and ideas about themselves, and to better grasp their autopathographical narrative through its externalization and articulation. And in the specific case of logopoiesis, they do so through the digital, procedural, and computational praxes involved in computer game design.

7.3.1 The role of catharsis in AGD

Catharsis is an ambiguous phenomenon in autopathographical game design worthy of further research. Consider some thoughts on catharsis by Greek philosopher Aristotle, philosopher Susanne K. Langer, and social psychologist Lev Vygotsky. In *Feeling and Form*, Langer (1953) argues that feeling and form are usually conceived of as polar opposites: “feeling is associated with spontaneity...and informality” while “form connotes formality, regulation...hence...*absence* of feeling” (p. 17). In her theory of art, Langer refutes this simplistic assumption with her claim that artists express feelings abstractly through artistic symbols; therefore, instances of “objective feelings” are depicted in art just as much as “subjective feelings”. She cites Otto Baensch, who in his 1923 essay “Art and Feeling”¹¹⁰ argues that “art, just like science, aims primarily to be ‘understood’” (as cited in Langer, 1953, p. 19). Langer suggests that artists mobilize the formal properties of an art form to create a certain ordering and arrangement of feeling unconnected to emotional release: “the relation of art to feeling is...something subtler than sheer catharsis or incitement” (p. 18). Langer (1942) claims that “the laws of emotional catharsis are natural laws, not artistic” (p. 184). For Aristotle, catharsis is a phenomenon whereby feelings of pity and fear in the viewer watching dramatic tragedy induce an emotional release, a purging of excess emotions. The cathartic formula of Aristotelian tragedy is “based on the sudden suspension of a steadily induced state of fear and pity” (Damasio, 1999, p. 59). According to Aristotle, the pity and fear that drama provokes is

¹¹⁰ Original German is “Kunst und Gefühl”.

ontologically different from pity and fear provoked in natural situations. In other words, if we watched a scene in a movie drama of a human being chased by a lion, we would feel excitement and tension, but the nature of this tension diverges from the tension one would feel if one were chased by an actual lion or saw somebody being chased by one in a natural environment. Vygotsky (1971)'s version of catharsis diverges from Aristotle's in that emotions are not purged via an art form's aesthetic effects on the audience; rather, emotions are dialectically sublimated into pleasure and resolved (p. 213). Games scholar Jesper Juul believes that tragedy invokes a loss of control that can only work in observer-based media, i.e., film, theatre, literature, which induce catharsis through the safe distance the medium engenders. Juul argues that gameplay, as opposed to tragic theatre, cinema, and novels, cannot engender catharsis because gameplay is immersive, not contemplative, and as such, "games do not purge emotions from us – they produce the emotions in the first place" (Juul, 2013). Juul says that catharsis requires some level of contemplation to dampen the immersion that creative forms engender for audiences.

Langer, for whom artistic expression shapes human emotion into form, endows the artist with wisdom and self-knowledge gained through their engagement with the formal properties of an art form. Her thesis builds an empowering portrait of the artist as someone who understands their emotions on the basis of their expressive ability to shape and structure feeling into artistic form: "feelings have definite forms, which become progressively articulated" (p. 92). In his *Theory of Emotions*, Jean-Paul Sartre (1962) proposes that emotions themselves are tools (*techne*) for crafting consciousness into form. Sartre theorizes that when we feel sadness, distress, or terror we resort to perceptive strategies such as "magical thinking" (i.e., irrational errors and fears) to mold into shape what we at that moment perceive to be a disordered world. From the phenomenological standpoint of the subject's embeddedness in an indeterminate world, human emotions invoke these perceptive strategies in order to control the perceived chaos around us.

The mental deliberation involved in formalizing autopathographical games are comparable to Langer's and Sartre's concepts of shaping feeling into form or consciousness and Vygotsky's concept of catharsis as a sublimation of pain into pleasure. In AGD, sublimation of pain into pleasure may be achieved through the authorship of game rules and the software

program's formal constraints¹¹¹. But catharsis may not be possible in AGD given the cool and self-observant position cultivated through the logico-analytical dimensions of computer game design. Alex claimed that authoring an autopathographical game based on intimate and personal experiences with mental illness “has moments of catharsis”; however, tempered that statement with this explanation:

For me, writing is very cathartic because it's not something I have to *craft* (emphasis added). I don't think. If I'm really emotional, if I'm in crisis, I'll write. And that's cathartic because I don't have to think about what I'm going to write. I just do it, whereas with game development it's not as easy to just do, you have to actually stop a little bit and *assess*.

For Alex, the possibility for immediate, spontaneous, and stream-of-consciousness creative expression that creative writing such as poetry and diary writing affords is far more cathartic than autopathographical game design, which requires careful assessment and deliberation. Autopathographical game design requires a much more strict and rigid plan, hence the strong form of logos inherent in this poetic form. The requirement for a systematic and methodical (i.e., logical) approach in computer game design may very well dampen the possibility for catharsis. The logico-analytical dimensions of computer game design are a strength that, despite not affording catharsis, feel therapeutic for AGD subjects engaging with them. Therefore, as Alex claimed, autopathographical game design *crafts* emotions encountered in the first-person perspective into a digital, computational, and procedural form. Thus, computer game design qua engineering of a system of rules and formal constraints involves a level of structural thinking that does not apply to creative-autopathographical forms such as diary and creative writing, the latter of which may provide better opportunities for purgation of emotions.

7.4 Formalizing the autopathographical narrative as a computer game

In AGD, study subjects formalize the autopathographical narrative as a digital game via programming praxis. They abstract their first-person experiences with mental illness, emotional trauma, and disability, which to some extent are already ordered as an autopathographical

¹¹¹ Discussed in an upcoming section on rule-design.

narrative through their intelligibility framework (IF)¹¹², into a computer game. They use a variety of technical and artistic processes specific to the formal properties of computer game design to render this abstraction. In a sense, they render complexity into simplified terms.

For Sam, distilling his fabulized self qua autopathographical narrative of social anxiety and ADHD via the structured processes of software development helps to communicate “the nuts and bolts of a personal experience”. Formalization in AGD adds another layer to subjects’ intelligibility framework (IF), a distillation method imparting self-insight on first-person experiences through its logopoetic affordances. Sam gave voice to the advantages of distilling a first-person experience into software code, explaining that writing software code in AGD allows him to view his experience through “a very simple lens”:

By taking a personal experience, especially a problematic one, and expressing it through code, I have to look at it objectively in numerical, rigid values and reproduce it as best as possible.

Thus, authoring a software program helps Sam gain an objective perspective on his experiences with debilitating anxiety through its affordances for precision. The requirements of software development forces Sam to implement his autopathographical narrative in stricter terms than he might use were he to frame that narrative in non-computational terms. “Ordinary concepts have none of the rigidity or precise dependence” (of those concepts) “built up systematically and strictly hierarchically in mathematics and computer science” (Hofstadter and Sander, 2013, p. 54). As such, subjects must find parsimonious ways to distil their complex experiences into a game prototype and create a minimalist virtual world using a system of rules and formal constraints, minimalist colours, and simple digital graphics. Consider a statement by philosopher of science Karl Popper (1935; 2002), who asserts that simplicity is “highly desirable” in science because its overall informational content is higher:

Simple statements, if knowledge is our object, are to be prized more highly than less simple ones because they tell us more; because their empirical content is greater; and because they are better testable. (p. 128)

¹¹² The intelligibility framework (IF) was discussed in the last chapter on *Fabulopoiesis*.

In other words, Popper argues that in the realm of knowledge, simplicity promotes better understanding. Similarly, AGD subjects harness parsimony via formalizing the autopathographical game in order to distil their narratives to a computer game prototype.

To understand how formalization as a process of simplifying first-person experiences matters to AGD designers, consider some theoretical insights from existentialist philosopher Jean-Paul Sartre and computer scientist Joseph Weizenbaum. The first-person experience is structured through emergent processes between human agents and the world. According to Sartre (1943; 1992), existence itself is irreducible to essences and cannot be abstracted; “the existent in fact cannot be reduced to a *finite* series of manifestations since each one of them is a relation to a subject constantly changing” (p. xlvii). The lived experience is thus continuous, indeterminate, always in flux. In discussing the pitfalls of instrumental reason, Weizenbaum (1976) affirms the irreducibility of the human subject in relation to some forms of scientific practice that harness instrumental reason: “an individual is dehumanized whenever he is treated as less than a whole person” (p. 266). He implies that the individual is dehumanized when the scientist rushes to ‘break down’ and quantify the human experience, suggesting that intangible and ‘soft’ human traits such as feeling, empathy, compassion, intuition, wisdom, and moral and ethical conscience cannot be reduced to ‘hard’ numbers and ‘cold’ calculations.

For AGD subjects, programming involves distilling and abstracting the subjective experience with illness, trauma, and disability in order to rebuild it as a software program. Yet subjects do not de-humanize themselves via the processes of formalization, as suggested by Sartre’s critique of abstraction and Weizenbaum’s critique of instrumental reason; rather, they gain resilience by discovering novel ways of understanding their first-person experiences via computational methods. As previously mentioned in the *Autopoiesis* chapter, the skills of craftsmanship involved in computer game design is aided by subjects’ willingness to reflect on one’s actions and behaviors, tell a personal story about pain and illness, and critically question oneself and one’s experiences. Autopathographical game design thus can become an ethical exercise in that subjects adopt methods of shifting perspective in order to be able to view themselves differently. As a form of adopting the third-person perspective, AGD logopoiesis can disambiguate the vagueness and indeterminacy that characterizes subjective experiences of

illness, trauma, and disability. In other words, AGD logopoiesis can remove uncertainty from subjective experiences with illness, trauma, and disability burdened by our emotional and mental appraisals of them. In this regard, Nussbaum (2001) implies that emotional pain is more ambiguous than physical pain because it is wrought with added layers of mental rumination. In a way, Nussbaum suggests that emotional distress is not as discrete as localized physical pain: “how simple life would be, if grief were only a pain in the leg, or jealousy but a very bad backache. Jealousy and grief torment us mentally; it is the thoughts we have about objects that are the source of agony – and, in other cases, delight” (p. 16).

Formalization in AGD appears to be also perceived as an intelligibility framework (IF), as a plan for gaining self-clarity, that the non-computational poetic arts do not engender. Consider Sam’s testimony, which compared the strengths of computer game design as a form of deliberate reckoning to the more immediate and spontaneous affordances of painting. Deliberation for Sam means carefully contemplating all the digital, computational, and procedural elements that comprise a computer game autopathography, which by extension allows him to deliberate on his own first-person experiences with ADHD and social anxiety. Sam made the game autopathography *Anxiety Corp.* about his ADHD experiences with social anxiety. He invoked the formal properties of computer game design and the fact that computer games must be structurally engineered and formalized in order to build a functional system and playable game:

I think games are excellent autopathographies partially because they’re literally a recreation of an experience and partially because they’re so deliberate. In making a game, it’s not as immediately expressive as a work of art where you can have brush strokes of different colours.

A game you need to spend a lot of time creating and honing and testing and debugging and perfecting. It forces you to look more analytically at your experiences and at what drives your experience because in order to recreate that experience for an audience, you need to have all of the contributing and causal factors in place.

You don’t necessarily get the understanding (with flowing artworks). I think (authoring) the mechanics and the rules (of a game) forces you to think very, very precisely about your problem and help you to identify the root causes and contributing factors.

You’re finding what are the core elements, what makes this problem tick.

Sam suggests that while painting or drawing affords immediate expression, the lack thereof in computer game design translates into an advantage. In the latter, the requirement for deliberation of the autopathographical narrative and designing an error-free software is a form of calculated reckoning, forcing the designer to adopt an analytical perspective on their mental health issues. Deliberation also trains the designer to recognize and discern the causal mechanisms behind the illness, “the root causes and contributing factors”, to invoke Goodman’s point again about training awareness to recognize differences. Sam claims that in contrast to painting or drawing, the logico-analytical dimensions of computer game design confine the autopathographical game designer’s thinking to forms of discrete intelligibility, allowing them to better grasp the mechanism driving their pain, discomfort, or anxiety. Computer game design as a discrete and deliberate method imparts to the designer the ability to discern, step-by-step, the hidden logics that define their subjective experiences with mental illness, emotional trauma, or disability. Sam provided the concrete example of an “anxiety meter” as a way of representing the mechanism of his social anxiety in his game:

An anxiety meter—it’s a very direct representation of ‘oh man, if I have a full-blown panic attack I’m going to lose my job, so if the anxiety meter gets to 100% you lose’. That’s a very, very literal recreation.

An anxiety meter is an example of the use of game metrics. There are two ways to interpret Sam’s anxiety meter. First, the anxiety meter expresses the author’s struggles with anxiety in the procedural and quantitative terms of measurable player progress—“an expressive process” (Wardrip-Fruin, 2009) that motivates the player to take control of his anxiety. The anxiety meter is an expressive process that encourages Sam to objectively face his struggles with anxiety and panic attacks. And second, the anxiety meter is a self-deprecating device demonstrating that the game author can take a light-hearted approach to his painful struggles with mental illness. As such, it is an example of deploying humour transgressively in autopathographical game design, whereby authoring gameplay in the form of ‘playing your social anxiety’ summons self-transformation.

Another way in which Sam abstracted his first-person experiences is through digital design processes that harness formalization. Sam’s choice of using “stealth mechanics” and “ray casting” relates to the game narrative of the main game player character, an office worker in a

Consider what Whitehead (1911; 2017) has to say about geometry, which helps us understand the mathematical and thus quantitative processes that undergird the design of ‘ray casting’. He writes that the “fundamental ideas of geometry are exactly the same as those of algebra; except that algebra deals with numbers and geometry with lines, angles, areas, and other geometrical entities” (p. 145). Moreover, objects in both algebra and geometry are “measurable” and “abstract”, despite the common feeling that when we compare numbers to space, space appears less abstract than numbers. “Geometry, viewed as a mathematical science...may be called the science of dimensional order” (p. 147), affirms Whitehead. Sam’s mathematical knowledge—he has a Bachelor of Science in Mathematics—helped him to author a virtual world in order to tell his autopathographical narrative dimensionally, via the properties of space, using geometric notation (see Figure 44 above). It is a process which renders his experience with social anxiety and ADHD via the spatial attributes of a game environment.

Given Sam’s appraisal of the aforementioned technical processes involved in authoring autopathographical games, it would be useful to comment on the aesthetics of mathematics that undergirds the above-mentioned “ray casting” process and programming praxis for some insight on why Sam and other AGD designers find these processes therapeutic, given their invocation of beauty and pleasure. In *The Study of Mathematics*, the logician Bertrand Russell (1907; 1985) describes the relationship between mathematics, art, and beauty, which is dependent on the exclusion of ornamentation and superfluous detail:

Mathematics, rightly viewed, possesses not only truth, but supreme beauty—a beauty cold and austere, like that of sculpture, without appeal to any part of our weaker nature, without the gorgeous trappings of painting or music, yet sublimely pure, and capable of a stern perfection such as only the greatest art can show. (p. 86)

Russell defines the aesthetics of mathematics according to a notion of beauty rooted in restraint and severity that mathematical formalization affords, a beauty whose strength lies in refusing to mobilize affect or pathos. According to mathematician and philosopher Gian-Carlo Rota (1997), “mathematical beauty” is a historically and culturally-mediated concept that is sometimes difficult to pinpoint. He suggests that there are several qualities of mathematical beauty found in mathematical theorems and proofs: brevity in expression, elegance in presentation, and an element of surprise in the statement, which stimulates insight. Russell compares literature to

mathematics in order to emphasize the heavy abstraction of the latter which distills mathematical expression to clarified essences: “literature embodies what is general in particular circumstances whose universal significance shines through their individual dress; but mathematics endeavours to present whatever is most general in its purity, without any irrelevant trappings” (p. 87). Rota says that mathematical aesthetics are also judged according to how well the statement is refined and polished: “lack of beauty is associated with lack of definitiveness.” Rota explains, albeit critically, that with mathematical praxis we enter a different, more ordered, and mechanistic world where we bypass the complexity and indeterminacy of ordinary living:

All talk of mathematical beauty is a copout from confronting the logic of enlightenment, a copout that is intended to keep our formal description of mathematics as close as possible to the description of a mechanism. This copout is a step in a cherished activity of mathematicians, that of building a perfect world immune from the messiness of the ordinary world, a world where what we think should be true turns out to be forever true, a world that is to be kept free from the disappointments, the ambiguities, the failures of that other world in which we are forced to live. (p. 182)

Rota ties mathematical abstraction, brevity, and resolution of form to aesthetics, which in the context of AGD might provide relief from the confusion and chaos of subjective experiences with mental illness, emotional trauma, and disability. Therefore, AGD proves to be emotionally economical: by discretizing first-person experiences through formalization processes, Sam and others convert an emotional experience into a highly abstract and densely symbolic form, which appears to silence the extraneous and distracting noise of messy living and diminish, if only for a moment, life’s difficult and complex moments.

7.5 AGD logopoiesis: the therapeutic role of digitality

The ontological nature of computer games as digital objects enables AGD study subjects to craft their autopathographical narratives in discrete terms using software code. Discreteness refers to three general properties of digitality: the differentiation of systems, the notational structure of systems, and the mathematical description of systems.

Consider Sam’s testimony again on the formal properties of digital design, invoking the differences between non-digital and digital forms of creative expression:

Rather than in the visual arts where it’s a very emotional and flowing artwork you’re creating, game design is almost the polar opposite, it’s deliberate. Meticulous even.

When Sam says “flowing”, he is invoking the analog properties of non-digital art forms such as painting and drawing in contrast to the discrete properties of digital game design. Sam implies that the discrete properties of computer games do not allow for the representation of ‘flowing emotions’ in the same way as painting. We can understand Sam’s comparison by turning to Nelson Goodman’s (1968) distinction between analog (continuous and non-differentiated) and digital (discontinuous and differentiated). According to Goodman, despite the fact that “analog instruments may offer greater sensitivity and flexibility” (p. 161), the distinction between digital and analog machines and systems is harder to define. For example, the continuity and density of analog forms does not necessarily suggest “complete lack of differentiation”. Goodman explains: “but density, while it implies, is not implied by complete lack of differentiation; and a system is analog only if dense” (p. 160).

Given Goodman’s theorization, when Sam invokes the analog qualities of “flowing” artworks, we ought not to assume that all forms of analog art are purely analog. According to art historian Meredith Hoy (2017), not all forms of “not mechanically applied” drawing and painting are analog. An example of a digital form of manual—human-generated (as opposed to computer-generated) drawing or painting is pointillism, a formal grammar whereby the artist must use small dots or points to constitute form, shape, line, and composition on a surface. Hoy claims that “we find again that in practice, when we are looking at materials and products of material processes, it becomes difficult to adhere to notionally ‘pure’ definitions of digital and analog, because pointillism tends nevertheless toward digitality in numerous ways; it is chromatically discrete, in its unblendedness, and also syntactically and semantically verges toward the disjunct.” (p. 62). Hoy is to some extent echoing Goodman’s claim that defining analog and digital in theory is harder due to the overlap introduced by their material application in praxis. Nevertheless, the general distinct properties between analog and digital form are important markers for ensuring certain qualitative experiences for AGD designers engaging in the digital practices of computer game design.

In *The Language of New Media*, Lev Manovich (2001) lays out the key properties of digitality, framing it in differentiated and discrete terms:

1. Numerical Representation

All new media objects, whether they are created from scratch on computers or converted

from analog media sources, are composed of digital code; they are numerical representations.

This has two key consequences:

1.1. New media object can be described formally (mathematically). For instance, an image or a shape can be described using a mathematical function.

1.2. New media object is a subject to algorithmic manipulation. For instance, by applying appropriate algorithms, we can automatically remove “noise” from a photograph, improve its contrast, locate the edges of the shapes, or change its proportions. In short, media becomes programmable. (p. 27)

According to Manovich, what makes ‘objects’ digital and programmable, whether they are data (i.e., an image of any kind) or a process (i.e., a game rule, a formal constraint of the software program) is their formally expressible dimensions, meaning they are mathematically, numerically, and algorithmically expressible. Goodman (1968) further exemplifies digitality by invoking its notational structure: “the dime-counter displaying numerals is a simple example of what is called a digital computer” (p. 159-60). Goodman explains that the real virtues of digital instruments are those of notational systems: definiteness and repeatability of readings (p. 161). He adds: “if (the system) is also unambiguous and syntactically and semantically disjoint, it will therefore be notational” (p. 161) As such, “notation refers to symbols invented for writing about something” (Dasgupta, 2016, p. 64) in concise and unambiguous forms—clear and distinct forms—such as numbers and decimals. For example, programming languages are partly a notational system of writing instructions for a computer. In other words, ‘notation’ is ontologically speaking, digital. Whitehead (1911; 2017) confirms the benefits of notation to logical reasoning qua mental deliberation: “our modern power of easy reckoning with decimal fractions is the almost miraculous result of the gradual discovery of a perfect notation” (p. 33).

In *Tools for Thought: The History and Future of Mind-Expanding Technology*, historian Harold Rheingold (2000) writes that George Boole brought the digital, grammatical, and logical realms together when he mathematized Aristotelian formal logic by translating syllogistic propositions into mathematical symbols. Boole performed a distillation of logical propositions in his invention of mathematical logic. Rheingold reminds us that even though digitality as a binary form invokes simplicity, the simple two-value binary system of 1’s and 0’s render as complex at

the end-user level in digital computing: “the inventors of the first computers learned that a simple system with only two values can weave very sophisticated computations indeed” (p. 24). To give an example of how digitality loosely connects to human thought, consider the theory of dualism put forward by philosopher and mathematician René Descartes. Descartes theorized the separation of mind and body based on the epistemological assumption that, according to Sartre (1962), “generally we can trust our ideas to be true representations of things, if we take care to concentrate upon *clear* and *distinct ideas* (emphasis added)” (p. 6). Descartes’s differentiation between body and mind wrests philosophical ideation from the indeterminate and continuous (analog) confines of emotional and corporeal vicissitude in order to achieve conceptual clarity, thus evoking a downright digital enterprise in his attempt to differentiate and discretize between body and mind.

Digitality is of great relevance to AGD logopoiesis. Digitality manifests as various digital design processes that AGD subjects engage in, such as authoring pixel art and digital compositing such as ‘ray casting’, a technique for simulating 3D space in 2D form¹¹⁴. Digital game design enables subjects to disambiguate their first-person experiences with illness, trauma, and disability, and dwell for a moment in the determinate and clarified terms offered by abstract binary expressions, such as an if/else conditional statement (which instantiates as a game rule and/or formal constraint of the software program), hexadecimal colour notation, and other forms of discrete and precise computational terms.

There are two types of digital authorship that subjects encounter as therapeutic due to their ability to engender the objective and analytical perspective. These are a) digital media design as the generation and implementation of digital art assets, i.e., audiovisual graphics such as game player characters, animations, images, background tiles, and so forth¹¹⁵; and b)

¹¹⁴ For more on this technique, see previous section discussing Sam’s use of ray casting in his game.

¹¹⁵ It is worth noting that some subjects retrieved some of their digital art assets from online sources and digital art libraries. These assets may be modified via image editing tools such as Photoshop and then implemented in their games. Other subjects generated their graphics ‘from scratch’. Some may ostensibly combine both practices.

programming and scripting¹¹⁶, which generate the algorithms, rules, procedures, formal constraints, and computational processes of the computer game qua functional software program. Programming is also used to generate digital processes that operationalize visual graphics via computational processes, such as procedurally-generated graphics automatically generated by an algorithm that is implemented and/or authored by the designer.

It is important to emphasize that digitality plays a role in a wide array of praxes in computer game design. Sam correctly framed the act of mental deliberation that AGD logopoiesis engenders both as a praxis for authoring data (e.g., images, animations, characters) and for authoring processes (e.g., algorithms, rules, formal constraints):

In the process of game design, you have to identify these little elements that can help you communicate that experience and whether that's a certain word that your co-worker chooses to use in dialogue or whether that's a sound, or whether that's a particular way you draw your character, all of these things are both a deliberate choice on the part of the developer and are subtle cues that communicate an experience more effectively.

Sam thus accounts for the detail-driven authorship of computer game design processes, including the generation of character dialogue, the style of drawing or modelling a game character (i.e., more 'realistic', cartoony, or whimsical), or the implementation of audio (a specific sound effect) as important to communicating the autopathographical narrative. These are the minute details that a developer must discern when developing digital systems.

Sam's point also speaks to the idea of differentiating between "data-intensive systems" and "process-intensive systems", to use game studies scholar Noah Wardrip-Fruin's (2009) helpful distinction of computer games. Data-intensive systems are those videogames that privilege the authorship of the audiovisual elements of a game. Wardrip-Fruin cites game designer Greg Costikyan who claims that 80% of labour and cost for a videogame is in the

¹¹⁶ Many subjects used scripting languages such as JavaScript to write their software code. Computer scientist Ronald P. Loui (2008) suggests that there are technical yet meaningful differences between a programming language and a scripting language; among them is the latter's ease-of-use for the designer in *rapid prototyping* contexts. Also, scripting languages abstract information at a much higher level in order to allow the designer to type shorter code. The term 'programming' is used here as a general term to denote *programming practice* or *coding practice* that includes the use of programming and scripting languages. See <http://www.cse.wustl.edu/~loui/praiseieee.html> for more.

creation of visual assets and graphics. Process-intensive systems, in contrast, privilege the authorship of the processual or “operational logics” of a game, such as algorithmically generated graphics, e.g., procedural content generation; rules; mechanics; “spatial logics”, e.g., collision detection, navigation in the game, and so forth (Wardrip-Fruin, 2009). Wardrip-Fruin highlights the aesthetic importance of the operational logics of computer games and interactive narratives by calling them “expressive processes”. Process-intensive systems are often taken for granted in game authorship, he suggests. Wardrip-Fruin argues that game designers and interactive storytellers ought to focus more on authoring process-intensive game systems in order to maximize creative experimentation with games as a ludic art form. Subjects emphasize the values of authoring both types of systems in engendering a third-person perspective, fostering self-understanding and promoting relief.

Vicky voiced the therapeutic benefits of designing the first type of system, the data-intensive system, in the detail-oriented digital design of graphics, namely the stress-relieving and calming powers of this digital praxis, which engenders the third-person perspective due to the precision involved in “working with the pixels”. The detail-intensive dimensions of digital authorship “quiets her (overactive) brain”. In fact, her experiences of designing “funny and cute” pixel art game player characters during the AGJ are described as “the most meaningful”, precisely due to the detail-oriented, precise nature of this activity:

What I found the most meaningful is the fact of working with the pixels and working in such a small form that is so exact. You’re zoomed in closely at the fine details and you’re creating a thing. To me, it was very absorbing. It sucked me in where my brain was quiet and I was just focused on the act of what I was doing. That doesn’t happen for me very often at all.

So, I found that to be very exciting and revealing and I realized that I like this. My brain quiets down while I do this and I can focus in on these tiny details and it can encompass me to the point that I’m not as anxious or prone to stress as much. It’s very difficult for me to find activities that do that.

Phenomenologically speaking, “zooming closely at the fine details” felt “exciting” and “revealing” to Vicky, which she said “sucks (her) in” and “encompasses” her. The ability to stand back and calm down by intensely focusing on the minute details of digital making is pleasantly absorbing and whisks her away from stressful thoughts and rumination. The detail-oriented nature of digital design immerses Vicky in the cool third-person perspective, imparting

some form of insight—as she said, she found the praxis “revealing”, although she did not state what of in particular. This acute attention to and preoccupation with digital details in a precise and exacting way, which engenders a more cool and detached standpoint soothes her over-stressed mind. Vicky’s experience of digital design praxis also evokes pointillistic methods. Pointillism as an artistic technique of formalization “emulates scientific objectivity” (p. 263) according to Maleuvre (2016)¹¹⁷. Hence digital design praxis such as the one described by Vicky bears the following similarity to pointillistic praxis—a concern for *pixels* in digital design methods can be compared to a concern for *points* in pointillistic methods in drawing and painting.

7.6 AGD logopoiesis: the therapeutic role of programming praxis

AGD as craft involves a process of formalizing the autopathographical narrative via a programming language. Via programming, AGD designers practice self-care by disengaging from disquieting emotions and alternatively, finding deliberate and measured ways to work with the discomfort of mental illness, trauma, and disability experiences. Several subjects including John, Alex, Fred, and Ellie emphasized the therapeutic effects of programming praxis. John explained why programming is therapeutic in general (irrespective of AGD):

When I’m thinking really intently upon a problem in programming, I find it kind of soothing and relaxing and it takes me away from thinking about emotional problems.

Programming allows John to escape his own subjective inner states, which can be rife with emotional turmoil. Programming is also an effective distancing device that takes John away from the vagaries of emotions. The problem-solving nature of programming praxis not only distances, but also soothes calamitous emotions. Brody echoes this feeling by stating that programming as a form of problem-solving is much more engaging than writing:

The reason is that I am not a writer, writing does not engage me. There’s not problems to be solved in the same way like there’s problems in programming.

¹¹⁷ In “The Art of Civilization: A Bourgeois History” Didier Maleuvre (2016) points out that Pointillism, Cubism and Constructivism in art “emulated scientific objectivity” (p. 263).

In other words, programming qua composing actions and behaviors differentiates computer game design from writing a book:

The constraints within the game really open it up beyond what the creative constraints on a book are. You're only working with words (when writing a book), where in a game, you're working with graphics as well so they can interact. So, you're working with behavior.

Moreover, the nature of programming as a measured, calculated, and deliberate praxis invites subjects to practice mindfulness when rendering their autopathographical narratives in the computer game, which Fred characterized as a "Zen garden". For example, Fred spoke of the therapeutic effects of algorithm design, both as a general practice and in the context of AGD and the Autopathographical Game Jam:

I do think there is a way to process difficult emotions through the technical aspects and I think that's honestly a major characteristic of my experience with the game jam.

Once you understand the process of coding and are able to create original algorithms, the process of coding can be quite meditative. While writing the code, I personally reach a state where I focus on nothing but the math I'm creating. I'd say it's comparable to maintaining a Zen garden. You enter with a design, you know what you want it too look like, and you have to figure out how to bridge the two in order to get the result you want. At least for me this process has helped me deal with difficult emotions by placing me into an analytic head space which allowed me to work out my emotions with a new perspective.

For Fred, the process of designing algorithms appears to engender analytical self-insight. Fred deployed AGD towards an understanding of how he healed himself from compounded psychological trauma. Fred suggests that the algorithmic thinking (a.k.a. computational thinking) involved in computer game design helps to shift his perception of his own tumultuous emotions into the analytical, third-person perspective, which can, in turn, manage and/or resolve those emotions. Furthermore, he compares the practice of writing algorithms in computer game design, an acquired skill, to maintaining a sense of emotional harmony and balance—a type of mental 'Zen garden'. That is, the praxis of writing algorithms feels meditative to him. For Fred, "using game development as a form of coping with emotions", "self-therapy", and "DIY therapy" points to how programming, algorithm design, and similar formalization processes involved in computer game design can be re-imagined as therapeutic tools.

To assess how and why programming languages may feel therapeutic to AGD designers, consider what they are and what they do. “A *programming language* is an artificial language with its own rules of syntax, used for expressing programs” (Denning and Martell, 2015, p. 84). Programming languages thus afford the composition of actions into computational processes (Dasgupta, 2016) that a computer can read and execute. Programming languages are a combination of linguistic and notational form; they are a hybrid of natural and formal languages: “both natural languages and formal languages were taken as models for programming languages” (Priestley, 2013, p. 186). Programming is a precise art, an art of disambiguation intended for the computer to parse. Denning and Martell define a computer program as “an expression of an algorithm, encoded for execution on a machine” (p. 84), or more precisely:

A series of instructions arranged in a precise way to cause the machine to evaluate the desired function. Instructions and data are encoded as binary patterns stored in a memory. When fetched into a processor, instructions cause the hardware to transform input data into output data. (p. 60)

As such, programming in computer game design involves processes of formalization, which consist in determining the autopathographical narrative in precise terms such as notational and mathematical structures and adhering to a set of standards to ensure the functionality of the software program. A formalized structure is one whose elements, components, and properties are defined in a taxonomy of parameters and rules. As such, programmers formalize processes which are evolving, continuous, and indeterminate in their lived and concrete dimensions, such as first-person experiences, into a set of discrete procedures and rules encoded for execution by a computer and as a problem to be solved mechanically.

In *Notation as a Tool of Thought*, computer scientist Kenneth E. Iverson (1979) asserts that “programming languages, because they were designed for the purpose of directing computers, offer important advantages as tools of thought. Not only are they universal (general purpose), but they are also executable and unambiguous”. He adds that “the lack of ambiguity (offered by programming languages) makes possible precise thought experiments” (p. 445). Given that, according to computer scientist Edsger Dijkstra, “programming is one of the most difficult branches of applied mathematics” (as cited in Denning and Martell, 2015, p. 84). Whitehead (1911; 2017) has the following to say about the practical and intellectual use of

symbols in mathematics or notation proper, without which we would have to write in full the mathematical equations using words—a cumbersome and unwieldy process:

In mathematics...the symbolism is invariably an immense simplification...For it represents an analysis of the ideas of the subject and an almost pictorial representation of their relations to each other. (p. 33)

Iverson writes: “part of the suggestive power of a (notational or programming) language resides in the ability to represent identities in brief, general, and easily remembered forms”. He explains that, “as (Charles) Babbage (has) remarked...brevity facilitates reasoning. Brevity is achieved by subordinating detail...” (p. 447). Whitehead (1911; 2017) similarly writes that “relieving the brain of all unnecessary work, a good notation sets it free to concentrate on more advanced problems” (p. 32).

Formalization might allow subjects to ‘disambiguate’ the autopathographical narrative into discrete code, a distancing technique which can feel therapeutic. Simply put, formalizing the autopathographical narrative may engender organized and ordered thinking. As mentioned, distilling the autopathographical narrative into computational expressions might allow subjects to quell uncomfortable emotions, disambiguate vague and indeterminate emotions, remove a level of uncertainty, and gain a more focused self-perspective. As a tool for rendering transparent the emotional opacity of anxiety and depression, programming as a logical and digital praxis may aid AGD designers to view themselves in a new light—in discrete numbers and values, hexadecimal code, if/else statements, and so forth.

Ellie offered a strikingly candid and rational description of why writing software code is superior to writing poetry for individuals struggling with mental health issues. Writing software code satisfies her objectives of creating functionality and usability, which gives her a feeling of accomplishment:

Writing code is more therapeutic than writing poetry because writing poetry is so incredibly difficult. You have to be so incredibly honest when you’re doing it and when you read it afterwards you’re going to get that uncomfortable feeling of touching on deeply personal subjects. Writing code, you get the satisfaction of, first of all, the code is meaningless in terms of human emotion, second of all, you get the satisfaction of having solved a problem...but when you do that, you just get that little boost.

Poetry is almost taking something away from you whereas writing code is almost adding something to the world that can be used.

There's just a feeling of purpose there...I really do feel like creative work is much more draining than technical work.

Ellie first framed therapeutics in relation to self-actualization, which seemed more important to her than healing emotional pain. She found that “technical work” such as writing code distances her from emotional pain in ways that writing poetry, which facilitates a direct and raw encounter with emotions, does not. Ellie invoked poetry's semiotic properties, i.e., the cultural norms that frame writing poetry as a deeply personal artistic praxis, while the semiotic properties of writing code frame this praxis through objective, third-person terminology. Ellie affirmed that “the code is meaningless in terms of human emotion”, suggesting that poetry reflects intimate first-person experiences whereas code does not. In other words, we can infer from her testimony (also suggested by Cole) that software code, unlike poetry, may not reflect the author's emotions; in fact, it may deflect the author's emotions because software code is comprised of symbol structures and a set of discrete computational expressions that are experienced by some subjects as objective and ‘devoid’ of emotions.

There is deeper nuance implicit in Ellie's rationale, which frames the instrumentality of programming as a technical form of ‘making’ towards satisfying practical and universal human needs of both the maker and the audience engaging with the author's work. Ellie suggests that writing software code might be therapeutic because it has a pragmatic dimension that supports the author's self-actualization through a recognition of its ability to engender functionality and usability in its service to others. She demotes writing poetry from its familiar pedestal of expressing emotions, and promotes writing code as a more practical approach for self-actualization.

Ellie praises the merits of programming for whisking the coder away from the calamities of emotional distress, a good invocation of programming's ability to foster the third-person perspective. Emotional calamities are likely to arouse pain and discomfort in the poet and immobilize them from action, making, and poesis. Programming invites the opposite from poetry, galvanizing the programmer to create a functional software program by “adding

something to the world that can be used”, which has a “feeling of purpose”. Writing software code also engenders a feeling of self-satisfaction and accomplishment for having “solved a problem”. Ellie suggests that problem-solving engages practical problems that carry significance for humans. She thus implicitly defends the fruitful aspects of implementing code into a software program executable by a computer, one that can be used by both computer and human in contributing to the world as tools of play, communication, creative expression, and so forth. And what a “boost” it is to be the author of those tools that give meaning to the everyday lives of human beings. Programming is restorative for the coder on account of its ability to engender movement and praxis (‘doing’); additionally, programming praxis helps others lead more meaningful lives through technology. The functional purpose of coding in giving back to the world in utilitarian and pragmatic ways eclipses the aesthetic objectives of poetry, which may further shore up emotional instability for the poet.

Ellie qualified her opposing experiences with programming and poetry with more detail. She explained how her life “made sense” when she moved into science and technology from the fine arts (she has training in the visual arts). Her self-description as a “starving and depressed artist” is a reference to the economic disparity between the sciences and the fine arts and the general instability of the latter as a personally satisfying and economically viable professional practice:

As soon as I moved over into science and technology, I found my life made a lot more sense. Because it wasn’t driven by emotion, it was driven by purpose and function, which I think is in itself therapeutic— if you’re suffering from (mental health issues). The stereotypical artist is the starving, depressed artist and I’ve been that person before. Do I prefer my life as an analytical science person? Absolutely, because I have a lot of things to keep my mind busy and stimulated and gives me a purpose. Now, I find things like art extremely draining on the soul.

Calling creative work and art “draining” and “driven by emotion” is a sensible explanation given the phenomenon of catharsis that may apply to some forms of artistic practice such as poetry writing, which can ostensibly become exhausting when poets engage with intense emotions. But also, Ellie speaks from direct experience with the creative industry and art world as economically unstable and emotionally volatile for the artist. The wage income of visual artists is low not only

in relation to software programmers, whose salary is considerably higher¹¹⁸, but to most other professions¹¹⁹. Ellie emphasized the income disparity between these two professions, but she also conveyed the difference in first- and third-person perspective that science/technology and the fine arts produce. In other words, the field of science and technology, of which software programming is one profession, enables Ellie to be an “analytical science person” and engage the objective standpoint, whereas the economic volatility of the fine arts profession engenders emotional turmoil, which is burdensome to the artist’s sense of self, i.e., “extremely draining on the soul”.

Ellie later modified her initial statement that writing code is a form of escaping difficult emotions to state that AGD is a beneficial therapeutic tool:

I think (autopathographical game design) is a very good way of therapeutically working through some problems that you might have, but also in a kind of like— sneaky (way) maybe? You can let your own sense of humour or light spin, imaginative spin on things. In that way, it also becomes helpful to you (the designer) and helpful to the game player.

AGD is a furtive method for engaging the emotionally volatile first-person perspective and “working through some problems”, precisely because it allows AGD designers to abstract the emotional calamity from their problems into the emotion-free symbol structures of software code in order to think through the structure of the problem. The *Twine* software application affords a form of game authorship that is primarily text-based using prose that is often structured and formalized through HTML, CSS, and Javascript. In its hybrid use of natural language and markup/programming languages, authoring an interactive narrative with *Twine* is the epitome of

¹¹⁸ “A Software Developer (in Canada) earns an average salary of C\$62,375 per year.”
http://www.payscale.com/research/CA/Job=Software_Developer/Salary Retrieved July 29, 2017.

¹¹⁹ “The average annual income for artists overall is \$23,500, or 26 percent lower than the average earnings for the labour force as a whole, despite the fact that over 40 percent of artists have completed university degrees, certificates or diplomas, compared with 22 percent for the overall labour force”. The average earnings of painters, sculptors and other visual artists is \$18,700 according to Statistics Canada, which is close to the cut-off point for being considered in the low-income bracket. Please see Neil & Neil Craig Associates (2010) for more:
<http://ccarts.ca/wp-content/uploads/2010/10/StatusoftheArtistReport1126101-Copy.pdf>

logos. *Twine* mobilizes both definitions of *logos* as [word, discourse, narrative] and [computation, reckoning]. Ellie confirmed that strong connection of *logos* to her text-based *Twine* game autopathography *This Is Your Awesome Sleep Machine*: “mostly, my game is an intellectual experience because it’s text”.

Finally, refactoring the software code is a sub-praxis of programming that some subjects find effective in the totality of the therapoetics of programming praxis. Refactoring is a process of organizing and streamlining your software code to ensure maximum code readability and clarity serving both the developer who has written the code and other developers who may work with the code in future iterations of the software application or other applications (i.e. code extensibility). Refactoring is a form of code maintenance. Fowler (1999) defines refactoring:

Refactoring is the process of changing a software system in such a way that it does not alter the external behaviour of the code yet improves its internal structure. It is a disciplined way to clean up code that minimizes the chances of introducing bugs. In essence when you refactor you are improving the design of the code after it has been written (p. xvi).

Alex stated that refactoring boosts his confidence and adds to his sense of accomplishment:

There is a lot of personal anxiety about having unfinished projects and having bad code. When I go back to a project and I can’t read it and I don’t know that moment of where do I go from here, refactoring definitely helps to process and make you feel more confident that I can finish this.

Having “bad code” for Alex signifies a project that may not be a fully functioning or workable software program. In this incomplete and unfinished stage, the project adds anxiety for Alex because he equates not being able to read software code to a state of mental disorganization. Hence, refactoring the code provides a feeling of order and spurs productivity for game designers whose mental states impact their ability to see a project to its completion. Refactoring praxis is a meaningful process for Alex because organizing your software code and making it modular and reusable for another project feels productive in both the short-term and long-term realm.

7.6.1 The role of logic in programming the autopathographical game

As a process of formalization, programming the autopathographical game invites a disambiguation of complex first-person experiences into a set of discrete computational

expressions. This builds a strong form of the third-person perspective—the objective observer position, which feels therapeutic for subjects. Programming the autopathographical game and the imaginary construct of the fabulized self requires the use of logical reasoning, a key ingredient engendering therapy. To assess the reasons behind logic’s therapeutic properties in AGD and computer game design in general, consider some study subject testimony.

Alex also testified to the role of logic in computer game design as a confidence boost for game designers struggling with mental illness. Alex, who has bipolar disorder, explained that programming praxis is a compelling epistemic tool for self-knowledge and self-actualization precisely due to its strong logical requirements:

I definitely learned a lot about myself when I started making games just as far as what I’m capable of, more on the doing side, less on the feeling side. Like on the left brain it felt great to be capable of doing these logical things in programming.

Ironically, Alex asserted that engaging the logical side or “the left brain” of programming praxis “felt great”¹²⁰. Furthermore, logical reasoning in the context of computer game software development endows him with a sense of accomplishment and productivity, which is “more on the doing side”, as well as the ability to modulate affect and emotion, which is “less on the feeling side”. He indicated that the logical reasoning involved in programming praxis boosts his self-confidence and self-assurance. Logical reasoning via programming praxis boosts the self-confidence and self-assurance of game designers with mental health issues whose tumultuous emotional experiences might derail their sense of well-being and damage their self-confidence. The ability to feel that he is capable of logical reasoning, despite experiencing severe mental health issues, is a huge asset for Alex in this regard, given that individuals with mental health problems are stigmatized as not being capable of logical and reasoned thinking. Computer game design as a logical praxis—a *logical making* (logopoiesis) is a form of self-actualization for Alex because it engenders in him feelings of capability and a sense of achievement.

¹²⁰ Alex *metaphorically* differentiates between the left-side (logical, analytical) brain and the right-side (creative, artistic) brain. This conception of the brain is common in popular culture but is otherwise a myth that science has debunked.

The therapeutic effects of logical thinking go further. Consider Alex's next statement, which qualifies how and why the role of logic is therapeutic in programming praxis:

I'm spending my energy on being frustrated about something valuable. I'm frustrated about an actual, tangible thing instead of this existential problem that I have, you know. I'm not worried about what I'm doing with my life, I'm worried about why doesn't this box turn green when I click this button, you know (laughs). And it's more valuable to be frustrated about something that you can fix, you know (laughs), or fix immediately because you can really easily get wrapped up in all these giant things. Doing logical left-brain stuff is really good...it's not as silly to process all of the big stuff but just to reground yourself on the real world.

The crux of Alex's testimony is tying the therapeutic role of logic to programming praxis, that is, ensuring the functionality of the computer game as a tangible object. Engaging the logical side of his brain in programming praxis connects to being "frustrated" about immediate and practical as opposed to existential concerns, which "reground" Alex "on the real world". Alex essentially claims that the deployment of logic in computer game and software development affords the designer the ability to bypass the existential angst that is part of mental illness by focusing on the material properties of software development, such as a concern for whether a button will turn green or not. This concern for finding applied solutions to pressing material problems feels restorative and therapeutic because it is solving immediate and practical 'real-world' problems. It is also helpful to turn to Denning and Martell's (2015) invocation of programming praxis as an intrinsically material praxis for solving problems through the formulation of step-by-step procedures and algorithms, framed as a general two-step process dependent on a physical machine:

Programming is the practice of encoding algorithms for execution on a machine, (consisting of) two parts:

1. Design an expression of a solution method for a problem and encode it for a machine.
2. Verify that a machine controlled by the program properly solves the problem. (p. 86)

Programming praxis thus ensures that software is designed according to a set of computing principles in order to 'run' on a physical computer (hardware system) that an end-user can successfully interact with. The direct engagement afforded by the materiality and applied nature of software development imparts a sense of rootedness for Alex because he can dwell in and

focus on the present moment as opposed to turning to endless and unproductive anxious rumination.

Consider also Cole's testimony, which gives voice to the therapeutic role of logic in programming praxis. Cole stated that for him, "coding can be very therapeutic", giving the following explanation as to the reasons behind programming's therapeutic effects:

Well, it's logical. We as humans strive to be logical but we're just so much more irrational than a computer...

For some people, it could be a very therapeutic process. There's a logical structure to (computer game design) and that's OK, that's predictable, it's not like a human, I don't have to deal with its emotions. So, you get the problem is only in myself because I can't get it to work and that's where it would become therapeutic...

Cole's testimony suggests that the logical nature of programming is a helpful tool in autopathographical game design as a way to deflect the designer's own calamitous emotions. In other words, unlike humans who are capable of expressing emotion fully, irrationally, and sometimes messily, software code does not directly express emotions, nor does it reflect emotions given the strong role logic plays in programming praxis. In its notational form, software code may be too symbolically abstract to be able to conjure affect, inviting programmers to take distance from their own emotions.

In order to understand the therapeutic role of logic in programming praxis for Cole and other subjects, consider first the role of logic in a brief selection of domains and applications. First of all, logical reasoning plays a constitutive role in programming praxis, which involves engineering computer games as digital objects and rule-based formal systems. Logic is intrinsically connected to computing: "a computing machine is a logic machine" (Davis, 2000). Logic connects to procedurality in several ways. Logic itself is a set of procedures governed by rules that impose order and structure on content; logic also manifests in processes operationalized via formal rules, such as in computer games. Logic is thus constitutive of games in a variety of ways. Philosopher Johan van Benthem (2014) claims that logic and games are ontologically connected. He offers two forms in which they are connected—"logic in games" and "logic as games"—suggesting that these two forms instantiate in ways such as the

deployment of strategic forms of reasoning in playing a game in the former and the use of “basic reasoning activities” in the latter (p. 2).

Logic is a conceptual tool. Logical reasoning evaluates a proposition as either true or false through an assessment of the connection or relationship of the premises to the conclusion (it does not assess the content of statements in an argument, only its form). “Logic is the study of the methods and principles used to distinguish good (correct) from bad (incorrect) reasoning” (Copi and Cohen, 1990, p. 3). Logical reasoning is an axiomatic tool for proving validity, and is thus a timeless and universal method despite its origins in Western thought. Furthermore, logic is a constitutive element of objectivity and the third-person perspective. Hacking (1979) writes that according to Frege, “laws of logic are laws of thought, not in the sense that they are laws about what men do in fact take for true but because they are ‘the most general laws, which prescribe universally the way in which one ought to think if one is to think at all’” (p. 288). Logic is the basis for the scientific method, philosophy, and language itself, allowing humans to form sentences in order to articulate their thoughts and ideas coherently. “Logic is the science of order” (Langer, 1967).

We can also view logic as a tool for regulating human thought processes. In his famous treatise, *An Investigation of the Laws of Thought*, logician George Boole (1854; 2005) connects the enterprise of logic to a form of ruled thinking. Boole suggests that humans must continually wrestle with psychological and emotional “laws of interference”:

The mathematical laws of reasoning are, properly speaking, the laws of *right* reasoning only, and their actual transgression is a perpetually recurring phenomenon (p. 317).

Boole suggests that logic is not always easy to practice due to our irrepressible fallibility as human beings living chaotic lives, feeling intense emotions, making errors, and falling ill. Sometimes humans implement logic with difficulty and great toil. Philosopher Michael Maurer (2010) suggests that according to Wittgenstein, logic is not easy to implement in natural language. Wittgenstein considered language the philosopher’s *Achilles heel*, ensnaring them in conceptual confusion, and hence saw philosophical praxis as the remedy or cure for achieving self-clarity:

Philosophical therapy is, in most cases, self-therapy. We are constantly in struggle with ourselves and our own inclination to transcend the limits of what can be uttered by language and still make sense...(Maurer, 2010, p. 3).

Maurer adds that the philosopher's task according to Wittgenstein is ultimately self-therapeutic: to disentangle the "spider web of our confused ideas" in order to communicate philosophical ideas with clarity and coherence.

In his book *The Puzzle Instinct*, Marcel Danesi (2002) explains the allure of logic to Enlightenment philosophers as a solution to human lawlessness and disorder for the likes of Thomas Hobbes, "who claimed that logic was the only attribute that kept human beings from regressing to wild beasts...and Descartes (who) also believed that logic was the only way to solve all human problems" (p. 141). This idealistic view of logic as a panacea for human disorder has been critiqued. In his book *Computer Power and Human Reason: From Judgment to Calculation*, Joseph Weizenbaum (1976) suggests that logic qua instrumental reason can be dangerous and detrimental for humanity unless supplemented with empathy, love, morality, compassion, and other similar intangible human qualities to offset its pitfalls. Boole is also careful to state that knowledge of correct logical reasoning is not the most important property of intellectual praxis, which also includes the perception of beauty and morality, emotion, and affection: "as truly, therefore, as the cultivation of the mathematical or deductive faculty is a part of intellectual discipline, so truly is it only a part" (p. 327).

As previously discussed, logic as *narrative logos* (narrative plot, action, discourse) is a primary element of the three-act structure of Western drama. In the *Poetics*, Aristotle argues that "most important of all is the structure of the incidents" (p. 62), invoking the causal and thus logical rules of dramatic action and plot. In his introduction to the *Poetics* (2006), translator Joe Sachs suggests that plot means the totality of narrative in Aristotle's terms:

The word 'plot' may suggest a skeletal framework of events (when Aristotle speaks of such a framework in Ch. 17, he calls it a *logos*) onto which a poet can impose an illusion of life, but stories are genuine wholes that already have a life of their own; this is precisely what Aristotle means when he says that the story is like the soul of the tragedy (1450a 38-39) (p. 4).

But consider a different interpretation. By assigning structural and hence logical power to drama

when he says that, “the Plot...is the first principle, and as it were, the soul of the tragedy”, Aristotle suggests that the plot delineates the dramatic action, giving form to a narrative much like the “chalk outline of a portrait” (p. 63). In analogizing plot to a “chalk outline”, Aristotle points to *logos* as that which undergirds narrative, delineating form as the crucial element of drama, which *builds* dramatic tension if the incidents are “arranged” correctly. In other words, the logical components are the invisible building blocks of drama and narrative. They may not be perceptible due to the rich layering of characters, pathos, performances, rhetorical, and visual style (e.g., décor and art direction) that constitutes the totality of narrative, but they nonetheless exist underneath this totality. To strip away those layers, as narratologist Vladimir Propp (1928;1968) does in his *Morphology of the Folktale*, the logical building blocks of narrative—character functions in Propp’s case—structure the dramatic action. “The basic unit, the narrative atom, is still the *function*, applied as in Propp, to actions and events which, when grouped in sequences, generate the narrative” (Bremond and Cancalon, 1980, p. 387). To distill the “arrangement of incidents” to discrete abstract symbols, we would still be left with the structure of the plot of narrative, which Aristotle uses the metaphor of a “chalk outline of a portrait” to suggest is immediately perceptible in its form.

Jerome Bruner (1986) states that “there is a heartlessness to logic” that does not apply to narrative contexts where endings may be more nuanced and complex than logic would allow:

There is a heartlessness to logic: one goes where one’s premises and conclusions and observations take one, give or take some of the blindnesses that even logicians are prone to...Paul Ricoeur argues that narrative is built upon concern for the human condition: stories reach sad or comic or absurd denouements, while theoretical arguments are simply conclusive or inconclusive” (p. 13-14).

Bruner alludes to the inherent binary nature of logic, the ‘conclusive/inconclusive’ and ‘true/false’ outcomes of logical deductive reasoning. Bruner adds that “the scientific...and logical stance...yields accounts of experience that are replicable, interpersonally amenable to calibration and easy correction. But much of experience is not of this order” (p. 110). Bruner’s last sentence invokes precisely why AGD subjects experience logical reasoning as therapeutic, namely that logic is their tool for abstracting the complexity of first-person experiences with illness, trauma, and disability into “symbol structures” (Newell and Simon, 1976)—a set of

computable symbols, rules, and formal constraints in the autopathographical game. At the phenomenological level, this process of simplification and abstraction in AGD appears to eliminate levels of uncertainty that designers may struggle with every day. The strong role of logic in programming praxis offers a respite from emotional adversity and confusion that, phenomenologically speaking, feels restorative.

7.7 AGD logopoiesis: the therapeutic role of procedurality

Consider next the specifically therapeutic role of procedurality in autopathographical game design. For example, Fred’s autopathographical puzzle game *Brain Buster*, which is inspired by his traumatic experiences, represents the “articulation of ideas through (rule-based) game systems”. Fred summarized his game concept in the following way:

The whole idea is that I wanted to mechanically represent how I got over my PTSD through talking to people over a phone.

Fred stated that the puzzle game genre “encapsulates perfectly” the steps of how he systematically overcame the emotional distress of his incurred trauma:

I chose a puzzle game because I tried to represent apprehension with my game. And I felt that the puzzle—the player would be given an audio log after they completed a puzzle and the challenge presented within the puzzle, I abstracted that idea as the challenge that I suffered when deciding whether or not I should call a friend. I felt that the format of a puzzle game would encapsulate that perfectly because you have to use logical reasoning and deduction and like, ‘do I actually want to do this, what will happen if this person doesn’t (respond to my call)?

His goal was to represent the procedural steps of how his communication with friends over the phone facilitated his healing. When he would call for emotional support, some of his friends returned his call and others did not. Fred mapped out the emotional support he sought as a step-by-step plan with player outcomes that mirrored his personal experiences at the time of his acquired emotional trauma—a friend responding to his request for help or not responding to that request. Through the design of a puzzle game, Fred attempted to analyze his deliberation process of “whether or not” to call one friend or another friend for support, and the possible outcomes of each if/else option, through the procedural properties of the game, i.e., the system of game rules. He used AGD as a way to look back, review, and better understand his healing process.

We can understand Fred's emphasis on the importance of rules in his puzzle game by reviewing some general definitions and properties of rules, such as their conditional nature. A rule is a general concept denoting a procedure, axiom, principle, governance, pattern, and/or law, depending on usage and context. As in Fred's attempt to discover the conditional outcome behind requesting emotional support from a friend, a rule is always *conditional*: if you perform action X, then Y occurs; if you fail to perform action X, then Z occurs. Rules in game design are unambiguous because they must be formally and strictly defined and they give games structure (Juul, 2011, p. 58). Juul suggests that in game design, we can implement the condition of the rule (using a programming language, for example), but we cannot represent the rule itself, which is realized in use. An example of the condition that defines a rule is the following Boolean if/else statement:

```
If (boolean condition) Then
    (consequent)
Else
    (alternative)
End If
```

The above rule is purely static as written on this page. But feed it to a computer using a programming language and it is executable as an interaction between the computer program and the user. As Dasgupta says, "a program requires a material medium (physical computer) for it to work" (Dasgupta, 2016, p. 63). As such, rules are constraints implemented in a given context such as a computer system for (human and non-human) agents to interact with. Rules as such become dynamic when humans interact with those rules in countless ways, e.g., using the rules of language grammar in every day conversation, respecting road rules to avoid an accident, abiding by state laws to maintain civil order and thus preventing harm or death to citizens (i.e. Hobbes' social contract), and following game rules to win the game. To break the rules is to deviate from the acceptable pattern and breach the commitment of the interaction. In a game, breaking the rules means breaking the procedural commitment between the game player and the game and inducing a negative consequence, penalty, or loss.

A rule is also a combination. An example of a combination is a syllogism (Berge, 1986), which is governed by rules. And "rules can be regarded as statements of mathematical logic"

(Abelson, Sussman and Sussman, 1996). Rules and constraints propel creativity according to writers who employ the French experimental writing technique of *Oulipo* (*L'Ouvroir de littérature potentielle*) invented in 1960. Oulipian poetics deploy formal constraints and combinatory play to mathematize narratives and creatively experiment with the formal and semiotic properties of literature. Oulipo is also a form of “ludic literature” according to Motte, Jr. (1986). “The Oulipian text is quite explicitly offered as a game, as a system of ludic exchange between author and reader” (Motte Jr., 1986, p. 20). Oulipo mathematizes narratives in order to create multiple possibilities for the reader. To mathematize means to employ mathematical logic, a form of symbolic logic whose symbols are formally expressible in mathematical terms.

In fact, Oulipo is not dissimilar from the text-based interactive narratives of *Twine*, except that the former did not make use of computer technology to implement their narratives. One of the founders of Oulipo, Raymond Queneau, suggests that Oulipo is a computational form of literature that could have been usefully aided by machines: “we regret having no access to machines” (as cited in Motte Jr., 1986). Queneau affirms, “the classical playwright who writes his tragedy observing a certain number of familiar rules is freer than the poet who writes that which comes into his head and who is the slave of other rules of which he is ignorant” (as cited in Bénabou, 1986, p. 35).

Thus, a set of game rules are static until a player dynamically engages with them in a system of cause and effect. Procedurality, i.e., the rule-based formal properties of games that engender its temporality, allows the game player to make a choice in the game and experience and feel the effects of that choice. The player thus embodies a game character via the rule-based system of a game. Wardrip-Fruin (2009) observes that for the player/user, formal constraints in interactive software systems can limit freedoms yet simultaneously allow for emancipatory experiences. He adds that even “limited expressive spaces” can be used in multifold ways, some of them liberating. As games scholar Graeme Kirkpatrick (2011) claims, videogames are closer to dance because they supply a “choreographic script for player actions”. This choreographic script is composed from the game rules and formal constraints that comprise gameplay. Nathan explained that people who play autopathographical games for the first time can say:

I’ve never had to make that choice in my life before now, now I know what it’s like to

make that choice. So, that's how (a game) will express a complex process (such as the first-person experience with illness, trauma, and disability).

Alternatively, the game can evoke what it is like not to be allowed to make a choice in one's life by withdrawing choice for the player. Therefore, designing game rules means either giving the player a choice or taking the choice away from the player, both equally compelling strategies depending on the narrative the autopathographical game is trying to tell and/or the autopathographical game designer's intentions.

7.7.1 Authoring game rules in AGD: towards symbolic overcoming and self-understanding

In AGD, the game designer authors and implements game rules and formal constraints in order to symbolically overcome their own challenges imposed by mental illness, emotional trauma, and disability, and create an inhabitable virtual world that communicates to the game player their first-person experiences. Rule-design in AGD also allows designers to create gameplay experiences that bring pleasure and delight to the designer in envisioning an array of choices and possibilities for the player, as well as lack of choices, which all structure gameplay.

Since Jesper Juul's (2011) assertion that gameplay is "mostly about having power and overcoming challenges", computer games have diversified in style and form, inviting unheard of forms of creative experimentation such as autopathographical game design. Consequently, not all games are competition-driven and reward-driven, such as several of the game autopathographies of this study, including Sam's game *Anxiety Corp.* and John's game about bipolar disorder and assisted suicide, *Untitled*, both of which have lose-lose outcomes, and Nathan's "de-motivational" game *ADHD Hero*. Despite not offering explicit forms of *agon* (competitive gameplay), positive rewards, and a clear-cut win-lose outcome, all of these games implement formal constraints and implicit rules that engage the player in an interactive experience of play. Juul also speaks to the pleasure that the game player derives from the game rules. He states that experienced game players disregard both the fictional (narrative) space of the game and the game's visual graphics in favour of the game as a system of rules (p. 139). For some players, engaging with game rules is a formative experience of gameplay and much more pleasurable than engaging with the game narrative or the game's evocative or high-production visual

graphics. Juul affirms that “rules are the most consistent source of player enjoyment in games” (p. 55). Galloway (2006) supports Juul’s claims by stating that “visual imagery is not what makes video games special” (p. 125).

John explains why designing the game rules in *Untitled*, his game autopathography on bipolar disorder and assisted suicide, gives him the *most* pleasure and satisfaction:

That’s actually *the most fun part for me* (emphasis added), is brainstorming and thinking of ways to get the person to think the way you’re thinking through game rules, through game mechanics, because it’s fun to try and think of how you could lure somebody into thinking the same way you are— you can’t put them in your shoes but you want to try and lure them into a similar situation psychologically, and that’s sort of a fun thing to think about.

Similar to the other game autopathographies in this study that are designed to cleverly control the player and withdraw their agency for maximum aesthetic effect, John’s game also sets up a false sense of agency and “tricks the player”:

At the start, it seems like a pretty cut and dry game where you’re just like ‘oh, collect points.’ So yeah, it’s supposed to be a little bit misleading...

Controlling the player’s gameplay experience is enjoyable for John and a simple method to actualize this design intention is through rules design. John’s invocation of fun also suggests that AGD is driven by his curiosity to explore his first-person experiences with bipolar disorder via explicitly ruled gameplay:

I wanted to see how I could translate (a dark game about bipolar disorder, assisted suicide, giving up, letting your character die and kill himself) into the rules of the game.

Thus, deploying specific rule design tactics in order to withdraw player agency and entertain how a player could experience the emotional toil of the designer’s experiences is pleasurable for John. More importantly, John is invoking the importance of the game’s *telos* emerging from the rule-design. John’s ultimate purpose for the game is not to lure or ensnare the player in a detrimental sense per se, but to galvanize them to contemplate his message about the uncontrollable nature of severe bipolar disorder and its one possible solution—euthanasia (assisted suicide).

AGD subject Martin cannot see the colour red. Martin creatively experimented with the distinct feeling of how he perceives the world in his game autopathography *Broken Lens*, using hexadecimal notation of color in combination with rule design:

That's in my code to show how somebody sees (like me). The mechanics of the game, the rules can demonstrate the troubles and the pressure of someone (who is color-blind), how normal tasks become harder for you.

In this game, non-color-blind players can inhabit the perception of a color-blind person like Martin. Martin emphasized the difference between the audiovisual component of the game, i.e., hexadecimal notation, which represents his inability to see red, and the procedural component of the game —its rules and mechanics, and how they allow the player to experience his color-blindness through a selection of tasks requiring the player to organize differently coloured items into separate bins. John and Martin both suggested that game rules can be used as a computational method for communicating designer experiences by encouraging players to enter the shoes of someone with a specific first-person experience. The designer's intention is to impart a deeper understanding of the phenomenological dimensions of illness in a given social environment.

John further elaborated that his choice for rule design is to implement procedurally generated game rules. Computer games that harness procedural generation deploy an algorithm that guides the game to self-generate its own data and process structures while interacting with inputs from a user. As such, objects in the system, including visuals, rules, and processes are generated mathematically —they are not pre-existing; rather, they emerge out of a process of instructions given to a computer program interacting with a user (player). In fact, “PCG refers to computer software that can create game content on its own, or together with one or many human players or designers” (Shaker, Togelius, & Nelson, 2016, p. 1) as well as “the algorithmical creation of game content with limited or indirect user input” (Alves, Coelho, & Nogueira, 2018). John explained procedural generation in his own words:

The rules will just keep playing themselves out. So, you could play (the game autopathography) for a minute or you could play it forever...And those are the type of games that I always try and design, the rules will never end. It will just keep building itself.

John's testimony also spoke to the fact that study subjects must work with various limitations imposed on their authorship process, such as those of the software and design tools they use to author their digital games; the constraints of the formal properties of computer games; the limitations of the Autopathographical Game Jam space; and the personal limitations of the AGD designer, such as their lack of skills, resources, etc.

Rule design also allows study subjects to symbolically transgress or overcome mental illness as a challenging experience. Juul states that for the player, "rules provide challenges that *require* ingenuity to overcome" (p. 55). Consider for a moment applying Caillois's (1958/2001) distinction between *ludus*, i.e., structured gameplay using explicit rules, and *paidia*, i.e., less ruled and more flexible play using implicit rules, to the autopathographical narrative. There is a palpable allusion to ludic play as a disciplinary device symbolically 'structuring' unstable emotions, and its opposite, paidiaic play, symbolically 'liberating' extreme self-disciplinary measures in the autopathographical subject. As such, game rules and formal constraints can be analogized to thoughts and emotions; they can be mapped symbolically to unruly emotions to 'tame them' or they can be mapped symbolically to over-disciplined thought processes to 'emancipate' them from excessive structure. Game rules and formal constraints in AGD may suggest possibilities, through the use of metaphor and analogy, of re-imagining the autopathographical subject's relationship to their pain, unease, and discomfort.

Autopathographical game design may enable study subjects to use their creative ingenuity to create rules in order to symbolically overcome or subvert their own physical and emotional limits. In the game autopathography, they can change or modify what they perceive to be an inability into a prowess; they can also emphasize a talent they possess that others are unaware of. The agency involved in re-imagining oneself via AGD as able and competent can be both irresistible and empowering.

To briefly illustrate the implications for the player of limitations implemented as formal constraints and implicit rules in virtual environments, it will be useful to briefly compare virtual environments to physical environments. Wardrip-Fruin (2009) suggests that software programs can both limit freedoms of users but also engender liberating experiences. For example, a virtual world allows players to manipulate various objects in its engine, perhaps even generate their own

digital content (draw, model, build, etc.)¹²¹ and/or collage new tableaux from an inventory of digital content (i.e., dress up and modify an avatar). A physical environment offers more freedom than a virtual environment in that objects are more versatile in terms of perception, sensation, and action. A physical rock can be used as a paper weight, it can be split into shards, coloured red, thrown, while a digital rock may not be able to perform in the same way unless those specific affordances are authored into the digital rock by the designer. Yet, a virtual environment may be less constrained than a physical environment insofar as it allows players certain freedoms that may not be accessible to them in physical spaces, e.g., no imposition of the vagaries of nature such as bad weather, the ability to conveniently enter and exit the virtual environment at the click of a mouse and possibly, the invitation to construct and customize the player's avatar or game character as per their desires.

In his game autopathography *Co-Op Defense*, Lukas's perspective on game rules is to frame their priority over other formal properties of computer game design in terms of their potential for healing the autopathographical author, or what he calls "self-hypnosis". His game rules resemble a prosthesis of the designer's subjectivity, aiding him in perceiving himself in a new light:

I think it's all about the game rules...this is a powerful tool, when you make the rules then you make a tool that enables for you to do that (express intimate and personal experiences)...The rules will drive (the story).

Contrary to Juul (2011), who claims that "it is *hard* to create a game about emotions because emotions are hard to implement in rules" (p. 20), Lukas suggested that game rules are indeed capable of expressing emotions. Lukas's intention is to discern how the player perceives the binary choice offered in his game of playing with friends or playing alone:

I will make an objective for (players), trying to create the scenario where they can play with friends or they can play alone. By looking at how they get into the objective, I will know if this is OK.

¹²¹ For user-generated content, see virtual worlds and sandbox games such as *Little Big Planet*, *Minecraft*, and *Second Life*.

He suggested that social expectations and stigma against shyness are too great and hence too burdensome for someone like him, who is a very shy person:

Because the social pressure on me, you cannot live alone, you cannot do anything alone. You need to get into a team, you need to get socialized, you need to network and that's the social pressure on our life. They reject people who are not socialized well.

As such, he gave voice to the feeling of being an outcast. Lukas explained that his game autopathography is not explicitly vocal about his experiences with social anxiety and extreme shyness—his wanting to be “alone”—despite the fact that he “does not want to be alone”.

I have a problem with social...so, I'm not sure if this is good or bad. So, I want to find out if it is OK to be working without friends?

His game “will contain the answer and I keep the question for myself”, which suggests that he chose to keep his autopathographical narrative that recount his struggles with shyness to himself. But Lukas emphasized that one of the objectives of his game autopathography is to demonstrate self-acceptance, and that “it is OK to be alone” in order to change and be with people.

When you have the answer and if the answer happens to be OK, then you encourage the social point of view, you encourage the new point of view. Then it can change.

He qualified this statement by explaining that his game's concealed outcome—it is OK to be alone/it is not OK to be alone?—which can be mapped to a generic win-lose outcome, is nevertheless supposed to encourage him to become a more social and collaborative individual whilst being shy:

And then gradually I try and to accept that as an answer to encourage me to coordinate to work even if I am alone.

Lukas stated that his gradual self-acceptance will happen via “self-hypnosis”. The game's other objective is to encourage him to become less shy whilst engendering self-acceptance:

When you hear an answer that's supportive you feel release in stress. Right. But, assuming that I expect the answer is OK, then it will reduce the stress...for working alone right now... at the moment...

Because I don't want to be this way forever (laughs)... But, still at least it helps at the moment. It helps right now...

Therefore, Lukas affirmed that his game tackles socially-engendered emotional pain by authoring rules that can provide a positive outcome for the player, which he calls a “positive answer” or “an answer that’s supportive”, in order to gradually help him change his outlook on his shyness.

AGD also allows the autopathographical game designer to reclaim their identity from medical diagnosis through the absence of explicit game rules and the authorship of implicit rules to engender restraints for the player. For example, Vicky claimed that for her, “a diagnosis is very restraining”. Vicky harnessed the absence of explicit game rules to communicate her experiences with mental illness diagnosis in a previous game autopathography she made at another game jam. That is, she omitted explicit game rules in the game. Given that their symbolic meaning in the context of AGD invokes an analogy between medical diagnosis and a personal limitation, she had chosen not to include explicit game rules in favour of including implicit rules as formal constraints for the player. Vicky discussed her previous computer game autopathography:

There's no rules whatsoever, there's no win state, there's no anything. It's just a pure, linear, interactive experience. You click when you're told to click and you just experience something very specific that I'm trying to impart on the player and that's the experience of being diagnosed. So, I didn't want to give too much choice because I wanted to place that restraint on the player the way that it feels restraining to have a diagnosis like that placed on you.

When Vicky says that “there's no rules whatsoever” in her game, she means to say that there are no explicit game rules governing agonistic gameplay. Of course, she had to design and implement the various formal constraints in the form of user-inputs: “you click when you are told to click”, as well as player choices, however minimal, in order to engineer a functional software program. In other words, rules are used as symbolic devices in order to communicate experiences of the designer, but also steer the experiences of the player by taking away their ability to choose from possibilities in the game.

To briefly conclude, AGD logopoiesis as a form of calculated making entails working with an array of technical and analytically-intensive praxes to implement the autopathographical narrative qua fabulized self into the formal and rule-based system of a game. These meticulous and precise praxes, which include designing digital graphics and writing algorithms, allow subjects to deliberate on their first-person experiences in measured ways and mobilize the advantages of computational thinking. Implementing the autopathographical narrative into a ‘clear and distinct’ digital/notational structure may also ironically liberate, perhaps momentarily, or for as long as one practices computer game design, from the ‘tumultuous confines’ of mental health struggles. To engineer one’s autopathographical narrative as a functional software application is to perhaps also cast off, for a moment, if not for longer, past or present emotional burdens. As Whitehead (1929;1978) insightfully explains, “objectification involves elimination...In the temporal world, it is the empirical fact that process entails loss: the past is present under an abstraction” (p. 340).

Chapter 8

Conclusion

8.1 The *therapoetics* of autopathographical game design

This dissertation has offered a data-driven theoretical analysis of the phenomenological ontology of autopathographical game design, drawing on a group of thirteen game designers' experiences during an autopathographical game jam. The configuration of the AGD-AGJ (autopathographical game design during an autopathographical game jam) can be understood as a balancing act among the first-person (subjective), the second-person (intersubjective), and the third-person (objective) perspectives achieved through the four dimensions of therapoiesis (a.k.a. the tetrad of therapoiesis)—autopoiesis, fabulopoiesis, logopoiesis, and sociopoiesis.

To summarize some gained insights from this study, the similarities between autopathographical writing and autopathographical game design are two therapoetic dimensions: autopoiesis, the self-reflexive processes of autopathographical authorship, and fabulopoiesis, the self-fabulization processes via the narrative devices of metaphor and analogy. The key difference between the two, which amplifies or heightens the therapeutic affordances of AGD-AGJ is logopoiesis, i.e., the computational, logical, digital, and procedural affordances of computer game design inviting deliberate and precise self-analyses. Importantly, through sociopoiesis, autopathographical game design during an autopathographical game jam has revealed itself to be an enriching and healing social experience that enabled subjects to share their autopathographical narratives while collectively making their games in the same physical space.

The foundations of autopathographical game design have also been theorized, such as the dialectical clash of the two semiospheres—the computer game and the autopathographical narrative—, and the deliciously pleasurable deployment of transgressive aesthetics for galvanizing the creative process of AGD. The fine structure of the AGD-AGJ process includes such distinct design processes as foregrounding and backgrounding the autopathographical narrative depending on context (i.e. AGJ as a discursive domain versus AGD as a non-discursive domain), inhabiting an auctorial posture in order to safely construct author identity, and authoring the fabulized self through an intelligibility framework (IF)—an heuristic space for solving the problem of how to render complex first-person experiences into an abstraction. The

IF then helps AGD subjects construct a structurally-articulated autopathographical narrative (SAN) in order to then implement it into a playable game.

Some brief observations about therapy may help situate game design therapoetics in relation to therapy as a general concept. Therapy is typically conceived of as a form of health care praxis such as medically-administered doctor's care, or the use of a remedy or cure for illness and disease. The Greek word for therapy is *therapeia* (θεραπεία), which translates to "curing, healing, service done to the sick" (Online Etymology Dictionary)¹²². However, the ancient Greeks mobilized a multi-dimensional therapeutic model of health care—an interconnected physical, intellectual, aesthetic, and spiritual care practice for body and soul, as suggested by Kleisiaris, Sfakianakis, & Papathanasiou (2014). For example, they claim that music and theater were deployed in the treatment of physical and mental illnesses in addition to improving human behavior. In the case of music, they explain that "it was believed that healing the soul through music also healed the body". If listening to music or viewing theater is therapeutic, then it follows that *making* music, theater, art, and in this study's case, games is *therapoetic*, a more apt term than therapeutics, given the integration of therapeutics and poetics involved in making games. As such, AGD is a *therapoetic* form of creative and artistic practice for the game author and game designer/developer.

Aristotle's concept of healing through catharsis could also use some clarification. The experience of catharsis, which may not have proven to be of much relevance to AGD designers, may have potential for study in another context. For example, playing one's game autopathography may produce cathartic moments for the AGD designer or seeing others play their game autopathography may also be cathartic for the AGD designer. As such, the cathartic and/or therapeutic effects of playtesting and playing game autopathographies is an area worthy of future study. In his discussion of catharsis, i.e., the purgation of the emotions of pity and fear, Aristotle discerned the therapeutic effects of art on audiences, specifically dramatic tragedy. Cavarnos (1979) interprets Aristotle's conception of catharsis as not necessarily producing a cure. Rather, it might offer "temporary relief" from emotional excess and irrationality (Cavarnos,

¹²²Retrieved from <http://www.etymonline.com/index.php?term=therapy>

1979, p. 57). “One will need such a catharsis again and again, from time to time, so long as the causes of undue or irrational pity and fear remain” (p. 57), claims Cavarnos. Moreover, Cavarnos states that catharsis relieves tension through two main methods, allopathy and homeopathy. The first removes tension by deploying a force that opposes it, such as calm and melodious music to purge tumultuous emotions, while the second expels emotional frenzy by a similar force, such as the passionate build-up of pity and fear in tragedy. It is thus important to frame AGD-AGJ’s therapoetic effects in a similar vein as producing forms of temporary relief such as *escape from* and/or a *confrontation with* the emotional excesses of mental health troubles, as well as presenting autopathographical game designers with various methods, tools, and processes (resources) for building resilience and better understanding of their subjective states, which may be clouded in ambiguity, negative rumination, excessive self-judgment, and tension.

8.2 Research contributions

The research contributions of this study can be divided into the following disciplinary areas: a) contributions to the study of authorship processes in Media Studies and game designer experiences in Game Studies; b) contributions to the mental health community of professional practitioners, scholars, and theorists studying the health applications of creative practice to a variety of contexts such as art therapy, serious games, and health informatics; c) contributions to autobiography and autopathography studies.

Consider first the study of authorship processes in Media Studies and game designer experiences in Game Studies. To this author’s knowledge, a phenomenological ontology of the therapoetic effects of AGD such as the one developed in this research study has not been conducted previously. Broadly speaking, Media Studies and Game Studies have emphasized either the study of a) media-as-text or game-as-text or artifact, b) user/player/viewer experiences (e.g., reception), deploying a variety of analytical methods, including close readings and close playings of media texts and/or games to study the media form or game object, or its reception in the case of user experiences, often within a broader cultural, social, and/or political framework. These are all important and worthy forms of study. In the process of giving the above research topics important attention, media and games scholars may sideline the author’s or designer’s place or experience within any part of this framework, given how complex it is to qualitatively

study media authorship or game design processes in their minutiae and states of flux. As such, the designer's or author's intentions, motivations, hopes, dreams, inspirations, and subjective states driving authorship have been decentered within scholarly discourse on media and games. The micro-analysis of designer or author experiences are relegated to applied and technical domains such as computer science and other similar technical fields, which typically do not use a humanistic approach to study these issues. This study thus offers a multidisciplinary contribution that fills a research gap in media and game studies. Regardless of the researcher's epistemological angle, in this case a structural and phenomenological perspective, the designer ought to have an important position within those frameworks.

Some scholars still claim, in the spirit of Roland Barthes's (1967; 1977) seminal essay "The Death of the Author", that studying the author's place in the generation of a text or artifact is irrelevant to its reception by the public or game player. But this claim misses how important it is to study the design processes deployed in authoring gameplay experiences, and how those design processes emerge to structure a game as a creative work of art, which sometimes reflects deeper truths about not only the author, given the multiple authorial (auctorial) postures they adopt, but society and culture at large. This study also contributes to media and game studies by offering an epistemological lens on how embodied forms of knowledge and lived experience are rendered through the creative, collective, and structural processes involved in AGD-AGJ, another important dimension of the examination of authorship processes. Game jam scholars may thus find some useful material in this dissertation's case study and treatment of the Autopathographical Game Jam.

With regard to the wider community of medical professionals and applied researchers that study the intersection of games and mental health within the framework of either art therapy or serious games, this study mobilizes a humanistic approach to the understanding of autopathographical game design that can be applied to clinical, medical, or psychological contexts. Serious games are forms of gamified software applications used in a variety of non-entertainment contexts including clinical practice, education, and academic research. But perhaps importantly, applied researchers and medical professionals might extrapolate from this study or find ways to test this study's findings using their own discipline-specific scientific and social scientific frameworks and tools. This dissertation invites the medical community to conduct

further research on this study's topic, findings, and thesis, to prove and/or disprove this study's findings and to work with the presented material using their own research tools.

Autopathography and autobiography studies may find the dissertation relevant in terms of its presentation of unexplored and novel forms of interdisciplinary research. In particular, the tetrad of therapoiesis is offered as an interdisciplinary framework for such studies. The style of documentation of the AGJ case study and the collection of processual artifacts might also be useful for qualitative study. In terms of data analysis, they might find fruitful the integration of a structural and phenomenological approach to studying design processes from the perspectives of game designers and theorizing based on the findings, subject testimony, and artifacts.

8.3 Future directions

Autopathographical game authorship is a fertile topic for further study in the digital humanities, media studies, and game studies. We can conceive of multiple routes to expand on the research and thesis laid out in this dissertation, including the generation of multiple papers each exploring one game authorship process in greater depth and detail. A compelling theorization worth pursuing based on the data is a comprehensive ontology of computer game design that would take into account the transgressive aesthetics of AGD in pushing the classical definition of games and play in new directions. Below are some additional concepts for future study based on the existing data and theoretical analysis in this dissertation.

8.3.1 Player-designer experience study: the player's inhabitation of the *fabulized self*

One research possibility that was not pursued in this dissertation due to lack of data, time, and resources is a player-experience study of autopathographical games. This research project did not have the time and the resources to collect data during the two-day AGJ of subject experiences playing their game autopathographies or other designers' games during the brief playtesting sessions. The concept of the player "inhabiting the fabulized self" through gameplay would be a compelling continuation of this research and the next step for a truly in-depth phenomenological study. The point of this analysis would be to relate the ways in which the first-person experience is shaped by emergent gameplay. Since we may be examining the game designer's experiences of both designing and playing their game autopathographies

simultaneously, then the *game designer* qua *player* has to playtest their game in order to experience the game and its space of potentiality. To inhabit the fabulized self means to finally come full circle to the embodied and irreducible standpoint. First-person experience, in its existential being-ness and dynamic, evolving relationship to the world, is subject to the process of emergence. The process of inhabitation on the part of the author-designer can be a significant extension of this study because for an AGD designer, the experience that emerges from playing their own game autopathography may be quite novel and may provide insights that structural approaches to game design may not. As mentioned, this experience may also prove to be cathartic.

8.3.2 In-depth analysis of the video recordings of the Autopathographical Game Jam

Another possibility for additional analysis is to focus on the videos documenting the Autopathographical Game Jam event. These videos have not been explored as visual data in this dissertation; they only served as source for the recorded subject testimony that was transcribed to text. In this regard, the videos analyzed as visual data may yield rich findings of subject relationships in the game jam space and provide compelling insights into gestural and visual forms of communication and expression.

8.3.3 Towards a typology of AGD analogical structures and ‘renderables’

Another possible study would be for the researcher to generate multiple typologies from the existing data analyses and, perhaps, unanalyzed data. For example, the generation of a typology of analogical structures in AGD would involve an analysis of the use of metaphor and analogy in all thirteen game autopathographies in order to create a detailed study classifying types of analogy and metaphor in AGD. A second typology would focus on ‘renderables’ between the game autopathography, the prose autopathography, the art autopathography, and other creative autopathographical forms demonstrating how elements of formal grammar render or do not render from one creative autopathographical medium to another.

Consider briefly this issue. Subjects reveal how their first-person experiences with illness, disability, and trauma are rendered into autopathographical narratives in a computer game through processes of abstraction and symbolism, which demonstrates a translation process

at work defined by various processes of rendering and implementation. Several of the analytical chapters already include a comparative theoretical analysis between a variety of art and media forms generating a set of ‘renderables’, i.e., how and what kind of content the authors render into computer game form, given that what the author chooses to render or not render carries therapeutic implications. For example, subjects disclose facets of their first-person experiences in verbal testimony that are not rendered in the game itself, for purposes of self-care and safety. The concepts of rendering and translation are not the same given that the term translation is often applied to contexts of translating from one language to another. However, the fields of *Translation and Performance Studies* may offer a useful framework for pursuing a typology of processes involved in inhabiting the fabulized self—performing a narrative through gameplay or role-play, as actors, performers, and dancers perform narratives and characters in a wide variety of creative contexts. Consider briefly Derrida’s (1978) point about ‘translation’ in the novel which may apply to the translation or rendering process between different non-discursive art and play forms. He argues that pure translation is an impossible task and reconceptualizes translation as a *transformation* engendered by the difference (i.e., slippage, misalignment) between two or more representational systems: “in the limits to which it is possible or at least appears possible, translation practices the difference between signified and signifier” (Derrida, 1978). This insight may also apply to translation between a game concept or game narrative and gameplay performance. While narrative structures may translate imperfectly to game rules, what happens when they are performed and acted through the reciprocal causal relations of maneuvering avatars and moving objects in games, as players do? An analysis of the levels of slippage that might accrue in translating a narrative into gameplay and most importantly, a game performance might offer some interesting insights in this regard.

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Appendix I: Document— informal and semi-structured interview questions to study subjects during the Autopathographical Game Jam

For principal researcher only:

3 commitments related to theoretical framework:

1. What are the authorship processes and experiences of designers authoring the autopathographical game through game design?
2. Arrive at a new theoretical understanding between process and procedurality and how they aid autopathographical storytelling (the “how does it hurt?” in my title)
3. Games as process-based vehicles for procedural expression of the autopathography
4. How do designers employ ‘procedural rhetoric’ in a game?
5. Methodologically – emergent, invisible, and imperceptible processes (computational – processes instantiated in code and non-computational – lived experiences).

Expressive processing: “what processes express in their design - which may not be visible to audiences” (Wardrip-Fruin, 2009)

For assistants/facilitators: I have divided questions into Essential Questions, Group Questions, and Nonessential Questions to account for time limitations. Ask only the essential questions and group questions.

Please feel free to rephrase questions in your own words. The interview process should feel natural for both the interviewer and interviewee.

Warm-up questions are posed to participants if you perceive nervousness or apprehension. They are meant to relax the participant.

If participants address a specific issue not addressed in these questions, feel free to follow-up with a question so that they elaborate on that issue.

When recording the interviews on camera/audio, try not to have your own voice overlap the participant’s voice.

Important themes: self-discovery, self-creation, group-discovery, process/procedure, insight, experimentation, private, and public

Warm-up questions:

1. Tell me the premise of your game idea—what story are you telling and why is this story meaningful to you?
2. How are you experiencing the game jam right now?
3. What are you hoping to accomplish in this stage of the design process –what are your intentions and plans? Have you done some prep work at home—tell us more.

Essential questions (relate to game jam process and its immediacy, its public dimension):

1. How do you feel about telling a personal story about your experiences with [illness/disability/trauma] in a public setting such as a game jam? What are some limitations and constraints of designing personal games in a game jam space like this?
2. How would you characterize your experience right now of articulating a personal story about [illness/disability/trauma] through game design? A game jam?
3. Can you describe/show us the game rules you are creating in your autopathographical game now? How is your chosen premise expressed through the game rules? How do you feel about telling personal and introspective details through game rules and procedures? How can this be done in your opinion?
4. Tell me about a meaningful design decision you are making now or thinking of integrating in your game during this jam. How and why is this design decision important to you?
5. How do you feel about your authorship/design process so far? What are some meaningful decisions you have made so far in the game design that you want to emphasize for us?
6. Is there a social dimension to your game or just a personal dimension, or both? If you are including a social dimension, what statement, argument, or critique are you making about existing attitudes on health, illness, and disability?
7. What are some aesthetic (visual, pictorial, aural?), narrative, and/or technical design decisions you decided *not* to pursue today? Why did you not pursue them?
8. Are you open to having other game jammers playtest your game?
9. (Day 2) Has this game jam and process given you insight into anything that you didn't have before?

10. (Day 2) Have you discovered something about yourself, about this process, that you can comment about?

Questions for Group Discussions:

1. Any group discoveries or insights in this game jam? (can be written on Bristol board paper)
2. Can anyone comment on this process of designing a personal game in a public setting such as a game jam?
3. How is a game jam a vehicle for articulating deeply private or personal stories of illness, disability, trauma? Is this method a contradiction –telling personal stories in a group setting? How so or how not?
4. How is game design an expressive vehicle for challenging social attitudes about health-related topics such as disability, illness, emotional trauma, addiction, etc.?
5. Is there something transgressive (which violates boundaries) about making a game about a personal health issue? What is that thing which is transgressive?

Non-essential questions (researcher can ask in post-game jam interviews):

1. Why use game design for exploring your own lived experiences with illness, disability, trauma, etc.? What is risky about making a personal game?
2. Tell me about your process of conveying unspoken, difficult, personal feelings and experiences about illness, disability, trauma in a game, through the rules of your game? In a game jam?
3. Is there an advantage of procedural storytelling over non-procedural, non-computational storytelling processes for autopathographical narratives?
4. Has your game idea, premise, or story changed in the process of design? What story are you telling right now? Why has it changed?
5. Why would you use a game to tell your story? Why a game jam? Why not a blog? A diary? Etc.?
6. How do you feel about the process of authoring an autopathographical game? In a game jam? Would you prefer to be alone instead?
7. How do you find the game design process for autopathographical storytelling?
8. Are you being explicit or implicit in the way (form, style, aesthetics, etc.) You are expressing your personal story in the game design process? How so?

9. Is this game jam productive for you? Inspiring? Motivating? How?
10. What do you think will be the reaction of the player to your game? How do you envision the player experiencing your game?

Chronology of semi-structured, in-person, individual interview questions with study subjects after the AGJ:

1. How did you like designing an autopathographical game during the AGJ? Tell me more about this experience.
2. How has your game idea/narrative evolved over the process of making the game in a game jam?
3. Do you think game design is a good way to express an autopathography? Why or why not? How so? Would you choose digital game design to tell a personal story/experience/memoir about disability, illness, or trauma? If so, why?
4. I'm wondering if there are other terms you prefer instead of 'autopathography', 'story' or 'narrative'? How would you conceptualize your game?
5. What part of the game design *workflow* [game concept development, asset acquisition, and pre-production, game design or coding, playtesting/debugging] did you find the most meaningful? How was it meaningful for you?
6. Was there a specific overarching design strategy that you used in expressing your game idea during the AGJ?
7. Tell me about your experiences expressing your personal narrative using the rule-based system of the game. How do you make an autopathographical game using the rules of that game?
8. Some game studies theorists argue that games are really good at expressing complex processes. Do you agree with this statement –why or why not? Can you tell me what those complex processes are in your game, and how they are expressed in your game?
9. Do you think you told your story/idea well with your game? Why or why not?
10. Do you recall any meaningful moments or experiences during the AGJ process? What were they? How and why were they meaningful?
11. Can you recall any challenging experiences that emerged during the (AGJ) and/or autopathographical game design process?

12. Would you use game design in the future for an autopathography? Would you participate in another game jam? What would you do differently, if you were the researcher organizing the AGJ?
13. a) Do you think DIY game design is an *expressive* medium for telling an autobiographical story about illness, disability, trauma? How so? If not, why not?
 b) How did you feel about telling very intimate and introspective details using game design? How would you characterize the experience of articulating a very intimate story through game design?
14. Do you feel that you were being direct or indirect in the way (narrative, aesthetics, rules, mechanics, etc.) you told your personal story and details in the game? How so?
15. How can the design of autopathographical games, as a form of authoring the illness and trauma narrative, function as a vehicle for social justice?
16. Do you think autopathographical games can change social assumptions, stigma, and negative beliefs about [illness, pain, trauma, disability]? Why or why not? How so? What are some of those social assumptions in your opinion? How does a game change those assumptions?
17. Do you think there are risks associated with authoring an autopathographical game? What are those risks for the game designer/artist, for you, personally?
18. Do you feel a shift in yourself after making this autopathographical game, in relation to how you feel about your [disability/illness/trauma]?
19. How difficult do you think is it to express your personal story with [illness, pain, trauma, disability] in a game? Would you rather use another medium for this purpose?
20. How does the design process for autopathographical games encourage your creative expression and disclosure of your story? Do you think you were able to be creative in the autopathographical game design process? Why or why not? Was there something that limited your creativity or design intentions in the process?
21. Do you think there is a way to process (or not process) difficult emotions, pain/hurt, trauma, through coding, programming, problem-solving and all the “nitty-gritty” technical aspects of game development?
22. How has designing your game created a process of self-awareness for yourself?

Copyright Acknowledgements

Figure 1: Jesper Juul's (2011) classification of games: “on the borders of the classic game model” (Figure 2.10, p. 44). From: *Half-real: Video games between real rules and fictional worlds*. Cambridge, MA: The MIT Press.

Figure 23: M. C. Escher's drawing. Courtesy of Flickr user James Wang:
<https://creativecommons.org/licenses/by-nc-nd/2.0/>

Figure 32: Capture screen shot of *Twine* courtesy of Sandra Danilovic.

Figure 41: The Necker cube: This file is licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license. Retrieved from Wikipedia.

Figure 42: Image sources of *Robocop* (1987) and *Terminator: The Sarah Connor Chronicles* (TV Series, 2008-09): <http://www.popmatters.com/feature/hardcore-pov-hardcore-henry-the-pov-shot-and-lets-watch-cinema/P1/>

Figure 43: Example of visual overlays, a score card, and map in the videogame *Breakaway*. Image from: <http://www.gamefromscratch.com/post/2016/11/04/Amazon-Lumberyard-16-Beta-Released.aspx>